MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

3690 GOLD ROOM

DESIGN CALCULATION

AS BUILT

MD-216-3690-EG-CV-CAL-1371

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 306 :2007	Saudi Steel Structures Code
AISC LRFD	Building Code Requirement of Steel Structural Building
AISC Steel Design Guide 1 - 2nd Edition	Base Plate and Anchor Rod Design

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
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QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The document is prepared for design of Gold Room RCC Slab, Beam, Column & Foundations.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for design of Gold Room RCC Slab, Beam, Column & Foundations.
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
3D- Model and analysis have been performed in Staad-Pro Software.
Foundation & Suspended slab have been design using MS-Excel as Per SBC-303:2007, SBC-304:2007 & ACI 318-08.
Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

4 DESIGN OF RCC PORTION

4.1 Design — Design of RCC Str. and its components shall be designed in accordance with

[1] SBC-301-2007-Structural Loading & Forces.
[2] SBC-303-2007-Structural Soil & Foundations.
[3] SBC-304-2018-Structural Concrete Structures.
[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.
[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete
[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.
[7] ACI 318-08 Building Code Requirements.
[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

4.2 Materials

Compressive Strength of Concrete at 28th days (fc') = 35 MPa
Yield Strength of Steel (fy) = 420 MPa

4.3 Reference — Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 LOAD CALCULATONS

5.1 DEAD LOAD (DL)

Selfweight of structure is applied through Staad Pro.

Dead load on RCC Beam

Tie Beam	=	26.75 kN/m UDL	(Considering 200mm Thk. RCC Precast Wall)
On Beam at 918.95 Between Grid A to D	=	27.50 kN/m UDL	(Considering 200mm Thk. RCC Precast Wall)
On Beam at 918.95 Between Grid D to G	=	15.00 kN/m UDL	(Considering 200mm Thk. RCC Precast Wall)

Dead load of Floor Slab at 918.95

Slab Thickness 125 mm with metal deck = 3.72 kN/Sqm

Roof Slab

Hollow Core Slab (HCS) 150mm Thk	=	2.52 kN/m^2	
Roof Finish with 75 mm Concrete screed above HCS	=	1.55 kN/m^2	
Total Load	=	4.07 kN/m^2	
UDL on Roof Beam for 5.4m grid	=	22.0 kN/m	
	=	10.989 kN/m	(At Edge)
UDL on Roof Beam for 6m grid	=	24.4 kN/m	
	=	12.21 kN/m	(At Edge)

Dead Load From Stair

Selfweight of Grating	=	0.35 kN/m^2
Selfweight of stringer	=	0.33 kN/m^2
Stair width	=	1.2 m
Stringer Span	=	8 m
Stringer Length	=	8 m
Concentrated load on beam supporting stringer	=	1.2 x 8 x 0.35 / 4 + 0.33 x 8 / 2
	=	2.2 kN

5.2 LIVE LOAD (LL)

Live load on Staircase	=	4.00 kN/m^2	
Concentrated load on beam supporting stringer			
= 1.2 x 8 x 4 / 4	=	9.6 kN	

Live load of Floor Slab at 918.95	=	5.00 kN/m^2	
Roof Live Load	=	0.5 kN/m^2	
UDL on Roof Beam for 5.4m grid	=	2.7 kN/m	
	=	1.35 kN/m	(At Edge)
UDL on Roof Beam for 6m grid	=	3.0 kN/m	
	=	1.5 kN/m	(At Edge)

5.3 SUPERIMPOSED DEAD LOAD (SDL)

SLD1 For Equipments

Equip. Load at 918.950 as superimposed load	(Load Point P39 to P45) = (3.3+5)x6 =	50	kN
Equip. Load at 918.950 as superimposed load	(Load Point P37 & P38) = (1.25+7.5)x4 =	35	kN

For All equipment loads, Refer Attachment 1 - Mechanical General Arrangement drawings.

SLD2 for Momorail

Hoist 5T Capacity	60	kN
Hoist 5T Capacity	35	kN

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5.4 WIND LOAD (WL)

WIND LOAD CALCULATION

Wind Load Parameters (As per design Criteria) :

Basic wind speed, V_b	=	43	m/s	=	155	Km/h	
Exposure category	=	C	(For Process Plant Area)				
Building Category	=	II	(Ref. Table 1.6-1 : SBC 301-2007)				
Wind directionablity factor, K_d	=	0.85	(Ref. Table 6.4-1 : SBC 301-2007)				
Topography Factor, K_{zt}	=	1.00	(As The project is not located in isolated hills or escarpments)				
Importance Factor, I	=	1.00	(Ref. Table 6.5-1 : SBC 301-2007)				
Velocity Pressure Co-efficient	=	Kz	(Ref. Table 7.2-2 : SBC 301-2007)				
Gust Effect Factor, G	=	0.85	(For Rigid Structure)				
			(Ref. Section 7.2.7.1 : SBC 301-2007)				

Wind Velocity Pressure, qz = $0.0473 \times 10^{-3} K_z K_{zt} K_d V^2 I$ (kN/m^2)
(Ref. Eq. 7.2-12 : SBC 301-2007)

Height	kz	Pressure	Unit
5 m	0.86	0.831	kN/m^2
6 m	0.90	0.869	
8 m	0.95	0.918	
10 m	1.00	0.966	
12 m	1.04	1.005	

Force Co-efficient, C_f	=	1.30	Average Pressure on wall, qz	=	**0.92** kN/m^2
Wind Base Shear	=	475	kN		
Seismic Base Shear (From Analysis)	=	521	kN >> Wind Load, Hence Wind Analysis will not performed for this Building.		
(Summation of Support Reactions For Seismic Load Case)					

5.5 TEMPERATURE LOAD (TL)

There are two temperature Load case is considered that is TL1 & TL2

TL1 - For Temperature Difference + 25^0 C

TL2 - For Temperature Difference -25^0 C

5.6 EARTHQUAKE LOAD (EQL)

Seismic Design Category		=	A	
Site class		=	B	It is corresponds to Site Class 2 in Staad
Spectral response acceleration (1-sec)	S1	=	0.02	
Spectral response acceleration (.02-sec)	Ss	=	0.1	
Importance factor	I	=	1	(From Table 9.5, SBC 301:2007)
Response Reduction factor along X	Rx	=	2.5	(From Table 10.2, SBC 301:2007)
Response Reduction factor along Z	Rz	=	2.5	(From Table 10.2, SBC 301:2007)
Height of building	h	=	8	m
Time period along X	Tx	=	0.26	(eqn. 10.9.3.2-1, SBC 301:2007)
Time period along Z	Tz	=	0.26	(eqn. 10.9.3.2-1, SBC 301:2007)
Factor Fa	Fa	=	1	
Factor Fv	Fv	=	1	
SM1	Fa x S1	=	0.02	
SMS	Fa x Ss	=	0.1	
Design spectral acceleration (1-sec)	SD1 = 2/3 x SM1	=	0.0133	
Design spectral acceleration (0.2-sec)	SDS = 2/3 x SMS	=	0.0667	

The above parameters are used in the staad input to find the final earthquake load of the structure as per IBC-2003

5.7 LOAD COMBINATIONS

5.7.1 Factored Load Combination for Structural design

(As Per Section 2.3 of SBC 301-2007)

LOAD 201 1.4 (DL + SDL1)	(Eq. 2.3.2-1 of SBC 301)
LOAD 202 1.4 (DL + SDL2)	
LOAD 203 1.2 (DL + SDL1 + TL1) + 1.6 LL	(Eq. 2.3.2-2 of SBC 301)
LOAD 204 1.2 (DL + SDL2 + TL1) + 1.6 LL	
LOAD 205 1.2 (DL + SDL1 + TL2) + 1.6 LL	
LOAD 206 1.2 (DL + SDL2 + TL2) + 1.6 LL	
LOAD 215 1.2 (DL + SDL1) + 1.0 EL(+X) + 0.5 LL	(Eq. 2.3.2-5 of SBC 301)
LOAD 216 1.2 (DL + SDL1) + 1.0 EL(-X) + 0.5 LL	
LOAD 217 1.2 (DL + SDL1) + 1.0 EL(+Z) + 0.5 LL	
LOAD 218 1.2 (DL + SDL1) + 1.0 EL(-Z) + 0.5 LL	
LOAD 219 1.2 (DL + SDL2) + 1.0 EL(+X) + 0.5 LL	
LOAD 220 1.2 (DL + SDL2) + 1.0 EL(-X) + 0.5 LL	
LOAD 221 1.2 (DL + SDL2) + 1.0 EL(+Z) + 0.5 LL	
LOAD 222 1.2 (DL + SDL2) + 1.0 EL(-Z) + 0.5 LL	
LOAD 231 0.9 (DL + SDL1) + 1.0 EL(+X)	(Eq. 2.3.2-7 of SBC 301)
LOAD 232 0.9 (DL + SDL1) + 1.0 EL(-X)	
LOAD 233 0.9 (DL + SDL1) + 1.0 EL(+Z)	
LOAD 234 0.9 (DL + SDL1) + 1.0 EL(-Z)	
LOAD 235 0.9 (DL + SDL2) + 1.0 EL(+X)	
LOAD 236 0.9 (DL + SDL2) + 1.0 EL(-X)	
LOAD 237 0.9 (DL + SDL2) + 1.0 EL(+Z)	
LOAD 238 0.9 (DL + SDL2) + 1.0 EL(-Z)	

5.7.2 Load Combination for Servicibilty check

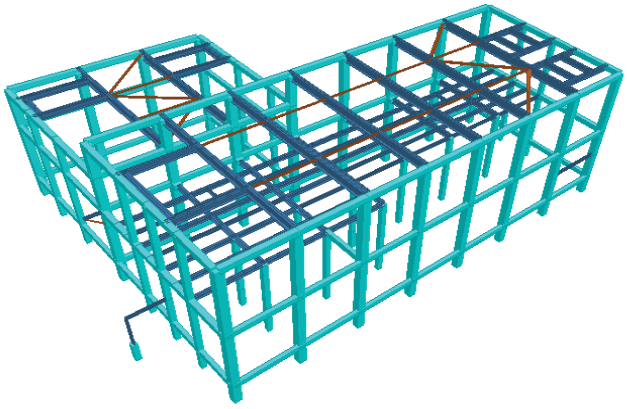
LOAD 101 (DL + SDL1)	(Eq. 2.4.1-1 of SBC 301)
LOAD 102 (DL + SDL2)	
LOAD 103 (DL + SDL1 + TL1) + LL	(Eq. 2.4.1-2 of SBC 301)
LOAD 104 (DL + SDL2 + TL1) + LL	
LOAD 105 (DL + SDL1 + TL2) + LL	
LOAD 106 (DL + SDL2 + TL2) + LL	
LOAD 115 (DL+ SDL1) + 0.75 LL + 0.525 EL(+X)	(Eq. 2.4.1-6 of SBC 301)
LOAD 116 (DL+ SDL1) + 0.75 LL + 0.525 EL(-X)	
LOAD 117 (DL+ SDL1) + 0.75 LL + 0.525 EL(+Z)	
LOAD 118 (DL+ SDL1) + 0.75 LL + 0.525 EL(-Z)	
LOAD 119 (DL+ SDL2) + 0.75 LL + 0.525 EL(+X)	
LOAD 120 (DL+ SDL2) + 0.75 LL + 0.525 EL(-X)	
LOAD 121 (DL+ SDL2) + 0.75 LL + 0.525 EL(+Z)	
LOAD 122 (DL+ SDL2) + 0.75 LL + 0.525 EL(-Z)	
LOAD 131 0.6 (DL+SDL1) + 0.7 EL(+X)	(Eq. 2.4.1-8 of SBC 301)
LOAD 132 0.6 (DL+SDL1) + 0.7 EL(-X)	
LOAD 133 0.6 (DL+SDL1) + 0.7 EL(+Z)	
LOAD 134 0.6 (DL+SDL1) + 0.7 EL(-Z)	
LOAD 135 0.6 (DL+SDL2) + 0.7 EL(+X)	
LOAD 136 0.6 (DL+SDL2) + 0.7 EL(-X)	
LOAD 137 0.6 (DL+SDL2) + 0.7 EL(+Z)	
LOAD 138 0.6 (DL+SDL2) + 0.7 EL(-Z)	

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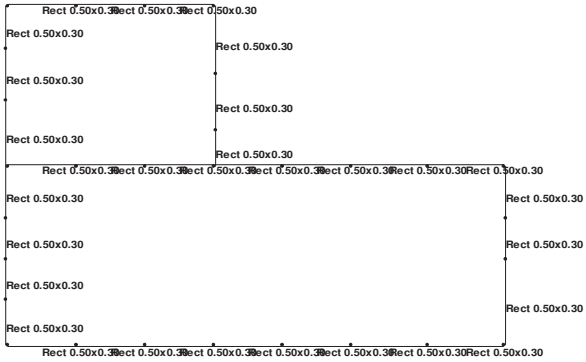
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6 GEOMETRY

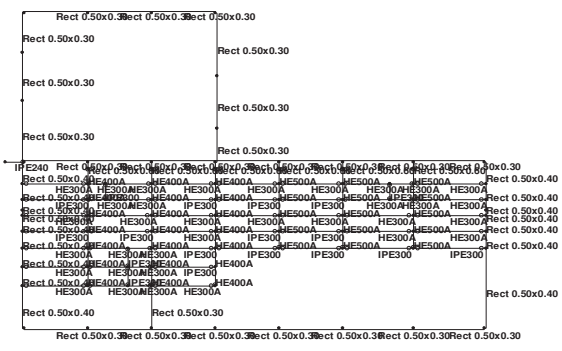
6.1 Geometry of GOLD ROOM



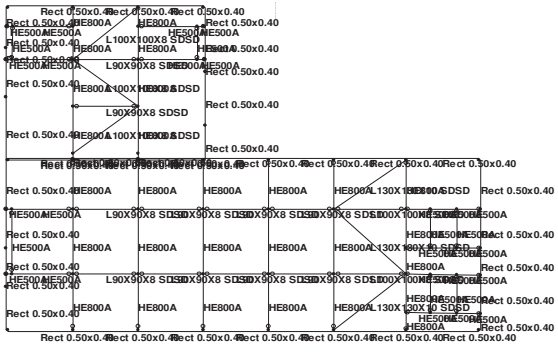
3D-View



Tie Beam Plan with Section



Floor Beam Plan with Section



Roof Beam Plan with Section

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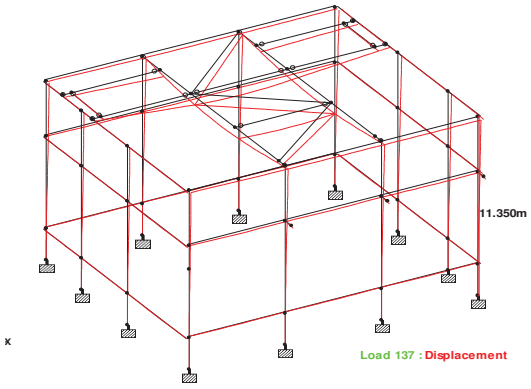
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7 ANALYSIS & RESULTS

For 3D analysis and results , refer Staad File - **GOLD ROOM CIVIL R0.STD**

7.1 Horizontal deflection of Column at RL(+)922.451 m

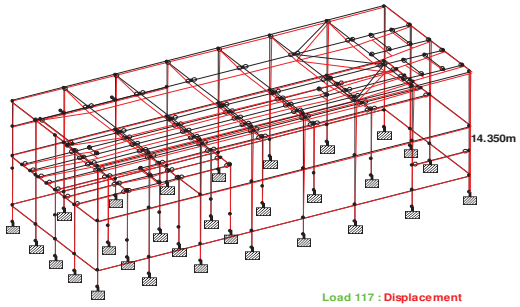
Hori. deflection of Column at RL(+)922.451 m = 8.72 mm under Load Case 137 for Node Number 122
Allowable deflection = 11350/500 = 22.70 Hence Safe (As per TABLE 1.4-1 SBC 301 2007)



Deflected Shape for load Case 137

7.2 Horizontal deflection of Column at RL(+)925.451 m

Hori. deflection of Column at RL(+)925.451 m = 17.61 mm under Load Case 117 for Node Number 140
Allowable deflection = 14350/500 = 28.70 Hence Safe (As per TABLE 1.4-1 SBC 301 2007)



Deflected Shape of Floor for load Case 117

8 DESIGN OF RCC ELEMENT

All Major components i.e. RCC Column, Beam have been design in Staad-Pro.
For Design Results, Refer Attachemnt 3 - Staad Output file.
Foundation & Suspended slab have been design using MS-Excel as Per SBC-303:2007, SBC-304:2007 & ACI 318-08.

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8.1 Design of Floor Slab - 125mm thick

Slab is designed as simply supported slab along shorter span

Span of slab	=	2.1	m	max span considered	Clear cover	=	20	mm	
Thickness of slab	=	0.125	m		Bar Dia	=	8	mm	
Grade of steel	=	420	N/mm2		Unit weight of concrete	=	24	KN/m3	
Grade of concrete	=	35	N/mm2		Live load on slab	=	5	KN/m2	
Mimimum thickness of one way slab	=	span/20		*Table 7.3.1.1 SBC 304*	β_1	=	0.8		
	=	0.105	m	Hence slab thickness okay					

Load calculation on slab

Dead load from slab	=	0.125 x 24	=	3	KN/m2		
Live load from slab	=			5	KN/m2		
Load combination	=	1.4DL + 1.7LL	=	1.4 x 3 + 1.7 x 5	=	12.7	KN/m2

Calculation of bending moment and shear force

BM at midspan	=	$wl^2/8$	=	7.0	KNm
SF at support	=	wl/2	=	13.3	KN

Calculation of flexural reinforcement

Effective Depth Along X = 101 mm

M = 7.00 KNm

$M_{lim} = \emptyset\, 0.85\, f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 69.65 KNm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

M_{lim} > M — Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\, f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 185.6 mm^2

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 7.6.1.1 SBC 304*

greater of or

A_{stmin} = 0.0014bd

A_{stmin} = 181.8 mm^2

A_{req} = 185.6 mm^2

Required Spacing = **270.9** mm

A_{prov} = 335.1 mm^2

Provide 8mm dia @ 150 c/c Top & bottom both directions

Shear Check

Reinforcement ratio	ρ_w	=	0.0033			Lesser of the following			
Effective Depth	d	=	101	mm	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$	
Shear Force	V_u	=	13.34	KN				*Table 22.5.5.1 SBC 304*	
Allowable shear strength	φVc	=	72.52	KN **O.K**		λ	=	1	Normal concrete
						Mu	=	7.00	KNm
						φ	=	0.75	for shear
						φVc	=	72.52	KN
					ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$	
						φVc	=	75.98	KN
					iii)	V_c	=	$0.29\lambda\sqrt{f_c'} b_w d$	
						φVc	=	129.96	KN

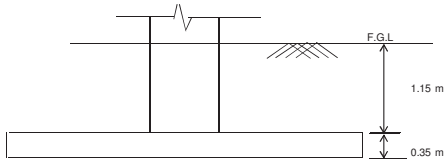
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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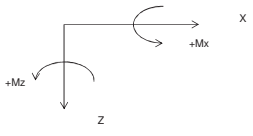
Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	30-05-2020	29-06-2020	28.11.2023	Date	30-05-2020	29-06-2020	28.11.2023
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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

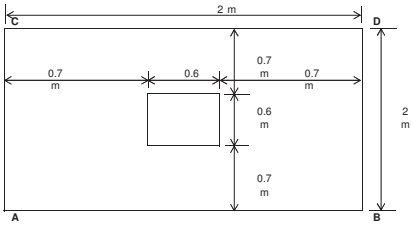
8.2 DESIGN OF FOUNDATION MKD. F1 :

(NODE NO. : 1, 4, 16, 32 & 39)

Net Safe Bearing Capacity of Soil =	40	Mton/m^2	
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	40 + 1.5 X 1.9 =	42.85	Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	2	m
Width of Foundation (Lz) =	2	m
Thickness of Base Raft =	0.35	m
Length of Column (lx) =	0.6	m
Width of Column (lz) =	0.6	m

Area , A = Lx*Lz
= 4 m^2
Zxx = Lx*Lz^2/6
= 1.333 m^3
Zzz = Lz*Lx^2/6
= 1.333 m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	2X2X0.35X2.5 =	3.5
Column =	0.6X0.6X1.15X2.5 =	1.035
Soil =	((2X2)-(0.6X0.6))X1.15X1.9 =	7.953
	Total =	12.488

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK :

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	1	120 DL + SDL	**2.885**	39.241	0.816	0.3	0.086	-3.299
Min Fx	39	119 DL + SDL	**-3.163**	71.679	-4.483	-3.632	0.107	4.406
Max Fy	39	117 DL + SDL	-2.365	**72.062**	-5.538	-9.447	-0.207	1.185
Min Fy	1	133 0.6 (DL +	1.265	**17.963**	-0.505	-4.72	0.219	0.082
Max Fz	4	118 DL + SDL	-2.078	47.523	**4.318**	6.303	-0.057	0.783
Min Fz	39	117 DL + SDL	-2.365	72.062	**-5.538**	-9.447	-0.207	1.185
Max Mx	4	134 0.6 (DL +	-1.138	29.863	3.222	**7.413**	-0.057	0.143
Min Mx	39	117 DL + SDL	-2.365	72.062	-5.538	**-9.447**	-0.207	1.185
Max My	32	135 0.6 (DL +	0.05	26.941	-0.617	-0.745	**0.594**	4.122
Min My	32	132 0.6 (DL +	2.25	31.35	-0.256	0.263	**-0.489**	-4.791
Max Mz	39	131 0.6 (DL +	-2.472	42.517	-2.599	-1.756	0.245	**5.056**
Min Mz	32	136 0.6 (DL +	2.251	31.371	-0.261	0.243	-0.489	**-4.795**

SLIDING CHECK:

L/C	(Wf+Fy) (Mton)	(Fx^2+ Fz^2)^0.5	Frictional coeff. (μ)	Resisting Force	F.O.S	Factor Limit	Remarks
120	51.729	2.998	0.5	25.865	8.63	1.5	**O.K.**
119	84.167	5.487	0.5	42.084	7.67	1.5	**O.K.**
117	84.55	6.022	0.5	42.275	7.02	1.5	**O.K.**
133	30.451	1.362	0.5	15.226	11.18	1.5	**O.K.**
118	60.011	4.792	0.5	30.006	6.26	1.5	**O.K.**
117	84.55	6.022	0.5	42.275	7.02	1.5	**O.K.**
134	42.351	3.417	0.5	21.176	6.2	1.5	**O.K.**
117	84.55	6.022	0.5	42.275	7.02	1.5	**O.K.**
135	39.429	0.619	0.5	19.715	31.85	1.5	**O.K.**
132	43.838	2.265	0.5	21.919	9.68	1.5	**O.K.**
131	55.005	3.587	0.5	27.503	7.67	1.5	**O.K.**
136	43.859	2.266	0.5	21.93	9.68	1.5	**O.K.**

OVERTURNING CHECK:

L/C : 133 0.6 (DL + SDL1) + 0.7 EL(+Z)

Max Upword Force =	0	Mton			
Self Wt. of Foundation =	12.488	Mton			
Overturning Moment =	0 X 2/2 + 4.72 =		4.72	Mton-m	
Resisting Moment =	12.488 X 0.9 X 2/2 =		11.239	Mton-m	
F.O.S =	2.38	>.	2	**O.K.**	

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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

BASE PRESSURE CHECK:

L/C	(Wf+Fy)/A (Mton/m²)	Mx/Zxx (Mton/m²)	Mz/Zzz (Mton/m²)	BASE PRESSURE AT POINT A	B	C	D	Remarks
120	12.932	0.225	2.475	15.632	10.682	15.182	10.232	**O.K.**
119	21.042	2.725	3.305	27.072	20.462	21.622	15.012	**O.K.**
117	21.138	7.087	0.889	29.114	27.336	14.94	13.162	**O.K.**
133	7.613	3.541	0.062	11.216	11.092	4.134	4.01	**O.K.**
118	15.003	4.728	0.587	20.318	19.144	10.862	9.688	**O.K.**
117	21.138	7.087	0.889	29.114	27.336	14.94	13.162	**O.K.**
134	10.588	5.561	0.107	16.256	16.042	5.134	4.92	**O.K.**
117	21.138	7.087	0.889	29.114	27.336	14.94	13.162	**O.K.**
135	9.857	0.559	3.092	13.508	7.324	12.39	6.206	**O.K.**
132	10.96	0.197	3.594	14.751	7.563	14.357	7.169	**O.K.**
131	13.751	1.317	3.793	18.861	11.275	16.227	8.641	**O.K.**
136	10.965	0.182	3.597	14.744	7.55	14.38	7.186	**O.K.**
			MAX =	29.114	27.336	21.622	15.012	

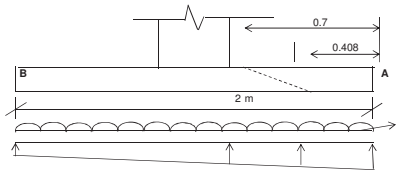
SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN :

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	1	220 1.2 (DL +	**3.951**	48.214	0.974	0.546	0.163	-5.881
Min Fx	39	219 1.2 (DL +	**-4.349**	85.64	-5.279	-3.874	0.305	7.605
Max Fy	39	201 1.4 (DL+S	-3.331	**94.197**	-6.363	-4.935	-0.143	1.527
Min Fy	1	233 0.9 (DL +	1.902	**27.269**	-0.687	-6.737	0.312	0.09
Max Fz	4	218 1.2 (DL +	-2.39	58.332	**5.793**	11.055	-0.091	0.627
Min Fz	39	217 1.2 (DL +	-2.829	86.34	**-7.269**	-14.827	-0.291	1.468
Max Mx	4	218 1.2 (DL +	-2.39	58.332	5.793	**11.055**	-0.091	0.627
Min Mx	39	217 1.2 (DL +	-2.829	86.34	-7.269	**-14.827**	-0.291	1.468
Max My	32	219 1.2 (DL +	0.725	55.343	-1.14	-1.25	**0.877**	5.71
Min My	32	232 0.9 (DL +	3.296	46.867	-0.396	0.36	**-0.695**	-6.868
Max Mz	39	215 1.2 (DL +	-4.346	85.693	-5.312	-4.084	0.298	**7.608**
Min Mz	32	220 1.2 (DL +	3.87	61.672	-0.632	0.162	-0.67	**-7.028**

STRESS CALCULATION FOR STRENGTH DESIGN :

L/C	(Wf+Fy)/A (Mton/m²)	Mx/Zx (Mton/m²)	Mz/Zz (Mton/m²)	BASE PRESSURE AT POINT A	B	C	D
220	15.176	0.41	4.412	19.998	11.174	19.178	10.354
219	24.532	2.906	5.705	33.143	21.733	27.331	15.921
201	26.671	3.702	1.146	31.519	29.227	24.115	21.823
233	9.939	5.054	0.068	15.061	14.925	4.953	4.817
218	17.705	8.293	0.47	26.468	25.528	9.882	8.942
217	24.707	11.123	1.101	36.931	34.729	14.685	12.483
218	17.705	8.293	0.47	26.468	25.528	9.882	8.942
217	24.707	11.123	1.101	36.931	34.729	14.685	12.483
219	16.958	0.938	4.284	22.18	13.612	20.304	11.736
232	14.839	0.27	5.152	20.261	9.957	19.721	9.417
215	24.545	3.064	5.707	33.316	21.902	27.188	15.774
220	18.54	0.122	5.272	23.934	13.39	23.69	13.146

PRESSURE DISTRIBUTION ALONG X - DIRECTION :



L/C	Pressure at Point A (Mton/m²)	Pressure at Point B (Mton/m²)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
220	19.998	11.174	16.91	18.2	4.654	1.51
219	33.143	21.733	29.15	30.82	8.023	2.633
201	31.519	29.227	30.72	31.05	7.103	2.395
233	15.061	14.925	15.01	15.03	2.949	1.001
218	26.468	25.528	26.14	26.28	5.789	1.959
217	36.931	34.729	36.16	36.48	8.424	2.844
218	26.468	25.528	26.14	26.28	5.789	1.959
217	36.931	34.729	36.16	36.48	8.424	2.844
219	22.18	13.612	19.18	20.43	5.174	1.688
232	20.261	9.957	16.65	18.16	4.804	1.548
215	33.316	21.902	29.32	30.99	8.065	2.647
220	23.934	13.39	20.24	21.78	5.717	1.857
				MAX =	8.424	2.844

ALONG X - DIRECTION :

Mu = 8.424 Mton-m

Φ*Kn = Mu/bd² deff. = 292 mm

Mu/bd² = 8.424X10^7/(1000X292^2) = 0.988 N/mm² < 6.9 N/mm²

Pt = 0.0027

Min. Pt = 0.0018 Reqd. Ast = 0.0027X1000X292/100 = 7.88 cm²

Provided **16** mm @ **150** C/C Provide Ast = 13.4 cm² at Bottom **O.K.**

Total Provide Pt = 0.0046

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ONE-WAY SHEAR CHECK

Vu = 2.844 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0046X2.844X10^4X292/8.424/10^7)X1000X292/10^4

= 27.865 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0046)X1000X292/10^4

= 29.923 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X292/10^4

= 50.097 Mton

HENCE, Vc = 27.865 Mton

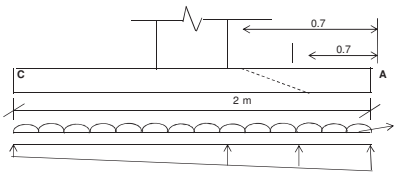
Φt*Vc = 0.75X27.865 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 20.899 Mton > **2.844Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(27.865X10^4+0.66X35^0.5X1000X292)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 87.601 Mton > **Vu= 2.844Mton , CROSS SECTION O.K.**

PRESSURE DISTRIBUTION ALONG Z - DIRECTION :



L/C	Pressure at Point A (Mton/m²)	Pressure at Point C (Mton/m²)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
220	19.998	19.178	19.71	19.71	4.197	3.862
219	33.143	27.331	31.11	31.11	7.702	5.329
201	31.519	24.115	28.93	28.93	7.395	4.372
233	15.061	4.953	11.52	11.52	3.519	-0.609
218	26.468	9.882	20.66	20.66	6.684	-0.089
217	36.931	14.685	29.14	29.14	9.571	0.487
218	26.468	9.882	20.66	20.66	6.684	-0.089
217	36.931	14.685	29.14	29.14	9.571	0.487
219	22.18	20.304	21.52	21.52	4.792	4.026
232	20.261	19.721	20.07	20.07	4.245	4.025
215	33.316	27.188	31.17	31.17	7.763	5.261
220	23.934	23.69	23.85	23.85	5.128	5.028
				MAX =	9.571	5.329

ALONG Z - DIRECTION :

Mu = 9.571 T-m

Φ*Kn = Mu/bd² deff. = 292 mm

Mu/bd² = 9.571X10^7/(1000X292^2) = 1.123 N/mm² < 6.9 N/mm²

Pt = 0.0030

Min. Pt = 0.0018 Reqd. Ast = 0.003X1000X292/100 = 8.76 cm²

Provided **16** mm @ **150** C/C Provide Ast = 13.4 cm² at Bottom **O.K.**

Total Provide Pt = 0.0046

ONE-WAY SHEAR CHECK

Vu = 5.329 T

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0046X5.329X10^4X292/9.571/10^7)X1000X292/10^4

= 28.011 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0046)X1000X292/10^4

= 29.923 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X292/10^4

= 50.097 Mton

HENCE, Vc = 28.011 Mton

Φt*Vc = 0.75X28.011 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 21.008 Mton > **5.329Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(28.011X10^4+0.66X35^0.5X1000X292)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 87.612 Mton > **Vu= 5.329Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 4 m²

Perimeter (bo) = 2 X ((0.6 + 0.292) + (0.6 + 0.292)) = 3.568 m

β = 1

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 29.114 Mton/m²

Vu = ((4 - (0.6X0.6))X29.114

= 105.975 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X3.568X10^3X292/10^4

= 203.402 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X3.568X10^3X292/10^4

= 314.349 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+30X292/(3.568X10^3))X1X35^0.5X(3.568X10^3)X292/10^4

= 227.92 Mton

HENCE, Vc = 203.402 Mton

Φt*Vn = Φt*Vc = 0.75X203.402 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

152.552 Mton > **105.975Mton , Hence O.K.**

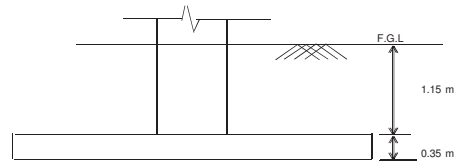
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

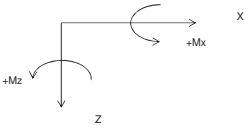
Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	30-05-2020	29-06-2020	28.11.2023	Date	30-05-2020	29-06-2020	28.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT
Prepared By	SNM	SNM	SNM	Prepared By	SNM	SNM	SNM
Checked	SJM	SJM	SJM	Checked	SJM	SJM	SJM
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

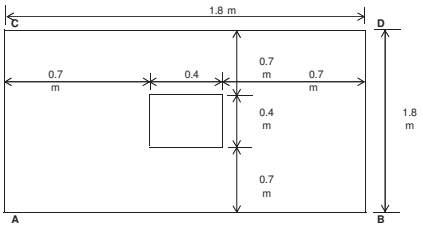
DESIGN OF FOUNDATION MKD. F2, F3 & F3A:

(NODE NO. : 5 to 8, 17, 20 to 31, 192 & 207)

Net Safe Bearing Capacity of Soil =	40	Mton/m^2	
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	40 + 1.5 X 1.9 =	42.85	Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	1.8	m
Width of Foundation (Lz) =	1.8	m
Thickness of Base Raft =	0.35	m
Length of Column (lx) =	0.4	m
Width of Column (lz) =	0.4	m
Area , A = Lx*Lz		
=	3.24	m^2
Zxx = Lx*Lz^2/6		
=	0.972	m^3
Zzz = Lz*Lx^2/6		
=	0.972	m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	1.8X1.8X0.35X2.5 =	2.835
Column =	0.4X0.4X1.15X2.5 =	0.46
Soil =	((1.8X1.8)-(0.4X0.4))X1.15X1.9 =	6.73
	Total =	10.025

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK :

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	21	132 0.6 (DL +	**0.424**	28.028	0.111	0.136	-0.002	-1.805
Min Fx	27	135 0.6 (DL +	**-0.405**	34.194	1.724	0.989	0.028	1.704
Max Fy	22	103 DL + SDL	0.007	**69.277**	-1.621	-4.181	-0.019	-0.055
Min Fy	31	118 DL + SDL	-0.002	**-0.475**	0.097	0.731	0	0.013
Max Fz	27	122 DL + SDL	-0.011	60.058	**3.938**	3.072	0.004	-0.007
Min Fz	20	117 DL + SDL	0.022	54.053	**-2.674**	-2.961	-0.01	-0.157
Max Mx	27	138 0.6 (DL +	-0.021	32.951	3.391	**3.227**	0.007	0.053
Min Mx	192	117 DL + SDL	0.011	42.047	-2.077	**-6.149**	-0.031	-0.089
Max My	30	131 0.6 (DL +	-0.205	10.675	0.15	0.402	**0.087**	1.295
Min My	28	132 0.6 (DL +	0.342	27.791	0.252	0.22	**-0.106**	-1.563
Max Mz	21	135 0.6 (DL +	-0.388	27.985	-0.067	-0.091	-0.008	**1.706**
Min Mz	21	132 0.6 (DL +	0.424	28.028	0.111	0.136	-0.002	**-1.805**

SLIDING CHECK:

L/C	(Wf+Fy) (Mton)	(Fx^2+ Fz^2)^0.5	Frictional coeff. (μ)	Resisting Force	F.O.S	Factor Limit	Remarks
132	38.053	0.438	0.5	19.027	43.44	1.5	**O.K.**
135	44.219	1.771	0.5	22.11	12.48	1.5	**O.K.**
103	79.302	1.621	0.5	39.651	24.46	1.5	**O.K.**
118	9.55	0.097	0.5	4.775	49.23	1.5	**O.K.**
122	70.083	3.938	0.5	35.042	8.9	1.5	**O.K.**
117	64.078	2.674	0.5	32.039	11.98	1.5	**O.K.**
138	42.976	3.391	0.5	21.488	6.34	1.5	**O.K.**
117	52.072	2.077	0.5	26.036	12.54	1.5	**O.K.**
131	20.7	0.254	0.5	10.35	40.75	1.5	**O.K.**
132	37.816	0.425	0.5	18.908	44.49	1.5	**O.K.**
135	38.01	0.394	0.5	19.005	48.24	1.5	**O.K.**
132	38.053	0.438	0.5	19.027	43.44	1.5	**O.K.**

OVERTURNING CHECK:

L/C : 118 DL + SDL1 + 0.75 LL + 0.525 EL(-Z)

Max Upword Force =	0.475	Mton			
Self Wt. of Foundation =	10.025	Mton			
Overturning Moment =	0.475 X 1.8/2 + 0.731 =		1.159	Mton-m	
Resisting Moment =	10.025 X 0.9 X 1.8/2 =		8.12	Mton-m	
F.O.S =	7.01	>.	2		**O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	30-05-2020	29-06-2020	28.11.2023	Date	30-05-2020	29-06-2020	28.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT
Prepared By	SNM	SNM	SNM	Prepared By	SNM	SNM	SNM
Checked	SJM	SJM	SJM	Checked	SJM	SJM	SJM
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

BASE PRESSURE CHECK:

L/C	(Wf+Fy)/A (Mton/m²)	Mx/Zxx (Mton/m³)	Mz/Zzz (Mton/m³)	BASE PRESSURE AT POINT A	B	C	D	Remarks
132	11.745	0.14	1.857	13.742	10.028	13.462	9.748	**O.K.**
135	13.648	1.017	1.753	16.418	12.912	14.384	10.878	**O.K.**
103	24.476	4.301	0.057	28.834	28.72	20.232	20.118	**O.K.**
118	2.948	0.752	0.013	3.713	3.687	2.209	2.183	**O.K.**
122	21.631	3.16	0.007	24.798	24.784	18.478	18.464	**O.K.**
117	19.777	3.046	0.162	22.985	22.661	16.893	16.569	**O.K.**
138	13.264	3.32	0.055	16.639	16.529	9.999	9.889	**O.K.**
117	16.072	6.326	0.092	22.49	22.306	9.838	9.654	**O.K.**
131	6.389	0.414	1.332	8.135	5.471	7.307	4.643	**O.K.**
132	11.672	0.226	1.608	13.506	10.29	13.054	9.838	**O.K.**
135	11.731	0.094	1.755	13.58	10.07	13.392	9.882	**O.K.**
132	11.745	0.14	1.857	13.742	10.028	13.462	9.748	**O.K.**
			MAX =	28.834	28.72	20.232	20.118	

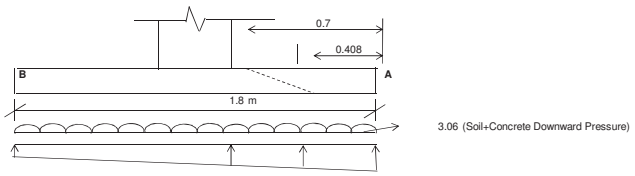
SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN :

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	21	216 1.2 (DL +	**0.621**	59.798	0.156	0.192	-0.007	-2.625
Min Fx	27	231 0.9 (DL +	**-0.577**	51.631	2.548	1.429	0.04	2.427
Max Fy	22	203 1.2 (DL +	0.009	**91.345**	-2.161	-5.574	-0.025	-0.074
Min Fy	31	203 1.2 (DL +	0.002	**-0.655**	-0.033	-0.257	0.007	-0.013
Max Fz	27	222 1.2 (DL +	-0.027	69.278	**5.708**	5.009	0.009	0.045
Min Fz	20	217 1.2 (DL +	0.031	62.983	**-4.203**	-4.78	-0.012	-0.222
Max Mx	27	222 1.2 (DL +	-0.027	69.278	5.708	**5.009**	0.009	0.045
Min Mx	192	217 1.2 (DL +	0.016	43.3	-2.538	**-7.916**	-0.042	-0.125
Max My	30	231 0.9 (DL +	-0.293	16.01	0.225	0.598	**0.124**	1.85
Min My	28	216 1.2 (DL +	0.493	56.922	0.426	0.351	**-0.162**	-2.245
Max Mz	21	231 0.9 (DL +	-0.551	42.364	-0.084	-0.118	-0.012	**2.428**
Min Mz	21	216 1.2 (DL +	0.621	59.798	0.156	0.192	-0.007	**-2.625**

STRESS CALCULATION FOR STRENGTH DESIGN :

L/C	(Wf+Fy)/A (Mton/m²)	Mx/Zx (Mton/m³)	Mz/Zz (Mton/m³)	BASE PRESSURE AT POINT A	B	C	D
216	21.55	0.198	2.701	24.449	19.047	24.053	18.651
231	19.03	1.47	2.497	22.997	18.003	20.057	15.063
203	31.287	5.735	0.076	37.098	36.946	25.628	25.476
203	2.892	0.264	0.013	3.169	3.143	2.641	2.615
222	24.476	5.153	0.046	29.675	29.583	19.369	19.277
217	22.533	4.918	0.228	27.679	27.223	17.843	17.387
222	24.476	5.153	0.046	29.675	29.583	19.369	19.277
217	16.458	8.144	0.129	24.731	24.473	8.443	8.185
231	8.035	0.615	1.903	10.553	6.747	9.323	5.517
216	20.663	0.361	2.31	23.334	18.714	22.612	17.992
231	16.169	0.121	2.498	18.788	13.792	18.546	13.55
216	21.55	0.198	2.701	24.449	19.047	24.053	18.651

PRESSURE DISTRIBUTION ALONG X - DIRECTION :



L/C	Pressure at Point A (Mton/m²)	Pressure at Point B (Mton/m²)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
216	24.449	19.047	22.35	23.22	5.583	1.848
231	22.997	18.003	21.05	21.87	5.203	1.722
203	37.098	36.946	37.04	37.06	8.349	2.835
203	3.169	3.143	3.16	3.16	0.028	0.01
222	29.675	29.583	29.64	29.65	6.526	2.217
217	27.679	27.223	27.5	27.58	6.061	2.055
222	29.675	29.583	29.64	29.65	6.526	2.217
217	24.731	24.473	24.63	24.67	5.326	1.807
231	10.553	6.747	9.07	9.69	2.078	0.672
216	23.334	18.714	21.54	22.29	5.26	1.745
231	18.788	13.792	16.85	17.66	4.17	1.372
216	24.449	19.047	22.35	23.22	5.583	1.848
				MAX =	8.349	2.835

ALONG X - DIRECTION :

Mu = 8.349 Mton-m

$\Phi^*Kn = Mu/bd^2$ deff. = 292 mm

Mu/bd^2 = 8.349X10^7/(1000X292^2) = 0.979 N/mm^2 < 6.9 N/mm^2

Pt = 0.0026

Min. Pt = 0.0018 Reqd. Ast = 0.0026X1000X292/100 = 7.59 cm^2

Provided **16** mm @ **150** C/C Provide Ast = 13.4 cm^2 at Bottom **O.K.**

Total Provide Pt = 0.0046

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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	1
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Prepared By	SNM	SNM	SNM	Prepared By	SNM	SNM	SNM
Checked	SJM	SJM	SJM	Checked	SJM	SJM	SJM
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

ONE-WAY SHEAR CHECK

Vu = 2.835 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0046X2.835X10^4X292/8.349/10^7)X1000X292/10^4

= 27.866 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0046)X1000X292/10^4

= 29.923 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X292/10^4

= 50.097 Mton

HENCE, Vc = 27.866 Mton

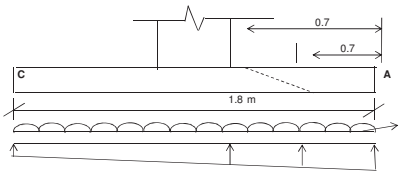
Φt*Vc = 0.75X27.866 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 20.9 Mton **> 2.835Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(27.866X10^4+0.66X35^0.5X1000X292)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 87.601 Mton **> Vu= 2.835Mton , CROSS SECTION O.K.**

PRESSURE DISTRIBUTION ALONG Z - DIRECTION :



L/C	Pressure at Point A (Mton/m²)	Pressure at Point C (Mton/m²)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
216	24.449	24.053	24.3	24.3	5.265	5.103
231	22.997	20.057	21.85	21.85	5.072	3.871
203	37.098	25.628	32.64	32.64	9.067	4.384
203	3.169	2.641	2.96	2.96	0.061	-0.155
222	29.675	19.369	25.67	25.67	7.175	2.967
217	27.679	17.843	23.85	23.85	6.657	2.641
222	29.675	19.369	25.67	25.67	7.175	2.967
217	24.731	8.443	18.4	18.4	6.343	-0.307
231	10.553	9.323	10.07	10.07	1.915	1.412
216	23.334	22.612	23.05	23.05	5.014	4.719
231	18.788	18.546	18.69	18.69	3.869	3.771
216	24.449	24.053	24.3	24.3	5.265	5.103
				MAX =	9.067	5.103

ALONG Z - DIRECTION :

Mu = 9.067 T-m

Φ*Kn = Mu/bd² deff. = 292 mm

Mu/bd² = 9.067X10^7/(1000X292^2) = 1.063 N/mm² < 6.9 N/mm²

Pt = 0.0029

Min. Pt = 0.0018 Reqd. Ast = 0.0029X1000X292/100 = 8.47 cm²

Provided **16** mm @ **150** C/C Provide Ast = 13.4 cm² at Bottom **O.K.**

Total Provide Pt = 0.0046

ONE-WAY SHEAR CHECK

Vu = 5.103 T

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0046X5.103X10^4X292/9.067/10^7)X1000X292/10^4

= 28.015 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0046)X1000X292/10^4

= 29.923 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X292/10^4

= 50.097 Mton

HENCE, Vc = 28.015 Mton

Φt*Vc = 0.75X28.015 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 21.011 Mton **> 5.103Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(28.015X10^4+0.66X35^0.5X1000X292)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 87.612 Mton **> Vu= 5.103Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 3.24 m²

Perimeter (bo) = 2 X ((0.4 + 0.292) + (0.4 + 0.292)) = 2.768 m

β = 1

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 28.834 Mton/m²

Vu = ((3.24 - (0.4X0.4))X28.834

= 88.809 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X2.768X10^3X292/10^4

= 157.796 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X2.768X10^3X292/10^4

= 243.867 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+30X292/(2.768X10^3))X1X35^0.5X(2.768X10^3)X292/10^4

= 204.979 Mton

HENCE, Vc = 157.796 Mton

Φt*Vn = Φt*Vc = 0.75X157.796 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

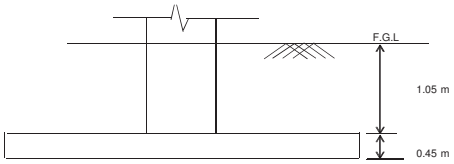
118.347 Mton **> 88.809Mton , Hence O.K.**

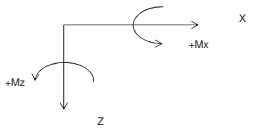
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

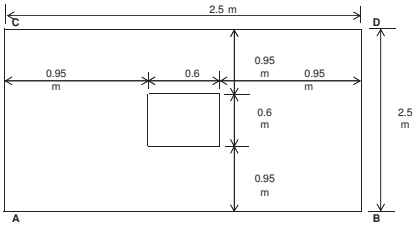
Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	30-05-2020	29-06-2020	28.11.2023	Date	30-05-2020	29-06-2020	28.11.2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT
Prepared By	SNM	SNM	SNM	Prepared By	SNM	SNM	SNM
Checked	SJM	SJM	SJM	Checked	SJM	SJM	SJM
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

DESIGN OF FOUNDATION MKD. F4: (NODE NO. : 5 to 8, 17, 20 to 31, 192 & 207)

Net Safe Bearing Capacity of Soil =	40	Mton/m^2	
Depth of Foundation from FGL =	1.5	m	
Gross Allowable Bearing Capacity of Soil =	40 + 1.5 X 1.9 =	42.85	Mton/m^2
Compressive Strength of Concrete , fc' =	35	N/mm^2	
Yield Strength of Main Reinf. , fy =	420	N/mm^2	
Yield Strength of Shear Reinf. , fyt =	420	N/mm^2	
Strength Reduction Factor of Main Reinf. , Φ =	0.9		
Strength Reduction Factor of Shear Reinf. , Φt =	0.75		
Clear Cover =	50	mm	
Density of Concrete =	2.5	Mton/m^3	
Density of Dry Soil =	1.9	Mton/m^3	
Density of Saturated Soil =	2	Mton/m^3	
Density of Water =	1	Mton/m^3	







FOUNDATION DIMENSION :

Length of Foundation (Lx) =	2.5	m
Width of Foundation (Lz) =	2.5	m
Thickness of Base Raft =	0.45	m
Length of Column (lx) =	0.6	m
Width of Column (lz) =	0.6	m
Area , A = Lx*Lz		
=	6.25	m^2
Zxx = Lx*Lz^2/6		
=	2.604	m^3
Zzz = Lz*Lx^2/6		
=	2.604	m^3

WT. OF FOUNDATION (Wf) :

ITEM	DRY CONDITION (Mton)	
Raft =	2.5X2.5X0.45X2.5 =	7.031
Column =	0.6X0.6X1.05X2.5 =	0.945
Soil =	((2.5X2.5)-(0.6X0.6))X1.05X1.9 =	11.751
	Total =	19.727

SUMMARY OF STAAD OUTPUT FOR STABILITY & STRESS CHECK :

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	9	120 DL + SDL	**2.689**	87.796	-1.986	-0.737	-0.148	-5.166
Min Fx	10	131 0.6 (DL +	**-2.407**	69.898	0.561	2.193	-0.158	6.19
Max Fy	10	104 DL + SDL	-0.195	**133.808**	1.325	4.937	-0.007	-0.023
Min Fy	2	132 0.6 (DL +	1.262	**35.986**	-0.22	0.214	0.106	-3.554
Max Fz	15	118 DL + SDL	0.298	119.931	**2.112**	8.817	0.05	-0.47
Min Fz	9	121 DL + SDL	1.696	84.657	**-2.977**	-4.644	-0.105	-1.123
Max Mx	10	118 DL + SDL	-0.277	126.906	2.11	**9.796**	-0.131	0.247
Min Mx	37	117 DL + SDL	0.592	75.462	-0.462	**-8.349**	0.079	-0.1
Max My	35	117 DL + SDL	-0.242	73.169	-0.161	-5.588	**0.501**	0.556
Min My	9	118 DL + SDL	1.544	85.626	-1.002	2.791	**-0.478**	-0.597
Max Mz	10	131 0.6 (DL +	-2.407	69.898	0.561	2.193	-0.158	**6.19**
Min Mz	14	132 0.6 (DL +	2.662	60.792	0.527	1.274	0.144	**-6.539**

SLIDING CHECK:

L/C	(Wf+Fy) (Mton)	(Fx^2+ Fz^2)^0.5	Frictional coeff. (μ)	Resisting Force	F.O.S	Factor Limit	Remarks
120	107.523	3.343	0.5	53.762	16.08	1.5	**O.K.**
131	89.625	2.472	0.5	44.813	18.13	1.5	**O.K.**
104	153.535	1.339	0.5	76.768	57.33	1.5	**O.K.**
132	55.713	1.281	0.5	27.857	21.75	1.5	**O.K.**
118	139.658	2.133	0.5	69.829	32.74	1.5	**O.K.**
121	104.384	3.426	0.5	52.192	15.23	1.5	**O.K.**
118	146.633	2.128	0.5	73.317	34.45	1.5	**O.K.**
117	95.189	0.751	0.5	47.595	63.38	1.5	**O.K.**
117	92.896	0.291	0.5	46.448	159.62	1.5	**O.K.**
118	105.353	1.841	0.5	52.677	28.61	1.5	**O.K.**
131	89.625	2.472	0.5	44.813	18.13	1.5	**O.K.**
132	80.519	2.714	0.5	40.26	14.83	1.5	**O.K.**

OVERTURNING CHECK:

L/C : 132 0.6 (DL + SDL1) + 0.7 EL(-X)

Max Upword Force =	0	Mton			
Self Wt. of Foundation =	19.727	Mton			
Overturning Moment =	0 X 2.5/2 + 0.214 =		0.214	Mton-m	
Resisting Moment =	19.727 X 0.9 X 2.5/2 =		22.193	Mton-m	
F.O.S =	103.71	>.	2	**O.K.**	

Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	30-05-2020	29-06-2020	28.11.2023	Date	30-05-2020	29-06-2020	28.11.2023
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Checked	SJM	SJM	SJM	Checked	SJM	SJM	SJM
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

BASE PRESSURE CHECK:

L/C	(Wf+Fy)/A (Mton/m²)	Mx/Zxx (Mton/m²)	Mz/Zzz (Mton/m²)	BASE PRESSURE AT POINT A	B	C	D	Remarks
120	17.204	0.283	1.984	19.471	15.503	18.905	14.937	O.K.
131	14.34	0.842	2.377	17.559	12.805	15.875	11.121	O.K.
104	24.566	1.896	0.009	26.471	26.453	22.679	22.661	O.K.
132	8.914	0.082	1.365	10.361	7.631	10.197	7.467	O.K.
118	22.345	3.386	0.18	25.911	25.551	19.139	18.779	O.K.
121	16.701	1.783	0.431	18.915	18.053	15.349	14.487	O.K.
118	23.461	3.762	0.095	27.318	27.128	19.794	19.604	O.K.
117	15.23	3.206	0.038	18.474	18.398	12.062	11.986	O.K.
117	14.863	2.146	0.214	17.223	16.795	12.931	12.503	O.K.
118	16.856	1.072	0.229	18.157	17.699	16.013	15.555	O.K.
131	14.34	0.842	2.377	17.559	12.805	15.875	11.121	O.K.
132	12.883	0.489	2.511	15.883	10.861	14.905	9.883	O.K.
			MAX =	27.318	27.128	22.679	22.661	

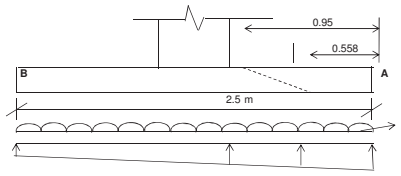
SUMMARY OF STAAD OUTPUT FOR STRENGTH DESIGN :

Design Condition	Node	L/C	Fx (Mton)	Fy (Mton)	Fz (Mton)	Mx (Mton-m)	My (Mton-m)	Mz (Mton-m)
Max Fx	14	216 1.2 (DL +	**4.075**	129.198	1.257	2.808	0.21	-9.645
Min Fx	10	215 1.2 (DL +	**-3.51**	146.759	1.382	5.193	-0.232	8.829
Max Fy	10	202 1.4 (DL+S	-0.242	**167.601**	1.152	4.992	0.005	0.02
Min Fy	2	232 0.9 (DL +	1.788	**53.99**	-0.331	0.308	0.154	-5.053
Max Fz	15	218 1.2 (DL +	0.315	137.725	**2.972**	14.589	0.078	-0.36
Min Fz	9	221 1.2 (DL +	2.065	98.767	**-4.226**	-8.146	0.02	-1.512
Max Mx	15	218 1.2 (DL +	0.315	137.725	2.972	**14.589**	0.078	-0.36
Min Mx	37	217 1.2 (DL +	0.694	89.617	-0.946	**-13.161**	0.097	-0.057
Max My	36	217 1.2 (DL +	-0.24	86.541	-0.771	-11.877	**0.677**	0.649
Min My	9	218 1.2 (DL +	1.779	100.66	-0.484	5.984	**-0.684**	-0.513
Max Mz	10	231 0.9 (DL +	-3.446	104.837	0.843	3.305	-0.226	**8.843**
Min Mz	14	216 1.2 (DL +	4.075	129.198	1.257	2.808	0.21	**-9.645**

STRESS CALCULATION FOR STRENGTH DESIGN :

L/C	(Wf+Fy)/A (Mton/m²)	Mx/Zx (Mton/m²)	Mz/Zz (Mton/m²)	BASE PRESSURE AT POINT A	B	C	D
216	23.828	1.078	3.704	28.61	21.202	26.454	19.046
215	26.638	1.994	3.391	32.023	25.241	28.035	21.253
202	29.972	1.917	0.008	31.897	31.881	28.063	28.047
232	11.795	0.118	1.94	13.853	9.973	13.617	9.737
218	25.192	5.603	0.138	30.933	30.657	19.727	19.451
221	18.959	3.128	0.581	22.668	21.506	16.412	15.25
218	25.192	5.603	0.138	30.933	30.657	19.727	19.451
217	17.495	5.054	0.022	22.571	22.527	12.463	12.419
217	17.003	4.561	0.249	21.813	21.315	12.691	12.193
218	19.262	2.298	0.197	21.757	21.363	17.161	16.767
231	19.93	1.269	3.396	24.595	17.803	22.057	15.265
216	23.828	1.078	3.704	28.61	21.202	26.454	19.046

PRESSURE DISTRIBUTION ALONG X - DIRECTION :



L/C	Pressure at Point A (Mton/m²)	Pressure at Point B (Mton/m²)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
216	28.61	21.202	25.79	26.96	12.351	4.14
215	32.023	25.241	29.45	30.51	13.817	4.657
202	31.897	31.881	31.89	31.89	12.988	4.481
232	13.853	9.973	12.38	12.99	5.286	1.761
218	30.933	30.657	30.83	30.87	12.582	4.337
221	22.668	21.506	22.23	22.41	8.953	3.07
218	30.933	30.657	30.83	30.87	12.582	4.337
217	22.571	22.527	22.55	22.56	8.784	3.029
217	21.813	21.315	21.62	21.7	8.493	2.922
218	21.757	21.363	21.61	21.67	8.454	2.91
231	24.595	17.803	22.01	23.08	10.468	3.501
216	28.61	21.202	25.79	26.96	12.351	4.14
				MAX =	13.817	4.657

ALONG X - DIRECTION :

Mu = 13.817 Mton-m

Φ*Kn = Mu/bd² deff. = 392 mm

Mu/bd² = 13.817X10^7/(1000X392^2) = 0.899 N/mm² < 6.9 N/mm²

Pt = 0.0024

Min. Pt = 0.0018 Reqd. Ast = 0.0024X1000X392/100 = 9.41 cm²

Provided **16** mm @ **150** C/C Provide Ast = 13.4 cm² at Bottom **O.K.**

Total Provide Pt = 0.0034

ONE-WAY SHEAR CHECK

Vu = 4.657 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0034X4.657X10^4X392/13.817/10^7)X1000X392/10^4

= 37.405 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0034)X1000X392/10^4

= 39.371 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X392/10^4

= 67.254 Mton

HENCE, Vc = 37.405 Mton

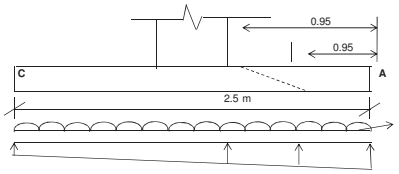
Φt*Vc = 0.75X37.405 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 28.054 Mton **> 4.657Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(37.405X10^4+0.66X35^0.5X1000X392)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 117.601 Mton **> Vu= 4.657Mton , CROSS SECTION O.K.**

PRESSURE DISTRIBUTION ALONG Z - DIRECTION :



3.12 (Soil+Concrete Downward Pressure)

L/C	Pressure at Point A (Mton/m²)	Pressure at Point C (Mton/m²)	Pressure Column Face	Pressure d-dist.from Column Face	B.M. at Column Face (Mton-m)	S.F. at d-dist.from Col. Face (Mton)
216	28.61	26.454	27.79	27.79	11.749	10.128
215	32.023	28.035	30.51	30.51	13.498	10.498
202	31.897	28.063	30.44	30.44	13.424	10.54
232	13.853	13.617	13.76	13.76	4.871	4.694
218	30.933	19.727	26.67	26.67	13.833	5.405
221	22.668	16.412	20.29	20.29	9.536	4.831
218	30.933	19.727	26.67	26.67	13.833	5.405
217	22.571	12.463	18.73	18.73	9.933	2.331
217	21.813	12.691	18.35	18.35	9.477	2.616
218	21.757	17.161	20.01	20.01	8.936	5.479
231	24.595	22.057	23.63	23.63	9.981	8.072
216	28.61	26.454	27.79	27.79	11.749	10.128
				MAX =	13.833	10.54

ALONG Z - DIRECTION :

Mu = 13.833 T-m

Φ*Kn = Mu/bd² deff. = 392 mm

Mu/bd² = 13.833X10^7/(1000X392^2) = 0.9 N/mm² < 6.9 N/mm²

Pt = 0.0024

Min. Pt = 0.0018 Reqd. Ast = 0.0024X1000X392/100 = 9.41 cm²

Provided **16** mm @ **150** C/C Provide Ast = 13.4 cm² at Bottom **O.K.**

Total Provide Pt = 0.0034

ONE-WAY SHEAR CHECK

Vu = 10.54 T

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0034X10.54X10^4X392/13.833/10^7)X1000X392/10^4

= 37.782 Mton

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0034)X1000X392/10^4

= 39.371 Mton

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X35^0.5X1000X392/10^4

= 67.254 Mton

HENCE, Vc = 37.782 Mton

Φt*Vc = 0.75X37.782 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 28.337 Mton **> 10.54Mton , Hence O.K.**

Φt*(Vc+0.66*fc'^0.5*b*d.eff) = 0.75X(37.782X10^4+0.66X35^0.5X1000X392)/10^4 *[SBC:304-CR-18, CL. No. 22.5.1.2, Page-164]*

= 117.629 Mton **> Vu= 10.54Mton , CROSS SECTION O.K.**

TWO-WAY SHEAR CHECK

Area (A) = 6.25 m²

Perimeter (bo) = 2 X ((0.6 + 0.392) + (0.6 + 0.392)) = 3.968 m

β = 1

αs = 30 (Edge Column) *[SBC:304-CR-18, CL. No. 22.6.5.3, Page-168]*

Pmax = 27.318 Mton/m²

Vu = ((6.25 - (0.6X0.6))X27.318

= 160.903 Mton

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.6.5.2, Page-175]*

[a] = 0.33*λ*fc'^0.5*bo*d.eff λ = 1

= 0.33X1X35^0.5X3.968X10^3X392/10^4

= 303.673 Mton

[b] = 0.17*(1+2/β)*λ*fc'^0.5*bo*d.eff

= 0.17X(1+2/1)X1X35^0.5X3.968X10^3X392/10^4

= 469.312 Mton

[c] = 0.083*(2+αs*d.eff/b_o)*λ*fc'^0.5*bo*d.eff

= 0.083X(2+30X392/(3.968X10^3))X1X35^0.5X(3.968X10^3)X392/10^4

= 379.12 Mton

HENCE, Vc = 303.673 Mton

Φt*Vn = Φt*Vc = 0.75X303.673 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

227.755 Mton **> 160.903Mton , Hence O.K.**

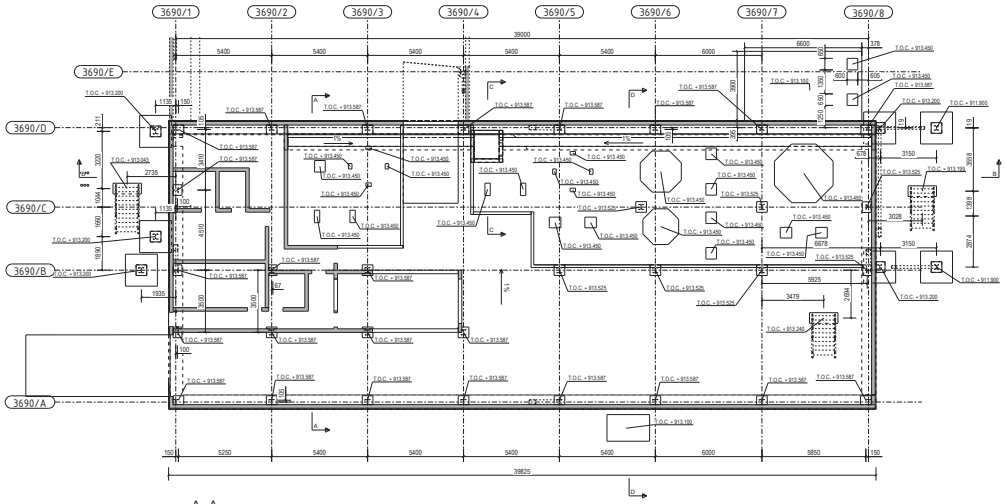
Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01			Outotec / L&T Revision No.:	0	1	2	
Date	30-05-2020	29-06-2020	28.11.2023			Date	30-05-2020	29-06-2020	28.11.2023	
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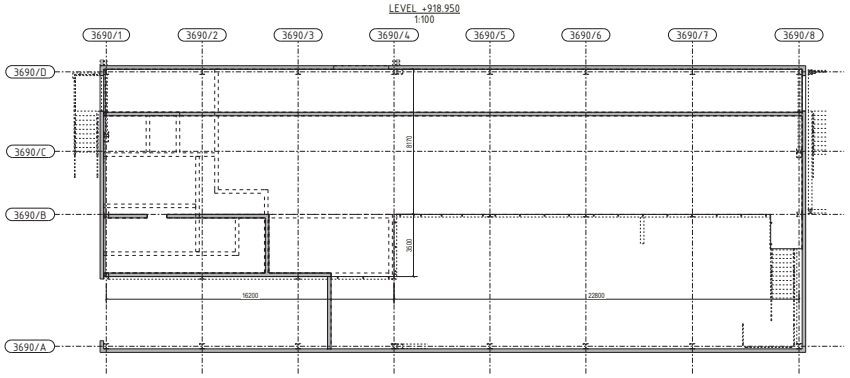
ATTACHMENT 1 : MECHANICAL GENERAL ARRANGEMENT DRAWING

DRAWING NUMBER: MD-216-3690-EG-CV-GAD-1001 A01

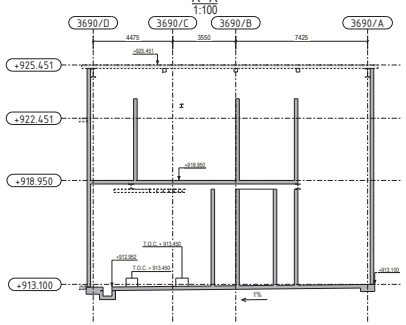
LEVEL +913.100
1:100



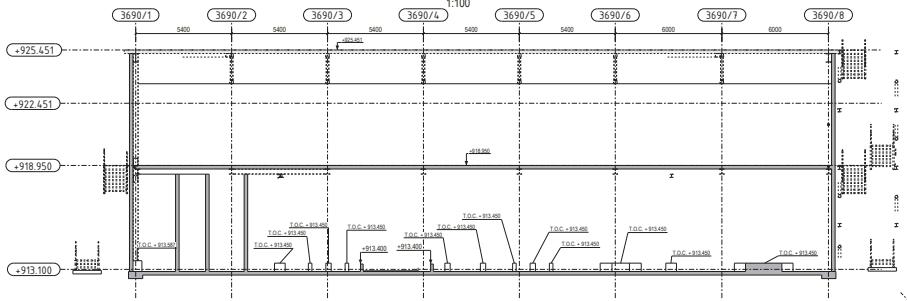
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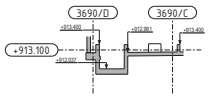
A - A
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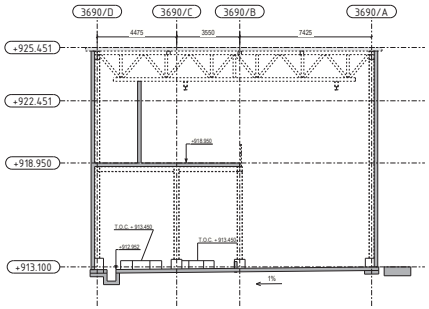
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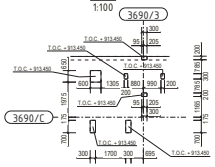
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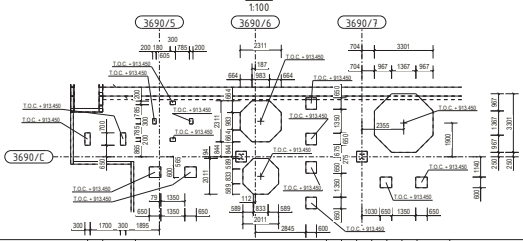
D - D
1:100



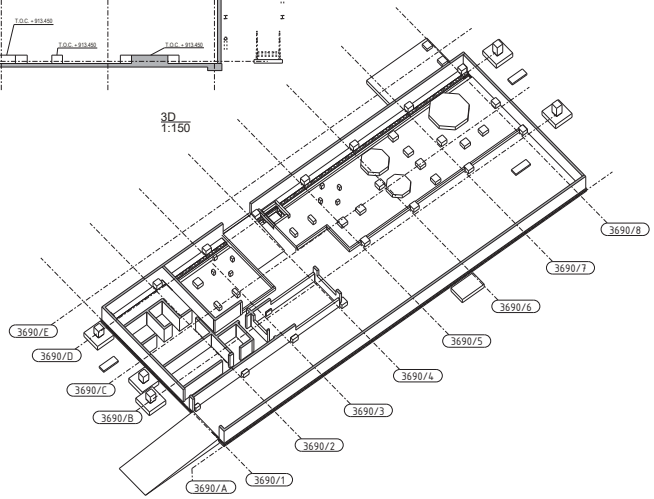
DET 1
1:100



DET 2
1:100



3D
1:150



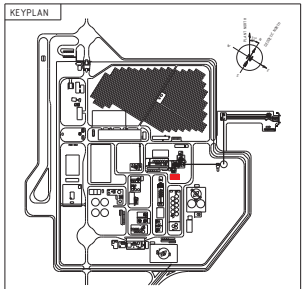
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3690 = RL(+)913.000 m

CHECKERED PLATE
GRATING
GRATING - FRP

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3690-EG-ST-GAD-1002	PLAN AT EL +922.251, SECTIONS A-A, B-B AND C-C	MD-216-3690-EG-ST-GAD-1001	PLAN AT EL +925.251, SECTIONS A-A - E-E
MD-216-3690-EG-CV-GAD-1002	PLAN AT EL +913.100	MD-216-3690-EG-ME-GAD-1001	PLAN AT EL +913.100
MD-216-3690-EG-ME-GAD-1002	SECTION B-B	MD-216-3690-EG-ME-GAD-1004	PLAN AT EL +913.100
MD-216-3690-EG-ME-GAD-1005	PLAN AT EL +913.100	MD-216-3690-EG-ME-GAD-1006	SECTION A-A
MD-216-9000-EG-ST-CRT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3690-EG-ME-GAD-1006	PLAN AT EL +918.950 & +925.450

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	02.04.2020	ISSUED FOR APPROVAL	Ravindra	Mallick	Sudhir	Shaikh	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company

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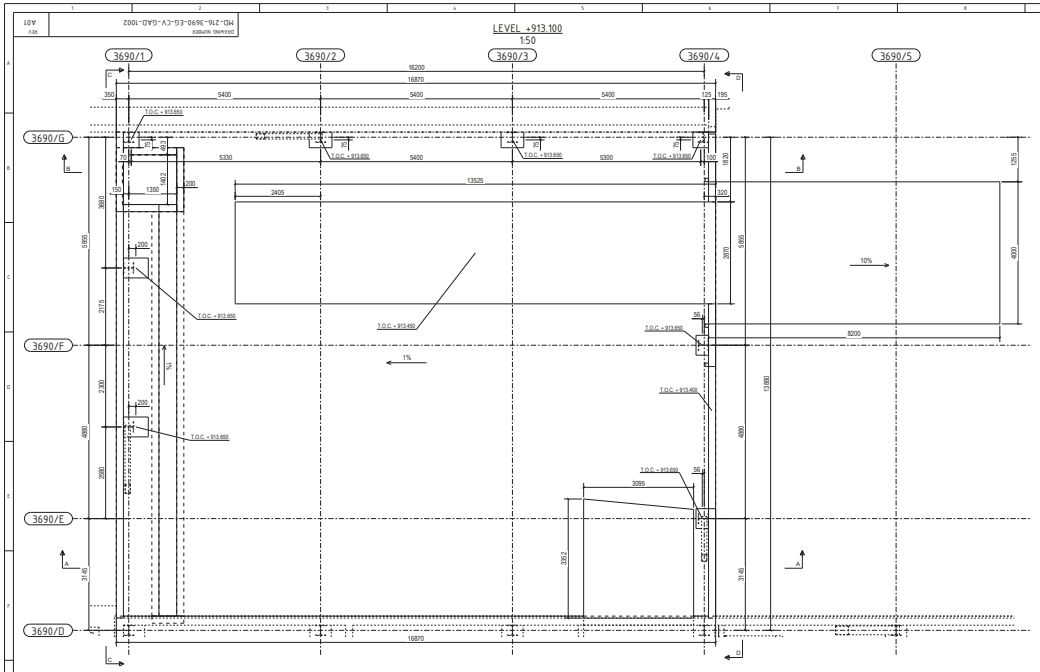
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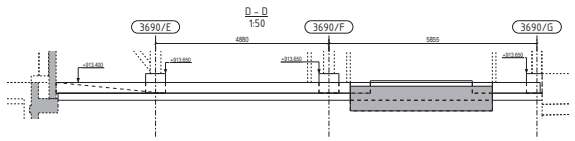
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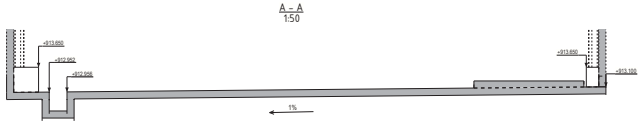
DRAWN	Ajit MK	02.04.2020
CHECKED	Mallick	02.04.2020
ENGINEER	Sudhir	02.04.2020
APPROVED	Shaikh	02.04.2020
EMPLOYER		
SCALE	1:100	A0

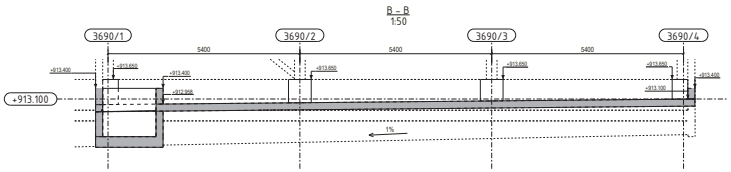
MANSOURAH - MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
GOLD ROOM AND ELECTROWINNING
PLAN AT EL. +913.100

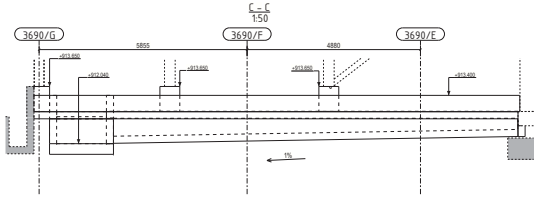
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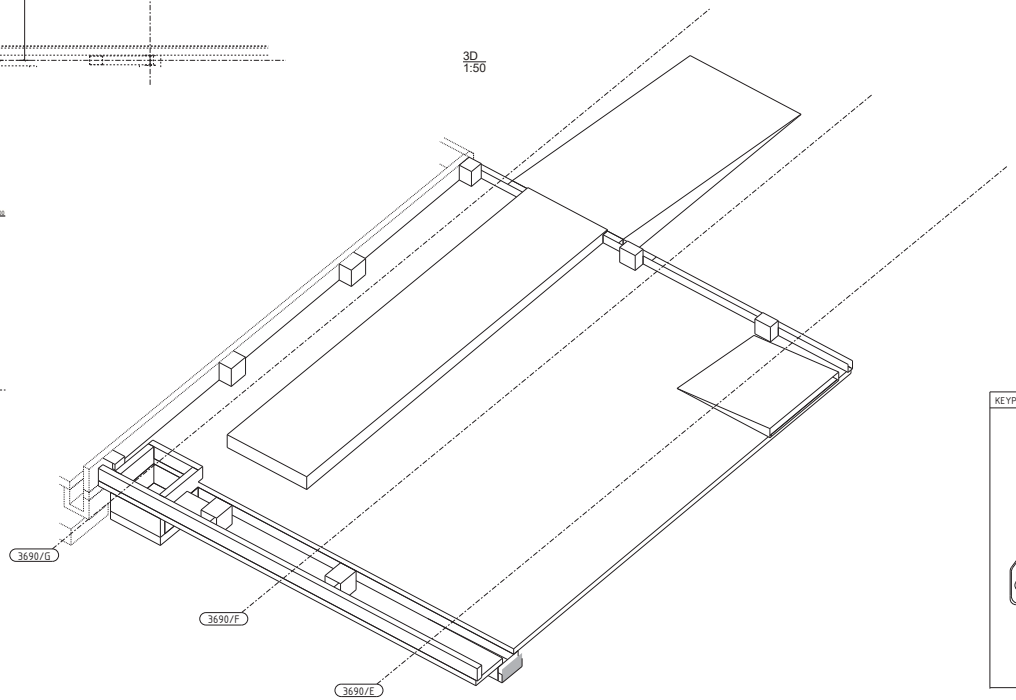












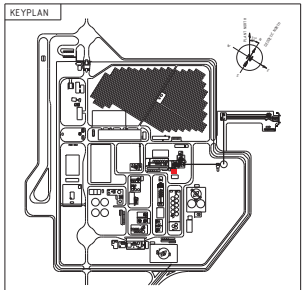
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3620 = RL(+)913.100 m

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ACID RESISTANT COATING
CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
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KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE
MD-216-3690-EG-ST-GAD-1002	PLAN AT EL +922.051, SECTIONS A-A, B-B AND C-C				
MD-216-3690-EG-CV-GAD-1001	PLAN AT EL +910.100	MD-216-3690-EG-ST-GAD-1001	PLAN AT EL +905.201, SECTIONS A-A , E-E		
MD-216-3690-EG-ME-GAD-1007	SECTION B-B	MD-216-3690-EG-ME-GAD-1008	PLAN AT EL +913.100		
MD-216-3690-EG-ME-GAD-1005	PLAN AT EL +913.100	MD-216-3690-EG-ME-GAD-1006	SECTION A-A		
MD-216-1000-EG-ST-CRT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3690-EG-ME-GAD-1004	PLAN AT EL +918.850 & +915.450		

REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01 02.04.2020 ISSUED FOR APPROVAL	Ashish	Mehul	Lutfu	Soulia	

REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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OUTOTEC DRAWING NUMBER: OU100668231 — REVISION: 0

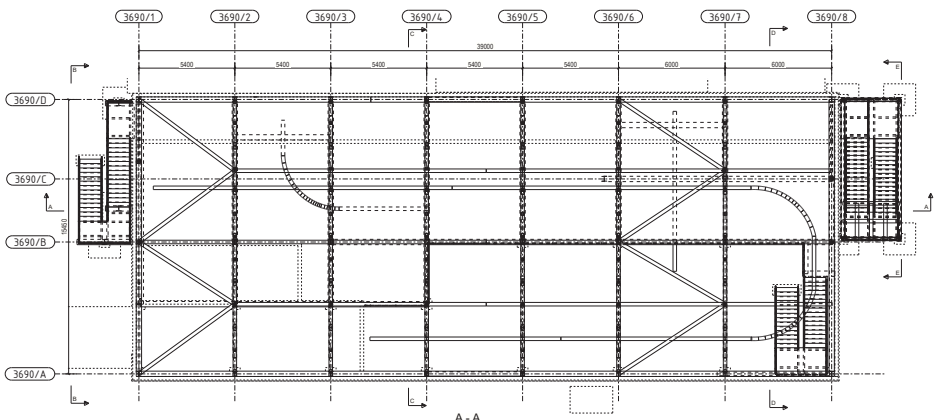
DRAWN	Ashish	02.04.2020
CHECKED	Mehul	02.04.2020
ENGINEER	Lutfu	02.04.2020
APPROVED	Soulia	02.04.2020
EMPLOYER		
SCALE	1:50	A0

MANSOURAH – MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
GOLD ROOM AND ELECTROWINNING
PLAN AT EL. +913.100

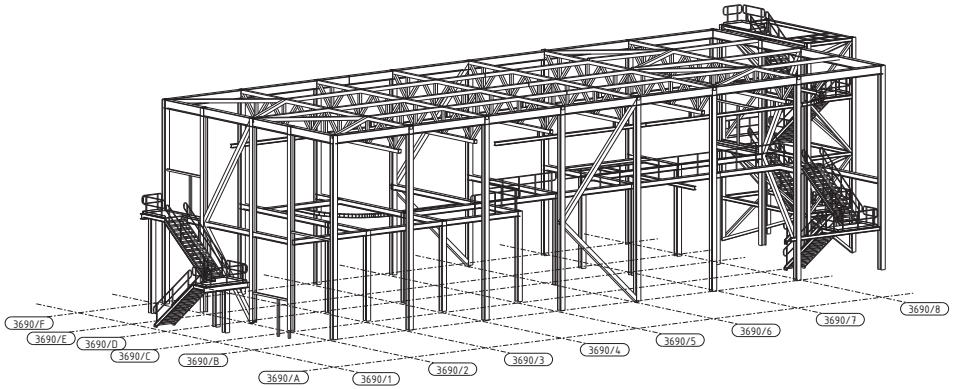
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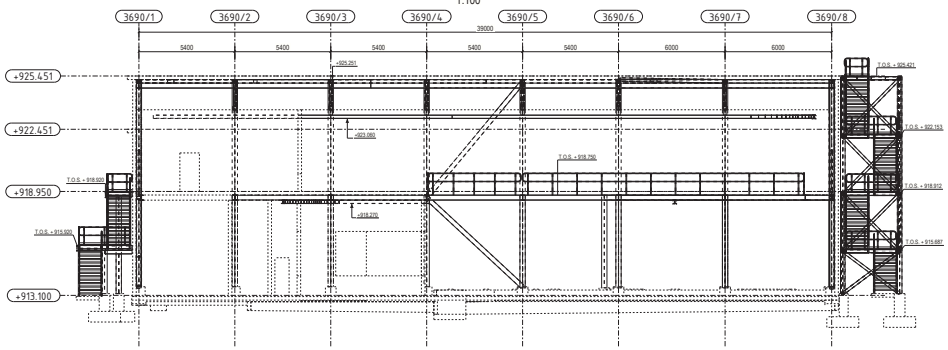
LEVEL +925.251
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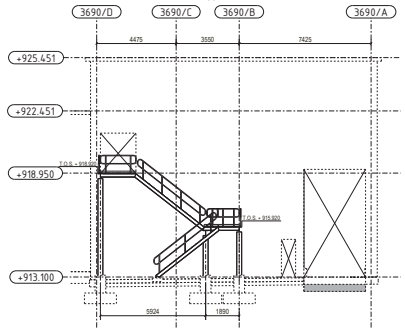
3D
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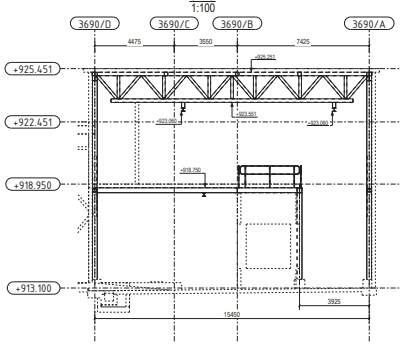
A - A
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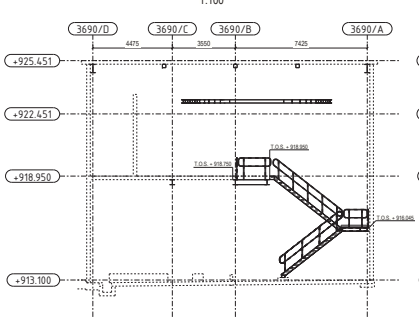
B - B
1:100



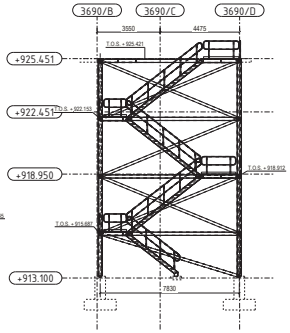
C - C
1:100



D - D
1:100



E - E
1:100



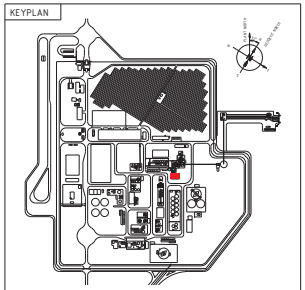
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3690 = RL(+)913.000 m

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ACID RESISTANT COATING
CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
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5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
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KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3690-EG-ST-GAD-1002	PLAN AT EL +922.251, SECTIONS A-A, B-B AND C-C	MD-216-3690-EG-CV-GAD-1002	PLAN AT EL +913.100
MD-216-3690-EG-CV-GAD-1001	PLAN AT EL +913.100	MD-216-3690-EG-ME-GAD-1008	PLAN AT EL +913.100
MD-216-3690-EG-ME-GAD-1007	SECTION B-B	MD-216-3690-EG-ME-GAD-1009	PLAN AT EL +913.100
MD-216-3690-EG-ME-GAD-1005	PLAN AT EL +913.100	MD-216-3690-EG-ME-GAD-1006	SECTION A-A
MD-216-9000-EG-ST-CRT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3690-EG-ME-GAD-1004	PLAN AT EL +918.950 & +915.450

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	CMP
A01	03.01.2020	ISSUED FOR APPROVAL	Krishn	Mehul	Lutfu	Soulia	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	CMP

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LARSEN & TOUBRO DRAWING NUMBER: 900707_GAU01_EE14_CTA05_36901 — REVISION: 0
OUTOTEC DRAWING NUMBER: OU100668224 — DOCUMENT ID — REVISION: 0

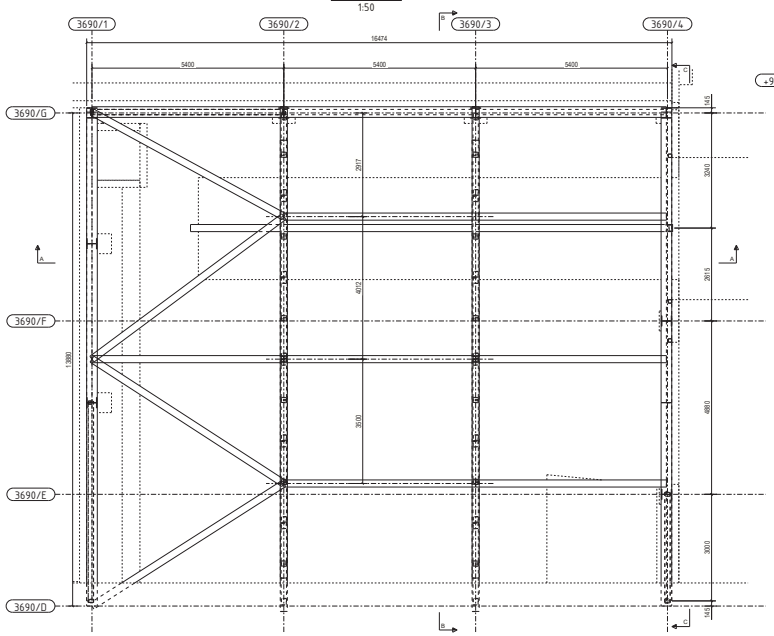
DRAWN	Nikhil	17.01.2020
CHECKED	Mehul	17.01.2020
ENGINEER	Lutfu	17.01.2020
APPROVED	Soulia	17.01.2020
EMPLOYER		

MANSOURAH – MASSARAH GOLD PROJECT
STEEL STRUCTURE GUIDE DRAWING
GOLD ROOM AND ELECTROWINNING
PLAN AT EL. +925.251, SECTIONS A-A – E-E

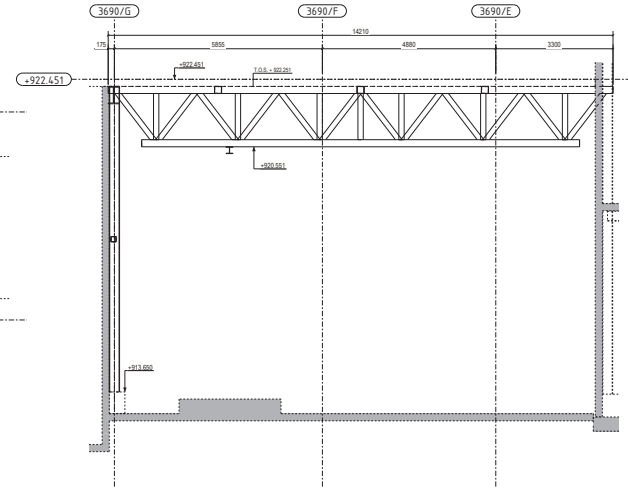
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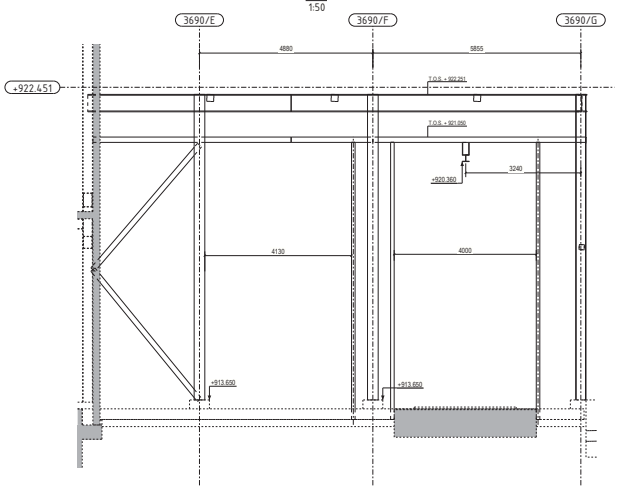
LEVEL +922.251
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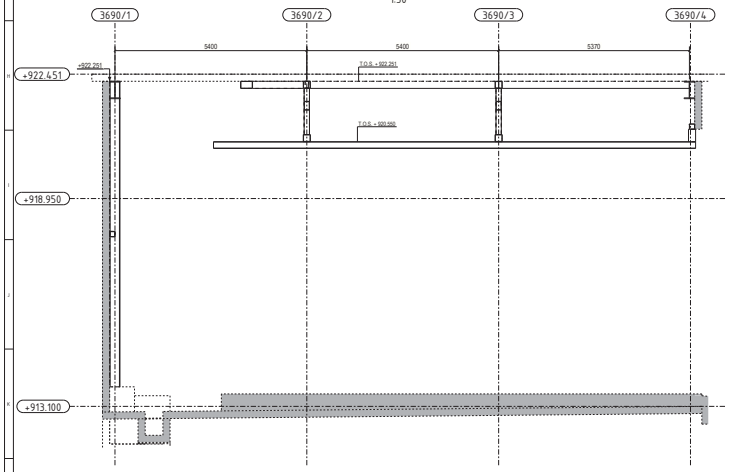
B – B
1:50



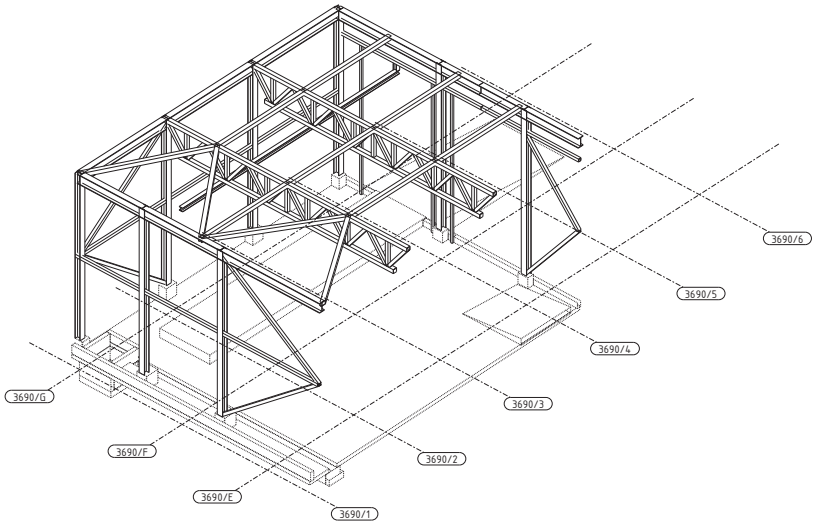
C – C
1:50



A – A
1:50



3D
1:75



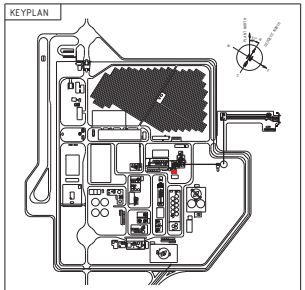
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3620 = RL(+)913.100 m

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ACID RESISTANT COATING
CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
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3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
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6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE
MD-216-3690-EG-ST-GAD-1001	PLAN AT EL. +925.251, SECTIONS A–A & E-E				
MD-216-3690-EG-CY-GAD-1001	PLAN AT EL. +910.100	MD-216-3690-EG-CY-GAD-1002	PLAN AT EL. +913.100		
MD-216-3690-EG-ME-GAD-1017	SECTION B-B	MD-216-3690-EG-ME-GAD-1008	PLAN AT EL. +913.100		
MD-216-3690-EG-ME-GAD-1035	PLAN AT EL. +910.100	MD-216-3690-EG-ME-GAD-1006	SECTION A-A		
MD-216-1000-EG-ST-CRT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3690-EG-ME-GAD-1005	PLAN AT EL. +918.850 & +925.450		

REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01 02.04.2020 ISSUED FOR APPROVAL	Ravish	Manoj	Sudhir	Shanthi	

REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company شركة معادن السعودية

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LARSEN & TOUBRO DRAWING NUMBER: 900707_GAU01_EE14_CTA05_36902 — REVISION: 0

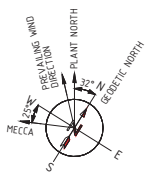
OUTOTEC DRAWING NUMBER: OU100668225 — DOCUMENT ID — REVISION: 0

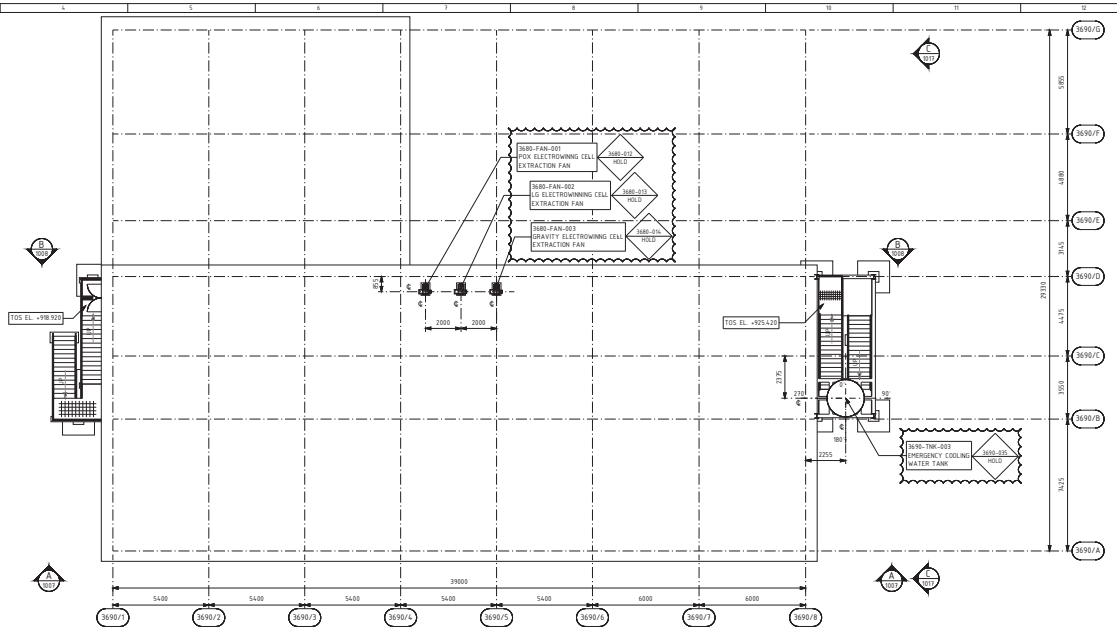
DRAWN	Ad MA	02.04.2020
CHECKED	MaLai	02.04.2020
ENGINEER	LuPhi	02.04.2020
APPROVED	SivaDa	02.04.2020
EMPLOYER		
SCALE	1:50	A0

MANSOURAH – MASSARAH GOLD PROJECT
STEEL STRUCTURE GUIDE DRAWING
GOLD ROOM AND ELECTROWINNING
PLAN AT EL. +922.251, SECTIONS A-A, B-B AND C-C

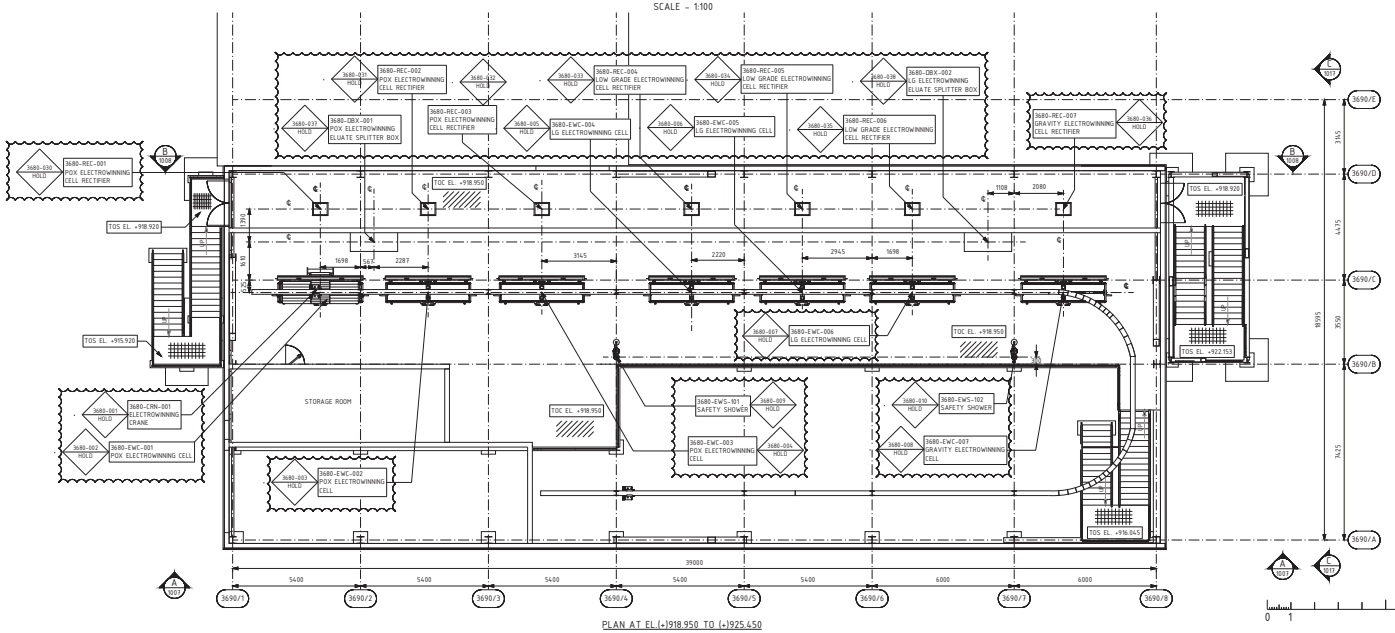
DRAWING NUMBER: MD-216-3690-EG-ST-GAD-1002 — REV: A01

DRAWING NUMBER
MD-216-3690-EG-ME-GAD-1006
REV
A01





PLAN AT EL.(+)925.450
SCALE - 1:100



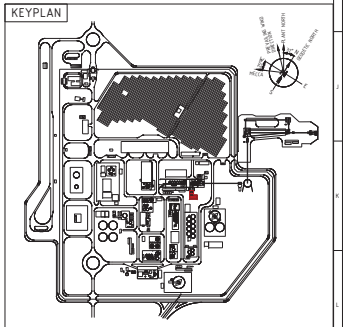
PLAN AT EL.(+)918.950 TO (+)925.450

0 1 10m

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ALKALI / ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METERS UNLESS OTHERWISE STATED.
2. FINISHED GROUND LEVEL (F.G.L.) FOR PROCESS AREA 3680 & 3690 = RL(+)912.600 m
3. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE. T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE. T.O.G. = TOP OF GRATING. HP = HIGHT POINT, LP = LOW POINT, T.O.R. = TOP OF RAIL.
4. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3690-EG-ME-GAD-1003	LOAD PLAN, PLAN AT EL. +913.100
MD-216-3600-EG-GM-LST-1010	HOLD LIST
MD-216-3690-EG-ME-GAD-1017	SECTION C-C
MD-216-3690-EG-ME-GAD-1008	SECTION B-B
MD-216-3690-EG-ME-GAD-1007	SECTION A-A
MD-216-3690-EG-ME-GAD-1005	PLAN AT EL. +913.100

DOCUMENT NUMBER	REFERENCE DOCUMENTS

REV	DATE	REVISION DESCRIPTION	GRN	CHK	ENG	APP	CMP
A01	17-04-2020	ISSUED FOR APPROVAL	GK	TV	SC	RL	

REV	DATE	REVISION DESCRIPTION	GRN	CHK	ENG	APP	CMP

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LARSEN & TOUBRO DRAWING NUMBER			REVISION
O19045_450_M_G_02006_002			0
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707__GAU01__EE14__MTL01__36902		OU100676823	0
DRAWN	GK	17-04-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	17-04-2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	SC	17-04-2020	GOLD ROOM, ELECTROWINNING & ICL
APPROVED	RL	17-04-2020	PLAN AT EL. +918.850 & EL. +925.450
EMPLOYER			
SCALE 1:75	A0		DRAWING NUMBER MD-216-3690-EG-ME-GAD-1006 REV A01

DRAWING NUMBER: MD-216-3690-EG-ME-GAD-1016
REV: A01

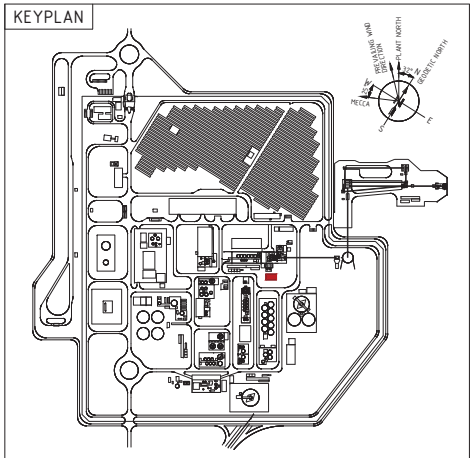
LOADING POINT	P1			P2			P3			P4.1/.../P4.4			P5.1/.../P5.2			P6			P7.1/.../P7.2			P8.1/.../P8.4			P9			P10			P11			P12			P13			P14			P15		
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]
1 DEAD LOAD EQUIPMENT	-	12		-	120		-	3		-	2.5		-	5		-	12		-	5		-	2.5		-	3		-	3		-	14		-	14		-	3		-	3		-	3	
2 DEAD LOAD (SECONDARY EQUIPMENT)																																												
3 OPERATION LOAD 1. FILLING																																												
4 OPERATION LOAD 2. PRESSURE																																												
5 OPERATION LOAD 3. TEMPERATURE																																												
6 OPERATION LOAD 4. DYNAMIC	-	18					-	5								-	18								-	5		-	5								-	5		-	5		-	5	
7 OPERATION LOAD 5. OTHERS																																												
8 TEST LOAD FILLING / EMERGENCY																																												
9 LIVE LOAD				-	425					-	10					-	20					-	10								-	90		-	90								
10 WIND ±X - DIRECTION																																												
11 WIND ±Y - DIRECTION																																												
12 EARTHQUAKE ±X - DIRECTION	-	-	0.6	-	-	21.3	-	-	-	-	-	0.5	-	-	1.0	-	-	0.6	-	-	1.0	-	-	0.5	-	-	-	-	-	4.5	-	-	4.5	-	-	-	-	-	-	-	-	-	-	-
13 EARTHQUAKE ±Y - DIRECTION	-	-	-	0.6	-	-	21.3	-	-	-	-	0.5	-	-	1.0	-	-	0.6	-	-	1.0	-	-	0.5	-	-	0.5	-	-	4.5	-	-	4.5	-	-	-	-	-	-	-	-	-	-	-
EQUIPMENT - NO.	3620-PMP-101			3620-PKG-001			3690-PMP-001			3690-TNK-001			3690-FLT-001			3690-PMP-101			3690-FLT-002			3690-TNK-002			3690-PMP-003			3690-PMP-002			3680-TNK-003			3680-TNK-002			3680-PMP-005			3680-PMP-006			3680-PMP-004		
DESCRIPTION	INTENSIVE CYANIDATION AREA SUMP PUMP			INTENSIVE CYANIDATION VENDOR PACKAGE			POX GOLD SLUDGE FILTER PUMP			POX GOLD SLUDGE TANK			POX GOLD SLUDGE FILTER			GOLD ROOM AREA SUMP PUMP			LG GOLD SLUDGE FILTER			LG GOLD SLUDGE TANK			LG GOLD SLUDGE FILTER PUMP			LG GOLD SLUDGE FILTER PUMP			POX ELUATE RETURN TANK			POX ELUATE RETURN TANK			LG ELUATE RETURN PUMP			LG ELUATE RETURN PUMP			POX ELUATE RETURN PUMP		

LOADING POINT	P16			P17			P18			P19			P20			P21			P22			P23			P24			P25			P26			P27			P28			P29			P30		
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]
1 DEAD LOAD EQUIPMENT	-	3		-	29		-	3		-	3		-	10		-	3		-	3		-	20		-	10		-	20		-	50		-	10		-	5		-	10		-	1.5	
2 DEAD LOAD (SECONDARY EQUIPMENT)																																												
3 OPERATION LOAD 1. FILLING																																												
4 OPERATION LOAD 2. PRESSURE																																												
5 OPERATION LOAD 3. TEMPERATURE																																												
6 OPERATION LOAD 4. DYNAMIC	-	5					-	5		-	5					-	5		-	5																									
7 OPERATION LOAD 5. OTHERS																																												
8 TEST LOAD FILLING / EMERGENCY																																												
9 LIVE LOAD				-	280																	-	70		-	20					-	100											
10 WIND ±X - DIRECTION															0.5																													
11 WIND ±Y - DIRECTION																		0.5																										
12 EARTHQUAKE ±X - DIRECTION	-	-	-	-	14	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	3.5	-	-	1	-	-	1.0	-	-	5	-	-	0.5	-	-	-	-	-	-	-	-
13 EARTHQUAKE ±Y - DIRECTION	-	-	-	-	-	14	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	3.5	-	-	1	-	-	1.0	-	-	5	-	-	0.5	-	-	-	-	-	-	-	-
EQUIPMENT - NO.	3680-PMP-003			3680-TNK-001			3680-PMP-001			3680-PMP-002			3690-CLT-001			3690-PMP-004			3690-PMP-005			3690-CRN-001			3690-FNC-002			-			3690-FNC-001			-			3690-MSC-006			3680-FLT-003			3680-REC-001		
DESCRIPTION	POX ELUATE RETURN PUMP			GRAVITY ELECTROWINNING LEACHATE TANK			LEACHATE PUMP			LEACHATE PUMP			FURNACE COOLING TOWER			FURNACE COOLING WATER PUMP			FURNACE COOLING WATER PUMP			GOLD ROOM JIB CRANE			SMELTING FURNACE			SMELTING FURNACE INVERTER			CALCINING OVEN			TRAY RACK			FLUX MIXER			SMELT FURNACE DUST COLLECTOR			POX ELECTROWINNING CELL RECTIFIER		

LOADING POINT	P31			P32			P33			P34			P35			P36			P37.1/.../P37.4			P38.1/.../P38.4			P39.1/.../P39.6			P40.1/.../P40.6			P41.1/.../P41.6		
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]
1 DEAD LOAD EQUIPMENT	-	1.5		-	1.5		-	1.5		-	1.5		-	1.5		-	1.5		-	1.25		-	0.25		-	3.3		-	3.3		-	3.3	
2 DEAD LOAD (SECONDARY EQUIPMENT)																																
3 OPERATION LOAD 1. FILLING																																
4 OPERATION LOAD 2. PRESSURE																																
5 OPERATION LOAD 3. TEMPERATURE																																
6 OPERATION LOAD 4. DYNAMIC																																
7 OPERATION LOAD 5. OTHERS																																
8 TEST LOAD FILLING / EMERGENCY																																
9 LIVE LOAD																			-	7.5		-	7.5		-	5		-	5		-	5	
10 WIND ±X - DIRECTION																																
11 WIND ±Y - DIRECTION																																
12 EARTHQUAKE ±X - DIRECTION	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0.4	-	-	0.4	-	-	0.25	-	-	0.25	-	-	0.25
13 EARTHQUAKE ±Y - DIRECTION	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0.4	-	-	0.4	-	-	0.25	-	-	0.25	-	-	0.25
EQUIPMENT - NO.	3680-REC-002			3680-REC-003			3680-REC-004			3680-REC-005			3680-REC-006			3680-REC-007			3680-DBX-001			3680-DBX-002			3680-EWC-001			3680-EWC-002			3680-EWC-003		
DESCRIPTION	POX ELECTROWINNING CELL RECTIFIER			POX ELECTROWINNING CELL RECTIFIER			LOW GRADE ELECTROWINNING CELL RECTIFIER			LOW GRADE ELECTROWINNING CELL RECTIFIER			LOW GRADE ELECTROWINNING CELL RECTIFIER			GRAVITY ELECTROWINNING CELL RECTIFIER			POX ELECTROWINNING ELUATE SPLITTER BOX			LG ELECTROWINNING ELUATE SPLITTER BOX			POX ELECTROWINNING CELL			POX ELECTROWINNING CELL			POX ELECTROWINNING CELL		

LOADING POINT	P42.1/.../P42.6			P43.1/.../P43.6			P44.1/.../P44.6			P45.1/.../P45.6			P46			P47			P48			P49		
LOAD CASE	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]
1 DEAD LOAD EQUIPMENT	-	3.3		-	3.3		-	3.3		-	3.3		-	3		-	3		-	3		-	12	
2 DEAD LOAD (SECONDARY EQUIPMENT)																								
3 OPERATION LOAD 1. FILLING																								
4 OPERATION LOAD 2. PRESSURE																								
5 OPERATION LOAD 3. TEMPERATURE																								
6 OPERATION LOAD 4. DYNAMIC																								
7 OPERATION LOAD 5. OTHERS																								
8 TEST LOAD FILLING / EMERGENCY																								
9 LIVE LOAD	-	5		-	5		-	5		-	5											-	80	
10 WIND ±X - DIRECTION															0.2			0.2			0.2			
11 WIND ±Y - DIRECTION																								
12 EARTHQUAKE ±X - DIRECTION	-	-	0.25	-	-	0.25	-	-	0.25	-	-	0.25	-	-	-	-	-	-	-	-	-	-	4	-
13 EARTHQUAKE ±Y - DIRECTION	-	-	-	0.25	-	-	0.25	-	-	0.25	-	-	0.25	-	-	-	-	-	-	-	-	-	-	4
EQUIPMENT - NO.	3680-EWC-004			3680-EWC-005			3680-EWC-006			3680-EWC-007			3680-FAN-001			3680-FAN-002			3680-FAN-003			3690-TNK-003		
DESCRIPTION	LG ELECTROWINNING CELL			LG ELECTROWINNING CELL			LG ELECTROWINNING CELL			GRAVITY ELECTROWINNING CELL			POX ELECTROWINNING CELL EXTRACTION FAN			LG ELECTROWINNING CELL EXTRACTION FAN			GRAVITY ELECTROWINNING CELL EXTRACTION FAN			EMERGENCY COOLING WATER TANK		

KEYPLAN



RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045_450_M_D_02003_001 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 __GAU01 __EE14 __MLD14 __36903 — DOCUMENT ID: OU100680654 — REVISION: 0

DRAWN	GK	08-04-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	08-04-2020	LOAD PLAN
ENGINEER	SC	08-04-2020	GOLD ROOM, ELECTROWINNING & ICL
APPROVED	RL	08-04-2020	LOAD TABLE
EMPLOYER			
SCALE	-	A1	DRAWING NUMBER: MD-216-3690-EG-ME-GAD-1016 — REV: A01

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DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-GM-LST-1010	HOLD LIST								
MD-216-3690-EG-ME-GAD-1004	LOAD PLAN, PLAN AT EL. +918.950 & EL. +925.450								
MD-216-3690-EG-ME-GAD-1003	LOAD PLAN, PLAN AT EL. +913.100	A01	08-04-2020	ISSUED FOR APPROVAL	GK	TV	SC	RL	

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01			Outotec / L&T Revision No.:	0	1	2	
Date	30-05-2020	29-06-2020	28.11.2023			Date	30-05-2020	29-06-2020	28.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT			Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT	
Prepared By	SNM	SNM	SNM			Prepared By	SNM	SNM	SNM	
Checked	SJM	SJM	SJM			Checked	SJM	SJM	SJM	
Approved	PRI	PRI	PRI			Approved	PRI	PRI	PRI	

ATTACHMENT 2 : STAAD INPUT FILE OF GOLD ROOM

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 20-Apr-20
JOB NAME MMGP/GOLD ROOM
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SNM
CHECKER NAME SJM
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 911.1 0; 2 5.4 911.1 0; 3 10.8 911.1 0; 4 16.2 911.1 0; 5 -0.15 911.1 3.68;
6 16.35 911.1 5.855; 7 -0.15 911.1 8.155; 8 16.35 911.1 10.735;
9 0 911.1 13.88; 10 5.4 911.1 13.88; 11 10.8 911.1 13.88; 12 16.2 911.1 13.88;
13 21.6 911.1 13.88; 14 27 911.1 13.88; 15 33 911.1 13.88; 16 39 911.1 13.88;
17 -0.15 911.1 18.355; 20 39.15 911.1 18.355; 21 -0.15 911.1 21.905;
22 5.4 911.1 21.905; 23 10.8 911.1 21.905; 24 21.6 911.1 21.905;
25 27 911.1 21.905; 26 33 911.1 21.905; 27 39.15 911.1 21.905;
28 -0.15 911.1 25.405; 29 5.4 911.1 25.405; 30 10.8 911.1 25.405;
31 16.2 911.1 25.405; 32 0 911.1 29.33; 33 5.4 911.1 29.33;
34 10.8 911.1 29.33; 35 16.2 911.1 29.33; 36 21.6 911.1 29.33;
37 27 911.1 29.33; 38 33 911.1 29.33; 39 39 911.1 29.33; 40 0 913.1 0;
41 5.4 913.1 0; 42 10.8 913.1 0; 43 16.2 913.1 0; 44 -0.15 913.1 3.68;
45 16.35 913.1 5.855; 46 -0.15 913.1 8.155; 47 16.35 913.1 10.735;
48 0 913.1 13.88; 49 5.4 913.1 13.88; 50 10.8 913.1 13.88; 51 16.2 913.1 13.88;
52 21.6 913.1 13.88; 53 27 913.1 13.88; 54 33 913.1 13.88; 55 39 913.1 13.88;
56 -0.15 913.1 18.355; 59 39.15 913.1 18.355; 60 -0.15 913.1 21.905;
61 5.4 913.1 21.905; 62 10.8 913.1 21.905; 63 21.6 913.1 21.905;
64 27 913.1 21.905; 65 33 913.1 21.905; 66 39.15 913.1 21.905;
67 -0.15 913.1 25.405; 68 5.4 913.1 25.405; 69 10.8 913.1 25.405;
70 16.2 913.1 25.405; 71 0 913.1 29.33; 72 5.4 913.1 29.33;
73 10.8 913.1 29.33; 74 16.2 913.1 29.33; 75 21.6 913.1 29.33;
76 27 913.1 29.33; 77 33 913.1 29.33; 78 39 913.1 29.33; 79 0 918.95 0;
80 5.4 918.95 0; 81 10.8 918.95 0; 82 16.2 918.95 0; 83 -0.15 918.95 3.68;
84 16.35 918.95 5.855; 85 -0.15 918.95 8.155; 86 16.35 918.95 10.735;
87 0 918.95 13.88; 88 5.4 918.95 13.88; 89 10.8 918.95 13.88;
90 16.2 918.95 13.88; 91 21.6 918.95 13.88; 92 27 918.95 13.88;
93 33 918.95 13.88; 94 39 918.95 13.88; 95 -0.15 918.95 18.355;
98 39.15 918.95 18.355; 99 -0.15 918.95 21.905; 100 5.4 918.95 21.905;
101 10.8 918.95 21.905; 102 21.6 918.95 21.905; 103 27 918.95 21.905;
104 33 918.95 21.905; 105 39.15 918.95 21.905; 106 -0.15 918.95 25.405;
107 5.4 918.95 25.405; 108 10.8 918.95 25.405; 109 16.2 918.95 25.405;
110 0 918.95 29.33; 111 5.4 918.95 29.33; 112 10.8 918.95 29.33;
113 16.2 918.95 29.33; 114 21.6 918.95 29.33; 115 27 918.95 29.33;
116 33 918.95 29.33; 117 39 918.95 29.33; 118 16.2 918.95 21.905;
119 0 922.45 13.88; 120 5.4 922.45 13.88; 121 10.8 922.45 13.88;
122 16.2 922.45 13.88; 127 0 922.45 0; 128 5.4 922.45 0; 129 10.8 922.45 0;
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130 16.2 922.45 0; 131 -0.15 922.45 3.68; 132 16.35 922.45 5.855;
133 -0.15 922.45 8.155; 134 16.35 922.45 10.735; 135 0 925.45 13.88;
136 5.4 925.45 13.88; 137 10.8 925.45 13.88; 138 16.2 925.45 13.88;
139 21.6 925.45 13.88; 140 27 925.45 13.88; 141 33 925.45 13.88;
142 39 925.45 13.88; 143 0 925.45 29.33; 144 5.4 925.45 29.33;
145 10.8 925.45 29.33; 146 16.2 925.45 29.33; 147 21.6 925.45 29.33;
148 27 925.45 29.33; 149 33 925.45 29.33; 150 39 925.45 29.33;
151 -0.15 925.45 18.355; 152 39.15 925.45 18.355; 153 -0.15 925.45 21.905;
154 39.15 925.45 21.905; 155 -0.15 925.45 25.405; 156 5.4 918.95 14.48;
157 10.8 918.95 14.48; 158 16.2 918.95 14.48; 159 21.6 918.95 14.48;
160 27 918.95 14.48; 161 33 918.95 14.48; 162 -0.15 918.95 15.905;
163 5.4 918.95 15.905; 164 10.8 918.95 15.905; 165 16.2 918.95 15.905;
166 21.6 918.95 15.905; 167 27 918.95 15.905; 168 33 918.95 15.905;
169 39.15 918.95 15.905; 170 -0.15 918.95 17.38; 171 5.4 918.95 17.38;
172 10.8 918.95 17.38; 173 16.2 918.95 17.38; 174 21.6 918.95 17.38;
175 27 918.95 17.38; 176 33 918.95 17.38; 177 39.15 918.95 17.38;
178 -0.15 918.95 18.855; 179 5.4 918.95 18.855; 180 10.8 918.95 18.855;
181 16.2 918.95 18.855; 182 21.6 918.95 18.855; 183 27 918.95 18.855;
184 33 918.95 18.855; 185 39.15 918.95 18.855; 186 -0.15 918.95 23.655;
187 5.4 918.95 23.655; 188 10.8 918.95 23.655; 189 16.2 918.95 23.655;
190 5.4 922.45 13.28; 191 10.8 922.45 13.28; 192 16.2 911.1 21.905;
193 16.2 913.1 21.905; 194 -0.15 918.95 20.33; 195 5.4 918.95 20.33;
196 10.8 918.95 20.33; 197 16.2 918.95 20.33; 198 21.6 918.95 20.33;
199 27 918.95 20.33; 200 33 918.95 20.33; 201 39.15 918.95 20.33;
202 -3.2 912.95 21.905; 203 -3.2 915.95 21.905; 204 -0.15 915.95 21.905;
205 -1.6 918.95 13.88; 206 0 917.45 13.88; 207 -3.2 911.1 21.905;
208 33 925.45 18.355; 209 33 925.45 24.255; 210 39.15 925.45 24.3782;
211 33 925.45 21.905; 212 33 925.45 27.755; 213 39.15 925.45 27.8768;
214 35 925.45 18.355; 215 34.9996 925.45 24.2954; 216 35 925.45 21.905;
217 34.9996 925.45 27.7954; 218 37.19 925.45 18.355;
219 37.1891 925.45 24.3396; 220 37.19 925.45 21.905;
221 37.1891 925.45 27.8396; 222 5.4 925.45 18.355; 223 -0.15 925.45 24.255;
224 5.4 925.45 24.255; 225 10.8 925.45 18.355; 226 10.8 925.45 24.255;
227 16.2 925.45 18.355; 228 16.2 925.45 24.255; 229 21.6 925.45 18.355;
230 21.6 925.45 24.255; 231 27 925.45 18.355; 232 27 925.45 24.255;
233 0.35 925.45 18.355; 234 0.35 925.45 24.255; 235 -0.0709239 922.45 1.74;
236 5.4 922.45 1.74; 237 -0.15 922.45 4.74; 238 5.4 922.45 4.74;
239 10.8 922.45 1.75; 240 16.2448 922.45 1.75; 241 10.8 922.45 4.75;
242 16.3217 922.45 4.75; 243 5.4 922.45 9.01; 244 10.8 922.45 9.015;
245 0.429076 922.45 1.74; 246 0.35 922.45 4.74; 247 15.7 922.45 1.75;
248 15.7 922.45 4.75; 249 33 916.025 29.33; 250 39 916.025 29.33;
251 31 918.95 15.905; 252 31 918.95 17.38; 253 6.9 918.95 15.905;
254 6.9 918.95 17.38; 255 8.8 918.95 25.405; 256 8.8 918.95 21.905;
257 8.8 918.95 23.655;
MEMBER INCIDENCES
1 1 40; 2 2 41; 3 3 42; 4 4 43; 5 5 44; 6 6 45; 7 7 46; 8 8 47; 9 9 48;
10 10 49; 11 11 50; 12 12 51; 13 13 52; 14 14 53; 15 15 54; 16 16 55; 17 17 56;
20 20 59; 21 21 60; 22 22 61; 23 23 62; 24 24 63; 25 25 64; 26 26 65; 27 27 66;
28 28 67; 29 29 68; 30 30 69; 31 31 70; 32 32 71; 33 33 72; 34 34 73; 35 35 74;
36 36 75; 37 37 76; 38 38 77; 39 39 78; 40 40 41; 41 41 42; 42 42 43; 43 48 49;

44 49 50; 45 50 51; 46 51 52; 47 52 53; 48 53 54; 49 54 55; 50 71 72; 51 72 73;
52 73 74; 53 74 75; 54 75 76; 55 76 77; 56 77 78; 57 40 44; 58 44 46; 59 46 48;
60 48 56; 61 56 60; 62 60 67; 63 67 71; 64 43 45; 65 45 47; 66 47 51; 67 55 59;
68 59 66; 69 66 78; 70 40 79; 71 41 80; 72 42 81; 73 43 82; 74 44 83; 75 45 84;
76 46 85; 77 47 86; 78 48 206; 79 49 88; 80 50 89; 81 51 90; 82 52 91;
83 53 92; 84 54 93; 85 55 94; 86 56 95; 89 59 98; 90 60 204; 91 61 100;
92 62 101; 93 63 102; 94 64 103; 95 65 104; 96 66 105; 97 67 106; 98 68 107;
99 69 108; 100 70 109; 101 71 110; 102 72 111; 103 73 112; 104 74 113;
105 75 114; 106 76 115; 107 77 249; 108 78 250; 109 79 80; 110 80 81;
111 81 82; 112 87 88; 113 88 89; 114 89 90; 115 90 91; 116 91 92; 117 92 93;
118 93 94; 119 110 111; 120 111 112; 121 112 113; 122 113 114; 123 114 115;
124 115 116; 125 116 117; 126 79 83; 127 83 85; 128 85 87; 129 87 162;
130 95 178; 131 99 186; 132 106 110; 133 82 84; 134 84 86; 135 86 90;
136 94 169; 137 98 185; 138 105 117; 139 88 156; 140 100 187; 141 89 157;
142 101 188; 143 90 158; 144 91 159; 145 92 160; 146 93 161; 147 106 107;
148 107 255; 149 108 109; 150 118 189; 151 118 102; 152 102 103; 153 103 104;
154 104 105; 155 87 119; 156 88 120; 157 89 121; 158 90 122; 159 91 139;
160 92 140; 161 93 141; 162 94 142; 163 79 127; 164 80 128; 165 81 129;
166 82 130; 167 83 131; 168 84 132; 169 85 133; 170 86 134; 171 127 128;
172 128 129; 173 129 130; 174 119 120; 175 120 121; 176 121 122; 177 127 235;
178 131 237; 179 133 119; 180 128 236; 181 129 239; 182 130 240; 183 132 134;
184 134 122; 185 119 135; 186 120 136; 187 121 137; 188 122 138; 193 110 143;
194 111 144; 195 112 145; 196 113 146; 197 114 147; 198 115 148; 199 116 149;
200 117 150; 201 95 151; 202 98 152; 203 99 153; 204 105 154; 205 106 155;
206 135 136; 207 136 137; 208 137 138; 209 138 139; 210 139 140; 211 140 141;
212 141 142; 213 143 144; 214 144 145; 215 145 146; 216 146 147; 217 147 148;
218 148 149; 219 149 150; 220 135 151; 221 151 153; 222 153 223; 223 155 143;
224 136 222; 225 137 225; 226 138 227; 227 139 229; 228 140 231; 229 141 208;
230 142 152; 231 152 154; 232 154 210; 233 156 163; 234 157 164; 235 158 165;
236 159 166; 237 160 167; 238 161 168; 239 162 170; 240 163 171; 241 164 172;
242 165 173; 243 166 174; 244 167 175; 245 168 176; 246 169 177; 247 162 163;
248 163 253; 249 164 165; 250 165 166; 251 166 167; 252 167 251; 253 168 169;
254 170 95; 255 171 179; 256 172 180; 257 173 181; 258 174 182; 259 175 183;
260 176 184; 261 177 98; 262 170 171; 263 171 254; 264 172 173; 265 173 174;
266 174 175; 267 175 252; 268 176 177; 269 178 194; 270 179 195; 271 180 196;
272 181 197; 273 182 198; 274 183 199; 275 184 200; 276 185 201; 277 178 179;
278 179 180; 279 180 181; 280 181 182; 281 182 183; 282 183 184; 283 184 185;
284 99 100; 285 100 256; 286 101 118; 287 186 106; 288 187 107; 289 188 108;
290 189 109; 291 186 187; 292 187 257; 293 188 189; 294 190 120; 295 191 121;
296 192 193; 297 193 118; 298 194 99; 299 195 100; 300 196 101; 301 197 118;
302 198 102; 303 199 103; 304 200 104; 305 201 105; 306 194 195; 307 195 196;
308 196 197; 309 197 198; 310 198 199; 311 199 200; 312 200 201; 313 202 203;
314 204 99; 315 203 204; 316 87 205; 317 206 87; 318 205 206; 319 202 207;
320 108 112; 321 208 211; 322 208 214; 323 209 212; 324 210 213; 325 209 215;
326 211 209; 327 211 216; 328 212 149; 329 213 150; 330 212 217; 331 214 218;
332 215 219; 333 216 220; 334 217 221; 335 218 152; 336 219 210; 337 220 154;
338 221 213; 339 214 216; 340 218 220; 341 215 217; 342 219 221; 343 222 224;
344 151 233; 345 223 155; 346 224 144; 347 223 234; 348 225 226; 349 222 225;
350 226 145; 351 224 226; 352 227 228; 353 225 227; 354 228 146; 355 226 228;
356 229 230; 357 227 229; 358 230 147; 359 228 230; 360 231 232; 361 229 231;

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362 232 148; 363 230 232; 364 231 208; 365 232 209; 366 141 231; 367 231 209;
368 209 148; 369 233 222; 370 234 224; 371 233 234; 372 235 131; 373 236 238;
374 235 245; 375 237 133; 376 238 243; 377 237 246; 378 239 241; 379 240 242;
380 241 244; 381 242 132; 382 239 247; 383 241 248; 384 243 190; 385 244 191;
386 238 241; 387 243 244; 388 129 238; 389 238 244; 390 244 190; 391 245 236;
392 246 238; 393 245 246; 394 247 240; 395 248 242; 396 247 248; 397 249 116;
398 250 117; 399 249 250; 400 251 168; 401 252 176; 402 251 252; 403 253 164;
404 254 172; 405 253 254; 406 255 108; 407 256 101; 408 257 188; 409 256 257;
410 257 255;
MEMBER RELEASE
147 TO 149 152 TO 154 180 181 224 TO 229 233 TO 238 247 TO 253 262 TO 268 -
277 TO 286 291 TO 293 306 TO 312 315 316 322 325 327 330 339 TO 342 344 347 -
349 351 353 355 357 359 361 363 TO 365 371 374 377 382 383 386 387 393 396 -
399 402 405 409 410 START MZ
147 149 151 TO 154 247 249 TO 251 253 262 264 TO 266 268 277 TO 284 -
288 TO 291 293 TO 295 302 TO 304 306 TO 312 315 328 335 TO 342 346 -
349 TO 351 353 TO 355 357 TO 359 361 TO 365 369 TO 371 386 387 391 TO 396 -
399 TO 410 END MZ
286 END MZ
151 START MZ
313 START MY
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 78.5
ALPHA 1.2e-005
DAMP 0.02
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 9 TO 16 32 TO 39 70 TO 73 78 TO 85 101 TO 108 155 TO 166 185 TO 188 -
193 TO 200 317 397 398 PRIS YD 0.6 ZD 0.6
5 TO 8 17 20 TO 31 74 TO 77 86 89 TO 100 167 TO 170 201 TO 205 296 297 314 -
319 PRIS YD 0.4 ZD 0.4
40 TO 69 PRIS YD 0.5 ZD 0.3
109 TO 128 133 TO 135 320 PRIS YD 0.6 ZD 0.3
129 TO 132 136 TO 138 171 TO 179 182 TO 184 206 TO 223 230 TO 232 239 246 -
254 261 269 276 287 298 305 324 329 345 372 375 379 381 PRIS YD 0.6 ZD 0.4
139 141 143 TO 146 294 295 PRIS YD 0.5 ZD 0.6
MEMBER PROPERTY EUROPEAN
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224 TO 229 321 323 326 328 343 346 348 350 352 354 356 358 360 -
362 TABLE ST HE800A
236 TO 238 243 TO 245 258 TO 260 273 TO 275 302 TO 304 322 325 327 -
330 TO 342 344 347 369 TO 371 374 377 382 383 391 TO 396 TABLE ST HE500A
147 TO 149 247 TO 253 263 267 268 277 TO 285 291 292 308 311 312 400 401 403 -
404 406 TO 408 TABLE ST HE300A
140 142 150 233 TO 235 240 TO 242 255 TO 257 270 TO 272 288 TO 290 -
299 TO 301 TABLE ST HE400A
151 TO 154 262 264 TO 266 286 293 306 307 309 310 399 402 405 409 -
410 TABLE ST IPE300
180 181 373 376 378 380 384 385 TABLE ST HE800A
313 315 316 TABLE ST IPE240
318 TABLE ST L65X65X6
366 TO 368 TABLE SD L130X130X10 SP 0.008
349 351 353 355 357 359 361 363 386 387 TABLE SD L90X90X8 SP 0.008
364 365 388 TO 390 TABLE SD L100X100X8 SP 0.008
CONSTANTS
BETA 90 MEMB 313
MATERIAL CONCRETE MEMB 1 TO 17 20 TO 86 89 TO 139 141 143 TO 146 155 TO 179 -
182 TO 188 193 TO 223 230 TO 232 239 246 254 261 269 276 287 294 TO 298 305 -
314 317 319 320 324 329 345 372 375 379 381 397 398
MATERIAL STEEL MEMB 140 142 147 TO 154 180 181 224 TO 229 233 TO 238 -
240 TO 245 247 TO 253 255 TO 260 262 TO 268 270 TO 275 277 TO 286 -
288 TO 293 299 TO 304 306 TO 313 315 316 318 321 TO 323 325 TO 328 -
330 TO 344 346 TO 371 373 374 376 TO 378 380 382 TO 396 399 TO 410
SUPPORTS
1 TO 17 20 TO 39 192 207 FIXED
******************
MEMBER TRUSS
318 366 TO 368 388 TO 390
MEMBER OFFSET
50 TO 56 119 TO 125 213 TO 219 START 0 0 0.15
50 TO 56 119 TO 125 213 TO 219 END 0 0 0.15
40 TO 49 109 TO 118 171 TO 173 206 TO 212 START 0 0 -0.15
40 TO 49 109 TO 118 171 TO 173 206 TO 212 END 0 0 -0.15
64 67 133 136 182 230 START 0.15 0 0
66 69 135 138 184 329 END 0.15 0 0
57 60 126 129 177 220 START -0.15 0 0
59 63 128 132 179 223 END -0.15 0 0
177 END -0.0790761 0 0
372 START -0.0790761 0 0
182 END 0.105167 0 0
379 START 0.105167 0 0
379 END 0.0283091 0 0
381 START 0.0283091 0 0
********************
DEFINE IBC 2003
SDS 0.067 SD1 0.013 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 4 PX 0.26 PZ 0.26
SELFWEIGHT 1
FLOOR WEIGHT
```

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YRANGE 918.95 918.95 FLOAD 4.38
MEMBER WEIGHT
180 181 224 TO 229 321 323 326 328 343 346 348 350 352 354 356 358 360 362 -
373 376 378 380 384 385 UNI 24.7
177 TO 179 182 TO 184 220 TO 223 230 TO 232 324 329 345 372 375 379 -
381 UNI 12.36
112 TO 125 129 TO 132 136 TO 139 141 TO 148 239 246 TO 254 261 269 276 284 -
285 287 289 298 305 320 400 403 406 407 UNI 30
40 TO 69 UNI 26.75
109 TO 111 126 TO 128 133 TO 135 UNI 15
315 316 CON 2.2 0.25
315 316 CON 2.2 1.425
129 315 CON 2.2 1.725
129 315 CON 2.2 2.95
399 CON 2.2 3
399 CON 2.2 4.2
154 399 CON 2.2 4.5
154 399 CON 2.2 5.7
***************
FLOOR WEIGHT
YRANGE 918.95 918.95 FLOAD 2.5
MEMBER WEIGHT
315 316 CON 4.8 0.25
315 316 CON 4.8 1.425
129 315 CON 4.8 1.725
129 315 CON 4.8 2.95
399 CON 4.8 3
399 CON 4.8 4.2
154 399 CON 4.8 4.5
154 399 CON 4.8 5.7
************
MEMBER WEIGHT
277 TO 283 CON 50 2.7
240 245 402 405 CON 17.5 0.725
***************
LOAD 11 SEISMIC LOAD Z (SLX)
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
***********
LOAD 12 SEISMIC LOAD Z (SLZ)
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
**************
LOAD 1 DEAD LOAD (DL)
SELFWEIGHT Y -1
ONEWAY LOAD
YRANGE 918.95 918.95 ONE -3.72 ZRANGE 13 26  GY
MEMBER LOAD
```

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180 181 224 TO 229 321 323 326 328 343 346 348 350 352 354 356 358 360 362 -
373 376 378 380 384 385 UNI GY -24.7
177 TO 179 182 TO 184 220 TO 223 230 TO 232 324 329 345 372 375 379 -
381 UNI GY -12.36
112 TO 125 129 TO 132 136 TO 139 141 143 TO 148 239 246 TO 254 261 269 276 -
284 285 287 298 305 320 400 403 406 409 410 UNI GY -27.5
40 TO 69 UNI GY -26.75
109 TO 111 126 TO 128 133 TO 135 UNI GY -15
315 316 CON GY -2.2 0.25
315 316 CON GY -2.2 1.425
315 CON GY -2.2 1.725
315 CON GY -2.2 2.95
129 399 CON GY -2.2 3
399 CON GY -2.2 4.2
154 399 CON GY -2.2 4.5
154 399 CON GY -2.2 5.7
********************
LOAD 2 LIVE LOAD (LL)
ONEWAY LOAD
YRANGE 918.95 918.95 ONE -5 ZRANGE 13 26  GY
MEMBER LOAD
180 181 224 TO 229 321 323 326 328 343 346 348 350 352 354 356 358 360 362 -
373 376 378 380 384 385 UNI GY -3
177 TO 179 182 TO 184 220 TO 223 230 TO 232 324 329 345 372 375 379 -
381 UNI GY -1.5
315 316 CON GY -9.6 0.25
315 316 CON GY -9.6 1.425
315 CON GY -9.6 1.725
315 CON GY -9.6 2.95
399 CON GY -9.6 3
399 CON GY -9.6 4.2
154 399 CON GY -9.6 4.5
154 399 CON GY -9.6 5.7
***************
LOAD 3 SUPERIMPOSED LOAD 1 (SDL1)
MEMBER LOAD
277 TO 283 CON GY -50 2.7
245 402 405 CON GY -17.5 0.725
***************
LOAD 4 SUPERIMPOSED LOAD 2 (SDL2)
MEMBER LOAD
321 343 348 352 356 360 CON GY -60 0.525001
373 396 CON GY -35 1.5
267 CON GY -35 3
404 CON GY -35 1.5
***************
LOAD 9 TEMERATURE LOAD 1
TEMPERATURE LOAD
1 TO 17 20 TO 86 89 TO 188 193 TO 405 TEMP 25
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LOAD 10 TEMERATURE LOAD 2
TEMPERATURE LOAD
1 TO 17 20 TO 86 89 TO 188 193 TO 405 TEMP -25
*************************
*Load Combination for servicibilty
LOAD COMB 101 DL+SDL1
1 1.0 3 1.0
LOAD COMB 102 DL+SDL2
1 1.0 4 1.0
LOAD COMB 103 DL + SDL1 + TL1 + LL
1 1.0 3 1.0 2 1.0
LOAD COMB 104 DL + SDL2 + TL1 + LL
1 1.0 4 1.0 2 1.0
LOAD COMB 105 DL + SDL1 + TL2 + LL
1 1.0 3 1.0 2 1.0
LOAD COMB 106 DL + SDL2 + TL2 + LL
1 1.0 4 1.0 2 1.0
LOAD COMB 115 DL + SDL1 + 0.75 LL + 0.525 EL(+X)
1 1.0 3 1.0 2 0.75 11 0.525
LOAD COMB 116 DL + SDL1 + 0.75 LL + 0.525 EL(-X)
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD COMB 117 DL + SDL1 + 0.75 LL + 0.525 EL(+Z)
1 1.0 3 1.0 2 0.75 12 0.525
LOAD COMB 118 DL + SDL1 + 0.75 LL + 0.525 EL(-Z)
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD COMB 119 DL + SDL2 + 0.75 LL + 0.525 EL(+X)
1 1.0 4 1.0 2 0.75 11 0.525
LOAD COMB 120 DL + SDL2 + 0.75 LL + 0.525 EL(-X)
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD COMB 121 DL + SDL2 + 0.75 LL + 0.525 EL(+Z)
1 1.0 4 1.0 2 0.75 12 0.525
LOAD COMB 122 DL + SDL2 + 0.75 LL + 0.525 EL(-Z)
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD COMB 131 0.6 (DL + SDL1) + 0.7 EL(+X)
1 0.6 3 0.6 11 0.7
LOAD COMB 132 0.6 (DL + SDL1) + 0.7 EL(-X)
1 0.6 3 0.6 11 -0.7
LOAD COMB 133 0.6 (DL + SDL1) + 0.7 EL(+Z)
1 0.6 3 0.6 12 0.7
LOAD COMB 134 0.6 (DL + SDL1) + 0.7 EL(-Z)
1 0.6 3 0.6 12 -0.7
LOAD COMB 135 0.6 (DL + SDL2) + 0.7 EL(+X)
1 0.6 4 0.6 11 0.7
LOAD COMB 136 0.6 (DL + SDL2) + 0.7 EL(-X)
1 0.6 4 0.6 11 -0.7
LOAD COMB 137 0.6 (DL + SDL2) + 0.7 EL(+Z)
1 0.6 4 0.6 12 0.7
LOAD COMB 138 0.6 (DL + SDL2) + 0.7 EL(-Z)
1 0.6 4 0.6 12 -0.7
****
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*Load Combination for Structural design
LOAD COMB 201 1.4 (DL+SDL1)
1 1.4 3 1.4
LOAD COMB 202 1.4 (DL+SDL2)
1 1.4 4 1.4
LOAD COMB 203 1.2 (DL + SDL1 + TL1) + 1.6 LL
1 1.2 3 1.2 2 1.6
LOAD COMB 204 1.2 (DL + SDL1 + TL2) + 1.6 LL
1 1.2 3 1.2 2 1.6
LOAD COMB 205 1.2 (DL + SDL2 + TL1) + 1.6 LL
1 1.2 4 1.2 2 1.6
LOAD COMB 206 1.2 (DL + SDL2 + TL2) + 1.6 LL
1 1.2 4 1.2 2 1.6
LOAD COMB 215 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
1 1.2 3 1.2 2 0.5 11 1.0
LOAD COMB 216 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD COMB 217 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
1 1.2 3 1.2 2 0.5 12 1.0
LOAD COMB 218 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD COMB 219 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
1 1.2 4 1.2 2 0.5 11 1.0
LOAD COMB 220 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD COMB 221 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
1 1.2 4 1.2 2 0.5 12 1.0
LOAD COMB 222 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD COMB 231 0.9 (DL + SDL1) + 1.0 EL(+X)
1 0.9 3 0.9 11 1.0
LOAD COMB 232 0.9 (DL + SDL1) + 1.0 EL(-X)
1 0.9 3 0.9 11 -1.0
LOAD COMB 233 0.9 (DL + SDL1) + 1.0 EL(+Z)
1 0.9 3 0.9 12 1.0
LOAD COMB 234 0.9 (DL + SDL1) + 1.0 EL(-Z)
1 0.9 3 0.9 12 -1.0
LOAD COMB 235 0.9 (DL + SDL2) + 1.0 EL(+X)
1 0.9 3 0.9 11 1.0
LOAD COMB 236 0.9 (DL + SDL2) + 1.0 EL(-X)
1 0.9 3 0.9 11 -1.0
LOAD COMB 237 0.9 (DL + SDL2) + 1.0 EL(+Z)
1 0.9 3 0.9 12 1.0
LOAD COMB 238 0.9 (DL + SDL2) + 1.0 EL(-Z)
1 0.9 3 0.9 12 -1.0
PERFORM ANALYSIS
*STEEL TAKE OFF LIST 1 TO 17 20 TO 86 89 TO 188 193 TO 408
*LOAD LIST 201 TO 206 215 TO 222 231 TO 238
PARAMETER 1
*CODE LRFD
```

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*BEAM 1 MEMB 140 142 147 TO 154 180 181 224 TO 229 233 TO 238 240 TO 245 247 -
*248 TO 253 255 TO 260 262 TO 268 270 TO 275 277 TO 286 288 TO 293 299 TO 304 -
*306 TO 313 315 316 318 321 TO 323 325 TO 328 330 TO 344 346 TO 371 373 374 -
*376 TO 378 380 382 TO 396 400 401 403 404 406 TO 408
*NSF 0.9 MEMB 140 142 147 TO 154 180 181 224 TO 229 233 TO 238 240 TO 245 247 -
*248 TO 253 255 TO 260 262 TO 268 270 TO 275 277 TO 286 288 TO 293 299 TO 304 -
*306 TO 313 315 316 318 321 TO 323 325 TO 328 330 TO 344 346 TO 371 373 374 -
*376 TO 378 380 382 TO 396 400 401 403 404 406 TO 408
*FU 41000 MEMB 140 142 147 TO 154 180 181 224 TO 229 233 TO 238 240 TO 245 -
*247 TO 253 255 TO 260 262 TO 268 270 TO 275 277 TO 286 288 TO 293 -
*299 TO 304 306 TO 313 315 316 318 321 TO 323 325 TO 328 330 TO 344 -
*346 TO 371 373 374 376 TO 378 380 382 TO 396 400 401 403 404 406 TO 408
*LY 0.5 MEMB 140 142 147 TO 154 233 TO 238 240 TO 245 247 TO 253 255 TO 260 -
*262 TO 268 270 TO 275 277 TO 286 288 TO 293 299 TO 304 306 TO 312 400 401 -
*403 404 406 TO 408
*LZ 15.35 MEMB 180 181 224 TO 229 321 323 326 328 343 346 348 350 352 354 356 -
*358 360 362 373 376 378 380 384 385
*LZ 7.5 MEMB 140 142 150 233 TO 238 240 TO 245 255 TO 260 270 TO 275 -
*288 TO 290 299 TO 304 320
*TRACK 0 MEMB 140 142 147 TO 154 180 181 224 TO 229 233 TO 238 240 TO 245 247 -
*248 TO 253 255 TO 260 262 TO 268 270 TO 275 277 TO 286 288 TO 293 299 TO 304 -
*306 TO 313 315 316 318 321 TO 323 325 TO 328 330 TO 344 346 TO 371 373 374 -
*376 TO 378 380 382 TO 396 400 401 403 404 406 TO 408
*KZ 0.85 MEMB 318 366 TO 368 388 TO 390
*KY 0.85 MEMB 318 366 TO 368 388 TO 390
*CHECK CODE MEMB 140 142 147 TO 154 180 181 224 TO 229 233 TO 238 240 TO 245 -
*247 TO 253 255 TO 260 262 TO 268 270 TO 275 277 TO 286 288 TO 293 -
*299 TO 304 306 TO 313 315 316 318 321 TO 323 325 TO 328 330 TO 344 -
*346 TO 371 373 374 376 TO 378 380 382 TO 396 399 TO 410
START CONCRETE DESIGN
CODE ACI
SKZ 2 MEMB 1 TO 17 20 TO 39 70 TO 86 89 TO 108 155 TO 170 185 TO 188 -
193 TO 205 296 297 313 314 317 319 397 398
SKY 2 MEMB 1 TO 17 20 TO 39 70 TO 86 89 TO 108 155 TO 170 185 TO 188 -
193 TO 205 296 297 313 314 317 319 397 398
DESIGN BEAM 40 TO 69 109 TO 139 141 143 TO 146 171 TO 179 182 TO 184 -
206 TO 223 230 TO 232 239 246 254 261 269 276 287 294 295 298 305 324 329 -
345 372 375 379 381
DESIGN COLUMN 1 TO 17 20 TO 39 70 TO 86 89 TO 108 155 TO 170 185 TO 188 193 -
194 TO 205 296 297 314 317 319 397 398
END CONCRETE DESIGN
FINISH
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01			Outotec / L&T Revision No.:	0	1	D01	
Date	30-05-2020	29-06-2020	28.11.2023			Date	30-05-2020	29-06-2020	28.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT			Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT	
Prepared By	SNM	SNM	SNM			Prepared By	SNM	SNM	SNM	
Checked	SJM	SJM	SJM			Checked	SJM	SJM	SJM	
Approved	PRI	PRI	PRI			Approved	PRI	PRI	PRI	

ATTACHMENT 3 : STAAD OUTPUT FILE OF GOLD ROOM

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAY 29, 2020                *
*           Time=    16:10:31                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\Maaden\Gold Room\STAAD\Gold Room Civil\GOLD ROOM CIVIL R0.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 20-APR-20
     4. JOB NAME MMGP/GOLD ROOM
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045
     7. JOB REV 0
     8. ENGINEER NAME SNM
     9. CHECKER NAME SJM
    10. APPROVED NAME PRI
    11. END JOB INFORMATION
    12. INPUT WIDTH 79
    13. UNIT METER KN
    14. JOINT COORDINATES
    15. 1 0 911.1 0; 2 5.4 911.1 0; 3 10.8 911.1 0; 4 16.2 911.1 0; 5 -0.15 911.1 3.68
    16. 6 16.35 911.1 5.855; 7 -0.15 911.1 8.155; 8 16.35 911.1 10.735
    17. 9 0 911.1 13.88; 10 5.4 911.1 13.88; 11 10.8 911.1 13.88; 12 16.2 911.1 13.88
    18. 13 21.6 911.1 13.88; 14 27 911.1 13.88; 15 33 911.1 13.88; 16 39 911.1 13.88
    19. 17 -0.15 911.1 18.355; 20 39.15 911.1 18.355; 21 -0.15 911.1 21.905
    20. 22 5.4 911.1 21.905; 23 10.8 911.1 21.905; 24 21.6 911.1 21.905
    21. 25 27 911.1 21.905; 26 33 911.1 21.905; 27 39.15 911.1 21.905
    22. 28 -0.15 911.1 25.405; 29 5.4 911.1 25.405; 30 10.8 911.1 25.405
    23. 31 16.2 911.1 25.405; 32 0 911.1 29.33; 33 5.4 911.1 29.33
    24. 34 10.8 911.1 29.33; 35 16.2 911.1 29.33; 36 21.6 911.1 29.33
    25. 37 27 911.1 29.33; 38 33 911.1 29.33; 39 39 911.1 29.33; 40 0 913.1 0
    26. 41 5.4 913.1 0; 42 10.8 913.1 0; 43 16.2 913.1 0; 44 -0.15 913.1 3.68
    27. 45 16.35 913.1 5.855; 46 -0.15 913.1 8.155; 47 16.35 913.1 10.735
    28. 48 0 913.1 13.88; 49 5.4 913.1 13.88; 50 10.8 913.1 13.88; 51 16.2 913.1 13.88
    29. 52 21.6 913.1 13.88; 53 27 913.1 13.88; 54 33 913.1 13.88; 55 39 913.1 13.88
    30. 56 -0.15 913.1 18.355; 59 39.15 913.1 18.355; 60 -0.15 913.1 21.905
    31. 61 5.4 913.1 21.905; 62 10.8 913.1 21.905; 63 21.6 913.1 21.905
    32. 64 27 913.1 21.905; 65 33 913.1 21.905; 66 39.15 913.1 21.905
    33. 67 -0.15 913.1 25.405; 68 5.4 913.1 25.405; 69 10.8 913.1 25.405
    34. 70 16.2 913.1 25.405; 71 0 913.1 29.33; 72 5.4 913.1 29.33
    35. 73 10.8 913.1 29.33; 74 16.2 913.1 29.33; 75 21.6 913.1 29.33
    36. 76 27 913.1 29.33; 77 33 913.1 29.33; 78 39 913.1 29.33; 79 0 918.95 0
    37. 80 5.4 918.95 0; 81 10.8 918.95 0; 82 16.2 918.95 0; 83 -0.15 918.95 3.68
    38. 84 16.35 918.95 5.855; 85 -0.15 918.95 8.155; 86 16.35 918.95 10.735
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39. 87 0 918.95 13.88; 88 5.4 918.95 13.88; 89 10.8 918.95 13.88
40. 90 16.2 918.95 13.88; 91 21.6 918.95 13.88; 92 27 918.95 13.88
41. 93 33 918.95 13.88; 94 39 918.95 13.88; 95 -0.15 918.95 18.355
42. 98 39.15 918.95 18.355; 99 -0.15 918.95 21.905; 100 5.4 918.95 21.905
43. 101 10.8 918.95 21.905; 102 21.6 918.95 21.905; 103 27 918.95 21.905
44. 104 33 918.95 21.905; 105 39.15 918.95 21.905; 106 -0.15 918.95 25.405
45. 107 5.4 918.95 25.405; 108 10.8 918.95 25.405; 109 16.2 918.95 25.405
46. 110 0 918.95 29.33; 111 5.4 918.95 29.33; 112 10.8 918.95 29.33
47. 113 16.2 918.95 29.33; 114 21.6 918.95 29.33; 115 27 918.95 29.33
48. 116 33 918.95 29.33; 117 39 918.95 29.33; 118 16.2 918.95 21.905
49. 119 0 922.45 13.88; 120 5.4 922.45 13.88; 121 10.8 922.45 13.88
50. 122 16.2 922.45 13.88; 127 0 922.45 0; 128 5.4 922.45 0; 129 10.8 922.45 0
51. 130 16.2 922.45 0; 131 -0.15 922.45 3.68; 132 16.35 922.45 5.855
52. 133 -0.15 922.45 8.155; 134 16.35 922.45 10.735; 135 0 925.45 13.88
53. 136 5.4 925.45 13.88; 137 10.8 925.45 13.88; 138 16.2 925.45 13.88
54. 139 21.6 925.45 13.88; 140 27 925.45 13.88; 141 33 925.45 13.88
55. 142 39 925.45 13.88; 143 0 925.45 29.33; 144 5.4 925.45 29.33
56. 145 10.8 925.45 29.33; 146 16.2 925.45 29.33; 147 21.6 925.45 29.33
57. 148 27 925.45 29.33; 149 33 925.45 29.33; 150 39 925.45 29.33
58. 151 -0.15 925.45 18.355; 152 39.15 925.45 18.355; 153 -0.15 925.45 21.905
59. 154 39.15 925.45 21.905; 155 -0.15 925.45 25.405; 156 5.4 918.95 14.48
60. 157 10.8 918.95 14.48; 158 16.2 918.95 14.48; 159 21.6 918.95 14.48
61. 160 27 918.95 14.48; 161 33 918.95 14.48; 162 -0.15 918.95 15.905
62. 163 5.4 918.95 15.905; 164 10.8 918.95 15.905; 165 16.2 918.95 15.905
63. 166 21.6 918.95 15.905; 167 27 918.95 15.905; 168 33 918.95 15.905
64. 169 39.15 918.95 15.905; 170 -0.15 918.95 17.38; 171 5.4 918.95 17.38
65. 172 10.8 918.95 17.38; 173 16.2 918.95 17.38; 174 21.6 918.95 17.38
66. 175 27 918.95 17.38; 176 33 918.95 17.38; 177 39.15 918.95 17.38
67. 178 -0.15 918.95 18.855; 179 5.4 918.95 18.855; 180 10.8 918.95 18.855
68. 181 16.2 918.95 18.855; 182 21.6 918.95 18.855; 183 27 918.95 18.855
69. 184 33 918.95 18.855; 185 39.15 918.95 18.855; 186 -0.15 918.95 23.655
70. 187 5.4 918.95 23.655; 188 10.8 918.95 23.655; 189 16.2 918.95 23.655
71. 190 5.4 922.45 13.28; 191 10.8 922.45 13.28; 192 16.2 911.1 21.905
72. 193 16.2 913.1 21.905; 194 -0.15 918.95 20.33; 195 5.4 918.95 20.33
73. 196 10.8 918.95 20.33; 197 16.2 918.95 20.33; 198 21.6 918.95 20.33
74. 199 27 918.95 20.33; 200 33 918.95 20.33; 201 39.15 918.95 20.33
75. 202 -3.2 912.95 21.905; 203 -3.2 915.95 21.905; 204 -0.15 915.95 21.905
76. 205 -1.6 918.95 13.88; 206 0 917.45 13.88; 207 -3.2 911.1 21.905
77. 208 33 925.45 18.355; 209 33 925.45 24.255; 210 39.15 925.45 24.3782
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83. 224 5.4 925.45 24.255; 225 10.8 925.45 18.355; 226 10.8 925.45 24.255
84. 227 16.2 925.45 18.355; 228 16.2 925.45 24.255; 229 21.6 925.45 18.355
85. 230 21.6 925.45 24.255; 231 27 925.45 18.355; 232 27 925.45 24.255
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87. 236 5.4 922.45 1.74; 237 -0.15 922.45 4.74; 238 5.4 922.45 4.74
88. 239 10.8 922.45 1.75; 240 16.2448 922.45 1.75; 241 10.8 922.45 4.75
89. 242 16.3217 922.45 4.75; 243 5.4 922.45 9.01; 244 10.8 922.45 9.015
90. 245 0.429076 922.45 1.74; 246 0.35 922.45 4.74; 247 15.7 922.45 1.75
91. 248 15.7 922.45 4.75; 249 33 916.025 29.33; 250 39 916.025 29.33
92. 251 31 918.95 15.905; 252 31 918.95 17.38; 253 6.9 918.95 15.905
93. 254 6.9 918.95 17.38; 255 8.8 918.95 25.405; 256 8.8 918.95 21.905
94. 257 8.8 918.95 23.655

95. MEMBER INCIDENCES
96. 1 1 40; 2 2 41; 3 3 42; 4 4 43; 5 5 44; 6 6 45; 7 7 46; 8 8 47; 9 9 48
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157. 398 250 117; 399 249 250; 400 251 168; 401 252 176; 402 251 252; 403 253 164
158. 404 254 172; 405 253 254; 406 255 108; 407 256 101; 408 257 188; 409 256 257
159. 410 257 255
160. MEMBER RELEASE
161. 147 TO 149 152 TO 154 180 181 224 TO 229 233 TO 238 247 TO 253 262 TO 268 -
162. 277 TO 286 291 TO 293 306 TO 312 315 316 322 325 327 330 339 TO 342 344 347 -
163. 349 351 353 355 357 359 361 363 TO 365 371 374 377 382 383 386 387 393 396 -
164. 399 402 405 409 410 START MZ
165. 147 149 151 TO 154 247 249 TO 251 253 262 264 TO 266 268 277 TO 284 -
166. 288 TO 291 293 TO 295 302 TO 304 306 TO 312 315 328 335 TO 342 346 -
167. 349 TO 351 353 TO 355 357 TO 359 361 TO 365 369 TO 371 386 387 391 TO 396 -
168. 399 TO 410 END MZ
169. 286 END MZ
170. 151 START MZ
171. 313 START MY
172. DEFINE MATERIAL START
173. ISOTROPIC STEEL
174. E 2.05E+008
175. POISSON 0.3
176. DENSITY 78.5
177. ALPHA 1.2E-005
178. DAMP 0.02
179. TYPE STEEL
180. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
181. ISOTROPIC CONCRETE
182. E 2.17185E+007
183. POISSON 0.17
184. DENSITY 25
185. ALPHA 1E-005
186. DAMP 0.05
187. TYPE CONCRETE
188. STRENGTH FCU 27579
189. END DEFINE MATERIAL
190. MEMBER PROPERTY AMERICAN
191. 1 TO 4 9 TO 16 32 TO 39 70 TO 73 78 TO 85 101 TO 108 155 TO 166 185 TO 188 -
192. 193 TO 200 317 397 398 PRIS YD 0.6 ZD 0.6
193. 5 TO 8 17 20 TO 31 74 TO 77 86 89 TO 100 167 TO 170 201 TO 205 296 297 314 -
194. 319 PRIS YD 0.4 ZD 0.4
195. 40 TO 69 PRIS YD 0.5 ZD 0.3
196. 109 TO 128 133 TO 135 320 PRIS YD 0.6 ZD 0.3
197. 129 TO 132 136 TO 138 171 TO 179 182 TO 184 206 TO 223 230 TO 232 239 246 -
198. 254 261 269 276 287 298 305 324 329 345 372 375 379 381 PRIS YD 0.6 ZD 0.4
199. 139 141 143 TO 146 294 295 PRIS YD 0.5 ZD 0.6
200. MEMBER PROPERTY EUROPEAN
201. 224 TO 229 321 323 326 328 343 346 348 350 352 354 356 358 360 -
202. 362 TABLE ST HE800A
203. 236 TO 238 243 TO 245 258 TO 260 273 TO 275 302 TO 304 322 325 327 -
204. 330 TO 342 344 347 369 TO 371 374 377 382 383 391 TO 396 TABLE ST HE500A
205. 147 TO 149 247 TO 253 263 267 268 277 TO 285 291 292 308 311 312 400 401 403 -
206. 404 406 TO 408 TABLE ST HE300A
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  207. 140 142 150 233 TO 235 240 TO 242 255 TO 257 270 TO 272 288 TO 290 -
  208. 299 TO 301 TABLE ST HE400A
  209. 151 TO 154 262 264 TO 266 286 293 306 307 309 310 399 402 405 409 -
  210. 410 TABLE ST IPE300
  211. 180 181 373 376 378 380 384 385 TABLE ST HE800A
  212. 313 315 316 TABLE ST IPE240
  213. 318 TABLE ST L65X65X6
  214. 366 TO 368 TABLE SD L130X130X10 SP 0.008
  215. 349 351 353 355 357 359 361 363 386 387 TABLE SD L90X90X8 SP 0.008
  216. 364 365 388 TO 390 TABLE SD L100X100X8 SP 0.008
  217. CONSTANTS
  218. BETA 90 MEMB 313
  219. MATERIAL CONCRETE MEMB 1 TO 17 20 TO 86 89 TO 139 141 143 TO 146 155 TO 179 -
  220. 182 TO 188 193 TO 223 230 TO 232 239 246 254 261 269 276 287 294 TO 298 305 -
  221. 314 317 319 320 324 329 345 372 375 379 381 397 398
  222. MATERIAL STEEL MEMB 140 142 147 TO 154 180 181 224 TO 229 233 TO 238 -
  223. 240 TO 245 247 TO 253 255 TO 260 262 TO 268 270 TO 275 277 TO 286 -
  224. 288 TO 293 299 TO 304 306 TO 313 315 316 318 321 TO 323 325 TO 328 -
  225. 330 TO 344 346 TO 371 373 374 376 TO 378 380 382 TO 396 399 TO 410
  226. SUPPORTS
  227. 1 TO 17 20 TO 39 192 207 FIXED
  228. *****************
  229. MEMBER TRUSS
  230. 318 366 TO 368 388 TO 390
  231. MEMBER OFFSET
  232. 50 TO 56 119 TO 125 213 TO 219 START 0 0 0.15
  233. 50 TO 56 119 TO 125 213 TO 219 END 0 0 0.15
  234. 40 TO 49 109 TO 118 171 TO 173 206 TO 212 START 0 0 -0.15
  235. 40 TO 49 109 TO 118 171 TO 173 206 TO 212 END 0 0 -0.15
  236. 64 67 133 136 182 230 START 0.15 0 0
  237. 66 69 135 138 184 329 END 0.15 0 0
  238. 57 60 126 129 177 220 START -0.15 0 0
  239. 59 63 128 132 179 223 END -0.15 0 0
  240. 177 END -0.0790761 0 0
  241. 372 START -0.0790761 0 0
  242. 182 END 0.105167 0 0
  243. 379 START 0.105167 0 0
  244. 379 END 0.0283091 0 0
  245. 381 START 0.0283091 0 0
  246. ********************
  247. DEFINE IBC 2003
  248. SDS 0.067 SD1 0.013 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 4 PX 0.26 PZ 0.26
  249. SELFWEIGHT 1
  250. FLOOR WEIGHT
**NOTE** about Floor/OneWay Loads/Weights.
  Please note that depending on the shape of the floor you may
  have to break up the FLOOR/ONEWAY LOAD into multiple commands.
  For details please refer to Technical Reference Manual
  Section 5.32.4.2 Note d and/or "5.32.4.3 Note f.

  251. YRANGE 918.95 918.95 FLOAD 4.38
  252. MEMBER WEIGHT
  253. 180 181 224 TO 229 321 323 326 328 343 346 348 350 352 354 356 358 360 362 -
  254. 373 376 378 380 384 385 UNI 24.7
  255. 177 TO 179 182 TO 184 220 TO 223 230 TO 232 324 329 345 372 375 379 -
  256. 381 UNI 12.36
```

```
257. 112 TO 125 129 TO 132 136 TO 139 141 TO 148 239 246 TO 254 261 269 276 284 -
258. 285 287 289 298 305 320 400 403 406 407 UNI 30
259. 40 TO 69 UNI 26.75
260. 109 TO 111 126 TO 128 133 TO 135 UNI 15
261. 315 316 CON 2.2 0.25
262. 315 316 CON 2.2 1.425
263. 129 315 CON 2.2 1.725
264. 129 315 CON 2.2 2.95
265. 399 CON 2.2 3
266. 399 CON 2.2 4.2
267. 154 399 CON 2.2 4.5
268. 154 399 CON 2.2 5.7
269. ***************
270. FLOOR WEIGHT
271. YRANGE 918.95 918.95 FLOAD 2.5
272. MEMBER WEIGHT
273. 315 316 CON 4.8 0.25
274. 315 316 CON 4.8 1.425
275. 129 315 CON 4.8 1.725
276. 129 315 CON 4.8 2.95
277. 399 CON 4.8 3
278. 399 CON 4.8 4.2
279. 154 399 CON 4.8 4.5
280. 154 399 CON 4.8 5.7
281. ************
282. MEMBER WEIGHT
283. 277 TO 283 CON 50 2.7
284. 240 245 402 405 CON 17.5 0.725
285. ***************
286. LOAD 11 SEISMIC LOAD Z (SLX)
287. IBC LOAD X 1
288. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**247**	**NUMBER OF MEMBERS**	**402**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**39**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 166/ 24/ 132 DOF
TOTAL PRIMARY LOAD CASES = 1, TOTAL DEGREES OF FREEDOM = 1248
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 165 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 14.3/ 20805.6 MB

```
************************************************************
*  IBC 2003 SEISMIC LOAD ALONG X  :                        *
*     CT =  0.028 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.610 T =  0.260 Tuser = 0.260                 *
*  TIME PERIOD USED (T) =  0.260                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.020 X     26392.81   *
*                       =       527.86  KN                 *
************************************************************
```

```
289. CHANGE
290. ***********
291. LOAD 12 SEISMIC LOAD Z (SLZ)
292. IBC LOAD Z 1
293. PERFORM ANALYSIS
```

```
************************************************************
*  IBC 2003 SEISMIC LOAD ALONG Z  :                        *
*     CT =  0.028 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.610 T =  0.260 Tuser = 0.260                 *
*  TIME PERIOD USED (T) =  0.260                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.020 X     26392.81   *
*                       =       527.86  KN                 *
************************************************************
```

```
294. CHANGE
295. **************
296. LOAD 1 DEAD LOAD (DL)
297. SELFWEIGHT Y -1
298. ONEWAY LOAD
299. YRANGE 918.95 918.95 ONE -3.72 ZRANGE 13 26  GY
300. MEMBER LOAD
```

```
301. 180 181 224 TO 229 321 323 326 328 343 346 348 350 352 354 356 358 360 362 -
302. 373 376 378 380 384 385 UNI GY -24.7
303. 177 TO 179 182 TO 184 220 TO 223 230 TO 232 324 329 345 372 375 379 -
304. 381 UNI GY -12.36
305. 112 TO 125 129 TO 132 136 TO 139 141 143 TO 148 239 246 TO 254 261 269 276 -
306. 284 285 287 298 305 320 400 403 406 409 410 UNI GY -27.5
307. 40 TO 69 UNI GY -26.75
308. 109 TO 111 126 TO 128 133 TO 135 UNI GY -15
309. 315 316 CON GY -2.2 0.25
310. 315 316 CON GY -2.2 1.425
311. 315 CON GY -2.2 1.725
312. 315 CON GY -2.2 2.95
313. 129 399 CON GY -2.2 3
314. 399 CON GY -2.2 4.2
315. 154 399 CON GY -2.2 4.5
316. 154 399 CON GY -2.2 5.7
317. ********************
318. LOAD 2 LIVE LOAD (LL)
319. ONEWAY LOAD
320. YRANGE 918.95 918.95 ONE -5 ZRANGE 13 26  GY
321. MEMBER LOAD
322. 180 181 224 TO 229 321 323 326 328 343 346 348 350 352 354 356 358 360 362 -
323. 373 376 378 380 384 385 UNI GY -3
324. 177 TO 179 182 TO 184 220 TO 223 230 TO 232 324 329 345 372 375 379 -
325. 381 UNI GY -1.5
326. 315 316 CON GY -9.6 0.25
327. 315 316 CON GY -9.6 1.425
328. 315 CON GY -9.6 1.725
329. 315 CON GY -9.6 2.95
330. 399 CON GY -9.6 3
331. 399 CON GY -9.6 4.2
332. 154 399 CON GY -9.6 4.5
333. 154 399 CON GY -9.6 5.7
334. **************
335. LOAD 3 SUPERIMPOSED LOAD 1 (SDL1)
336. MEMBER LOAD
337. 277 TO 283 CON GY -50 2.7
338. 245 402 405 CON GY -17.5 0.725
339. **************
340. LOAD 4 SUPERIMPOSED LOAD 2 (SDL2)
341. MEMBER LOAD
342. 321 343 348 352 356 360 CON GY -60 0.525001
343. 373 396 CON GY -35 1.5
344. 267 CON GY -35 3
345. 404 CON GY -35 1.5
346. **************
347. LOAD 9 TEMERATURE LOAD 1
348. TEMPERATURE LOAD
349. 1 TO 17 20 TO 86 89 TO 188 193 TO 405 TEMP 25
350. ********************
351. LOAD 10 TEMERATURE LOAD 2
352. TEMPERATURE LOAD
353. 1 TO 17 20 TO 86 89 TO 188 193 TO 405 TEMP -25
```

```
354. ***************************
355. *LOAD COMBINATION FOR SERVICIBILTY
356. LOAD COMB 101 DL+SDL1
357. 1 1.0 3 1.0
358. LOAD COMB 102 DL+SDL2
359. 1 1.0 4 1.0
360. LOAD COMB 103 DL + SDL1 + TL1 + LL
361. 1 1.0 3 1.0 2 1.0
362. LOAD COMB 104 DL + SDL2 + TL1 + LL
363. 1 1.0 4 1.0 2 1.0
364. LOAD COMB 105 DL + SDL1 + TL2 + LL
365. 1 1.0 3 1.0 2 1.0
366. LOAD COMB 106 DL + SDL2 + TL2 + LL
367. 1 1.0 4 1.0 2 1.0
368. LOAD COMB 115 DL + SDL1 + 0.75 LL + 0.525 EL(+X)
369. 1 1.0 3 1.0 2 0.75 11 0.525
370. LOAD COMB 116 DL + SDL1 + 0.75 LL + 0.525 EL(-X)
371. 1 1.0 3 1.0 2 0.75 11 -0.525
372. LOAD COMB 117 DL + SDL1 + 0.75 LL + 0.525 EL(+Z)
373. 1 1.0 3 1.0 2 0.75 12 0.525
374. LOAD COMB 118 DL + SDL1 + 0.75 LL + 0.525 EL(-Z)
375. 1 1.0 3 1.0 2 0.75 12 -0.525
376. LOAD COMB 119 DL + SDL2 + 0.75 LL + 0.525 EL(+X)
377. 1 1.0 4 1.0 2 0.75 11 0.525
378. LOAD COMB 120 DL + SDL2 + 0.75 LL + 0.525 EL(-X)
379. 1 1.0 4 1.0 2 0.75 11 -0.525
380. LOAD COMB 121 DL + SDL2 + 0.75 LL + 0.525 EL(+Z)
381. 1 1.0 4 1.0 2 0.75 12 0.525
382. LOAD COMB 122 DL + SDL2 + 0.75 LL + 0.525 EL(-Z)
383. 1 1.0 4 1.0 2 0.75 12 -0.525
384. LOAD COMB 131 0.6 (DL + SDL1) + 0.7 EL(+X)
385. 1 0.6 3 0.6 11 0.7
386. LOAD COMB 132 0.6 (DL + SDL1) + 0.7 EL(-X)
387. 1 0.6 3 0.6 11 -0.7
388. LOAD COMB 133 0.6 (DL + SDL1) + 0.7 EL(+Z)
389. 1 0.6 3 0.6 12 0.7
390. LOAD COMB 134 0.6 (DL + SDL1) + 0.7 EL(-Z)
391. 1 0.6 3 0.6 12 -0.7
392. LOAD COMB 135 0.6 (DL + SDL2) + 0.7 EL(+X)
393. 1 0.6 4 0.6 11 0.7
394. LOAD COMB 136 0.6 (DL + SDL2) + 0.7 EL(-X)
395. 1 0.6 4 0.6 11 -0.7
396. LOAD COMB 137 0.6 (DL + SDL2) + 0.7 EL(+Z)
397. 1 0.6 4 0.6 12 0.7
398. LOAD COMB 138 0.6 (DL + SDL2) + 0.7 EL(-Z)
399. 1 0.6 4 0.6 12 -0.7
400. ****
401. *LOAD COMBINATION FOR STRUCTURAL DESIGN
402. LOAD COMB 201 1.4 (DL+SDL1)
403. 1 1.4 3 1.4
404. LOAD COMB 202 1.4 (DL+SDL2)
405. 1 1.4 4 1.4
406. LOAD COMB 203 1.2 (DL + SDL1 + TL1) + 1.6 LL
407. 1 1.2 3 1.2 2 1.6
408. LOAD COMB 204 1.2 (DL + SDL1 + TL2) + 1.6 LL
409. 1 1.2 3 1.2 2 1.6
```

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 410. LOAD COMB 205 1.2 (DL + SDL2 + TL1) + 1.6 LL
 411. 1 1.2 4 1.2 2 1.6
 412. LOAD COMB 206 1.2 (DL + SDL2 + TL2) + 1.6 LL
 413. 1 1.2 4 1.2 2 1.6
 414. LOAD COMB 215 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
 415. 1 1.2 3 1.2 2 0.5 11 1.0
 416. LOAD COMB 216 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
 417. 1 1.2 3 1.2 2 0.5 11 -1.0
 418. LOAD COMB 217 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
 419. 1 1.2 3 1.2 2 0.5 12 1.0
 420. LOAD COMB 218 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
 421. 1 1.2 3 1.2 2 0.5 12 -1.0
 422. LOAD COMB 219 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
 423. 1 1.2 4 1.2 2 0.5 11 1.0
 424. LOAD COMB 220 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
 425. 1 1.2 4 1.2 2 0.5 11 -1.0
 426. LOAD COMB 221 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
 427. 1 1.2 4 1.2 2 0.5 12 1.0
 428. LOAD COMB 222 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
 429. 1 1.2 4 1.2 2 0.5 12 -1.0
 430. LOAD COMB 231 0.9 (DL + SDL1) + 1.0 EL(+X)
 431. 1 0.9 3 0.9 11 1.0
 432. LOAD COMB 232 0.9 (DL + SDL1) + 1.0 EL(-X)
 433. 1 0.9 3 0.9 11 -1.0
 434. LOAD COMB 233 0.9 (DL + SDL1) + 1.0 EL(+Z)
 435. 1 0.9 3 0.9 12 1.0
 436. LOAD COMB 234 0.9 (DL + SDL1) + 1.0 EL(-Z)
 437. 1 0.9 3 0.9 12 -1.0
 438. LOAD COMB 235 0.9 (DL + SDL2) + 1.0 EL(+X)
 439. 1 0.9 3 0.9 11 1.0
 440. LOAD COMB 236 0.9 (DL + SDL2) + 1.0 EL(-X)
 441. 1 0.9 3 0.9 11 -1.0
 442. LOAD COMB 237 0.9 (DL + SDL2) + 1.0 EL(+Z)
 443. 1 0.9 3 0.9 12 1.0
 444. LOAD COMB 238 0.9 (DL + SDL2) + 1.0 EL(-Z)
 445. 1 0.9 3 0.9 12 -1.0
 446. PERFORM ANALYSIS
**WARNING** LOAD BEYOND ITS LENGTH. FULL LENGTH ASSUMED. MEMB   129 CASE    1
            LENGTH=       2.02  LOAD POINT=       3.00


 447. *STEEL TAKE OFF LIST 1 TO 17 20 TO 86 89 TO 188 193 TO 408
 448. *LOAD LIST 201 TO 206 215 TO 222 231 TO 238
 449. PARAMETER 1
 450. *CODE LRFD
 451. *BEAM 1 MEMB 140 142 147 TO 154 180 181 224 TO 229 233 TO 238 240 TO 245 247 -
 452. *248 TO 253 255 TO 260 262 TO 268 270 TO 275 277 TO 286 288 TO 293 299 TO 304 -
 453. *306 TO 313 315 316 318 321 TO 323 325 TO 328 330 TO 344 346 TO 371 373 374 -
 454. *376 TO 378 380 382 TO 396 400 401 403 404 406 TO 408
 455. *NSF 0.9 MEMB 140 142 147 TO 154 180 181 224 TO 229 233 TO 238 240 TO 245 247 -
 456. *248 TO 253 255 TO 260 262 TO 268 270 TO 275 277 TO 286 288 TO 293 299 TO 304 -
 457. *306 TO 313 315 316 318 321 TO 323 325 TO 328 330 TO 344 346 TO 371 373 374 -
 458. *376 TO 378 380 382 TO 396 400 401 403 404 406 TO 408
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459. *FU 41000 MEMB 140 142 147 TO 154 180 181 224 TO 229 233 TO 238 240 TO 245 -
460. *247 TO 253 255 TO 260 262 TO 268 270 TO 275 277 TO 286 288 TO 293 -
461. *299 TO 304 306 TO 313 315 316 318 321 TO 323 325 TO 328 330 TO 344 -
462. *346 TO 371 373 374 376 TO 378 380 382 TO 396 400 401 403 404 406 TO 408
463. *LY 0.5 MEMB 140 142 147 TO 154 233 TO 238 240 TO 245 247 TO 253 255 TO 260 -
464. *262 TO 268 270 TO 275 277 TO 286 288 TO 293 299 TO 304 306 TO 312 400 401 -
465. *403 404 406 TO 408
466. *LZ 15.35 MEMB 180 181 224 TO 229 321 323 326 328 343 346 348 350 352 354 356 -
467. *358 360 362 373 376 378 380 384 385
468. *LZ 7.5 MEMB 140 142 150 233 TO 238 240 TO 245 255 TO 260 270 TO 275 -
469. *288 TO 290 299 TO 304 320
470. *TRACK 0 MEMB 140 142 147 TO 154 180 181 224 TO 229 233 TO 238 240 TO 245 247 -
471. *248 TO 253 255 TO 260 262 TO 268 270 TO 275 277 TO 286 288 TO 293 299 TO 304 -
472. *306 TO 313 315 316 318 321 TO 323 325 TO 328 330 TO 344 346 TO 371 373 374 -
473. *376 TO 378 380 382 TO 396 400 401 403 404 406 TO 408
474. *KZ 0.85 MEMB 318 366 TO 368 388 TO 390
475. *KY 0.85 MEMB 318 366 TO 368 388 TO 390
476. *CHECK CODE MEMB 140 142 147 TO 154 180 181 224 TO 229 233 TO 238 240 TO 245 -
477. *247 TO 253 255 TO 260 262 TO 268 270 TO 275 277 TO 286 288 TO 293 -
478. *299 TO 304 306 TO 313 315 316 318 321 TO 323 325 TO 328 330 TO 344 -
479. *346 TO 371 373 374 376 TO 378 380 382 TO 396 399 TO 410
480. START CONCRETE DESIGN
481. CODE ACI
482. SKZ 2 MEMB 1 TO 17 20 TO 39 70 TO 86 89 TO 108 155 TO 170 185 TO 188 -
483. 193 TO 205 296 297 313 314 317 319 397 398
484. SKY 2 MEMB 1 TO 17 20 TO 39 70 TO 86 89 TO 108 155 TO 170 185 TO 188 -
485. 193 TO 205 296 297 313 314 317 319 397 398
486. DESIGN BEAM 40 TO 69 109 TO 139 141 143 TO 146 171 TO 179 182 TO 184 -
487. 206 TO 223 230 TO 232 239 246 254 261 269 276 287 294 295 298 305 324 329 -
488. 345 372 375 379 381
```

ACI 318-11 BEAM NO. 40 DESIGN RESULTS

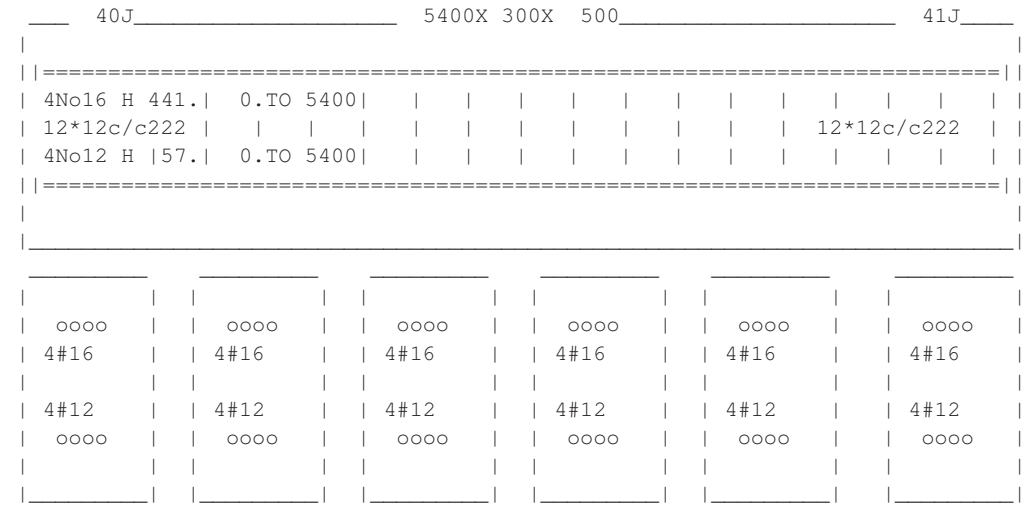
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	441.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 40 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 67.42 KNS Vc= 117.20 KNS Vs= 0.00 KNS
Tu= 0.44 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM

AT END SUPPORT - Vu= 70.65 KNS Vc= 116.23 KNS Vs= 0.00 KNS
Tu= 0.44 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM



ACI 318-11 BEAM NO. 41 DESIGN RESULTS

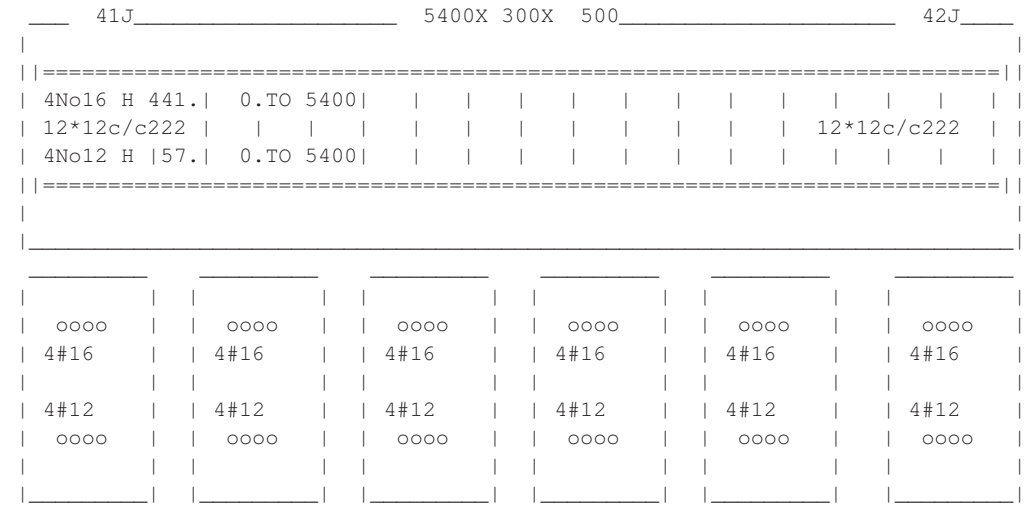
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	441.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 41 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 68.64 KNS Vc= 116.24 KNS Vs= 0.00 KNS
Tu= 0.04 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM

AT END SUPPORT - Vu= 69.43 KNS Vc= 116.04 KNS Vs= 0.00 KNS
Tu= 0.04 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM



ACI 318-11 BEAM NO. 42 DESIGN RESULTS

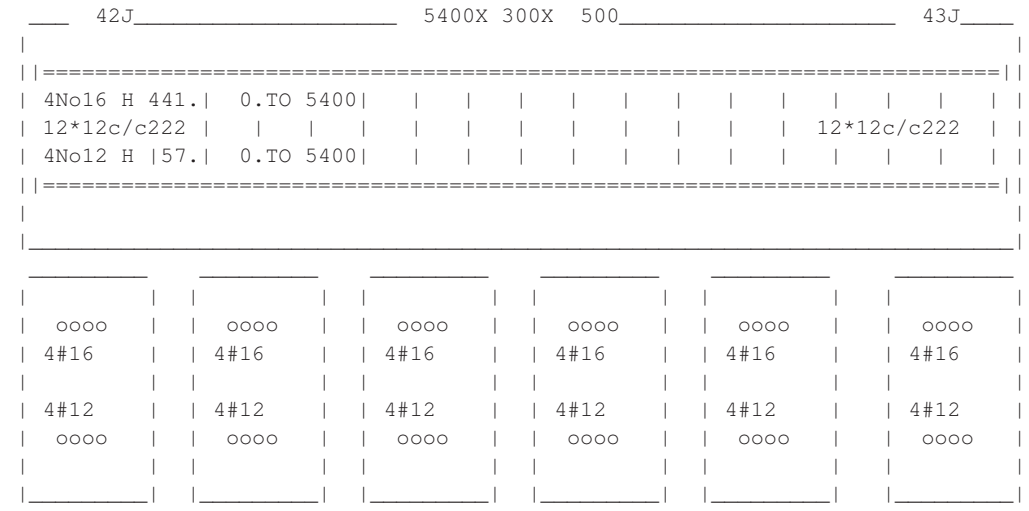
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	441.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 42 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 69.74 KNS Vc= 116.42 KNS Vs= 0.00 KNS
Tu= 0.95 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2264. MM

AT END SUPPORT - Vu= 68.34 KNS Vc= 116.83 KNS Vs= 0.00 KNS
Tu= 0.95 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2264. MM



ACI 318-11 BEAM NO. 43 DESIGN RESULTS

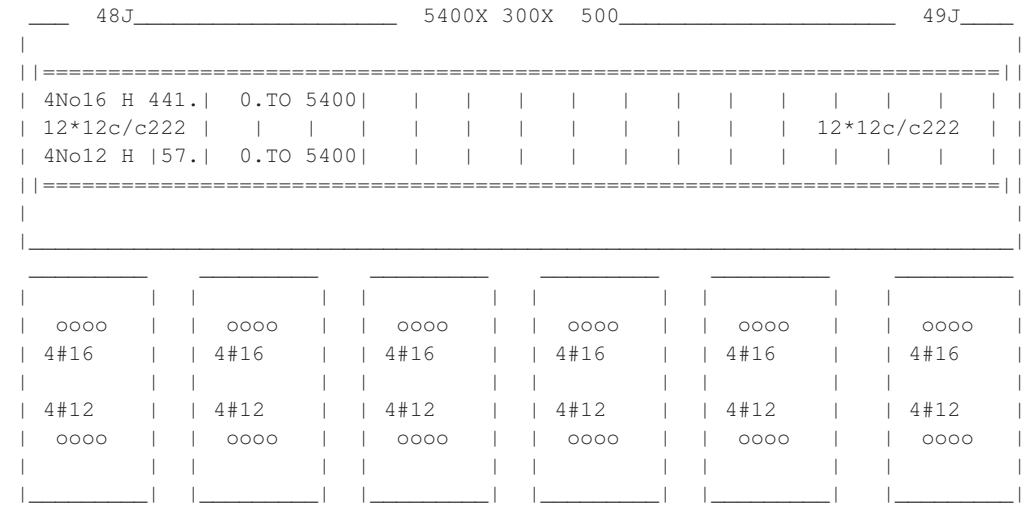
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	441.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 43 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 68.88 KNS Vc= 116.40 KNS Vs= 0.00 KNS
Tu= 0.66 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM

AT END SUPPORT - Vu= 69.19 KNS Vc= 116.32 KNS Vs= 0.00 KNS
Tu= 0.66 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM



ACI 318-11 BEAM NO. 44 DESIGN RESULTS

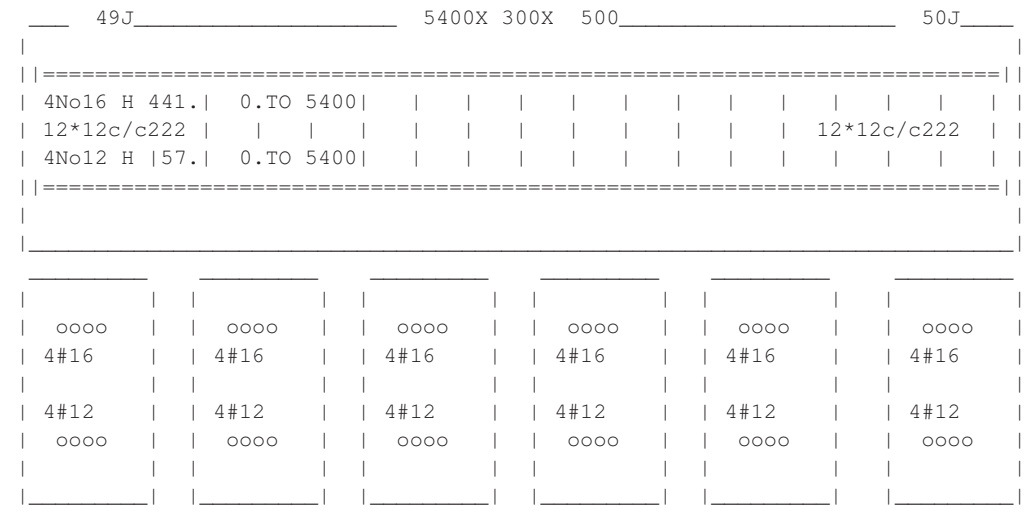
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	441.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 44 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 69.42 KNS Vc= 115.83 KNS Vs= 0.00 KNS
Tu= 0.15 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM

AT END SUPPORT - Vu= 68.65 KNS Vc= 116.01 KNS Vs= 0.00 KNS
Tu= 0.15 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM



ACI 318-11 BEAM NO. 45 DESIGN RESULTS

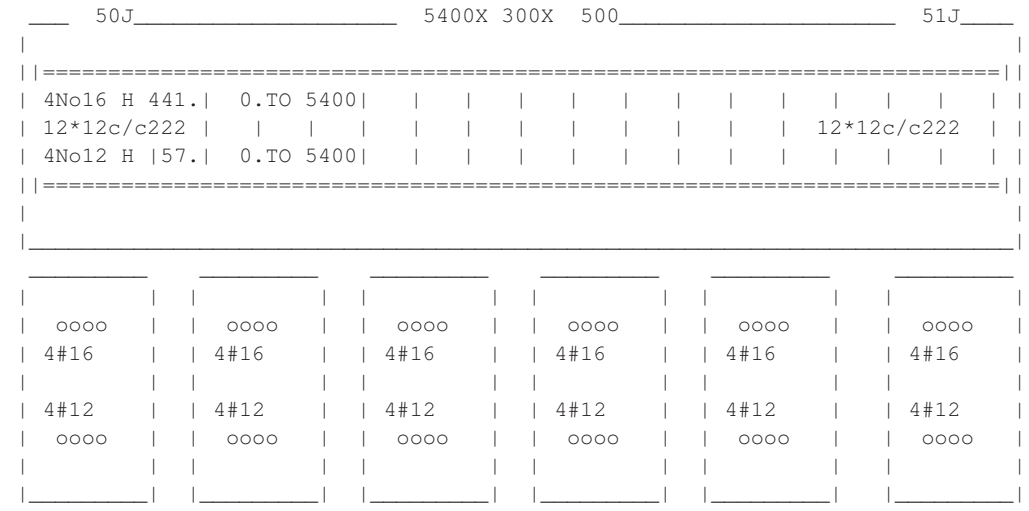
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	441.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 45 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 69.20 KNS Vc= 115.81 KNS Vs= 0.00 KNS
Tu= 0.52 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2264. MM

AT END SUPPORT - Vu= 68.87 KNS Vc= 115.88 KNS Vs= 0.00 KNS
Tu= 0.52 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2264. MM



ACI 318-11 BEAM NO. 46 DESIGN RESULTS

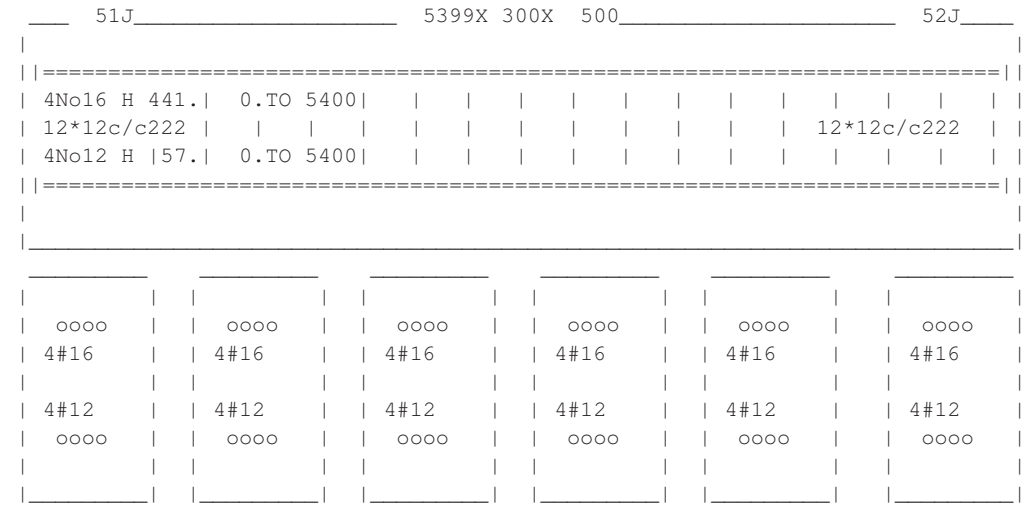
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	441.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 46 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 69.33 KNS Vc= 115.85 KNS Vs= 0.00 KNS
Tu= 0.21 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM

AT END SUPPORT - Vu= 68.74 KNS Vc= 115.99 KNS Vs= 0.00 KNS
Tu= 0.21 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM



ACI 318-11 BEAM NO. 47 DESIGN RESULTS

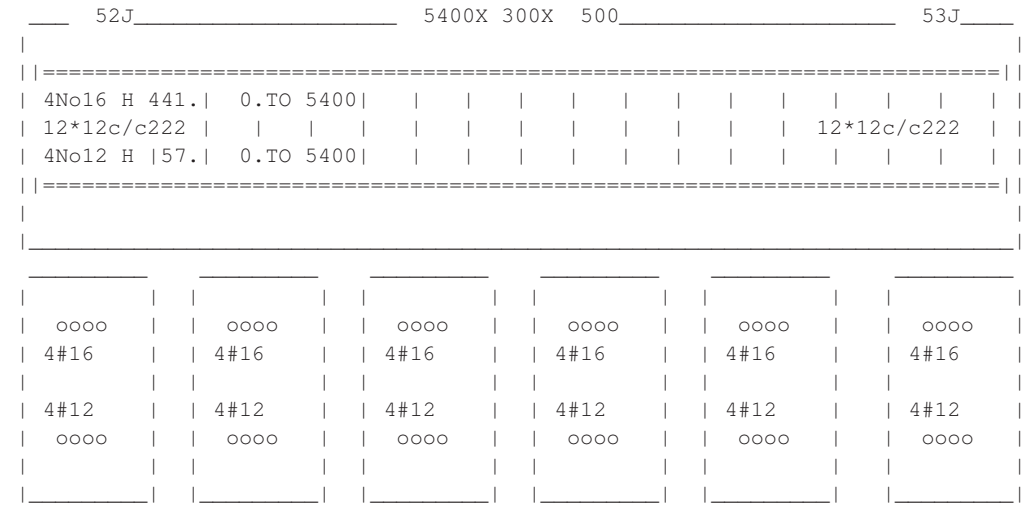
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	441.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 47 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 69.28 KNS Vc= 115.77 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2264. MM

AT END SUPPORT - Vu= 68.79 KNS Vc= 115.88 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2264. MM



ACI 318-11 BEAM NO. 48 DESIGN RESULTS

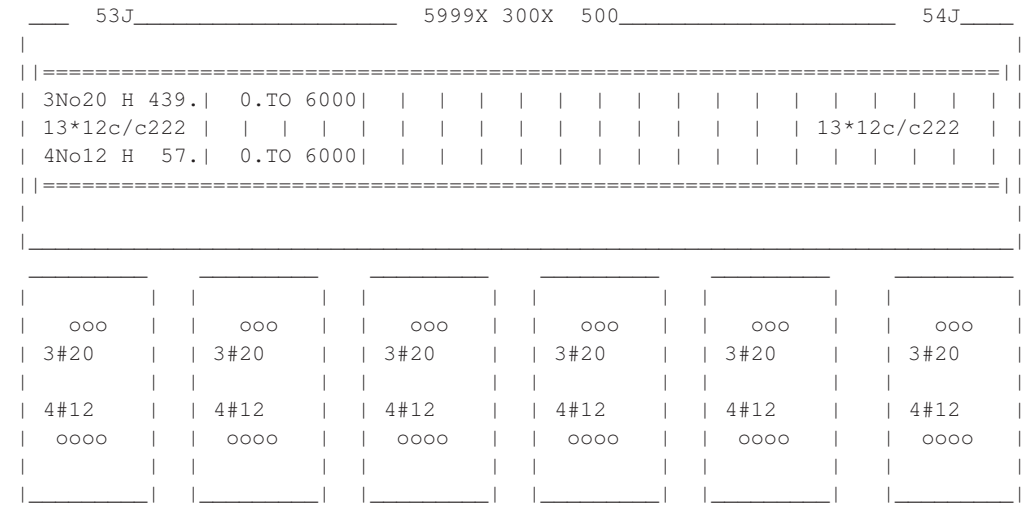
LEN - 6000. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	6000.	YES	YES
2	439.	3 - 20MM	0.	6000.	YES	YES

B E A M N O. 48 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 78.48 KNS Vc= 115.28 KNS Vs= 0.00 KNS
Tu= 0.04 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2563. MM

AT END SUPPORT - Vu= 77.89 KNS Vc= 115.38 KNS Vs= 0.00 KNS
Tu= 0.04 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2563. MM



ACI 318-11 BEAM NO. 49 DESIGN RESULTS

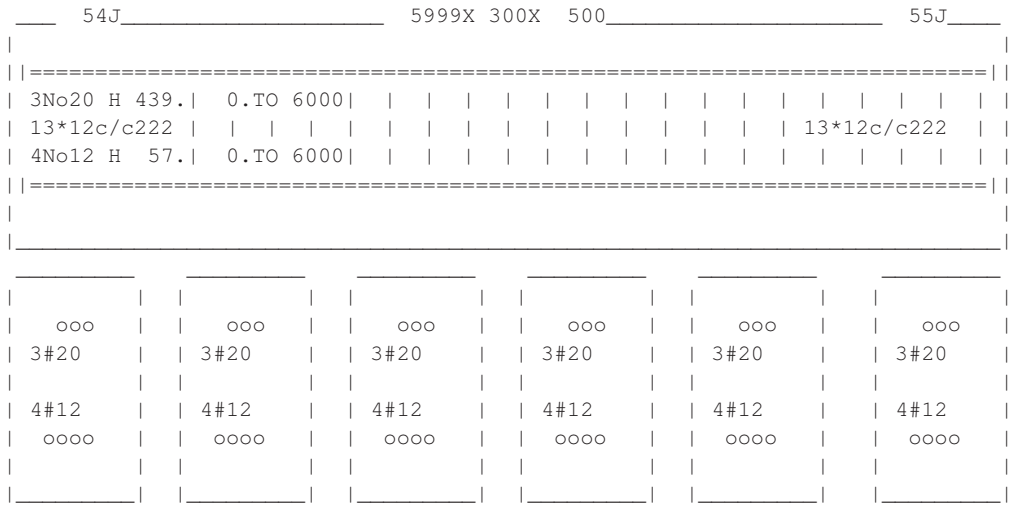
LEN - 6000. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	6000.	YES	YES
2	439.	3 - 20MM	0.	6000.	YES	YES

B E A M N O. 49 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 79.53 KNS Vc= 115.25 KNS Vs= 0.00 KNS
Tu= 1.11 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2563. MM

AT END SUPPORT - Vu= 76.84 KNS Vc= 115.77 KNS Vs= 0.00 KNS
Tu= 1.11 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2563. MM



ACI 318-11 BEAM NO. 50 DESIGN RESULTS

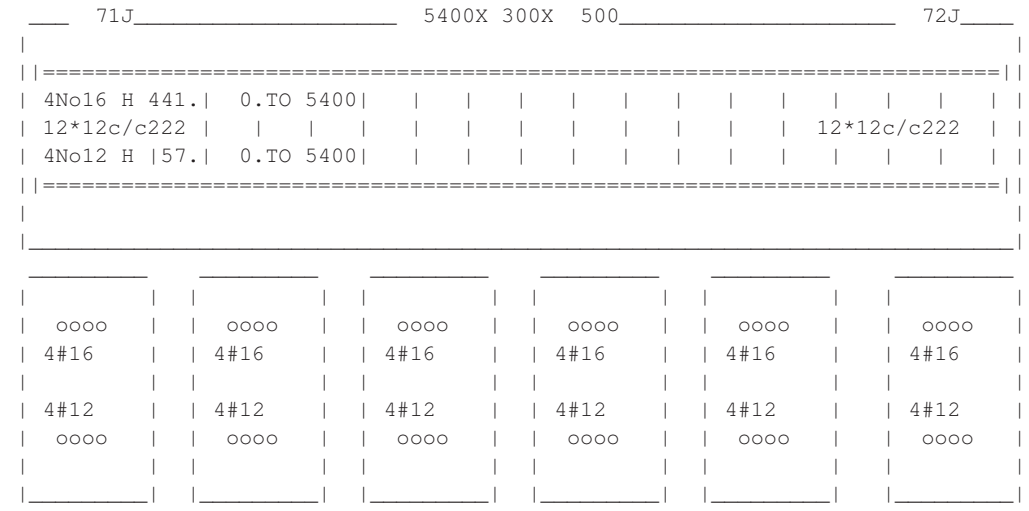
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	441.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 50 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 67.61 KNS Vc= 116.87 KNS Vs= 0.00 KNS
Tu= 0.60 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM

AT END SUPPORT - Vu= 70.47 KNS Vc= 116.09 KNS Vs= 0.00 KNS
Tu= 0.60 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM



ACI 318-11 BEAM NO. 51 DESIGN RESULTS

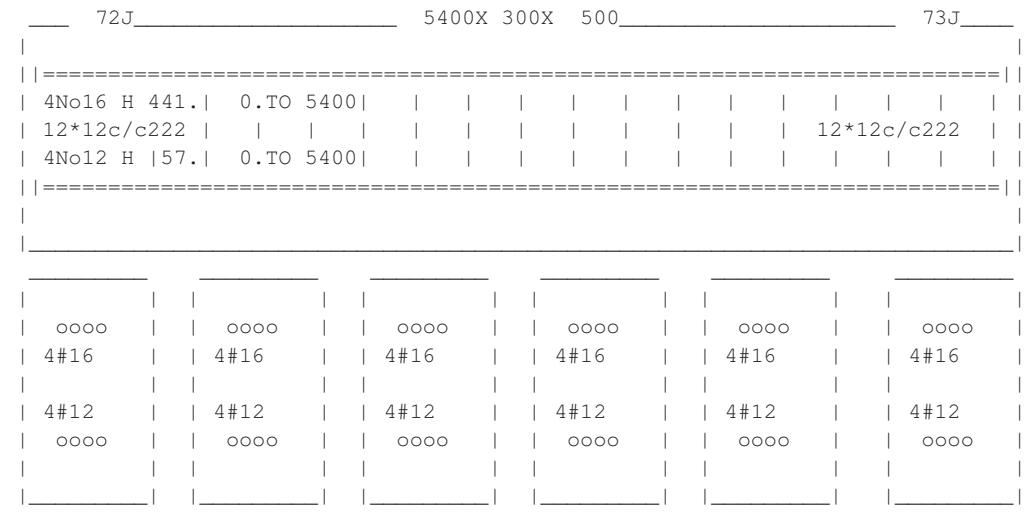
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	441.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 51 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 68.66 KNS Vc= 116.06 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM

AT END SUPPORT - Vu= 69.41 KNS Vc= 115.88 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM



ACI 318-11 BEAM NO. 52 DESIGN RESULTS

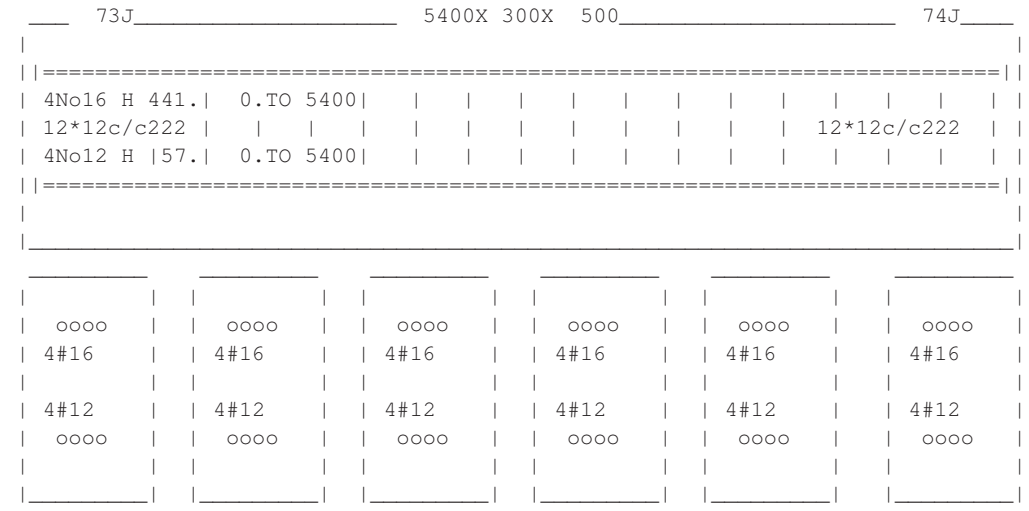
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	441.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 52 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 68.55 KNS Vc= 116.02 KNS Vs= 0.00 KNS
Tu= 0.20 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2264. MM

AT END SUPPORT - Vu= 69.53 KNS Vc= 115.80 KNS Vs= 0.00 KNS
Tu= 0.20 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2264. MM



ACI 318-11 BEAM NO. 53 DESIGN RESULTS

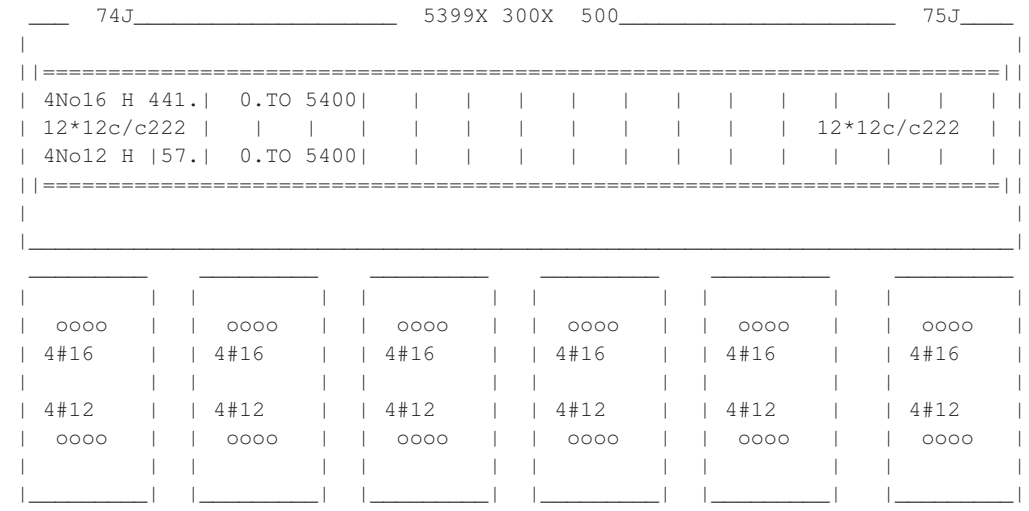
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	441.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 53 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 68.68 KNS Vc= 115.97 KNS Vs= 0.00 KNS
Tu= 0.29 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM

AT END SUPPORT - Vu= 69.39 KNS Vc= 115.81 KNS Vs= 0.00 KNS
Tu= 0.29 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2263. MM



ACI 318-11 BEAM NO. 54 DESIGN RESULTS

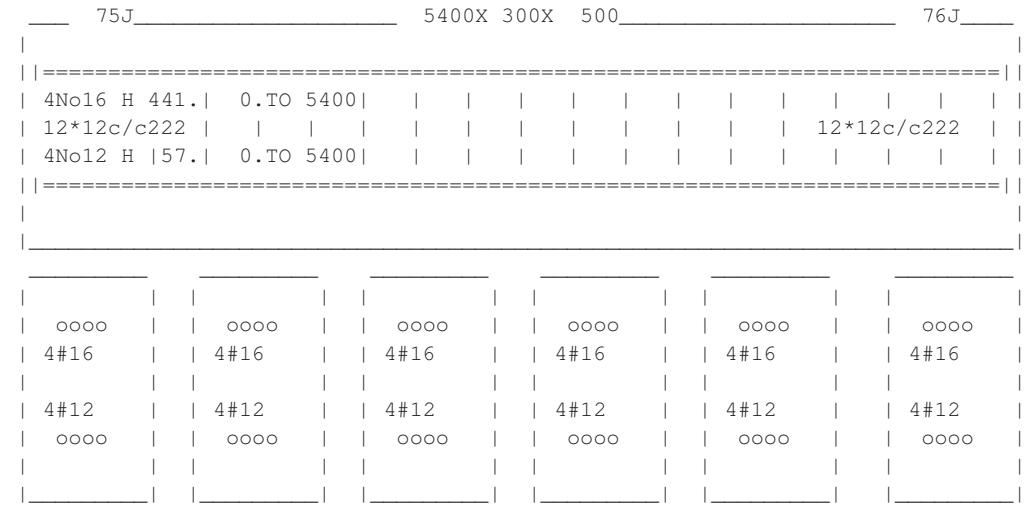
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	441.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 54 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 68.46 KNS Vc= 115.94 KNS Vs= 0.00 KNS
Tu= 0.11 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2264. MM

AT END SUPPORT - Vu= 69.62 KNS Vc= 115.68 KNS Vs= 0.00 KNS
Tu= 0.11 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2264. MM



ACI 318-11 BEAM NO. 55 DESIGN RESULTS

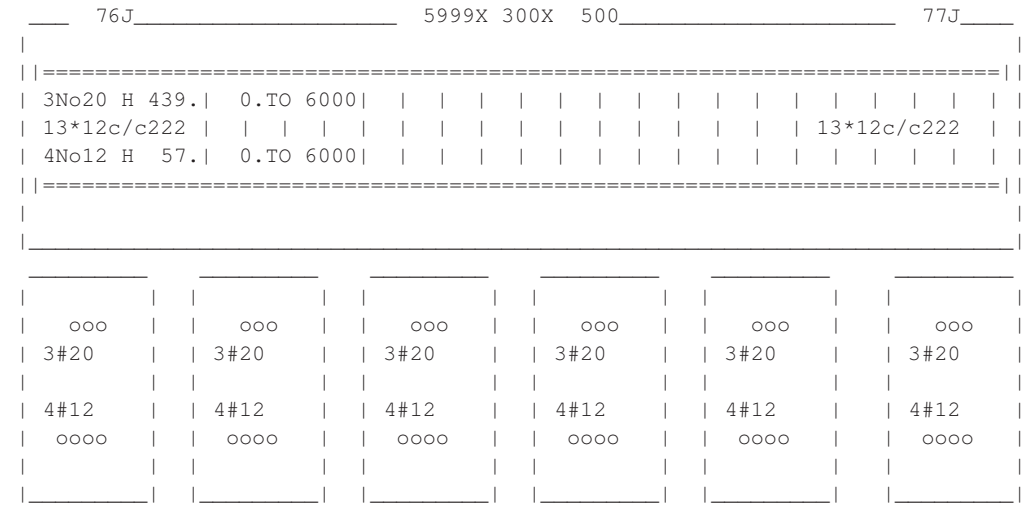
LEN - 6000. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	6000.	YES	YES
2	439.	3 - 20MM	0.	6000.	YES	YES

B E A M N O. 55 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 77.77 KNS Vc= 115.39 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2563. MM

AT END SUPPORT - Vu= 78.60 KNS Vc= 115.24 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2563. MM



ACI 318-11 BEAM NO. 56 DESIGN RESULTS

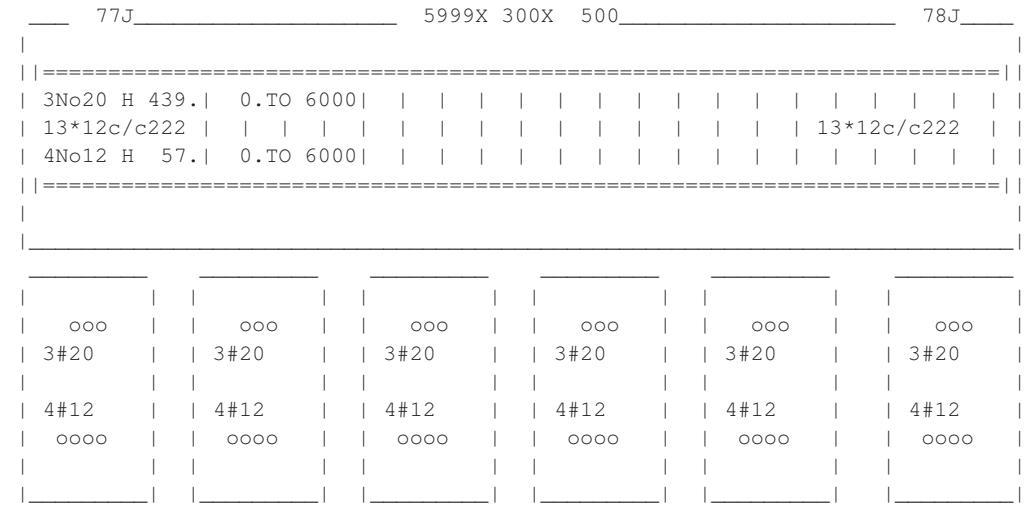
LEN - 6000. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	6000.	YES	YES
2	439.	3 - 20MM	0.	6000.	YES	YES

B E A M N O. 56 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 79.04 KNS Vc= 115.35 KNS Vs= 0.00 KNS
Tu= 1.41 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2563. MM

AT END SUPPORT - Vu= 77.34 KNS Vc= 115.68 KNS Vs= 0.00 KNS
Tu= 1.41 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2563. MM



ACI 318-11 BEAM NO. 57 DESIGN RESULTS

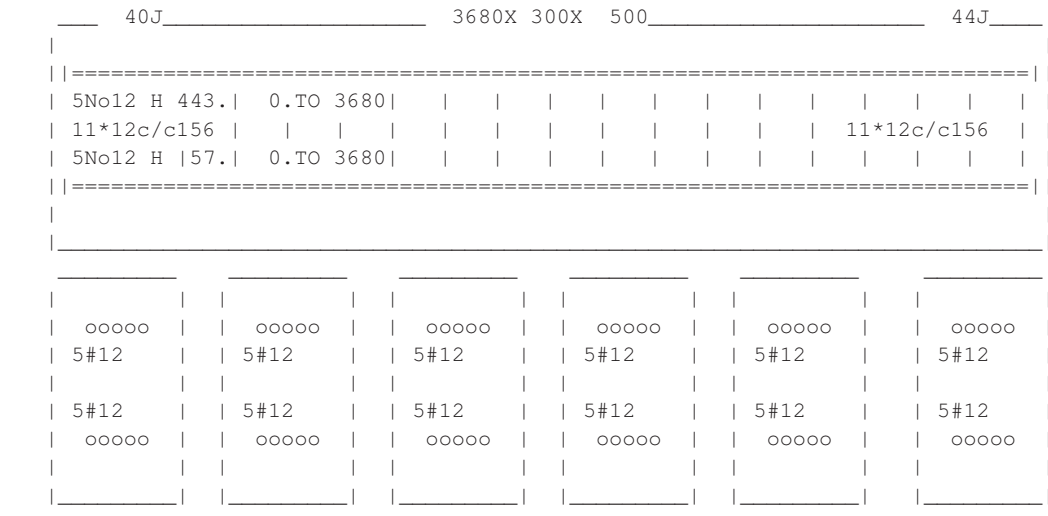
LEN - 3680. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	5 - 12MM	0.	3680.	YES	YES
2	443.	5 - 12MM	0.	3680.	YES	YES

B E A M N O. 57 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 34.95 KNS Vc= 49.27 KNS Vs= 0.00 KNS
Tu= 2.48 KN-MET Tc= NaN KN-MET Ts= 3.3 KN-MET LOAD 10
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1404. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 0.67 SQ.CM.

AT END SUPPORT - Vu= 34.95 KNS Vc= 49.27 KNS Vs= 0.00 KNS
Tu= 2.48 KN-MET Tc= NaN KN-MET Ts= 3.3 KN-MET LOAD 10
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1404. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 0.67 SQ.CM.



ACI 318-11 BEAM NO. 58 DESIGN RESULTS

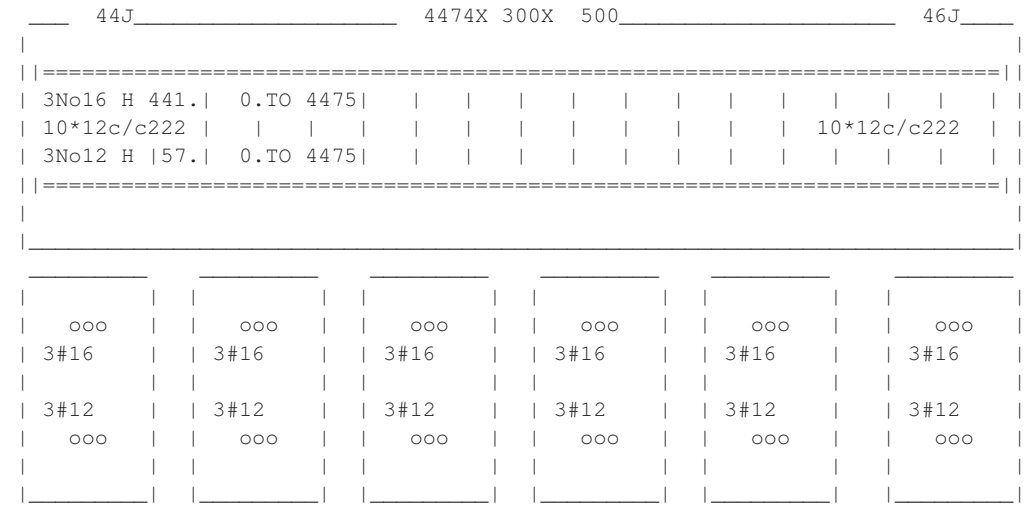
LEN - 4475. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	4475.	YES	YES
2	441.	3 - 16MM	0.	4475.	YES	YES

B E A M N O. 58 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 51.00 KNS Vc= 116.60 KNS Vs= 0.00 KNS
Tu= 0.13 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 1801. MM

AT END SUPPORT - Vu= 58.86 KNS Vc= 114.30 KNS Vs= 0.00 KNS
Tu= 0.13 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 1801. MM



ACI 318-11 BEAM NO. 59 DESIGN RESULTS

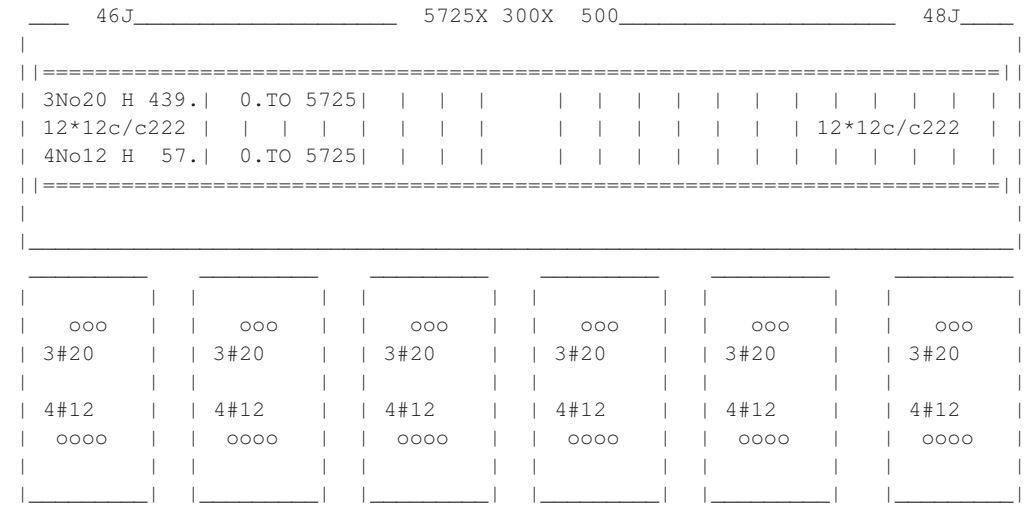
LEN - 5725. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5725.	YES	YES
2	439.	3 - 20MM	0.	5725.	YES	YES

B E A M N O. 59 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 73.04 KNS Vc= 116.50 KNS Vs= 0.00 KNS
Tu= 0.32 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2426. MM

AT END SUPPORT - Vu= 74.94 KNS Vc= 116.01 KNS Vs= 0.00 KNS
Tu= 0.32 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2426. MM



ACI 318-11 BEAM NO. 60 DESIGN RESULTS

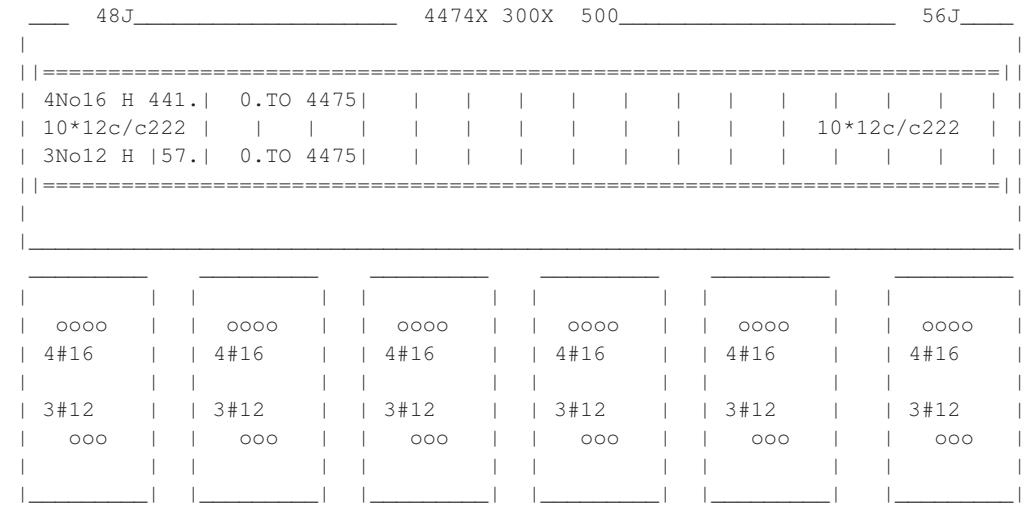
LEN - 4475. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	4475.	YES	YES
2	441.	4 - 16MM	0.	4475.	YES	YES

B E A M N O. 60 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 58.19 KNS Vc= 115.18 KNS Vs= 0.00 KNS
Tu= 0.49 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 1801. MM

AT END SUPPORT - Vu= 51.67 KNS Vc= 115.18 KNS Vs= 0.00 KNS
Tu= 0.49 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 1801. MM



ACI 318-11 BEAM NO. 61 DESIGN RESULTS

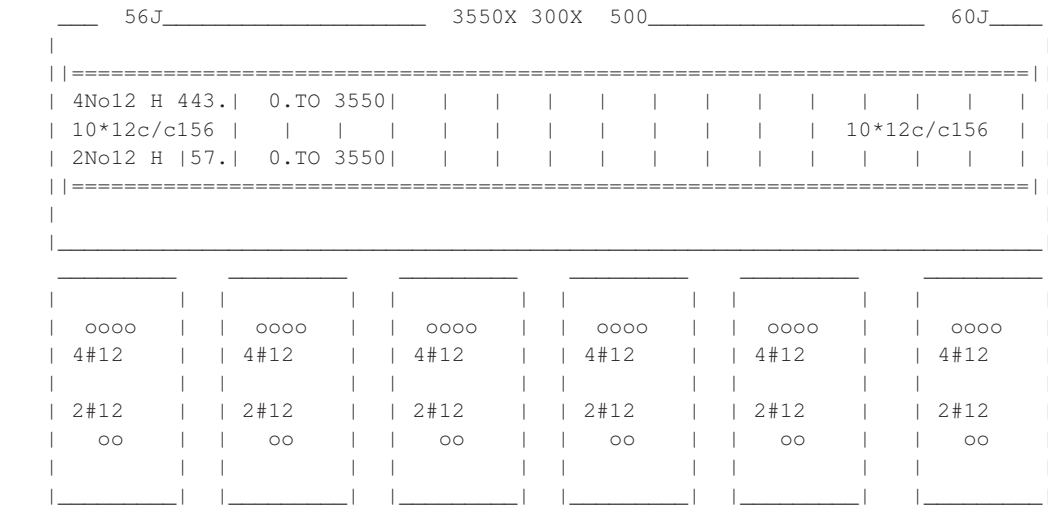
LEN - 3550. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	2 - 12MM	0.	3550.	YES	YES
2	443.	4 - 12MM	0.	3550.	YES	YES

B E A M N O. 61 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.70 KNS Vc= 24.59 KNS Vs= 0.00 KNS
Tu= 0.97 KN-MET Tc= NaN KN-MET Ts= 1.3 KN-MET LOAD 10
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1339. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 0.26 SQ.CM.

AT END SUPPORT - Vu= 7.70 KNS Vc= 24.59 KNS Vs= 0.00 KNS
Tu= 0.97 KN-MET Tc= NaN KN-MET Ts= 1.3 KN-MET LOAD 10
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1339. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 0.26 SQ.CM.



ACI 318-11 BEAM NO. 62 DESIGN RESULTS

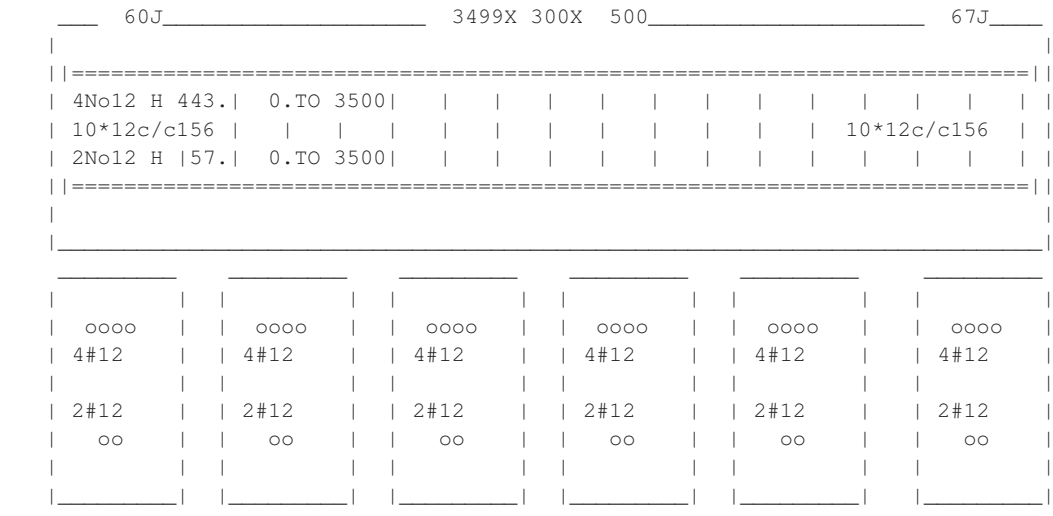
LEN - 3500. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	2 - 12MM	0.	3500.	YES	YES
2	443.	4 - 12MM	0.	3500.	YES	YES

B E A M N O. 62 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.58 KNS Vc= 33.94 KNS Vs= 0.00 KNS
Tu= 0.62 KN-MET Tc= NaN KN-MET Ts= 0.8 KN-MET LOAD 10
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1313. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 0.17 SQ.CM.

AT END SUPPORT - Vu= 12.58 KNS Vc= 33.94 KNS Vs= 0.00 KNS
Tu= 0.62 KN-MET Tc= NaN KN-MET Ts= 0.8 KN-MET LOAD 10
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1313. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 0.17 SQ.CM.



ACI 318-11 BEAM NO. 63 DESIGN RESULTS

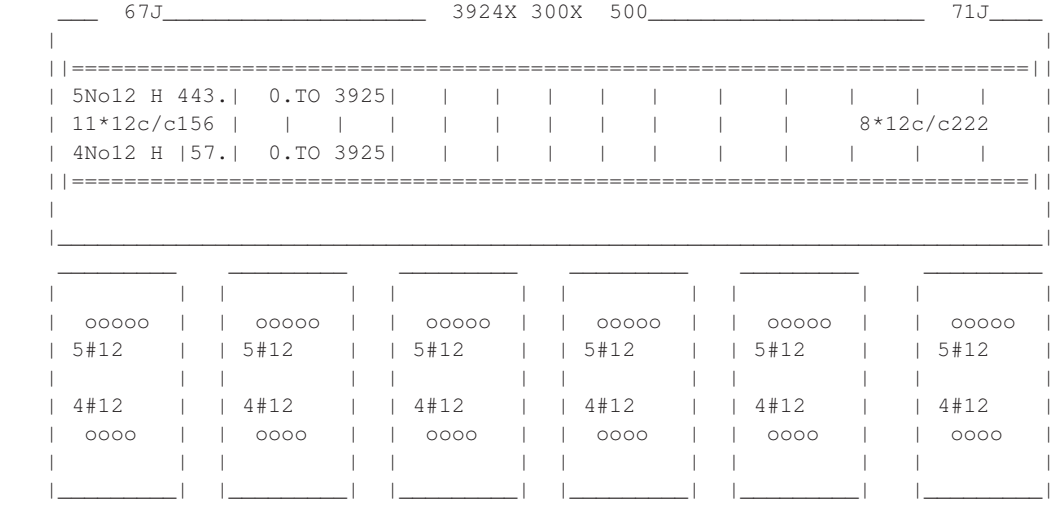
LEN - 3925. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	3925.	YES	YES
2	443.	5 - 12MM	0.	3925.	YES	YES

B E A M N O. 63 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 27.68 KNS Vc= 49.28 KNS Vs= 0.00 KNS
Tu= 3.12 KN-MET Tc= NaN KN-MET Ts= 4.2 KN-MET LOAD 10
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 156. MM C/C FOR 1526. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 0.85 SQ.CM.

AT END SUPPORT - Vu= 48.52 KNS Vc= 119.35 KNS Vs= 0.00 KNS
Tu= 0.77 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 1526. MM



ACI 318-11 BEAM NO. 64 DESIGN RESULTS

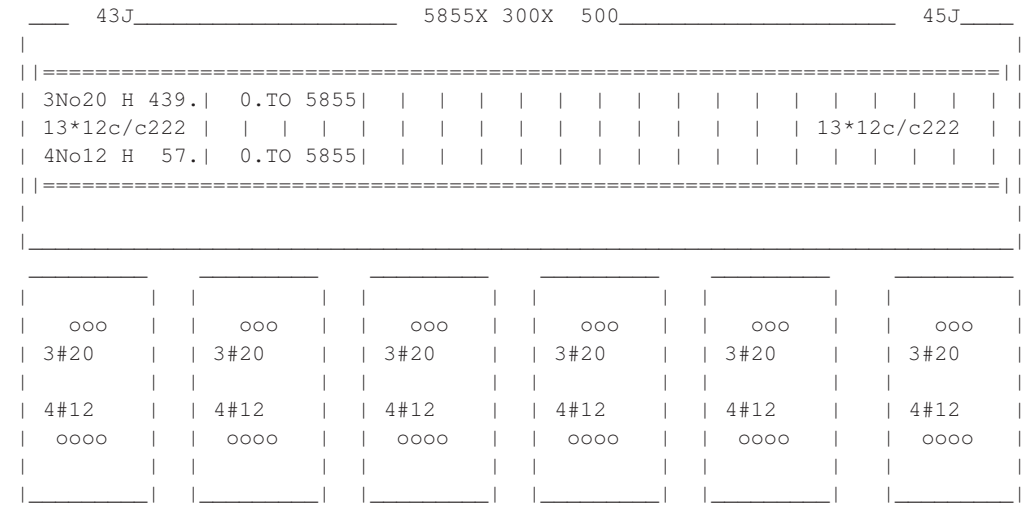
LEN - 5855. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5855.	YES	YES
2	439.	3 - 20MM	0.	5855.	YES	YES

B E A M N O. 64 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 75.76 KNS Vc= 116.27 KNS Vs= 0.00 KNS
Tu= 0.40 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2491. MM

AT END SUPPORT - Vu= 76.19 KNS Vc= 116.16 KNS Vs= 0.00 KNS
Tu= 0.40 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 2491. MM



ACI 318-11 BEAM NO. 65 DESIGN RESULTS

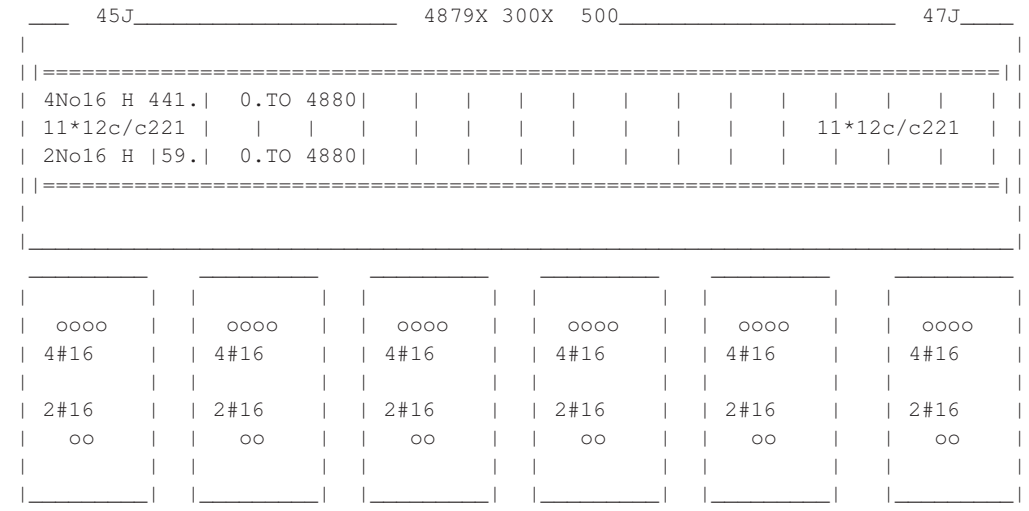
LEN - 4880. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	0.	4880.	YES	YES
2	441.	4 - 16MM	0.	4880.	YES	YES

B E A M N O. 65 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 63.85 KNS Vc= 114.94 KNS Vs= 0.00 KNS
Tu= 0.02 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 2003. MM

AT END SUPPORT - Vu= 58.37 KNS Vc= 116.82 KNS Vs= 0.00 KNS
Tu= 0.02 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 2003. MM



ACI 318-11 BEAM NO. 66 DESIGN RESULTS

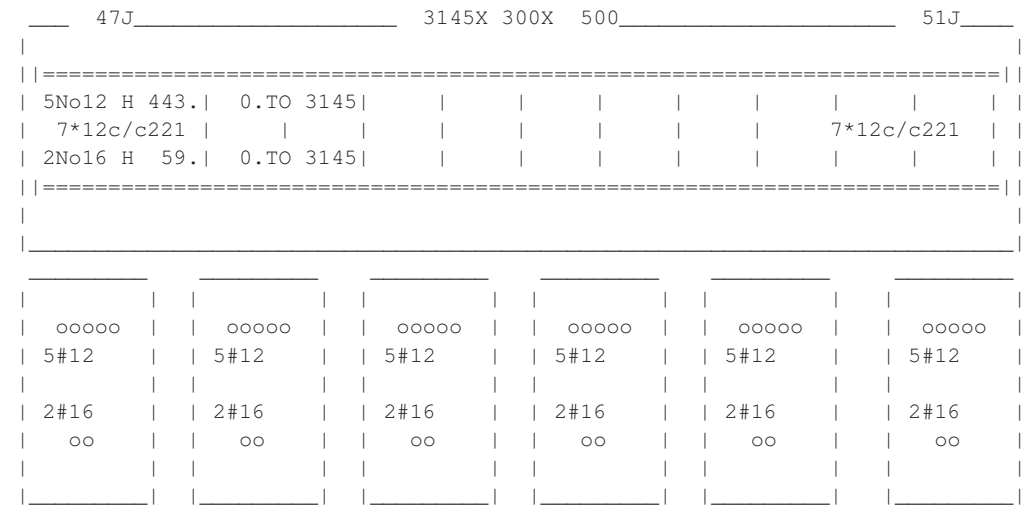
LEN - 3145. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	0.	3145.	YES	YES
2	443.	5 - 12MM	0.	3145.	YES	YES

B E A M N O. 66 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 45.62 KNS Vc= 118.65 KNS Vs= 0.00 KNS
Tu= 0.46 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 115
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 1136. MM

AT END SUPPORT - Vu= 61.24 KNS Vc= 114.79 KNS Vs= 0.00 KNS
Tu= 0.18 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 217
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 1136. MM



ACI 318-11 BEAM NO. 67 DESIGN RESULTS

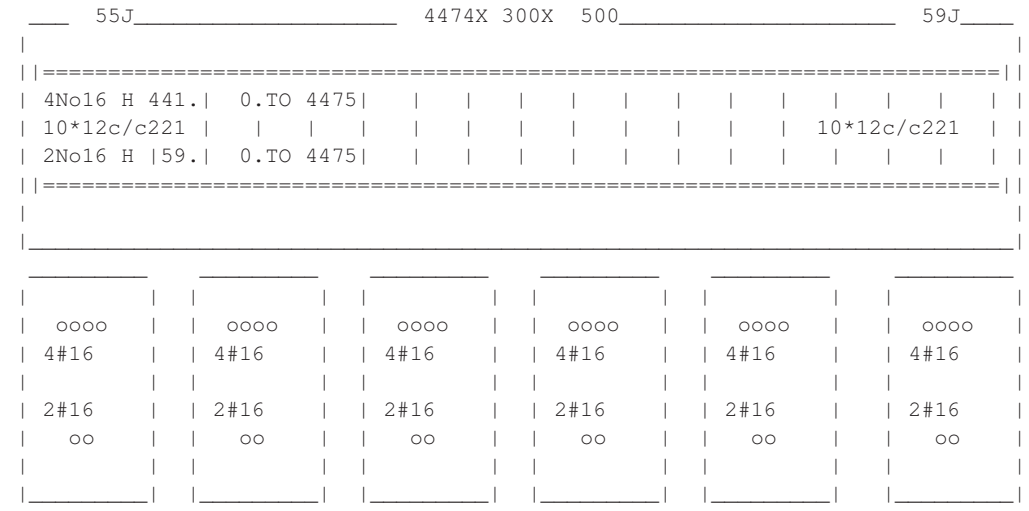
LEN - 4475. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	0.	4475.	YES	YES
2	441.	4 - 16MM	0.	4475.	YES	YES

B E A M N O. 67 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 55.15 KNS Vc= 117.00 KNS Vs= 0.00 KNS
Tu= 0.72 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 1801. MM

AT END SUPPORT - Vu= 54.71 KNS Vc= 117.22 KNS Vs= 0.00 KNS
Tu= 0.72 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 1801. MM



ACI 318-11 BEAM NO. 68 DESIGN RESULTS

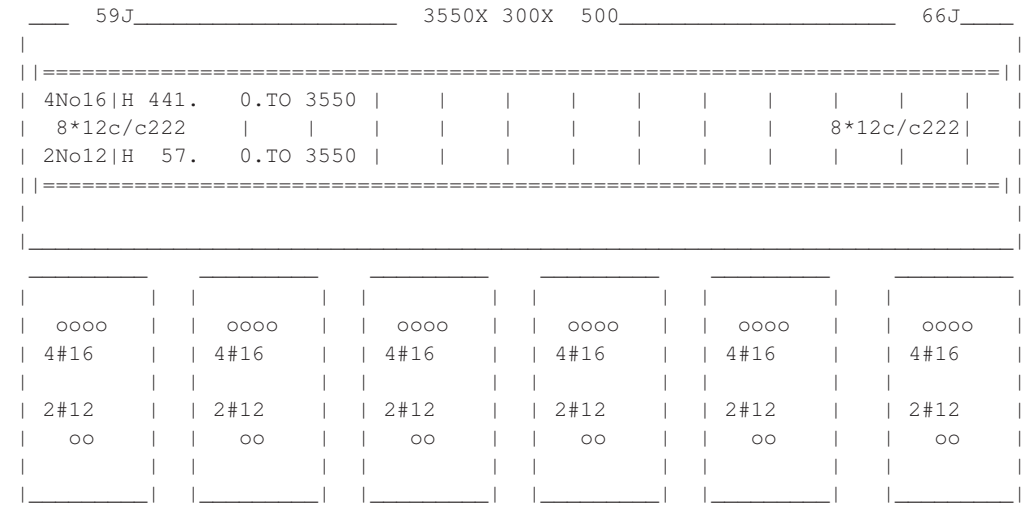
LEN - 3550. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	2 - 12MM	0.	3550.	YES	YES
2	441.	4 - 16MM	0.	3550.	YES	YES

B E A M N O. 68 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 51.32 KNS Vc= 115.16 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 218
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 1339. MM

AT END SUPPORT - Vu= 53.82 KNS Vc= 115.30 KNS Vs= 0.00 KNS
Tu= 0.06 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 222. MM C/C FOR 1339. MM



ACI 318-11 BEAM NO. 69 DESIGN RESULTS

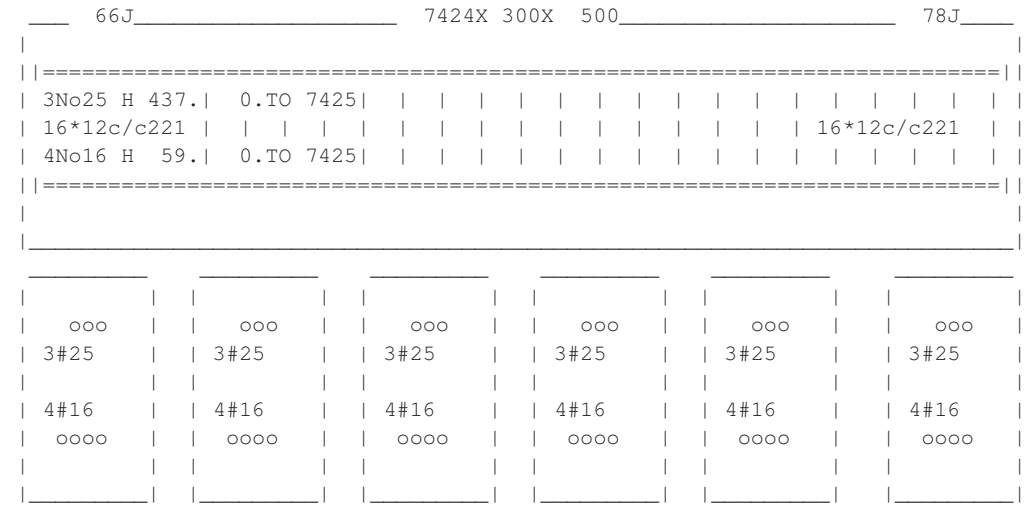
LEN - 7425. MM FY - 414. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	4 - 16MM	0.	7425.	YES	YES
2	437.	3 - 25MM	0.	7425.	YES	YES

B E A M N O. 69 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 134.99 KNS Vc= 116.58 KNS Vs= 63.40 KNS
Tu= 0.45 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 202
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 3276. MM

AT END SUPPORT - Vu= 144.97 KNS Vc= 115.34 KNS Vs= 77.96 KNS
Tu= 0.44 KN-MET Tc= 4.8 KN-MET Ts= 0.0 KN-MET LOAD 201
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 3276. MM



ACI 318-11 BEAM NO. 109 DESIGN RESULTS

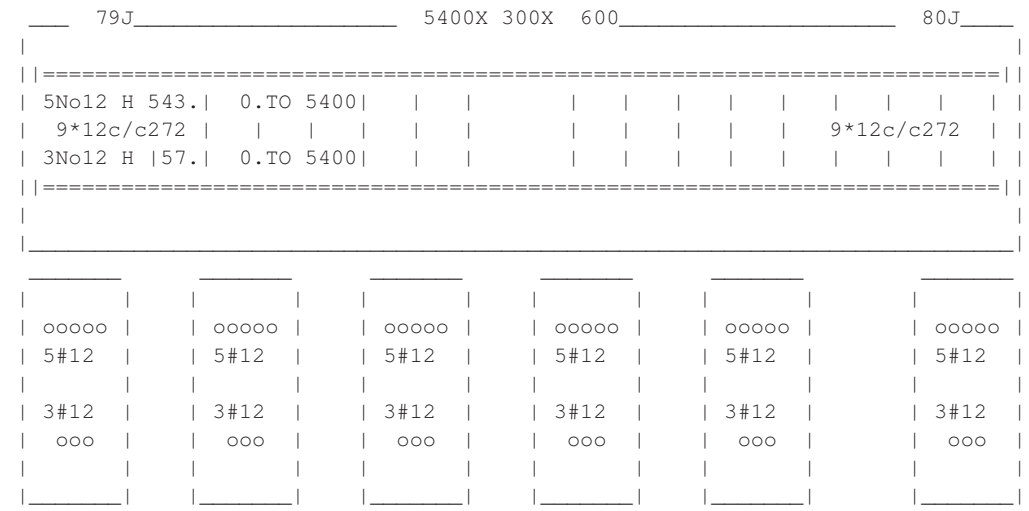
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	5 - 12MM	0.	5400.	YES	YES

B E A M N O. 109 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 58.01 KNS Vc= 141.78 KNS Vs= 0.00 KNS
Tu= 0.12 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 201
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2164. MM

AT END SUPPORT - Vu= 60.12 KNS Vc= 141.78 KNS Vs= 0.00 KNS
Tu= 0.12 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 201
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2164. MM



ACI 318-11 BEAM NO. 110 DESIGN RESULTS

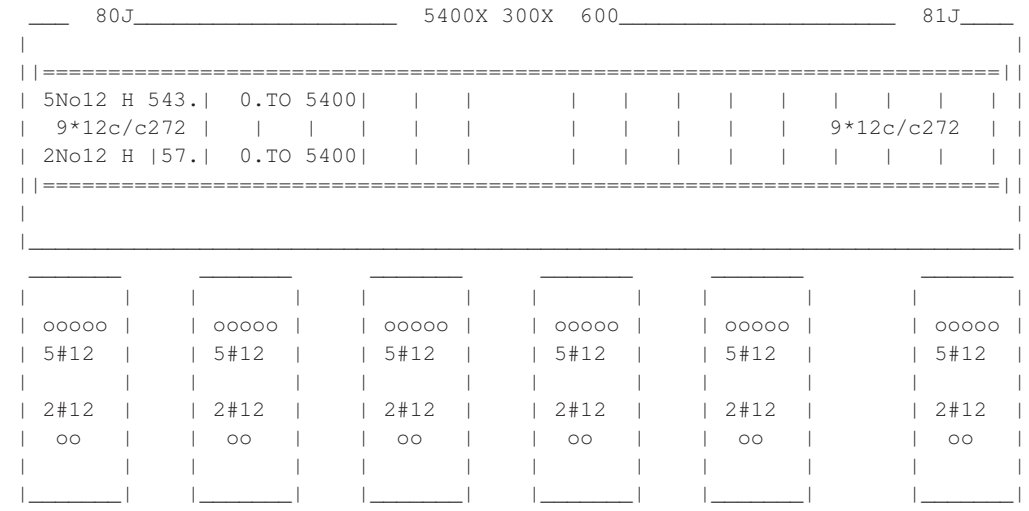
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	2 - 12MM	0.	5400.	YES	YES
2	543.	5 - 12MM	0.	5400.	YES	YES

B E A M N O. 110 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 57.90 KNS Vc= 141.50 KNS Vs= 0.00 KNS
Tu= 0.28 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 201
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2164. MM

AT END SUPPORT - Vu= 60.22 KNS Vc= 141.50 KNS Vs= 0.00 KNS
Tu= 0.28 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 201
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2164. MM



ACI 318-11 BEAM NO. 111 DESIGN RESULTS

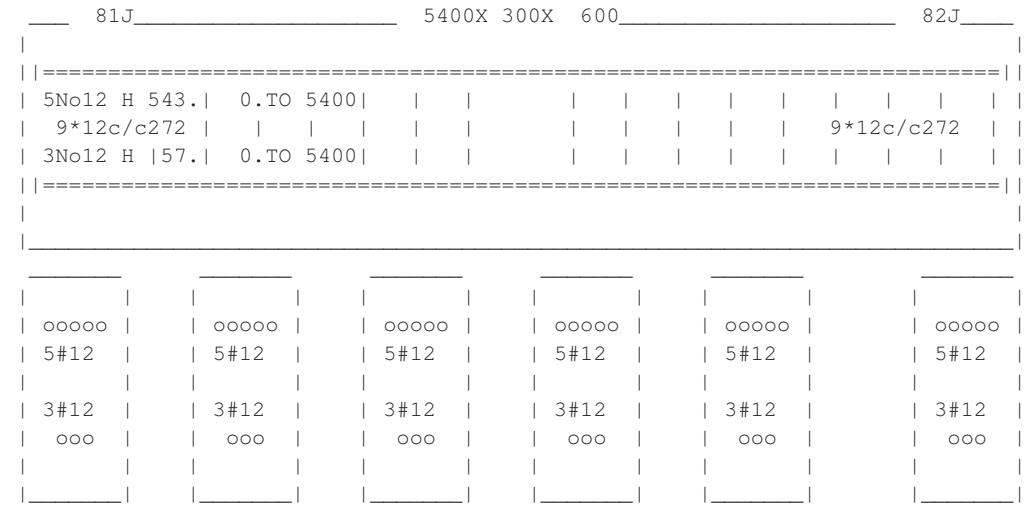
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	5 - 12MM	0.	5400.	YES	YES

B E A M N O. 111 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 57.96 KNS Vc= 142.02 KNS Vs= 0.00 KNS
Tu= 1.10 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 201
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2164. MM

AT END SUPPORT - Vu= 60.17 KNS Vc= 142.02 KNS Vs= 0.00 KNS
Tu= 1.10 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 201
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2164. MM



ACI 318-11 BEAM NO. 112 DESIGN RESULTS

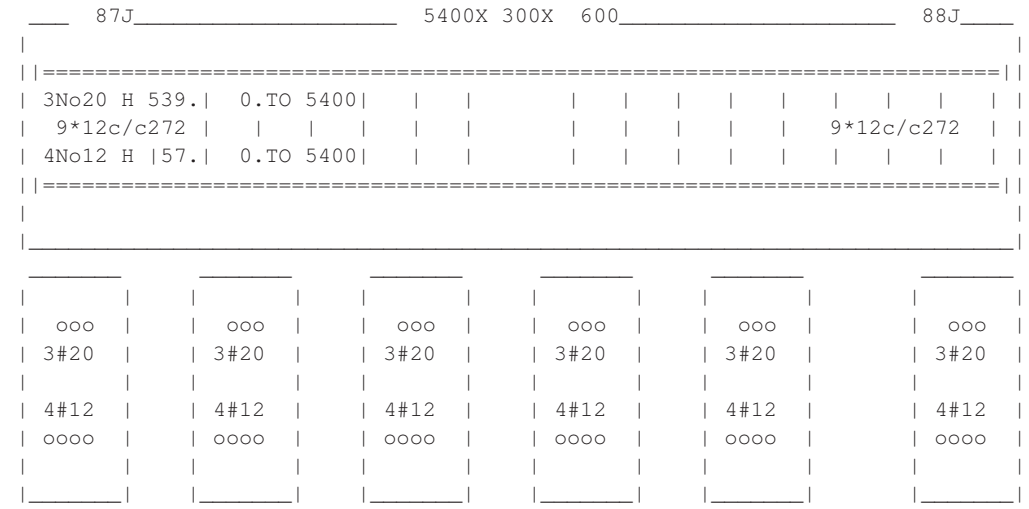
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	539.	3 - 20MM	0.	5400.	YES	YES

B E A M N O. 112 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 78.88 KNS Vc= 141.34 KNS Vs= 0.00 KNS
Tu= 1.61 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2164. MM

AT END SUPPORT - Vu= 75.99 KNS Vc= 141.34 KNS Vs= 0.00 KNS
Tu= 1.61 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2164. MM



ACI 318-11 BEAM NO. 113 DESIGN RESULTS

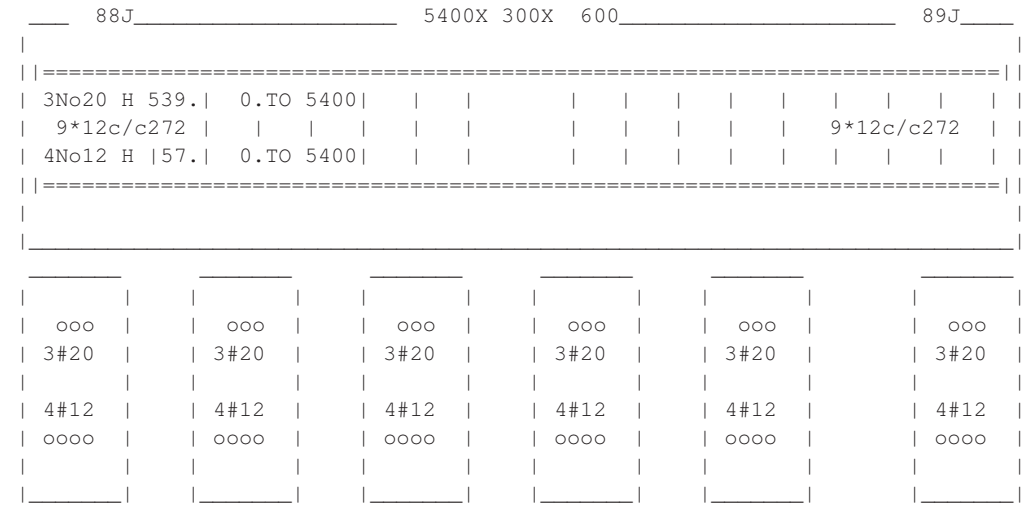
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	539.	3 - 20MM	0.	5400.	YES	YES

B E A M N O. 113 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 78.39 KNS Vc= 140.90 KNS Vs= 0.00 KNS
Tu= 0.04 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2164. MM

AT END SUPPORT - Vu= 76.38 KNS Vc= 140.90 KNS Vs= 0.00 KNS
Tu= 0.04 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2164. MM



ACI 318-11 BEAM NO. 114 DESIGN RESULTS

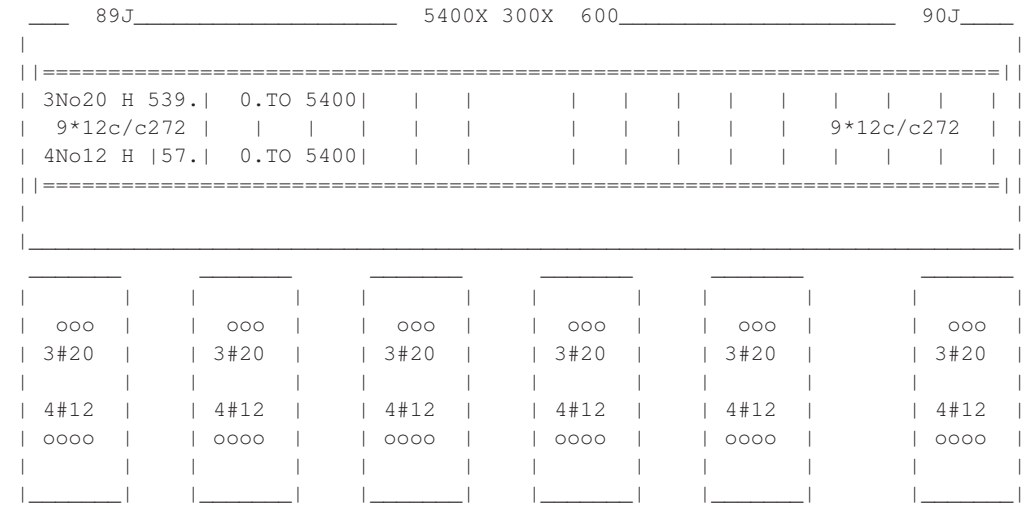
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	539.	3 - 20MM	0.	5400.	YES	YES

B E A M N O. 114 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 77.37 KNS Vc= 141.44 KNS Vs= 0.00 KNS
Tu= 0.68 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2164. MM

AT END SUPPORT - Vu= 77.39 KNS Vc= 141.44 KNS Vs= 0.00 KNS
Tu= 0.68 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2164. MM



ACI 318-11 BEAM NO. 115 DESIGN RESULTS

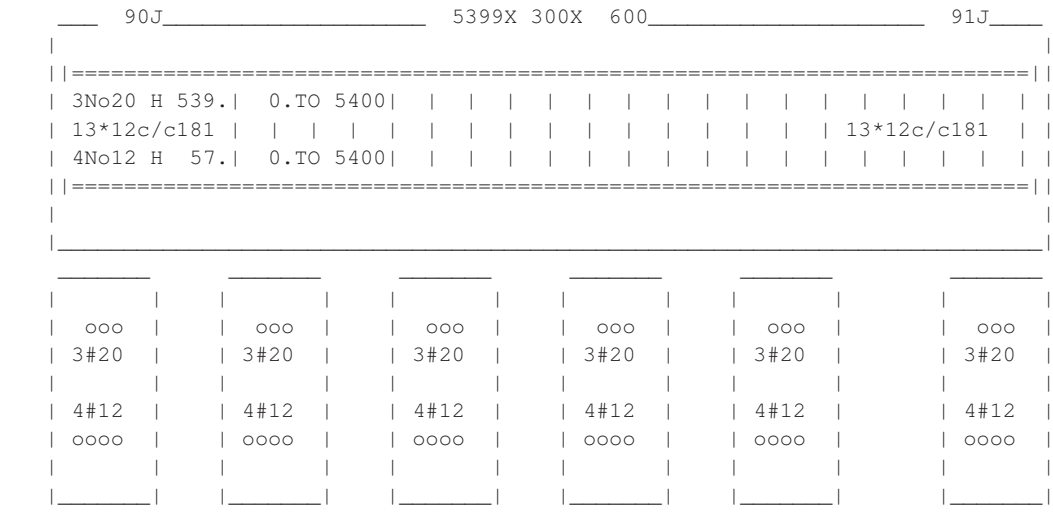
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	539.	3 - 20MM	0.	5400.	YES	YES

B E A M N O. 115 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 98.52 KNS Vc= 142.79 KNS Vs= 0.00 KNS
Tu= 8.82 KN-MET Tc= 5.9 KN-MET Ts= 11.8 KN-MET LOAD 217
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 181. MM C/C FOR 2163. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.23 SQ.CM.

AT END SUPPORT - Vu= 98.15 KNS Vc= 142.88 KNS Vs= 0.00 KNS
Tu= 8.82 KN-MET Tc= 5.9 KN-MET Ts= 11.8 KN-MET LOAD 217
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 181. MM C/C FOR 2163. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.23 SQ.CM.



ACI 318-11 BEAM NO. 116 DESIGN RESULTS

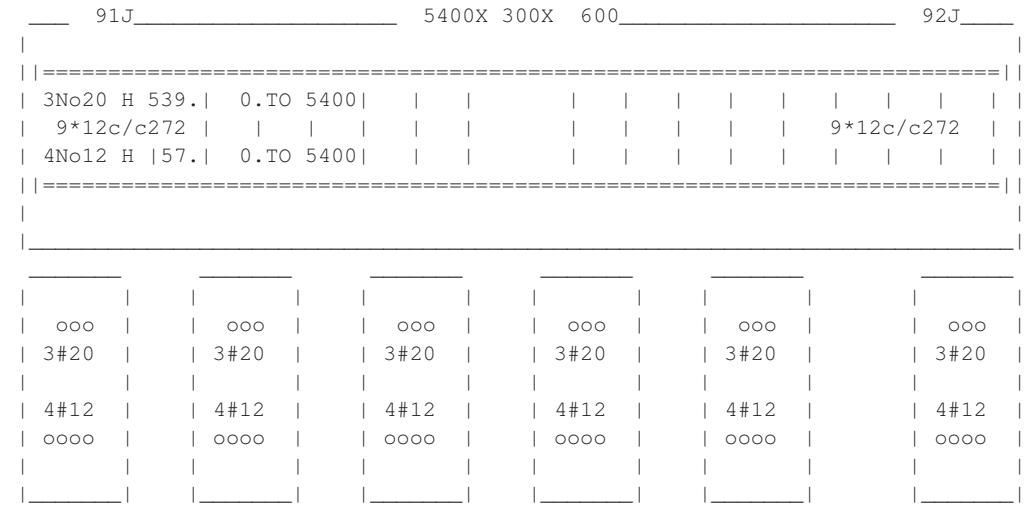
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5400.	YES	YES
2	539.	3 - 20MM	0.	5400.	YES	YES

B E A M N O. 116 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 77.75 KNS Vc= 142.27 KNS Vs= 0.00 KNS
Tu= 0.42 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2164. MM

AT END SUPPORT - Vu= 77.01 KNS Vc= 142.49 KNS Vs= 0.00 KNS
Tu= 0.42 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2164. MM



ACI 318-11 BEAM NO. 117 DESIGN RESULTS

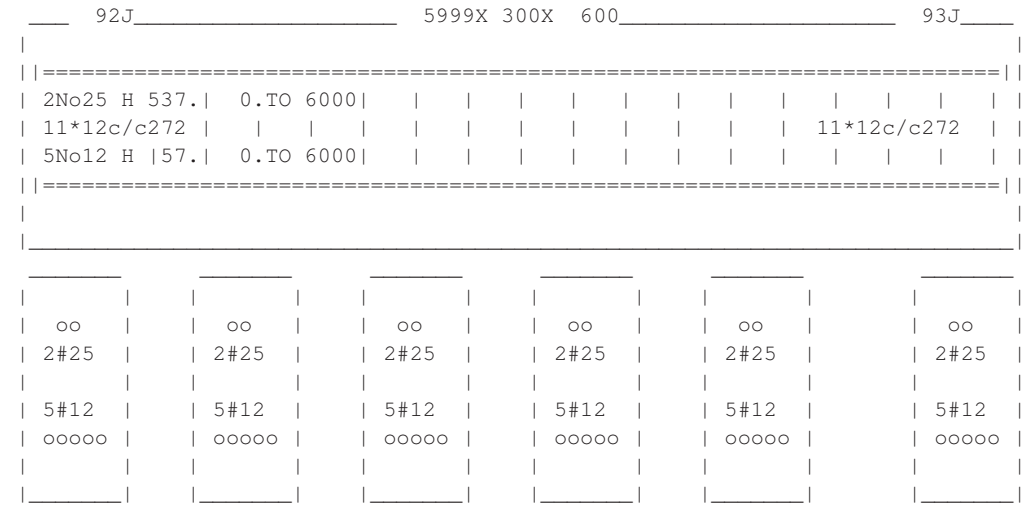
LEN - 6000. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	5 - 12MM	0.	6000.	YES	YES
2	537.	2 - 25MM	0.	6000.	YES	YES

B E A M N O. 117 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 88.20 KNS Vc= 143.26 KNS Vs= 0.00 KNS
Tu= 0.05 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2463. MM

AT END SUPPORT - Vu= 88.02 KNS Vc= 143.31 KNS Vs= 0.00 KNS
Tu= 0.05 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2463. MM



ACI 318-11 BEAM NO. 118 DESIGN RESULTS

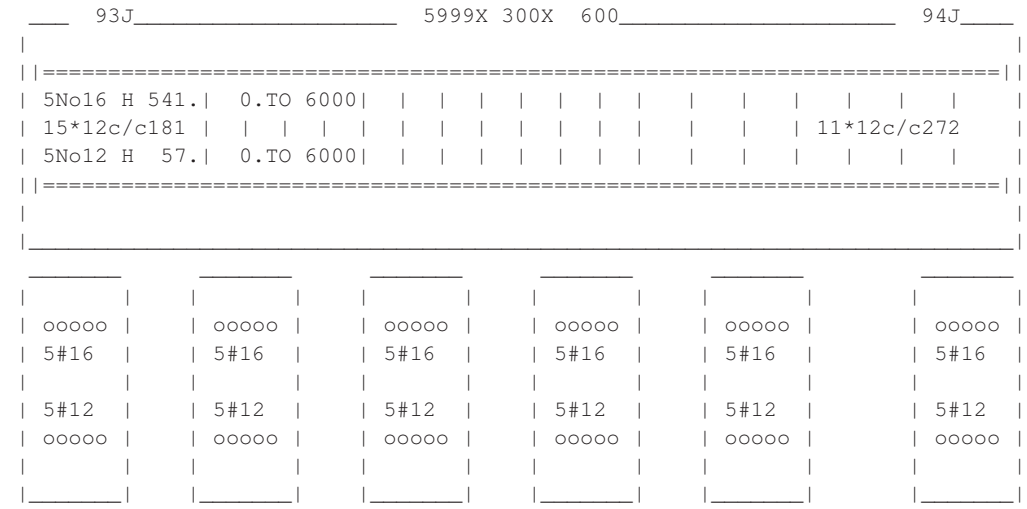
LEN - 6000. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	5 - 12MM	0.	6000.	YES	YES
2	541.	5 - 16MM	0.	6000.	YES	YES

B E A M N O. 118 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 116.95 KNS Vc= 143.28 KNS Vs= 12.65 KNS
Tu= 7.11 KN-MET Tc= 6.0 KN-MET Ts= 9.5 KN-MET LOAD 217
STIRRUPS ARE REQUIRED FOR SHEAR AND TORSION.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 181. MM C/C FOR 2463. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.80 SQ.CM.

AT END SUPPORT - Vu= 85.82 KNS Vc= 145.18 KNS Vs= 0.00 KNS
Tu= 2.01 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2463. MM



ACI 318-11 BEAM NO. 119 DESIGN RESULTS

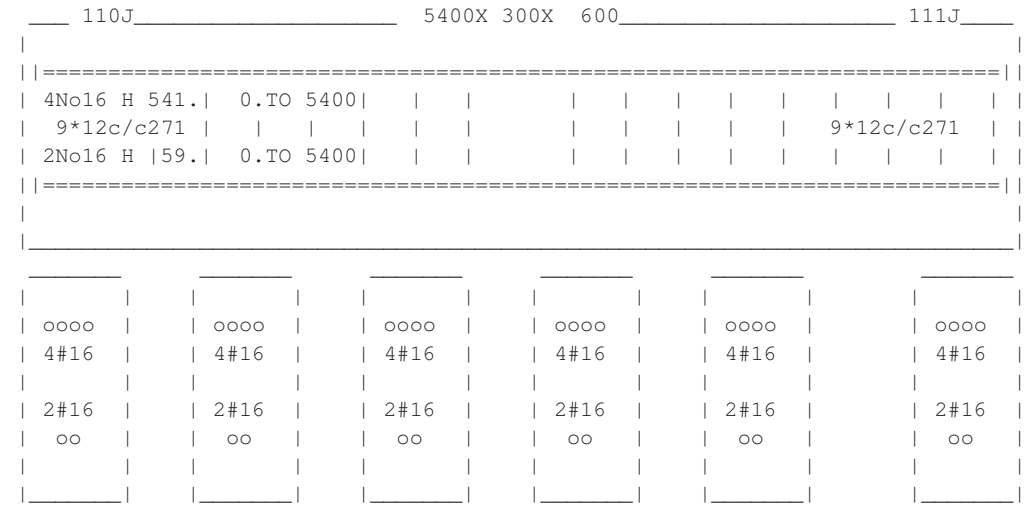
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	0.	5400.	YES	YES
2	541.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 119 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 66.81 KNS Vc= 144.07 KNS Vs= 0.00 KNS
Tu= 1.32 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 2164. MM

AT END SUPPORT - Vu= 71.66 KNS Vc= 141.92 KNS Vs= 0.00 KNS
Tu= 1.32 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 2164. MM



ACI 318-11 BEAM NO. 120 DESIGN RESULTS

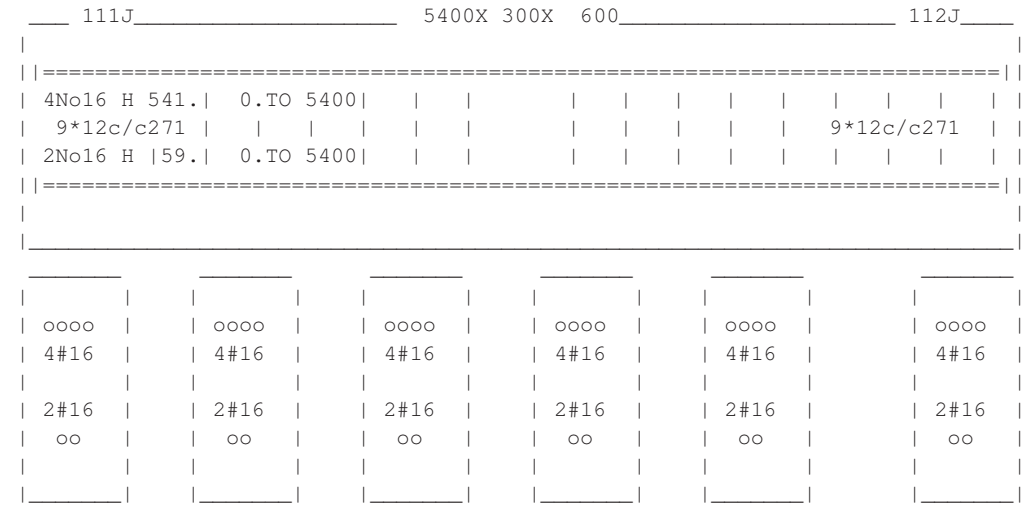
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	0.	5400.	YES	YES
2	541.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 120 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 68.83 KNS Vc= 141.34 KNS Vs= 0.00 KNS
Tu= 1.09 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 2164. MM

AT END SUPPORT - Vu= 69.63 KNS Vc= 141.10 KNS Vs= 0.00 KNS
Tu= 1.09 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 2164. MM



ACI 318-11 BEAM NO. 121 DESIGN RESULTS

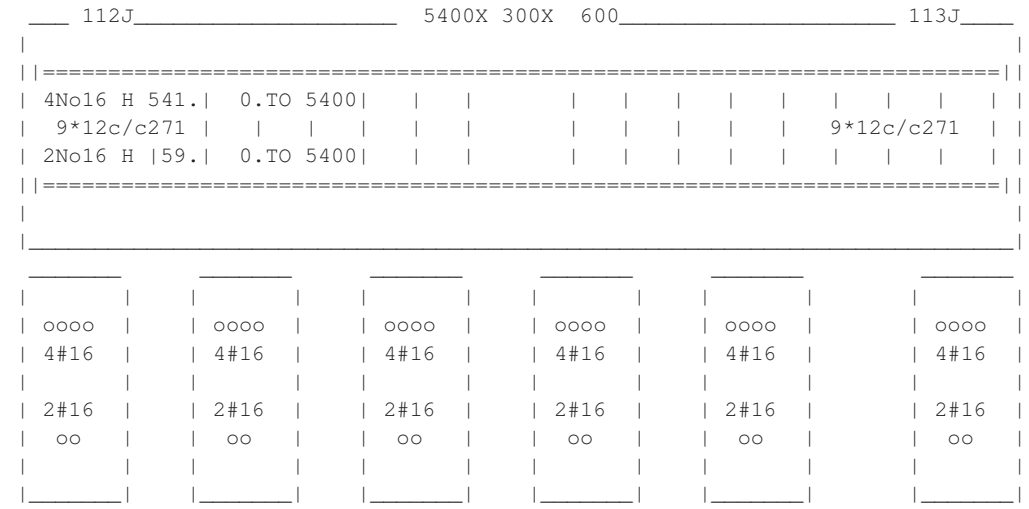
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	0.	5400.	YES	YES
2	541.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 121 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 68.01 KNS Vc= 141.66 KNS Vs= 0.00 KNS
Tu= 1.77 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 2164. MM

AT END SUPPORT - Vu= 70.46 KNS Vc= 140.92 KNS Vs= 0.00 KNS
Tu= 1.77 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 2164. MM



ACI 318-11 BEAM NO. 122 DESIGN RESULTS

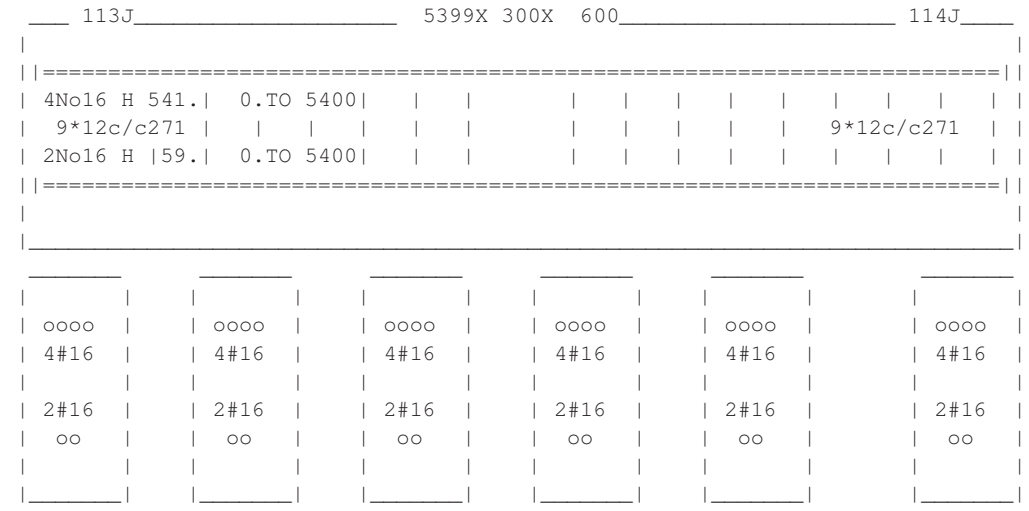
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	0.	5400.	YES	YES
2	541.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 122 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 68.45 KNS Vc= 141.55 KNS Vs= 0.00 KNS
Tu= 0.90 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 2163. MM

AT END SUPPORT - Vu= 70.01 KNS Vc= 141.08 KNS Vs= 0.00 KNS
Tu= 0.90 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 2163. MM



ACI 318-11 BEAM NO. 123 DESIGN RESULTS

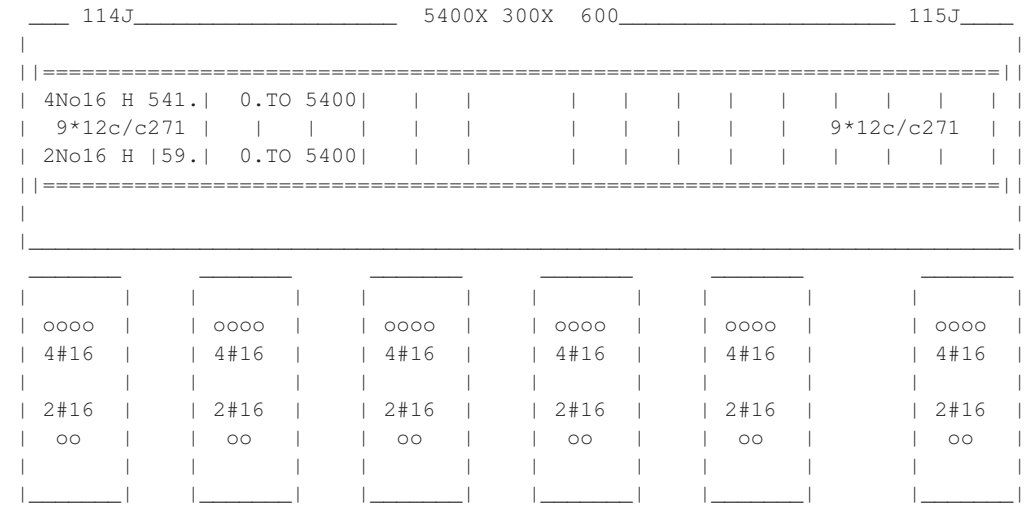
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	0.	5400.	YES	YES
2	541.	4 - 16MM	0.	5400.	YES	YES

B E A M N O. 123 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 67.67 KNS Vc= 141.50 KNS Vs= 0.00 KNS
Tu= 0.21 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 2164. MM

AT END SUPPORT - Vu= 70.80 KNS Vc= 140.61 KNS Vs= 0.00 KNS
Tu= 0.21 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 2164. MM



ACI 318-11 BEAM NO. 124 DESIGN RESULTS

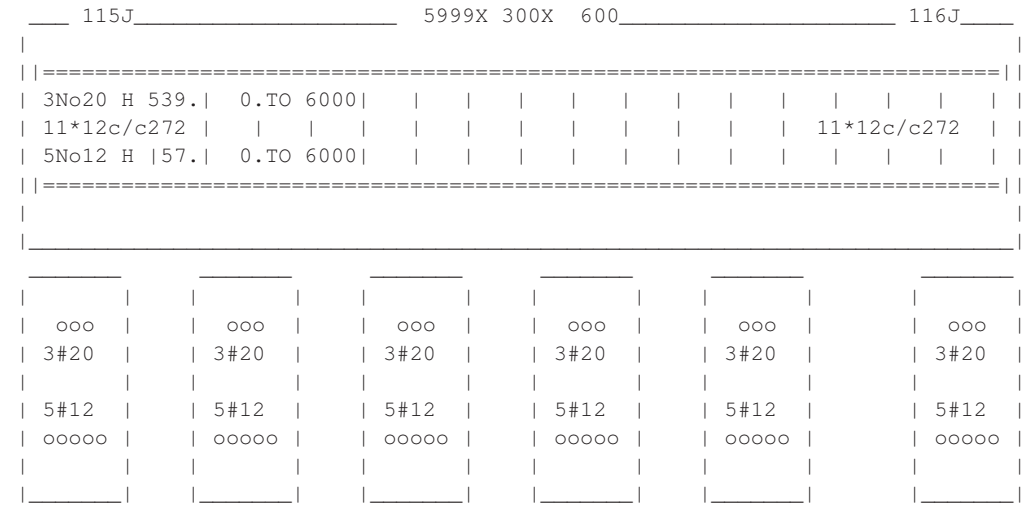
LEN - 6000. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	5 - 12MM	0.	6000.	YES	YES
2	539.	3 - 20MM	0.	6000.	YES	YES

B E A M N O. 124 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 77.44 KNS Vc= 142.67 KNS Vs= 0.00 KNS
Tu= 0.13 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2463. MM

AT END SUPPORT - Vu= 80.23 KNS Vc= 141.90 KNS Vs= 0.00 KNS
Tu= 0.13 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2463. MM



ACI 318-11 BEAM NO. 125 DESIGN RESULTS

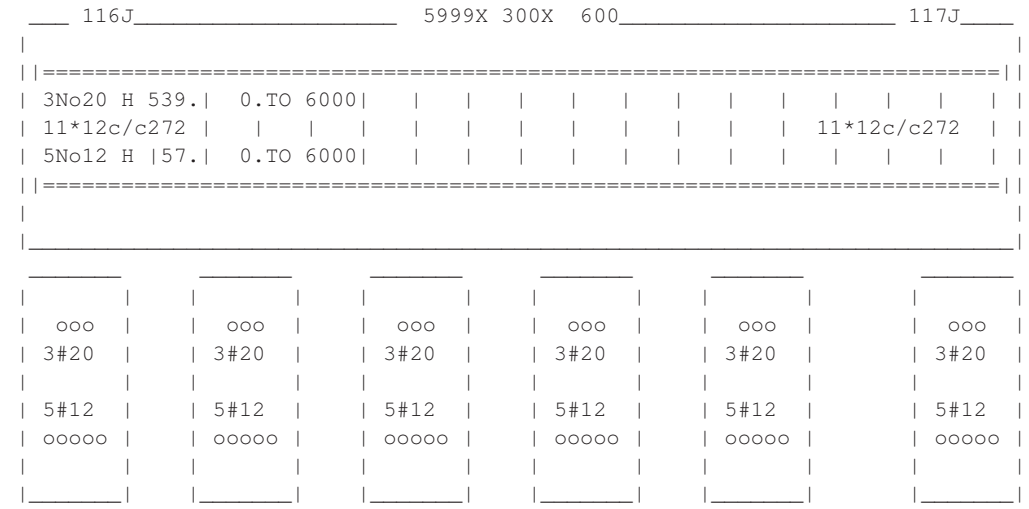
LEN - 6000. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	5 - 12MM	0.	6000.	YES	YES
2	539.	3 - 20MM	0.	6000.	YES	YES

B E A M N O. 125 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 80.00 KNS Vc= 142.89 KNS Vs= 0.00 KNS
Tu= 3.07 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2463. MM

AT END SUPPORT - Vu= 77.66 KNS Vc= 143.70 KNS Vs= 0.00 KNS
Tu= 3.07 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2463. MM



ACI 318-11 BEAM NO. 126 DESIGN RESULTS

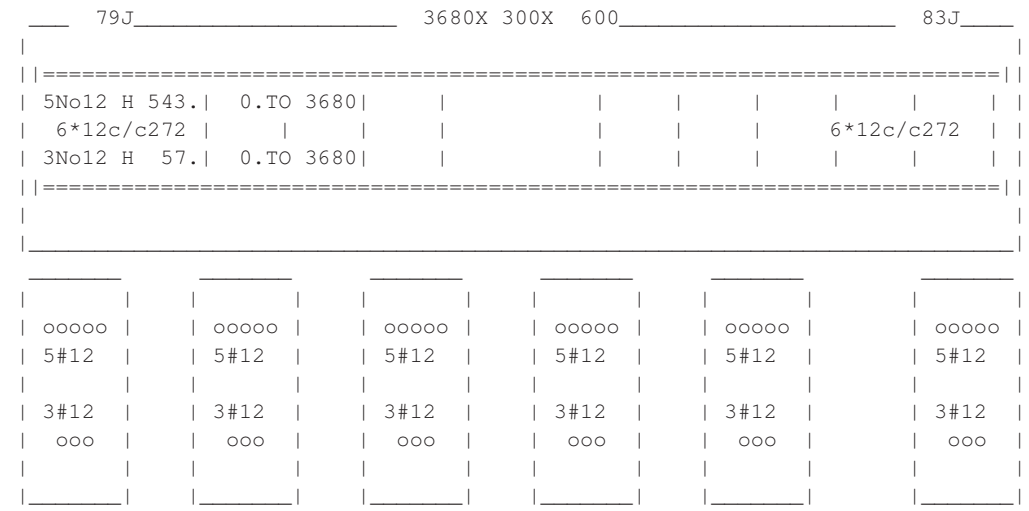
LEN - 3680. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	3680.	YES	YES
2	543.	5 - 12MM	0.	3680.	YES	YES

B E A M N O. 126 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 58.10 KNS Vc= 138.27 KNS Vs= 0.00 KNS
Tu= 0.62 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 218
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 1304. MM

AT END SUPPORT - Vu= 59.71 KNS Vc= 138.69 KNS Vs= 0.00 KNS
Tu= 1.14 KN-MET Tc= 5.7 KN-MET Ts= 0.0 KN-MET LOAD 217
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 1304. MM



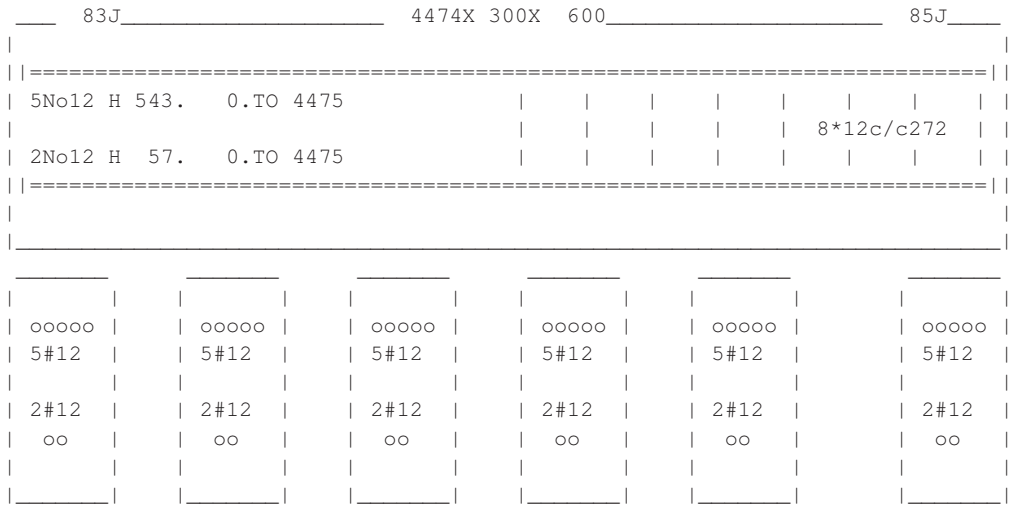
ACI 318-11 BEAM NO. 127 DESIGN RESULTS

LEN - 4475. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	2 - 12MM	0.	4475.	YES	YES
2	543.	5 - 12MM	0.	4475.	YES	YES

B E A M N O. 127 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 30.34 KNS Vc= 141.43 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 53.86 KNS Vc= 141.28 KNS Vs= 0.00 KNS
Tu= 0.26 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 217
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 1701. MM



ACI 318-11 BEAM NO. 128 DESIGN RESULTS

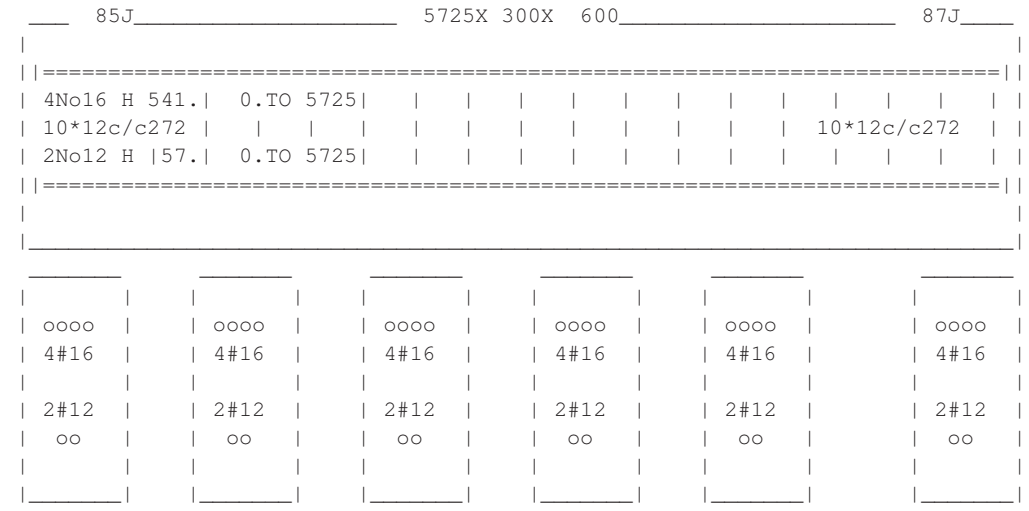
LEN - 5725. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	2 - 12MM	0.	5725.	YES	YES
2	541.	4 - 16MM	0.	5725.	YES	YES

B E A M N O. 128 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 55.77 KNS Vc= 141.38 KNS Vs= 0.00 KNS
Tu= 0.70 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 201
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2326. MM

AT END SUPPORT - Vu= 58.10 KNS Vc= 137.14 KNS Vs= 0.00 KNS
Tu= 0.28 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2326. MM



ACI 318-11 BEAM NO. 129 DESIGN RESULTS

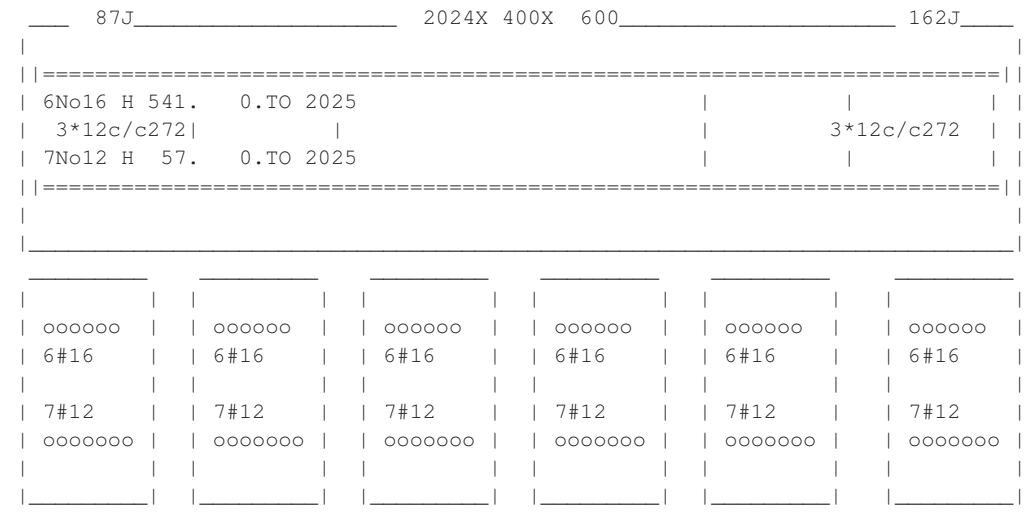
LEN - 2025. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	7 - 12MM	0.	2025.	YES	YES
2	541.	6 - 16MM	0.	2025.	YES	YES

B E A M N O. 129 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 119.98 KNS Vc= 198.25 KNS Vs= 0.00 KNS
Tu= 1.34 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 476. MM

AT END SUPPORT - Vu= 88.09 KNS Vc= 191.60 KNS Vs= 0.00 KNS
Tu= 1.34 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 476. MM



ACI 318-11 BEAM NO. 130 DESIGN RESULTS

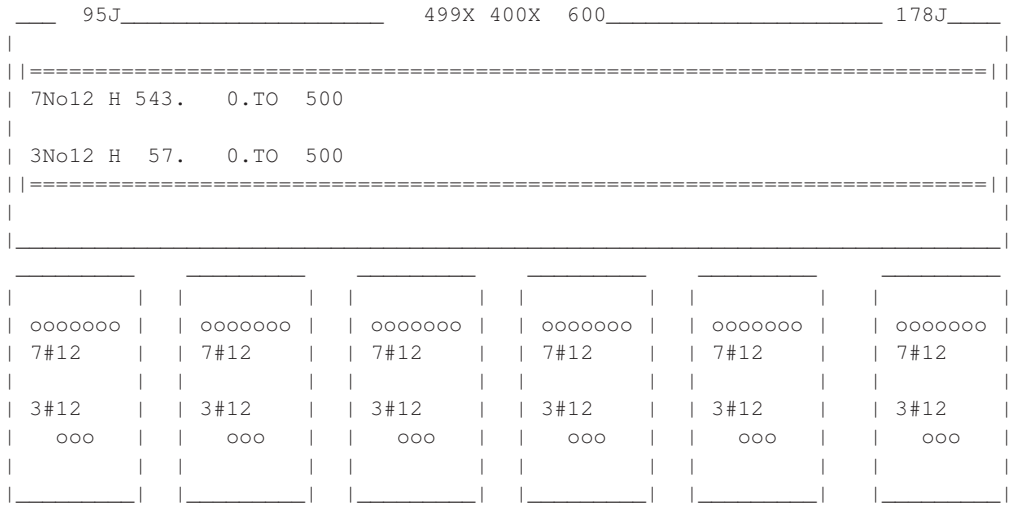
LEN - 500. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	500.	YES	YES
2	543.	7 - 12MM	0.	500.	YES	YES

B E A M N O. 130 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 130 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 130 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 131 DESIGN RESULTS

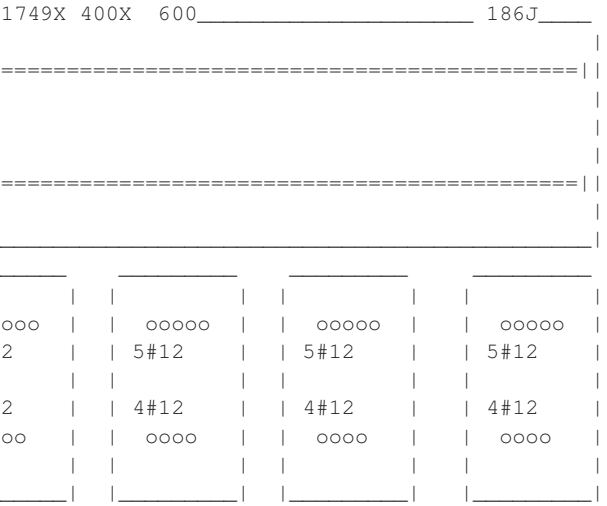
LEN - 1750. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	1750.	YES	YES
2	543.	5 - 12MM	0.	1750.	YES	YES

B E A M N O. 131 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 72.37 KNS Vc= 189.39 KNS Vs= 0.00 KNS
Tu= 1.10 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 201
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 338. MM

AT END SUPPORT - Vu= 28.35 KNS Vc= 189.52 KNS Vs= 0.00 KNS
Tu= 0.76 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 132 DESIGN RESULTS

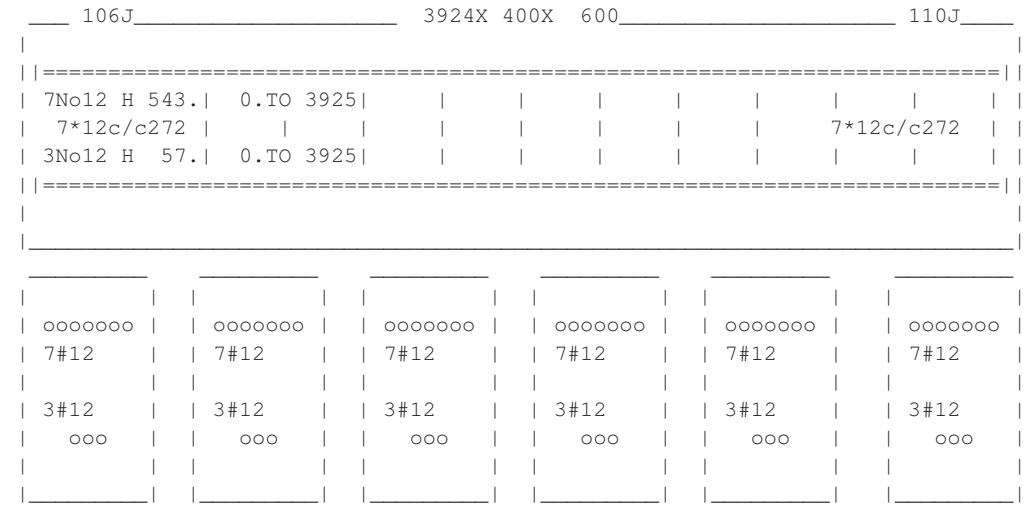
LEN - 3925. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	3925.	YES	YES
2	543.	7 - 12MM	0.	3925.	YES	YES

B E A M N O. 132 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 78.18 KNS Vc= 186.14 KNS Vs= 0.00 KNS
Tu= 2.86 KN-MET Tc= 9.2 KN-MET Ts= 0.0 KN-MET LOAD 218
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 1426. MM

AT END SUPPORT - Vu= 69.66 KNS Vc= 184.08 KNS Vs= 0.00 KNS
Tu= 2.44 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 1426. MM



ACI 318-11 BEAM NO. 133 DESIGN RESULTS

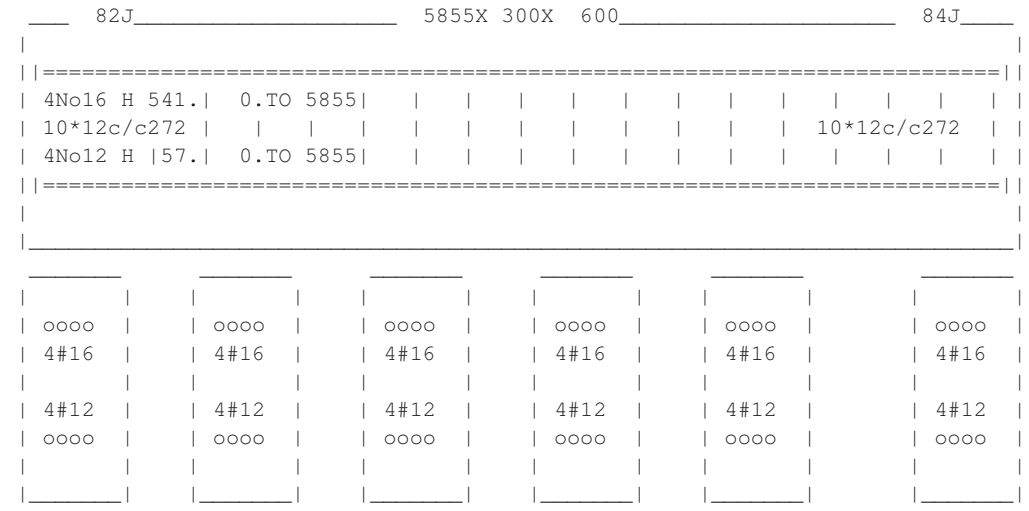
LEN - 5855. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5855.	YES	YES
2	541.	4 - 16MM	0.	5855.	YES	YES

B E A M N O. 133 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 53.93 KNS Vc= 139.94 KNS Vs= 0.00 KNS
Tu= 0.32 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 118
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2391. MM

AT END SUPPORT - Vu= 61.28 KNS Vc= 140.69 KNS Vs= 0.00 KNS
Tu= 0.12 KN-MET Tc= 5.8 KN-MET Ts= 0.0 KN-MET LOAD 117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 2391. MM



ACI 318-11 BEAM NO. 134 DESIGN RESULTS

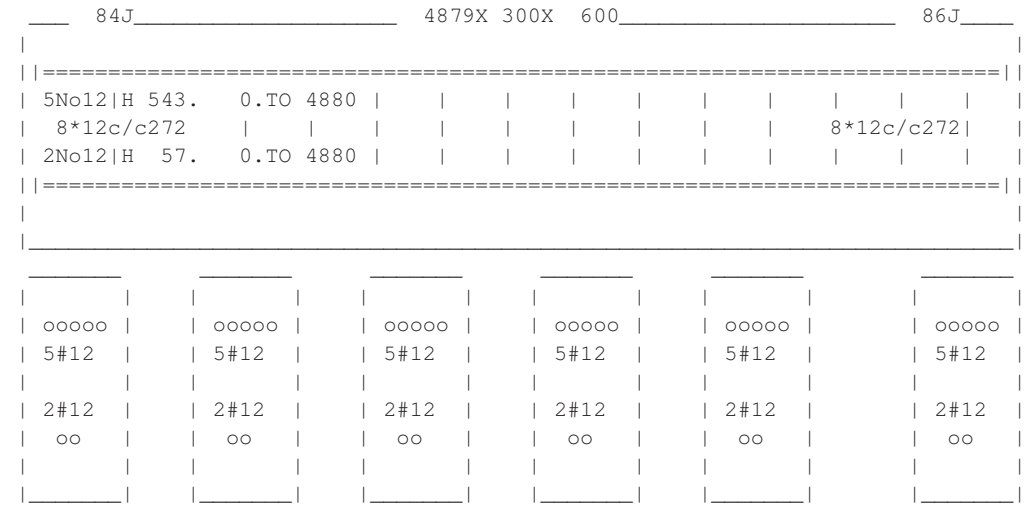
LEN - 4880. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	2 - 12MM	0.	4880.	YES	YES
2	543.	5 - 12MM	0.	4880.	YES	YES

B E A M N O. 134 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 55.22 KNS Vc= 138.04 KNS Vs= 0.00 KNS
Tu= 0.32 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 218
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 1903. MM

AT END SUPPORT - Vu= 55.24 KNS Vc= 139.47 KNS Vs= 0.00 KNS
Tu= 0.06 KN-MET Tc= 6.0 KN-MET Ts= 0.0 KN-MET LOAD 201
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 1903. MM



ACI 318-11 BEAM NO. 135 DESIGN RESULTS

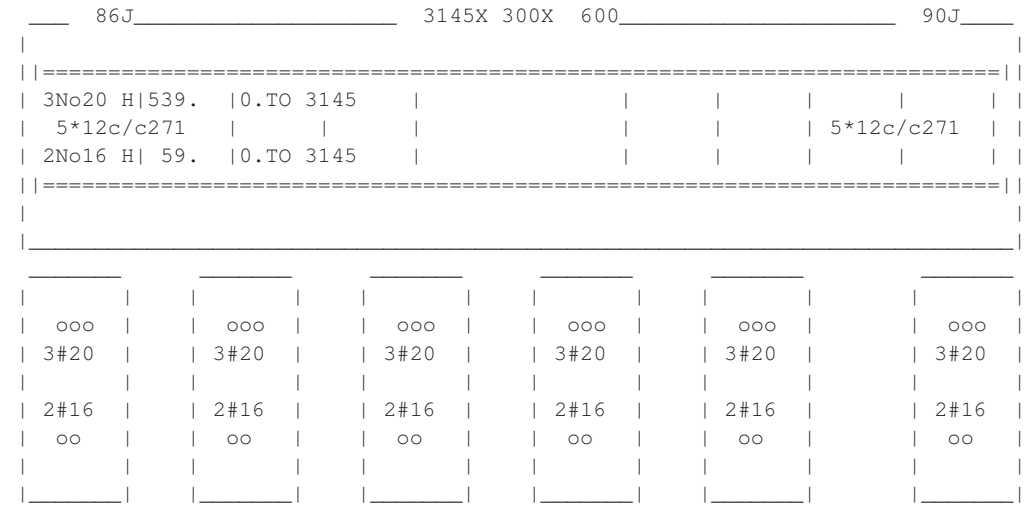
LEN - 3145. MM FY - 414. FC - 28. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	0.	3145.	YES	YES
2	539.	3 - 20MM	0.	3145.	YES	YES

B E A M N O. 135 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 60.38 KNS Vc= 141.34 KNS Vs= 0.00 KNS
Tu= 0.11 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 218
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 1036. MM

AT END SUPPORT - Vu= 57.73 KNS Vc= 138.04 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 6.2 KN-MET Ts= 0.0 KN-MET LOAD 117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 1036. MM



ACI 318-11 BEAM NO. 136 DESIGN RESULTS

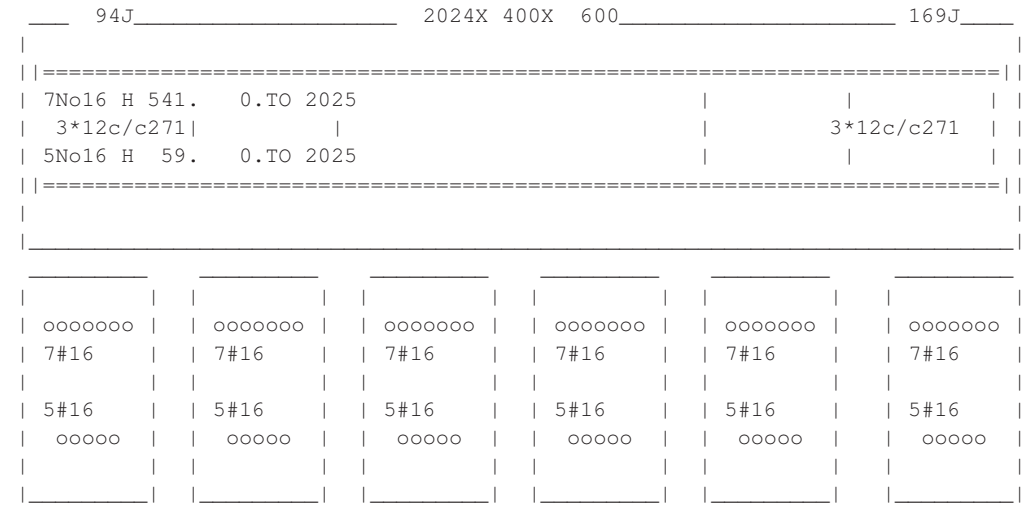
LEN - 2025. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	5 - 16MM	0.	2025.	YES	YES
2	541.	7 - 16MM	0.	2025.	YES	YES

B E A M N O. 136 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 118.34 KNS Vc= 188.86 KNS Vs= 0.00 KNS
Tu= 2.72 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 476. MM

AT END SUPPORT - Vu= 86.45 KNS Vc= 188.86 KNS Vs= 0.00 KNS
Tu= 2.72 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 476. MM



ACI 318-11 BEAM NO. 137 DESIGN RESULTS

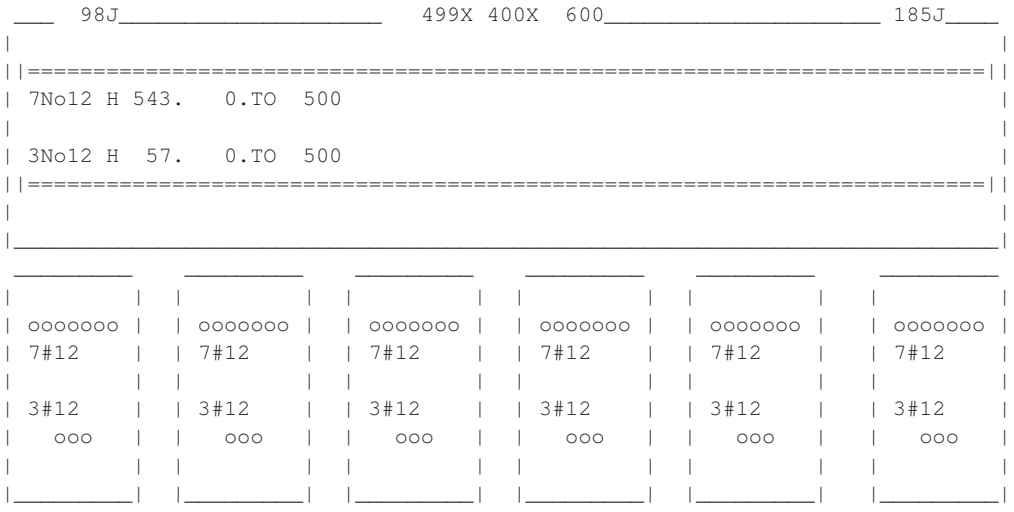
LEN - 500. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	500.	YES	YES
2	543.	7 - 12MM	0.	500.	YES	YES

B E A M N O. 137 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 137 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 137 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 138 DESIGN RESULTS

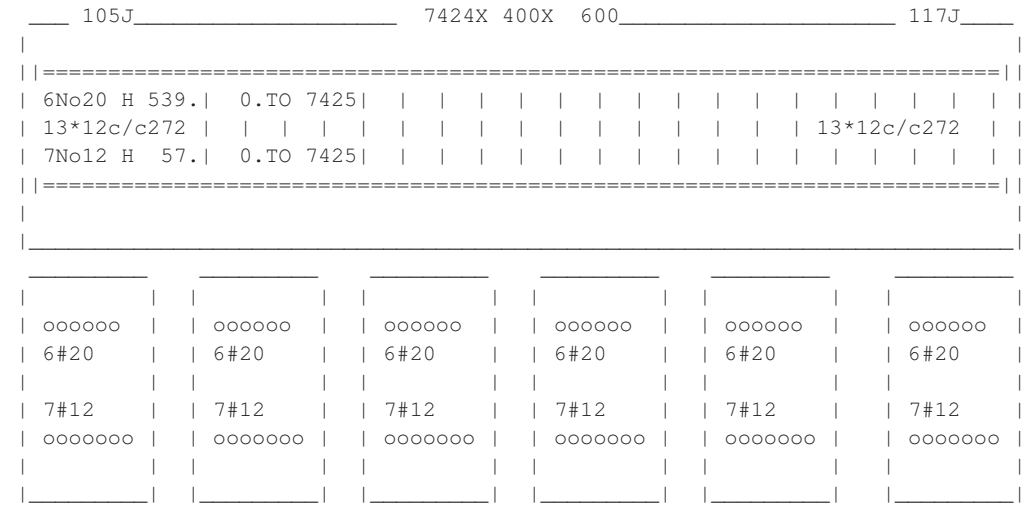
LEN - 7425. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	7 - 12MM	0.	7425.	YES	YES
2	539.	6 - 20MM	0.	7425.	YES	YES

B E A M N O. 138 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 98.82 KNS Vc= 189.72 KNS Vs= 0.00 KNS
Tu= 1.15 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 3176. MM

AT END SUPPORT - Vu= 113.97 KNS Vc= 189.72 KNS Vs= 0.00 KNS
Tu= 1.15 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 3176. MM



ACI 318-11 BEAM NO. 139 DESIGN RESULTS

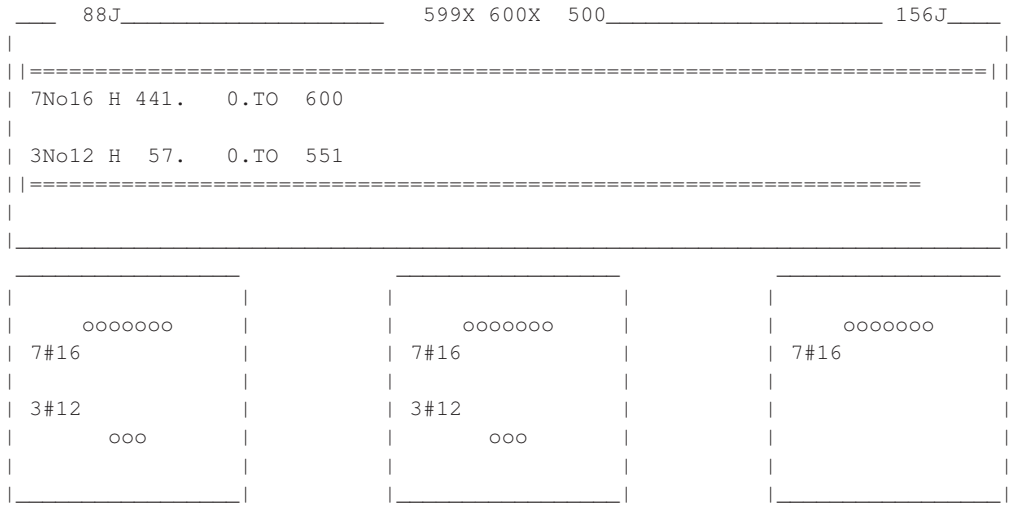
LEN - 600. MM FY - 414. FC - 28. MPA, SIZE - 600. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	551.	YES	NO
2	441.	7 - 16MM	0.	600.	YES	YES

B E A M N O. 139 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 139 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 139 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 141 DESIGN RESULTS

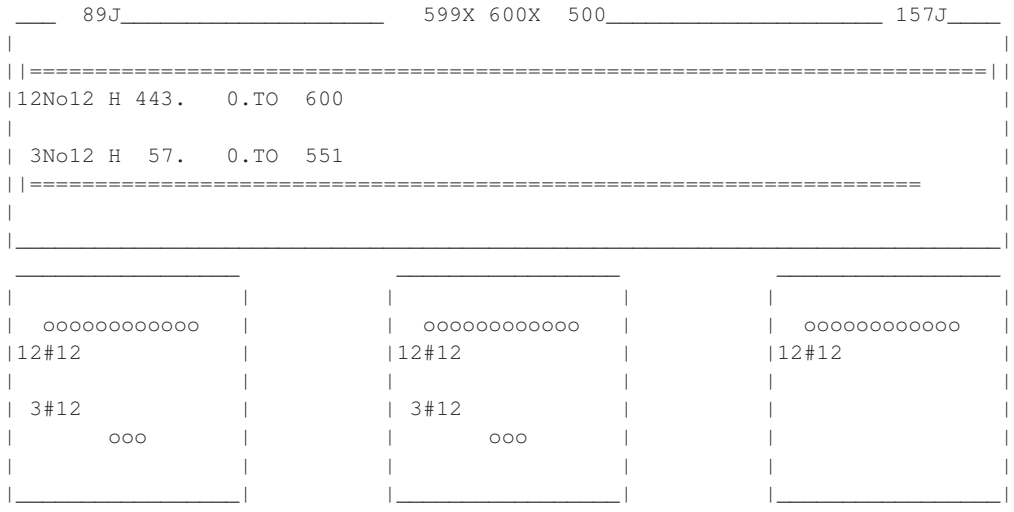
LEN - 600. MM FY - 414. FC - 28. MPA, SIZE - 600. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	551.	YES	NO
2	443.	12 - 12MM	0.	600.	YES	YES

B E A M N O. 141 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 141 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 141 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 143 DESIGN RESULTS

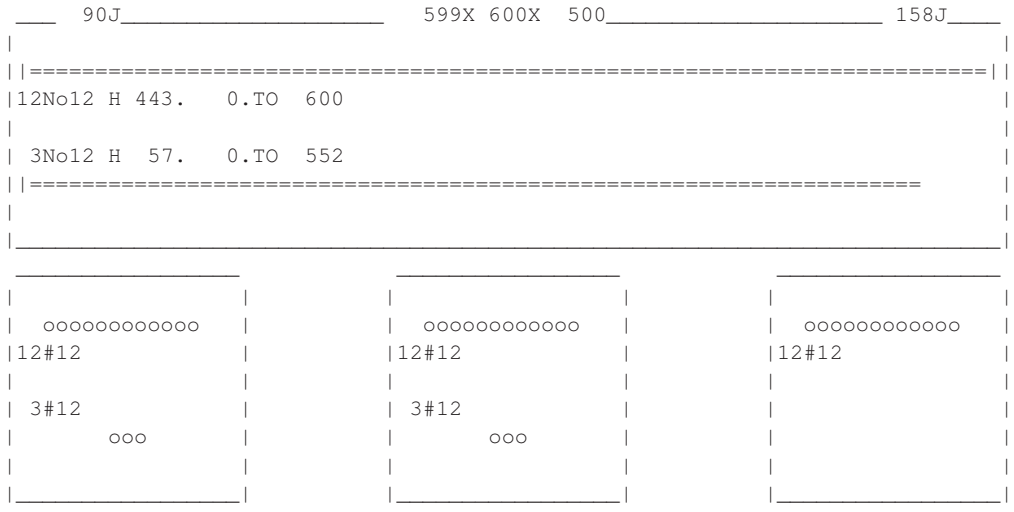
LEN - 600. MM FY - 414. FC - 28. MPA, SIZE - 600. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	552.	YES	NO
2	443.	12 - 12MM	0.	600.	YES	YES

B E A M N O. 143 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 143 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 143 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 144 DESIGN RESULTS

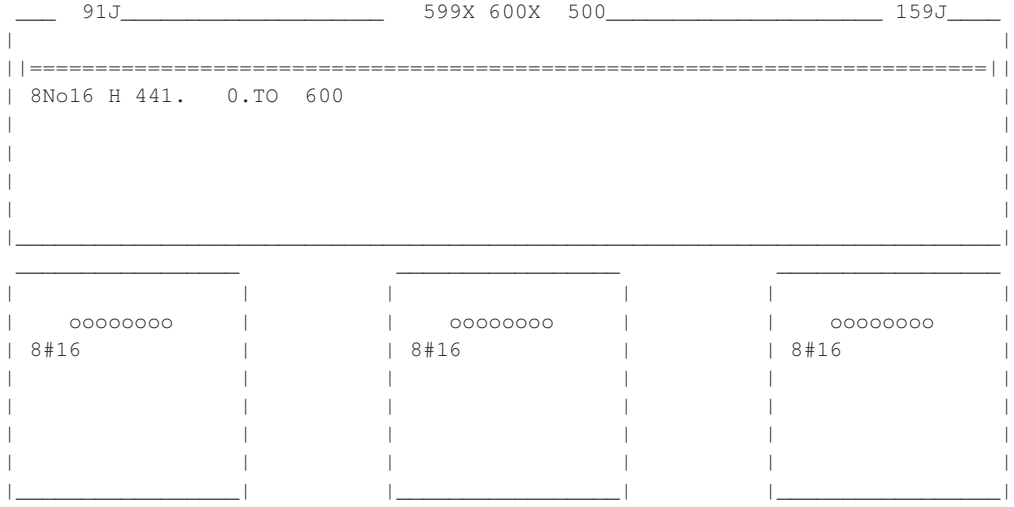
LEN - 600. MM FY - 414. FC - 28. MPA, SIZE - 600. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	441.	8 - 16MM	0.	600.	YES	YES

B E A M N O. 144 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 144 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 144 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 145 DESIGN RESULTS

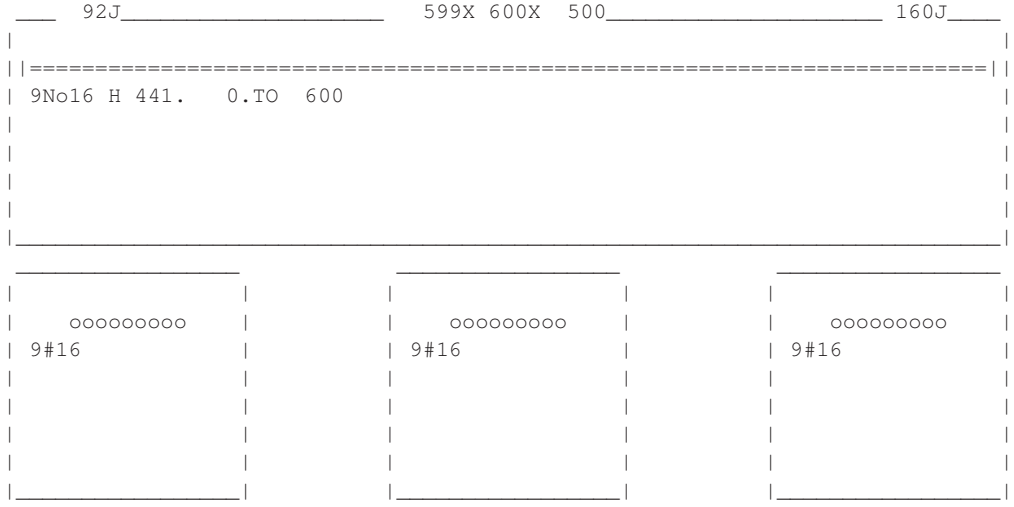
LEN - 600. MM FY - 414. FC - 28. MPA, SIZE - 600. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	441.	9 - 16MM	0.	600.	YES	YES

B E A M N O. 145 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 145 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 145 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 146 DESIGN RESULTS

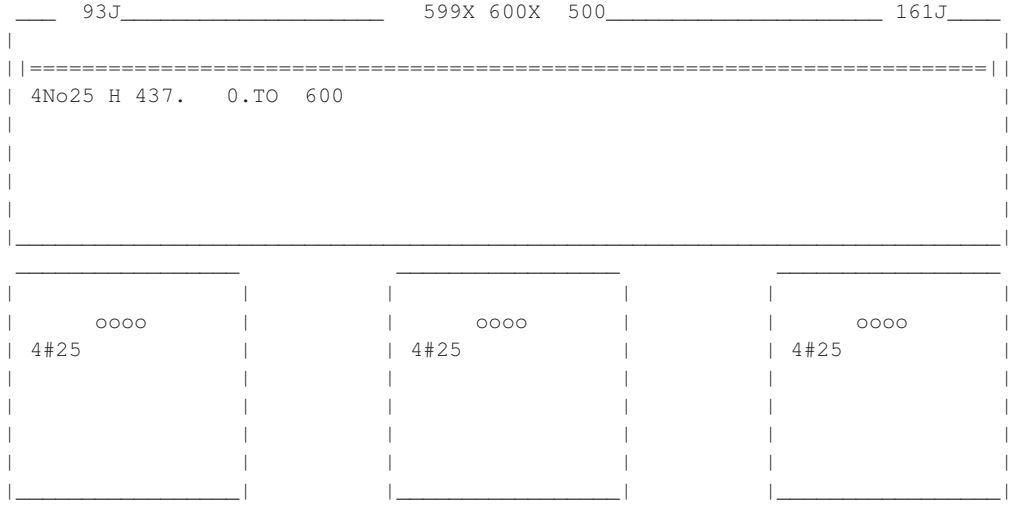
LEN - 600. MM FY - 414. FC - 28. MPA, SIZE - 600. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	437.	4 - 25MM	0.	600.	YES	YES

B E A M N O. 146 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 146 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 146 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



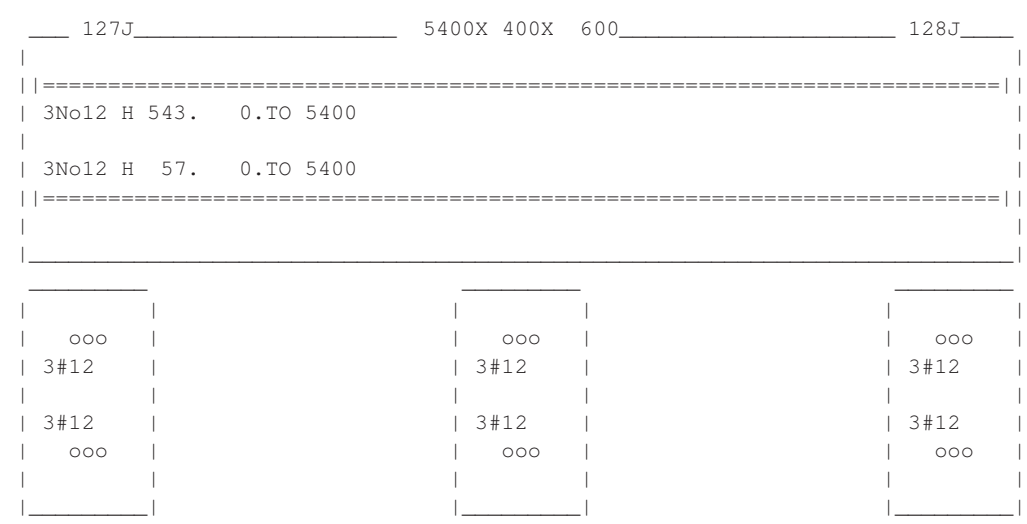
ACI 318-11 BEAM NO. 171 DESIGN RESULTS

LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	3 - 12MM	0.	5400.	YES	YES

B E A M N O. 171 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 14.15 KNS Vc= 182.03 KNS Vs= 0.00 KNS
Tu= 0.07 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.81 KNS Vc= 185.63 KNS Vs= 0.00 KNS
Tu= 0.07 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



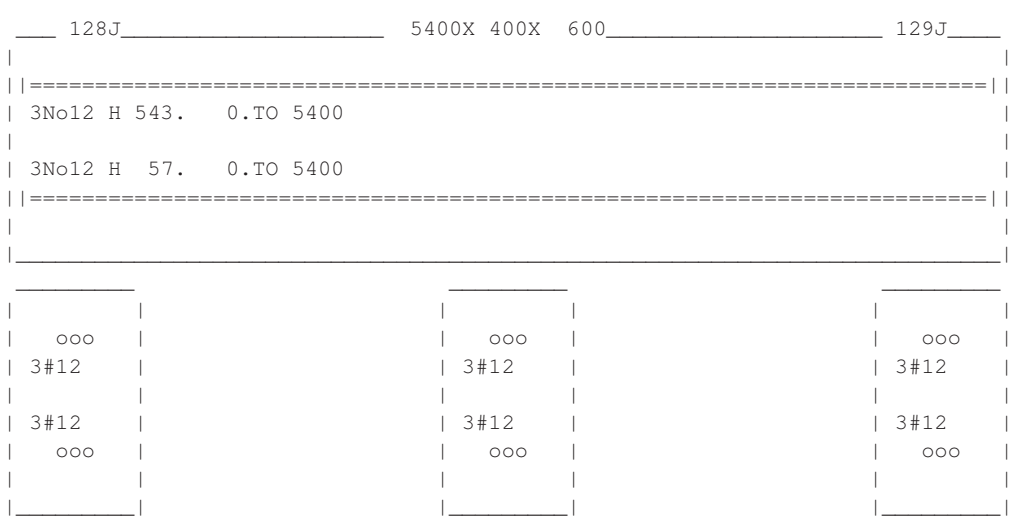
ACI 318-11 BEAM NO. 172 DESIGN RESULTS

LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	3 - 12MM	0.	5400.	YES	YES

B E A M N O. 172 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.42 KNS Vc= 184.71 KNS Vs= 0.00 KNS
Tu= 0.58 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.54 KNS Vc= 182.69 KNS Vs= 0.00 KNS
Tu= 0.58 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 173 DESIGN RESULTS

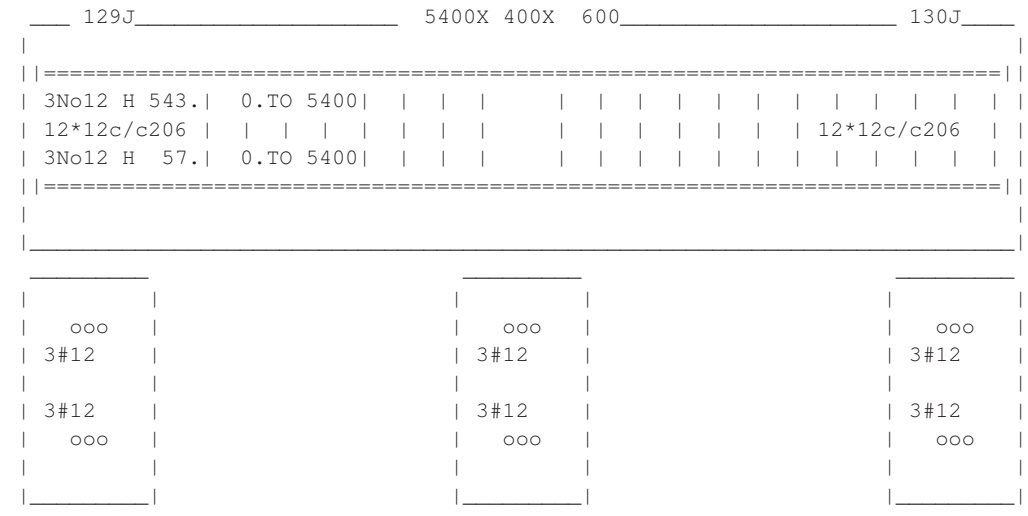
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	3 - 12MM	0.	5400.	YES	YES

B E A M N O. 173 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.72 KNS Vc= 184.84 KNS Vs= 0.00 KNS
Tu= 10.91 KN-MET Tc= 9.5 KN-MET Ts= 14.5 KN-MET LOAD 218
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 206. MM C/C FOR 2164. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.13 SQ.CM.

AT END SUPPORT - Vu= 17.43 KNS Vc= 181.77 KNS Vs= 0.00 KNS
Tu= 10.91 KN-MET Tc= 9.5 KN-MET Ts= 14.5 KN-MET LOAD 218
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 206. MM C/C FOR 2164. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.13 SQ.CM.



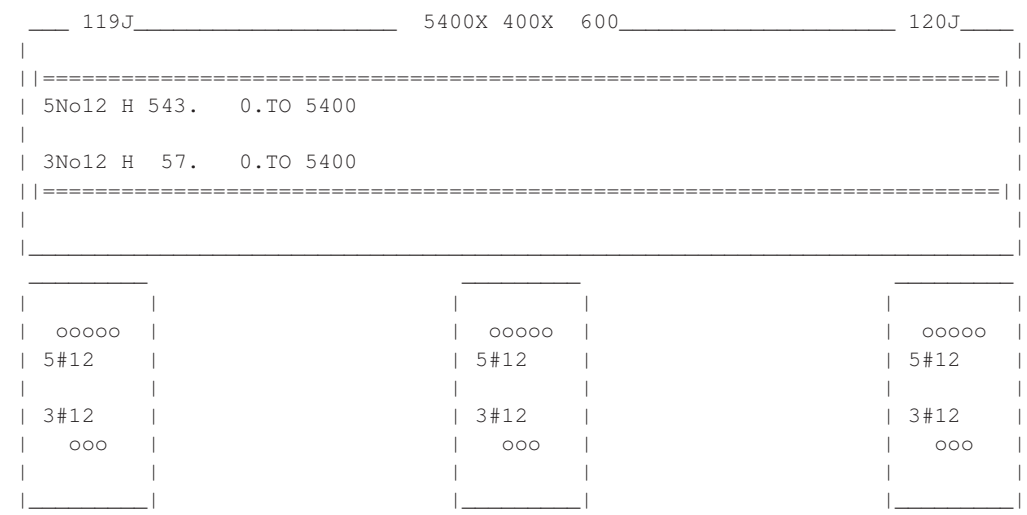
ACI 318-11 BEAM NO. 174 DESIGN RESULTS

LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	5 - 12MM	0.	5400.	YES	YES

B E A M N O. 174 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 18.93 KNS Vc= 182.72 KNS Vs= 0.00 KNS
Tu= 3.05 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 7.03 KNS Vc= 183.90 KNS Vs= 0.00 KNS
Tu= 3.05 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



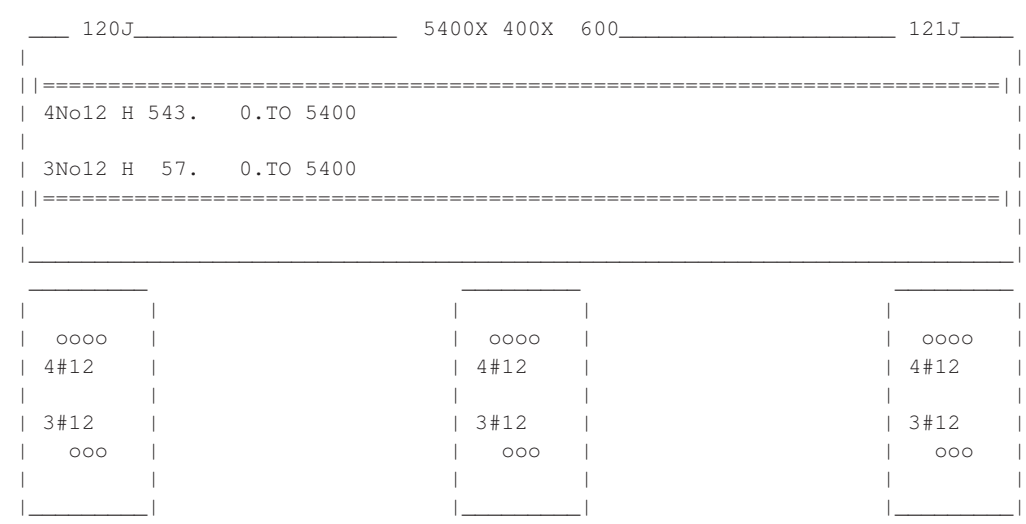
ACI 318-11 BEAM NO. 175 DESIGN RESULTS

LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	4 - 12MM	0.	5400.	YES	YES

B E A M N O. 175 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.29 KNS Vc= 189.30 KNS Vs= 0.00 KNS
Tu= 0.34 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.67 KNS Vc= 189.30 KNS Vs= 0.00 KNS
Tu= 0.34 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



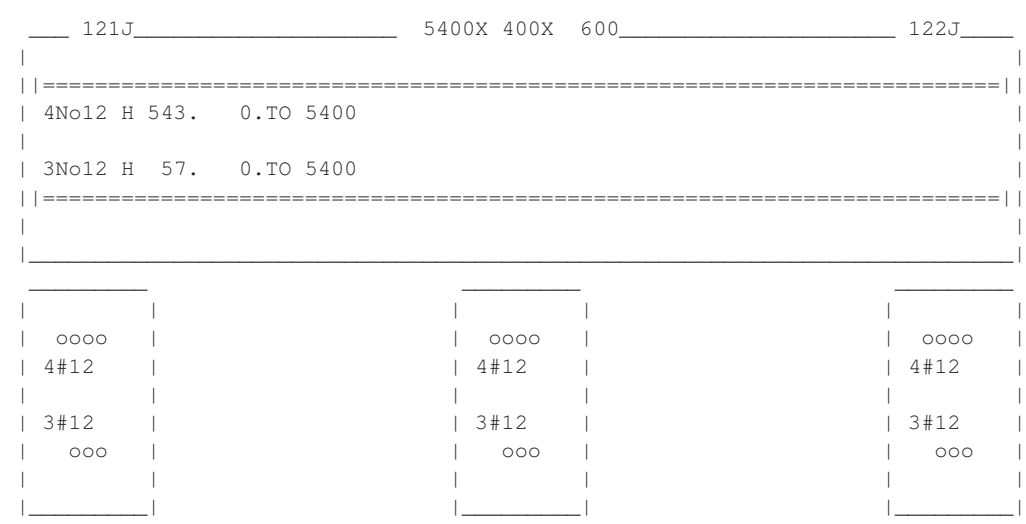
ACI 318-11 BEAM NO. 176 DESIGN RESULTS

LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	4 - 12MM	0.	5400.	YES	YES

B E A M N O. 176 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.69 KNS Vc= 189.60 KNS Vs= 0.00 KNS
Tu= 0.59 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.27 KNS Vc= 189.60 KNS Vs= 0.00 KNS
Tu= 0.59 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



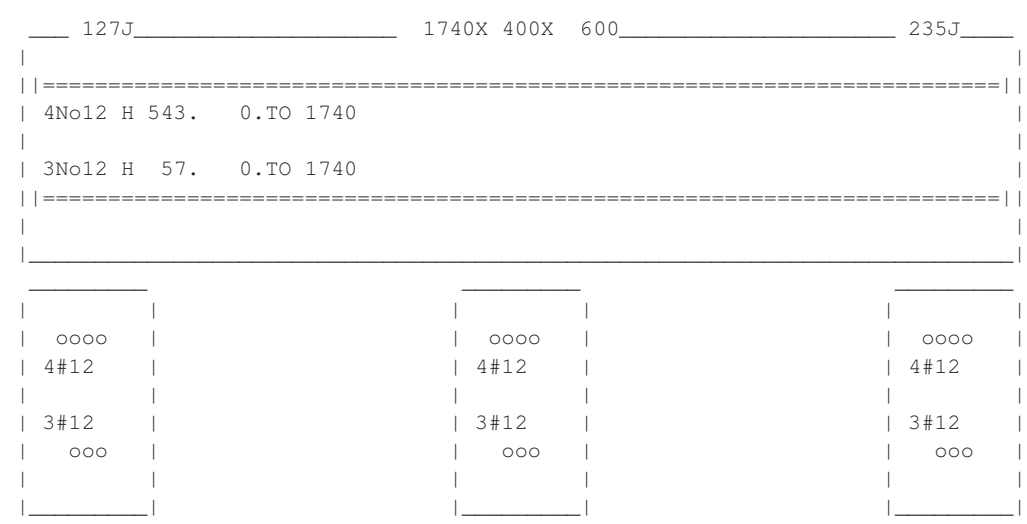
ACI 318-11 BEAM NO. 177 DESIGN RESULTS

LEN - 1740. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	1740.	YES	YES
2	543.	4 - 12MM	0.	1740.	YES	YES

B E A M N O. 177 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 25.40 KNS Vc= 333.39 KNS Vs= 0.00 KNS
Tu= 0.15 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.15 KNS Vc= 183.27 KNS Vs= 0.00 KNS
Tu= 0.15 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 178 DESIGN RESULTS

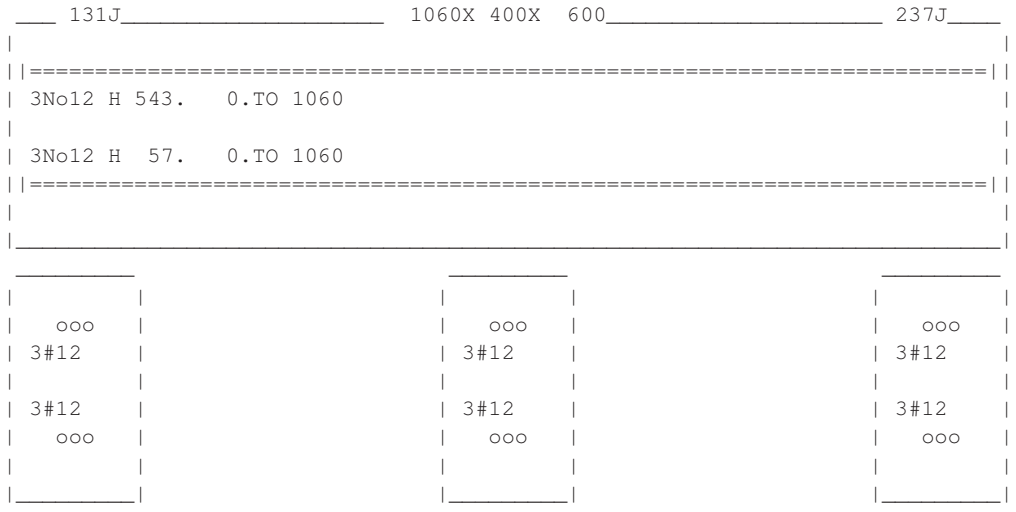
LEN - 1060. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	1060.	YES	YES
2	543.	3 - 12MM	0.	1060.	YES	YES

B E A M N O. 178 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 178 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 178 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



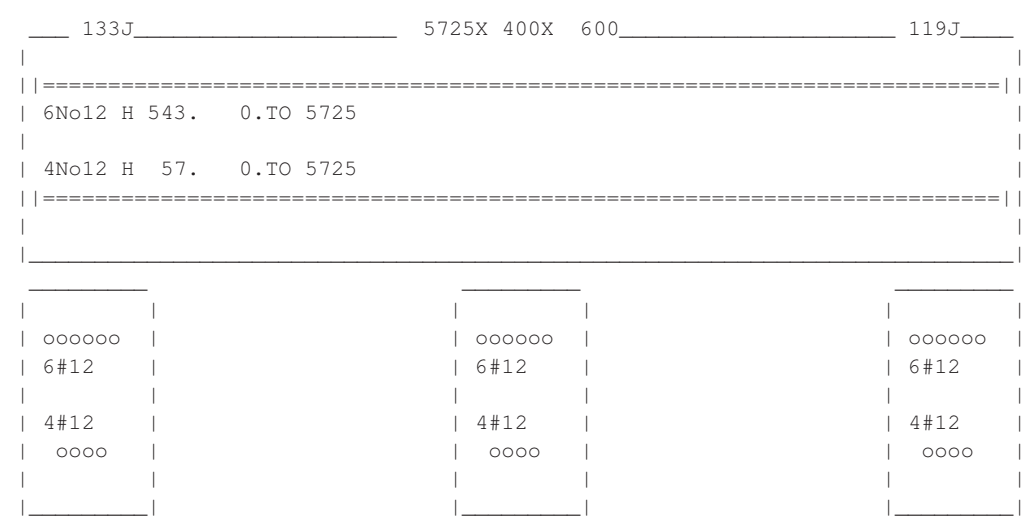
ACI 318-11 BEAM NO. 179 DESIGN RESULTS

LEN - 5725. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	5725.	YES	YES
2	543.	6 - 12MM	0.	5725.	YES	YES

B E A M N O. 179 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 42.99 KNS Vc= 188.23 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 42.42 KNS Vc= 188.76 KNS Vs= 0.00 KNS
Tu= 0.10 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 182 DESIGN RESULTS

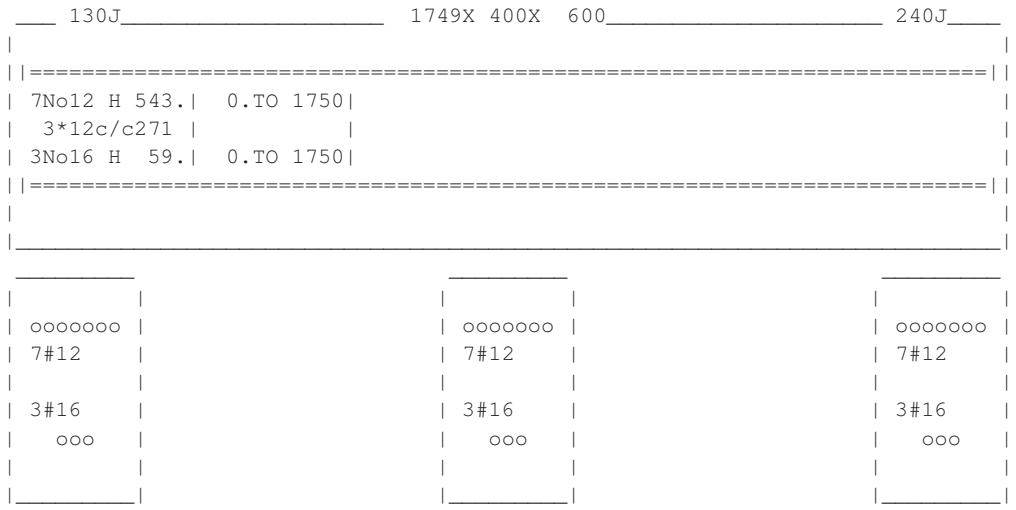
LEN - 1750. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	3 - 16MM	0.	1750.	YES	YES
2	543.	7 - 12MM	0.	1750.	YES	YES

B E A M N O. 182 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 83.32 KNS Vc= 211.45 KNS Vs= 0.00 KNS
Tu= 2.95 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 202
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 338. MM

AT END SUPPORT - Vu= 32.98 KNS Vc= 190.71 KNS Vs= 0.00 KNS
Tu= 0.61 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



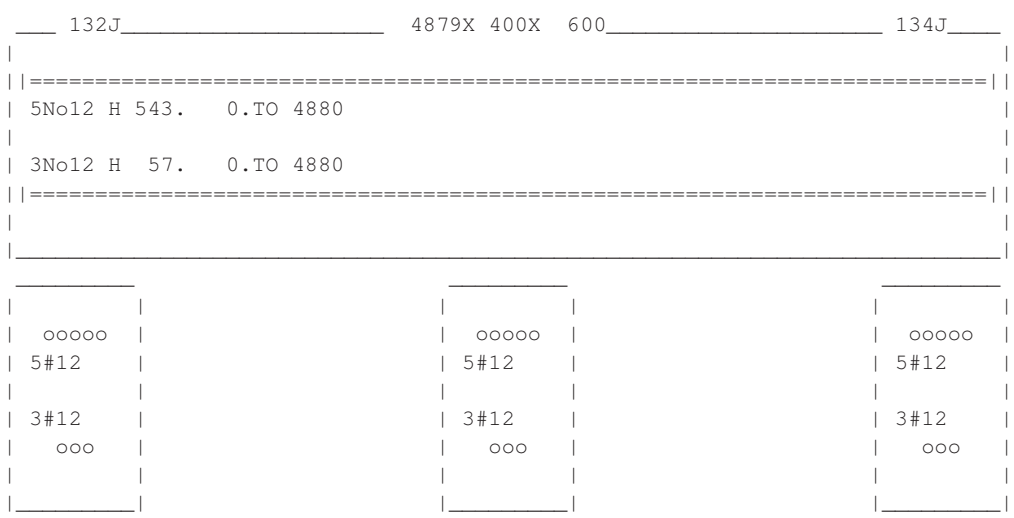
ACI 318-11 BEAM NO. 183 DESIGN RESULTS

LEN - 4880. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	4880.	YES	YES
2	543.	5 - 12MM	0.	4880.	YES	YES

B E A M N O. 183 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 33.32 KNS Vc= 189.20 KNS Vs= 0.00 KNS
Tu= 0.07 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 36.57 KNS Vc= 189.20 KNS Vs= 0.00 KNS
Tu= 0.07 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



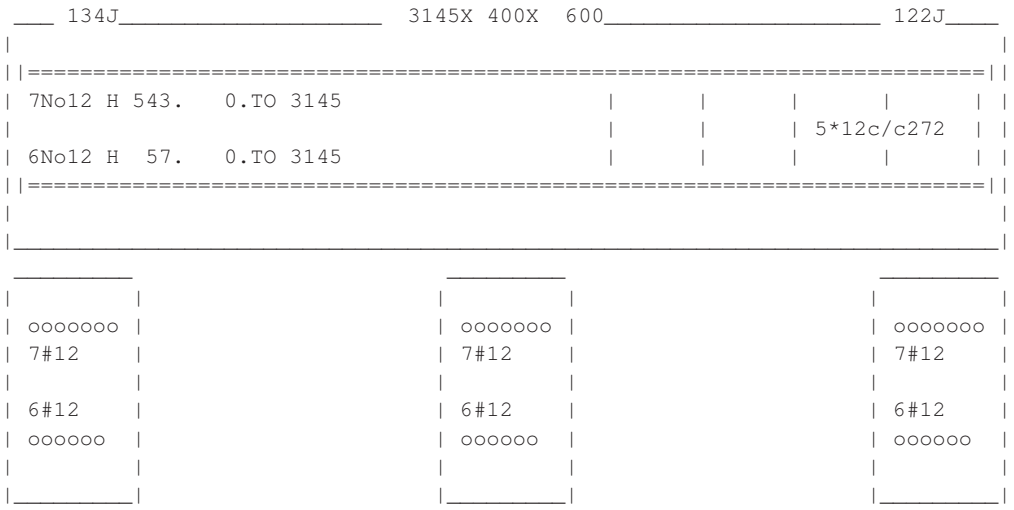
ACI 318-11 BEAM NO. 184 DESIGN RESULTS

LEN - 3145. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	6 - 12MM	0.	3145.	YES	YES
2	543.	7 - 12MM	0.	3145.	YES	YES

B E A M N O. 184 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 22.22 KNS Vc= 186.90 KNS Vs= 0.00 KNS
Tu= 0.49 KN-MET Tc= 9.3 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 70.89 KNS Vc= 172.73 KNS Vs= 0.00 KNS
Tu= 0.43 KN-MET Tc= 8.5 KN-MET Ts= 0.0 KN-MET LOAD 217
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 1036. MM



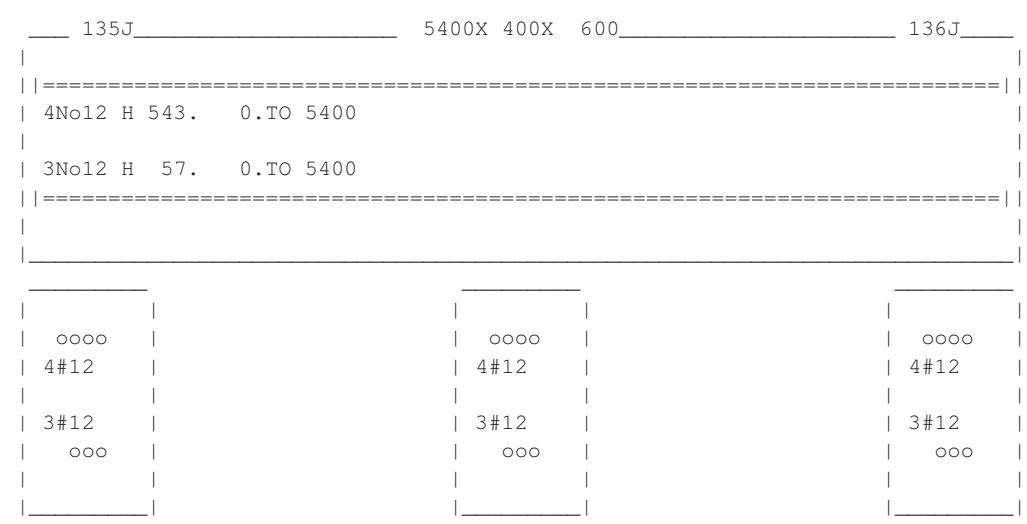
ACI 318-11 BEAM NO. 206 DESIGN RESULTS

LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	4 - 12MM	0.	5400.	YES	YES

B E A M N O. 206 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 17.32 KNS Vc= 181.96 KNS Vs= 0.00 KNS
Tu= 1.20 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 8.65 KNS Vc= 188.43 KNS Vs= 0.00 KNS
Tu= 1.20 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



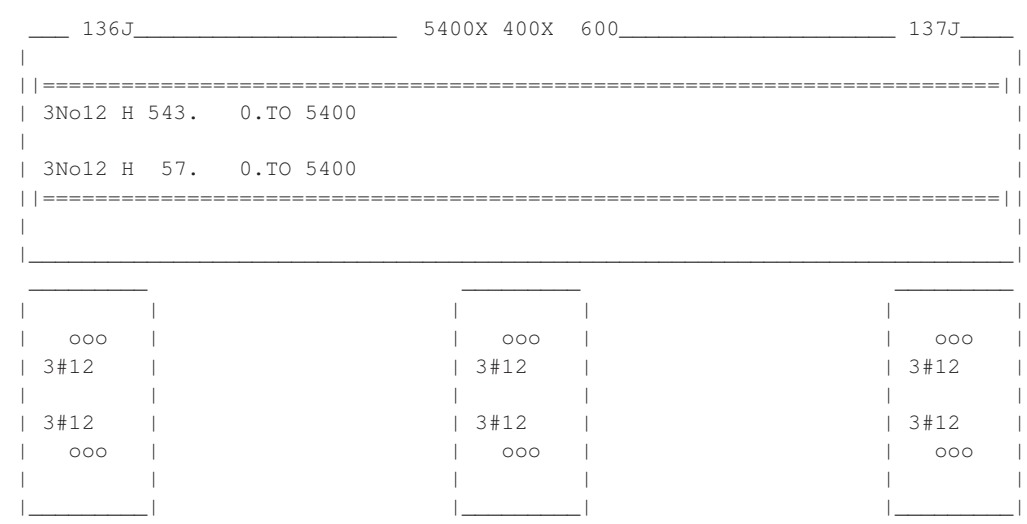
ACI 318-11 BEAM NO. 207 DESIGN RESULTS

LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	3 - 12MM	0.	5400.	YES	YES

B E A M N O. 207 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.85 KNS Vc= 188.20 KNS Vs= 0.00 KNS
Tu= 0.59 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.12 KNS Vc= 185.95 KNS Vs= 0.00 KNS
Tu= 0.59 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



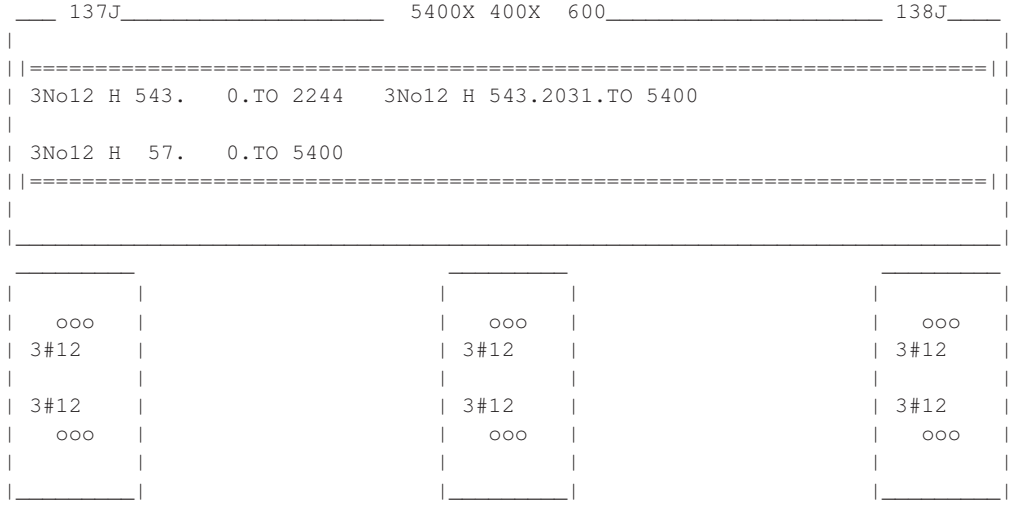
ACI 318-11 BEAM NO. 208 DESIGN RESULTS

LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	3 - 12MM	0.	2244.	YES	NO
3	543.	3 - 12MM	2031.	5400.	NO	YES

B E A M N O. 208 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.90 KNS Vc= 193.64 KNS Vs= 0.00 KNS
Tu= 2.16 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 14.06 KNS Vc= 182.36 KNS Vs= 0.00 KNS
Tu= 2.16 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



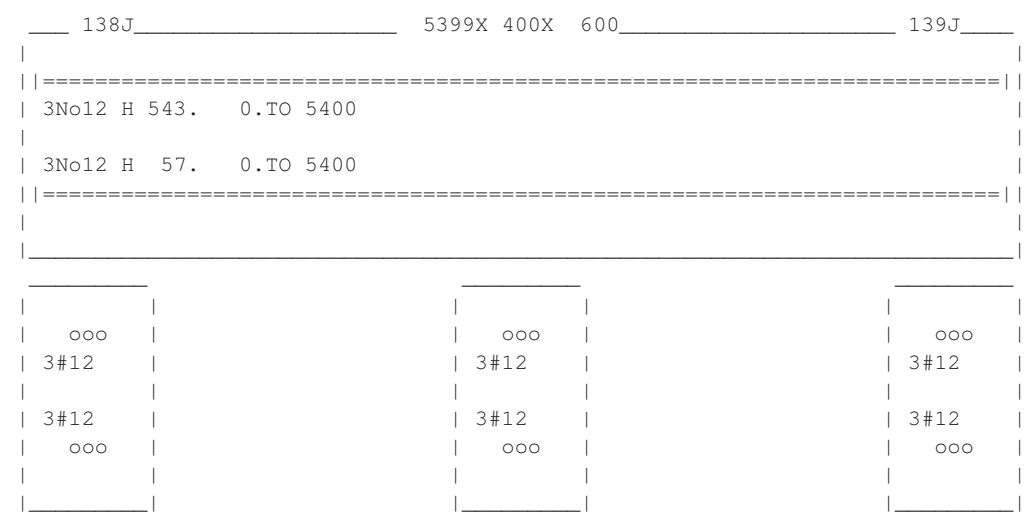
ACI 318-11 BEAM NO. 209 DESIGN RESULTS

LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	3 - 12MM	0.	5400.	YES	YES

B E A M N O. 209 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.15 KNS Vc= 182.42 KNS Vs= 0.00 KNS
Tu= 0.71 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.81 KNS Vc= 182.82 KNS Vs= 0.00 KNS
Tu= 0.71 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



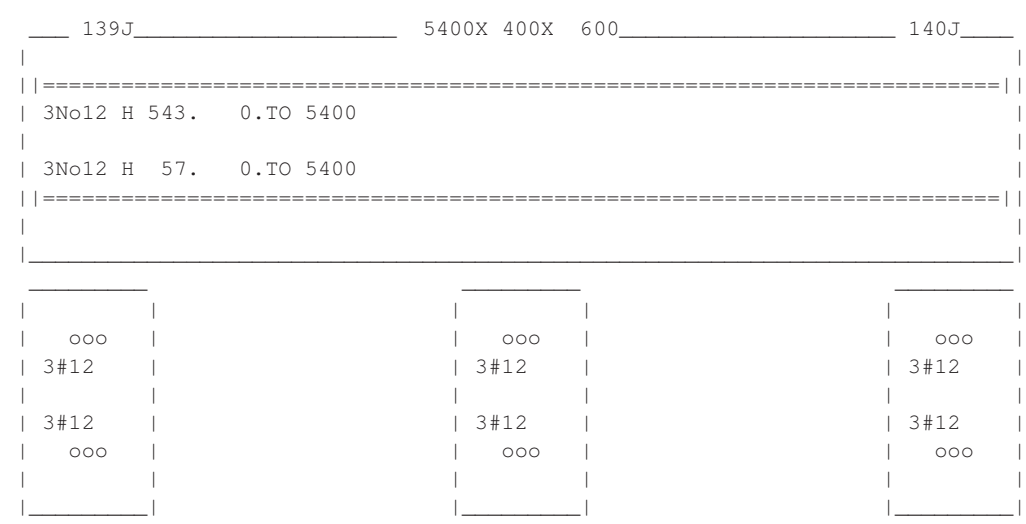
ACI 318-11 BEAM NO. 210 DESIGN RESULTS

LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	3 - 12MM	0.	5400.	YES	YES

B E A M N O. 210 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.50 KNS Vc= 182.33 KNS Vs= 0.00 KNS
Tu= 0.21 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.46 KNS Vc= 183.96 KNS Vs= 0.00 KNS
Tu= 0.21 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



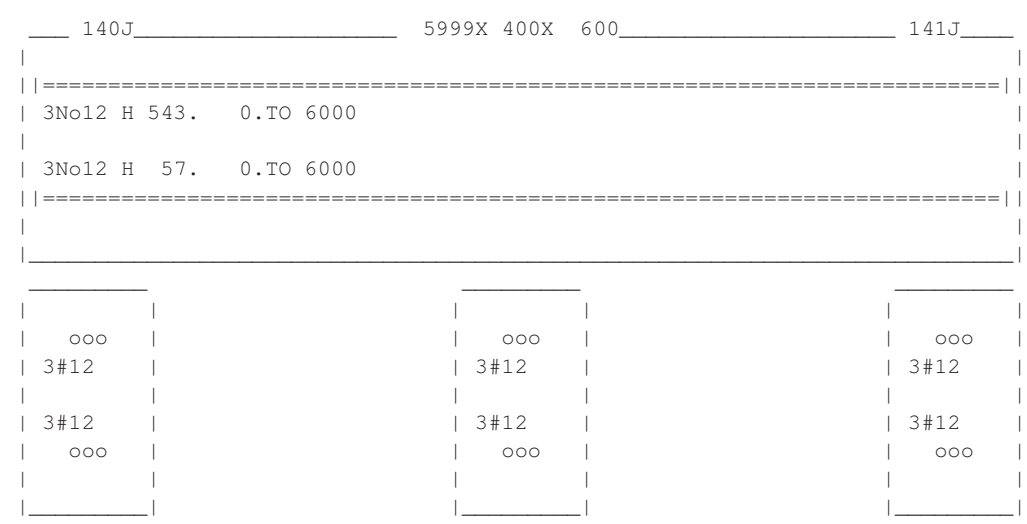
ACI 318-11 BEAM NO. 211 DESIGN RESULTS

LEN - 6000. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	6000.	YES	YES
2	543.	3 - 12MM	0.	6000.	YES	YES

B E A M N O. 211 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 16.23 KNS Vc= 182.62 KNS Vs= 0.00 KNS
Tu= 0.47 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.33 KNS Vc= 335.77 KNS Vs= 0.00 KNS
Tu= 0.47 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



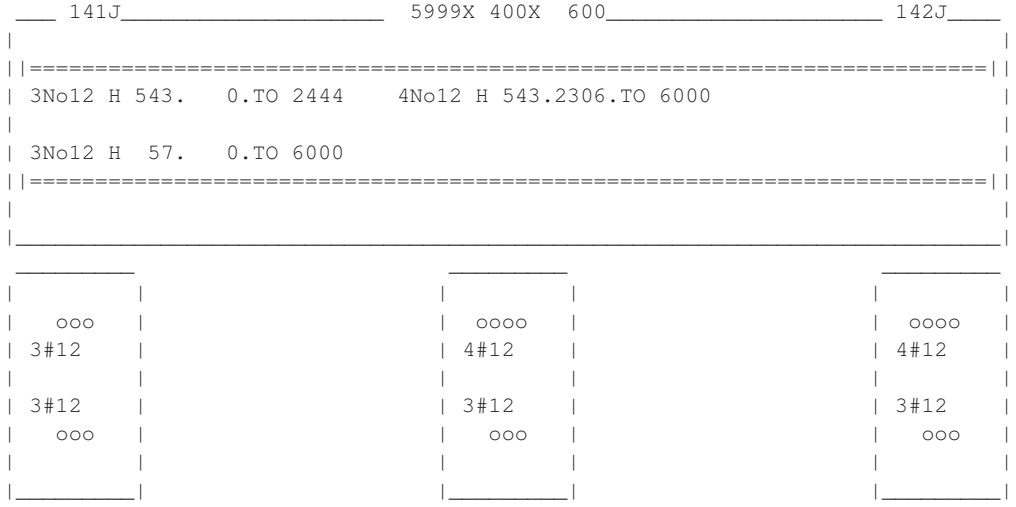
ACI 318-11 BEAM NO. 212 DESIGN RESULTS

LEN - 6000. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	6000.	YES	YES
2	543.	3 - 12MM	0.	2444.	YES	NO
3	543.	4 - 12MM	2306.	6000.	NO	YES

B E A M N O. 212 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.20 KNS Vc= 208.13 KNS Vs= 0.00 KNS
Tu= 0.83 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 17.36 KNS Vc= 181.64 KNS Vs= 0.00 KNS
Tu= 0.83 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 213 DESIGN RESULTS

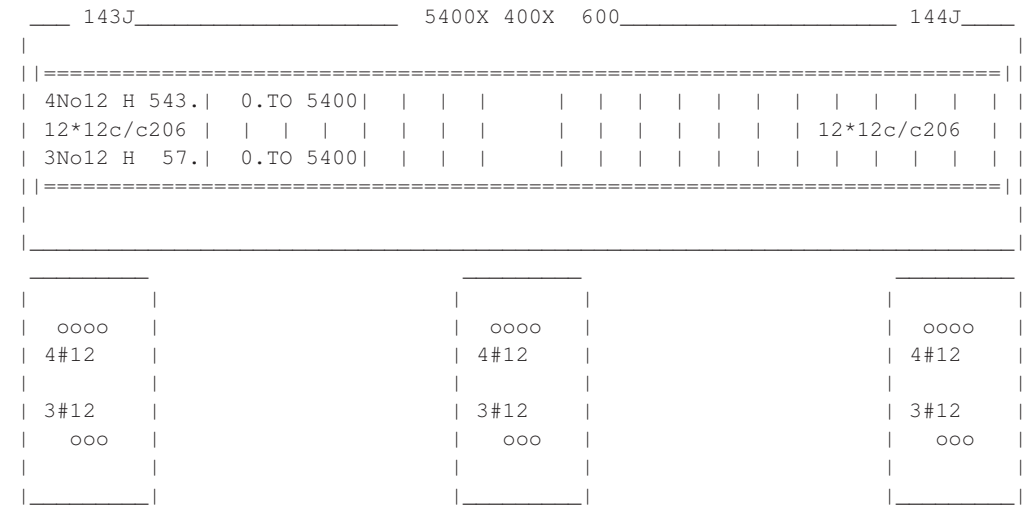
LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	4 - 12MM	0.	5400.	YES	YES

B E A M N O. 213 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 16.46 KNS Vc= 189.50 KNS Vs= 0.00 KNS
Tu= 12.06 KN-MET Tc= 9.4 KN-MET Ts= 16.1 KN-MET LOAD 217
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 206. MM C/C FOR 2164. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.36 SQ.CM.

AT END SUPPORT - Vu= 14.69 KNS Vc= 189.50 KNS Vs= 0.00 KNS
Tu= 12.06 KN-MET Tc= 9.4 KN-MET Ts= 16.1 KN-MET LOAD 217
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 206. MM C/C FOR 2164. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.36 SQ.CM.



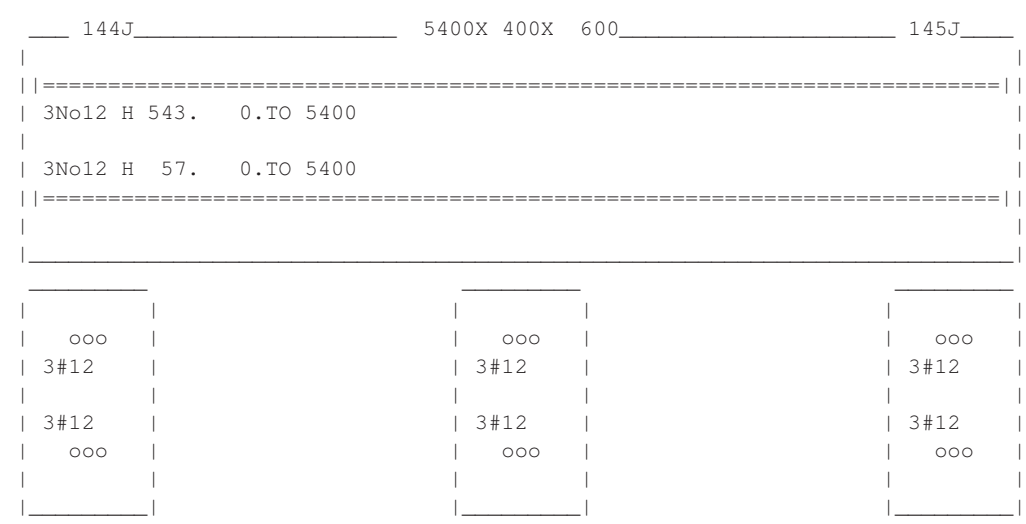
ACI 318-11 BEAM NO. 214 DESIGN RESULTS

LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	3 - 12MM	0.	5400.	YES	YES

B E A M N O. 214 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.35 KNS Vc= 183.96 KNS Vs= 0.00 KNS
Tu= 2.91 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.62 KNS Vc= 182.53 KNS Vs= 0.00 KNS
Tu= 2.91 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



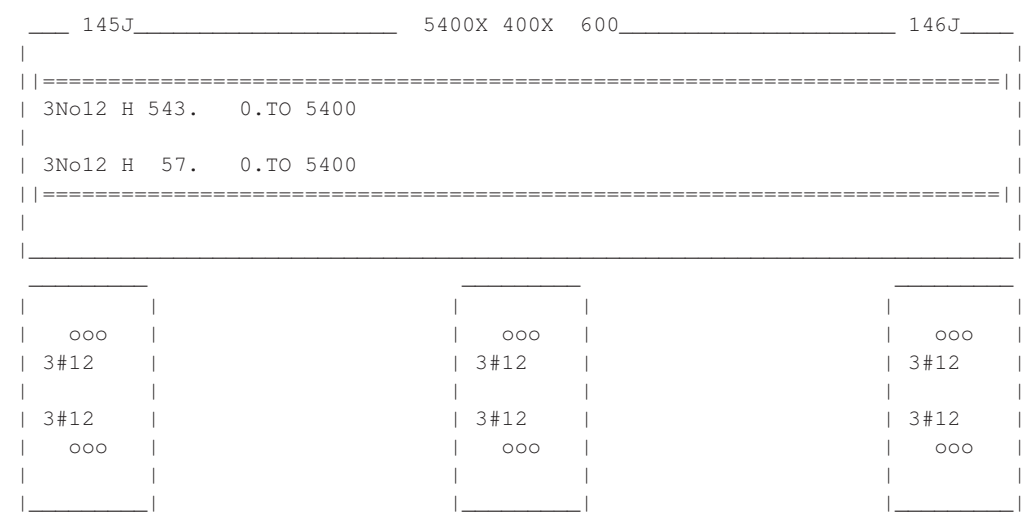
ACI 318-11 BEAM NO. 215 DESIGN RESULTS

LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	3 - 12MM	0.	5400.	YES	YES

B E A M N O. 215 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.64 KNS Vc= 183.65 KNS Vs= 0.00 KNS
Tu= 1.09 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 14.33 KNS Vc= 181.80 KNS Vs= 0.00 KNS
Tu= 1.09 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



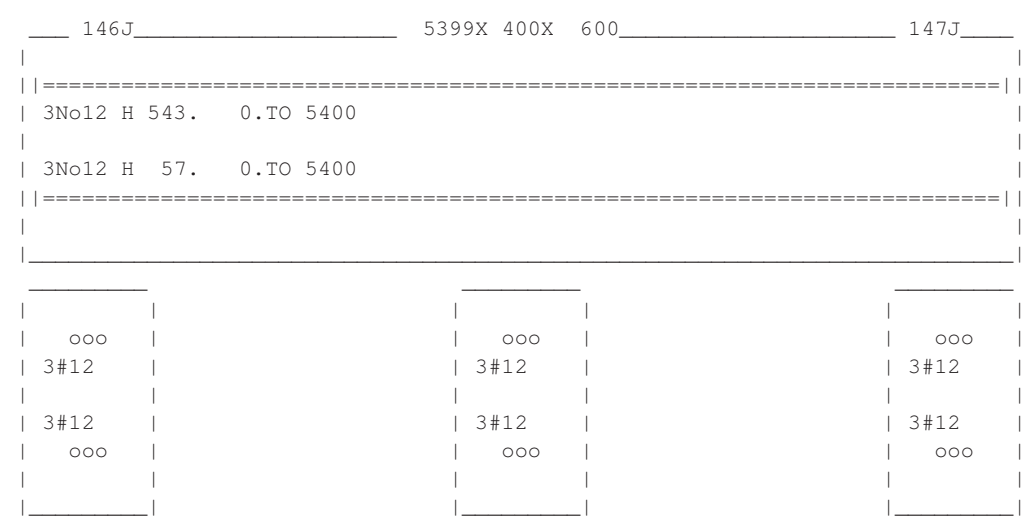
ACI 318-11 BEAM NO. 216 DESIGN RESULTS

LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	3 - 12MM	0.	5400.	YES	YES

B E A M N O. 216 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.49 KNS Vc= 189.61 KNS Vs= 0.00 KNS
Tu= 1.23 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.47 KNS Vc= 189.61 KNS Vs= 0.00 KNS
Tu= 1.23 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



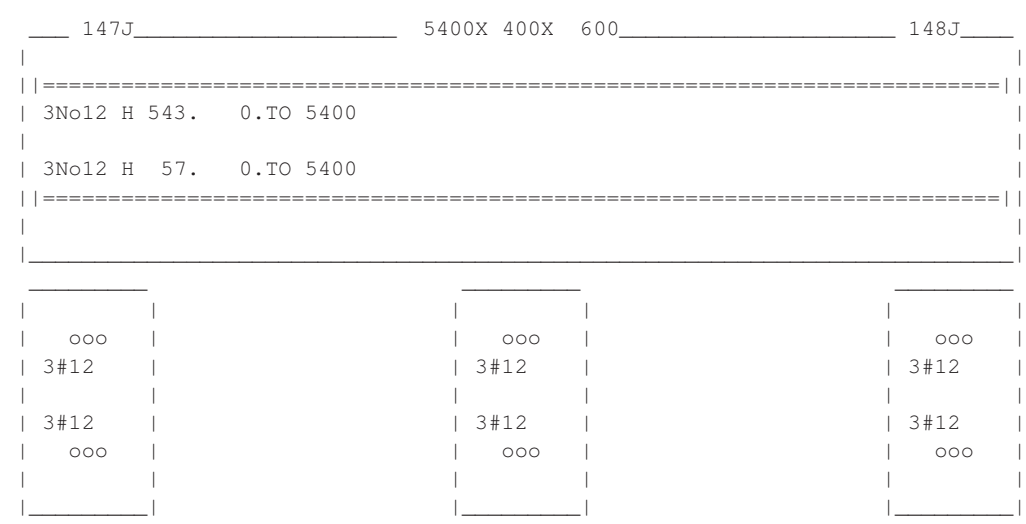
ACI 318-11 BEAM NO. 217 DESIGN RESULTS

LEN - 5400. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	5400.	YES	YES
2	543.	3 - 12MM	0.	5400.	YES	YES

B E A M N O. 217 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.14 KNS Vc= 189.16 KNS Vs= 0.00 KNS
Tu= 0.16 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.82 KNS Vc= 189.16 KNS Vs= 0.00 KNS
Tu= 0.16 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



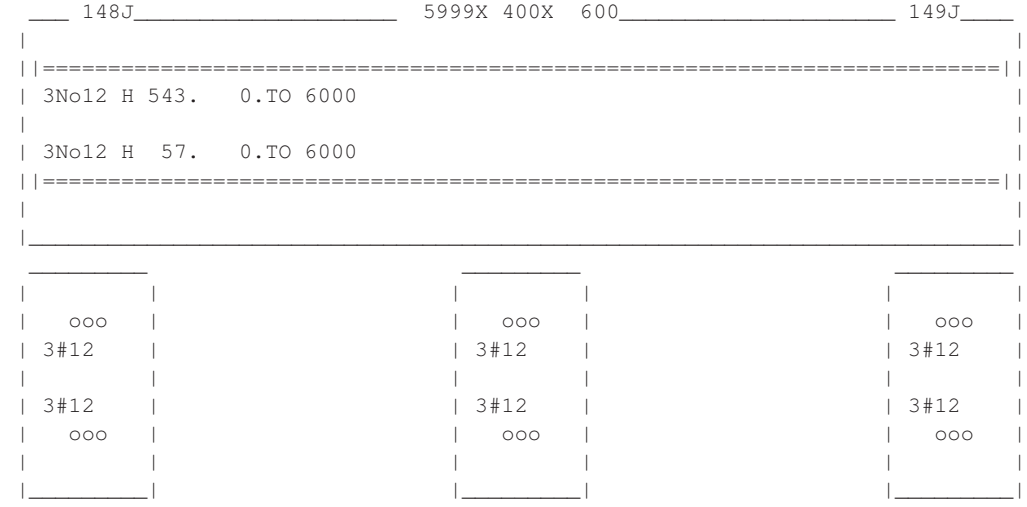
ACI 318-11 BEAM NO. 218 DESIGN RESULTS

LEN - 6000. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	6000.	YES	YES
2	543.	3 - 12MM	0.	6000.	YES	YES

B E A M N O. 218 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 14.84 KNS Vc= 189.67 KNS Vs= 0.00 KNS
Tu= 0.92 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 14.73 KNS Vc= 189.67 KNS Vs= 0.00 KNS
Tu= 0.92 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 219 DESIGN RESULTS

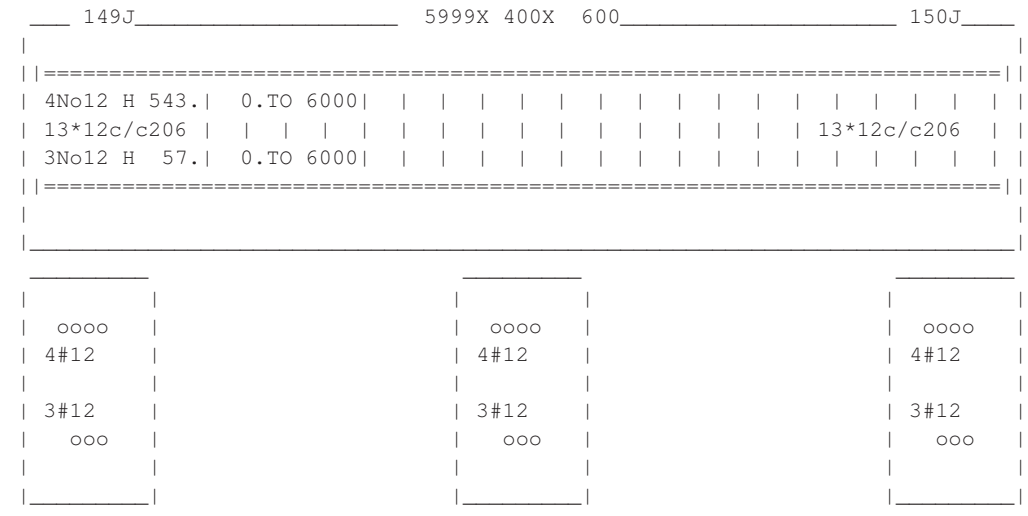
LEN - 6000. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	6000.	YES	YES
2	543.	4 - 12MM	0.	6000.	YES	YES

B E A M N O. 219 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.43 KNS Vc= 187.48 KNS Vs= 0.00 KNS
Tu= 17.45 KN-MET Tc= 9.3 KN-MET Ts= 23.3 KN-MET LOAD 217
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 206. MM C/C FOR 2463. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.42 SQ.CM.

AT END SUPPORT - Vu= 23.04 KNS Vc= 187.48 KNS Vs= 0.00 KNS
Tu= 17.45 KN-MET Tc= 9.3 KN-MET Ts= 23.3 KN-MET LOAD 217
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 206. MM C/C FOR 2463. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 3.42 SQ.CM.



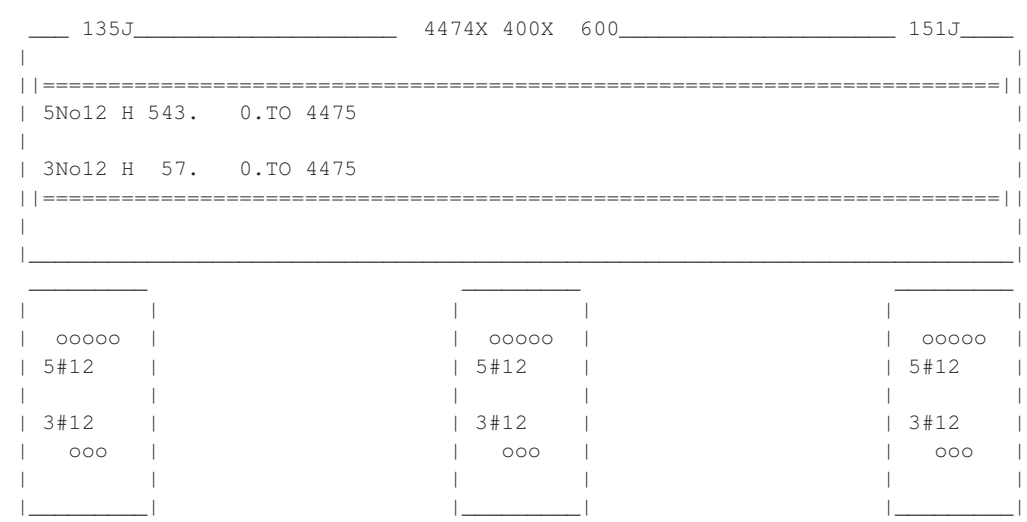
ACI 318-11 BEAM NO. 220 DESIGN RESULTS

LEN - 4475. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	4475.	YES	YES
2	543.	5 - 12MM	0.	4475.	YES	YES

B E A M N O. 220 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 32.74 KNS Vc= 190.31 KNS Vs= 0.00 KNS
Tu= 0.60 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 29.73 KNS Vc= 204.58 KNS Vs= 0.00 KNS
Tu= 0.60 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



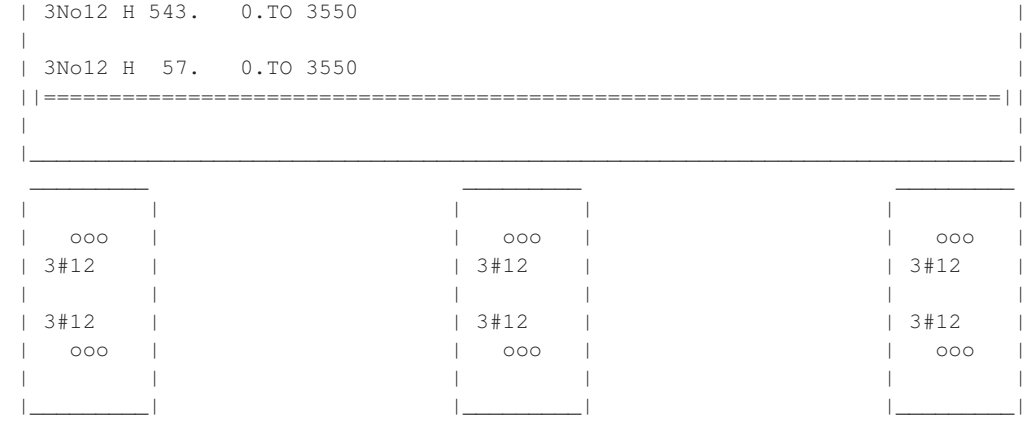
ACI 318-11 BEAM NO. 221 DESIGN RESULTS

LEN - 3550. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	3550.	YES	YES
2	543.	3 - 12MM	0.	3550.	YES	YES

B E A M N O. 221 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 21.77 KNS Vc= 185.89 KNS Vs= 0.00 KNS
Tu= 0.14 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 23.71 KNS Vc= 183.81 KNS Vs= 0.00 KNS
Tu= 0.14 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



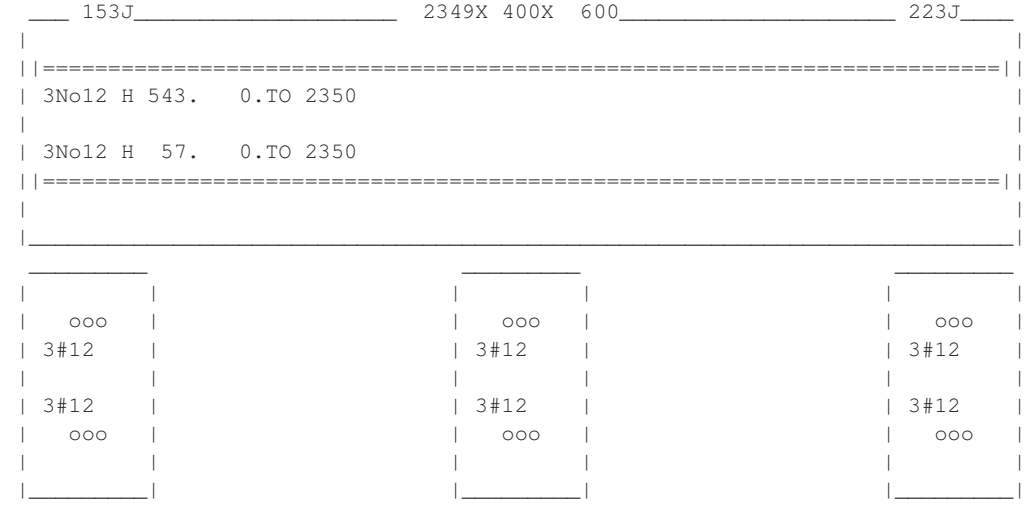
ACI 318-11 BEAM NO. 222 DESIGN RESULTS

LEN - 2350. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	2350.	YES	YES
2	543.	3 - 12MM	0.	2350.	YES	YES

B E A M N O. 222 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 26.92 KNS Vc= 189.39 KNS Vs= 0.00 KNS
Tu= 0.13 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 3.47 KNS Vc= 180.58 KNS Vs= 0.00 KNS
Tu= 0.13 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



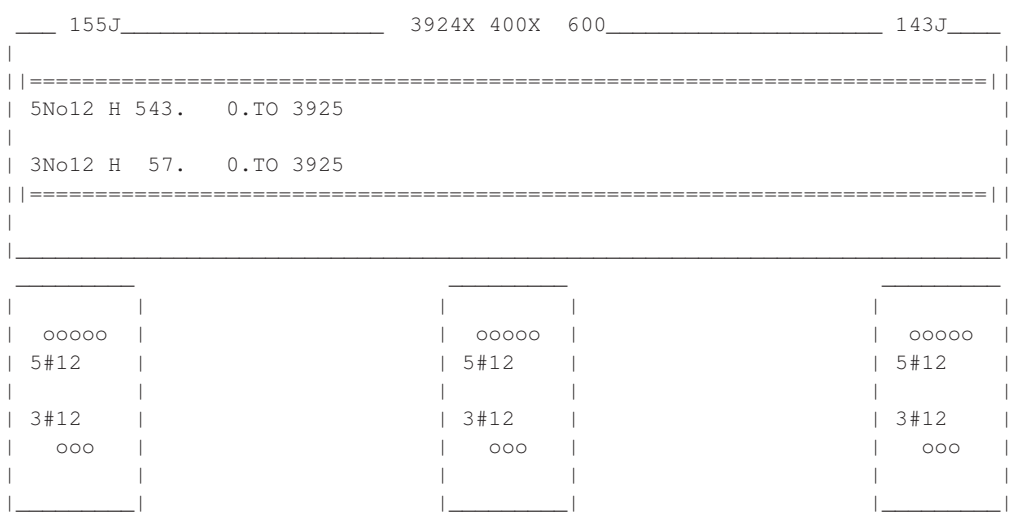
ACI 318-11 BEAM NO. 223 DESIGN RESULTS

LEN - 3925. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	3925.	YES	YES
2	543.	5 - 12MM	0.	3925.	YES	YES

B E A M N O. 223 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 21.22 KNS Vc= 227.08 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 31.14 KNS Vc= 185.36 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



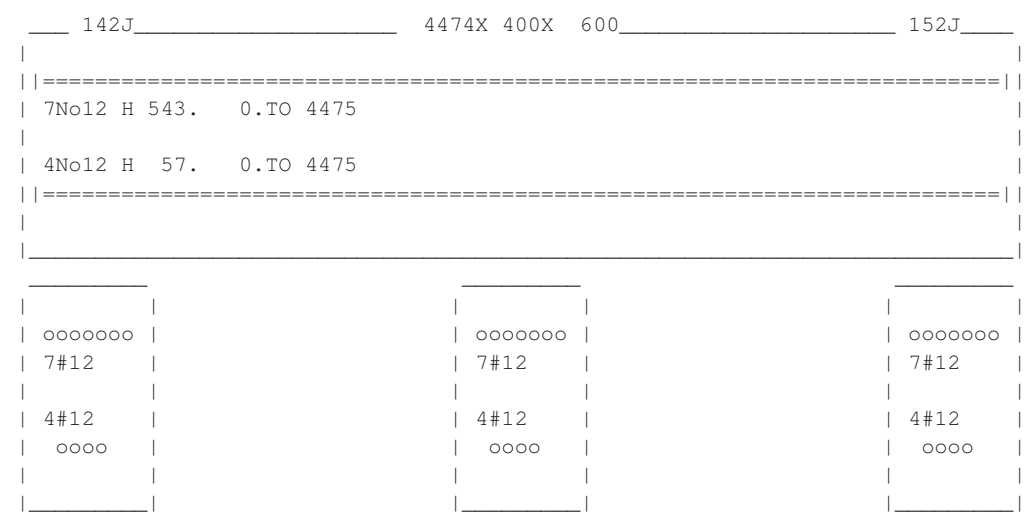
ACI 318-11 BEAM NO. 230 DESIGN RESULTS

LEN - 4475. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	4475.	YES	YES
2	543.	7 - 12MM	0.	4475.	YES	YES

B E A M N O. 230 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 31.55 KNS Vc= 239.14 KNS Vs= 0.00 KNS
Tu= 0.91 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 30.91 KNS Vc= 295.61 KNS Vs= 0.00 KNS
Tu= 0.91 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



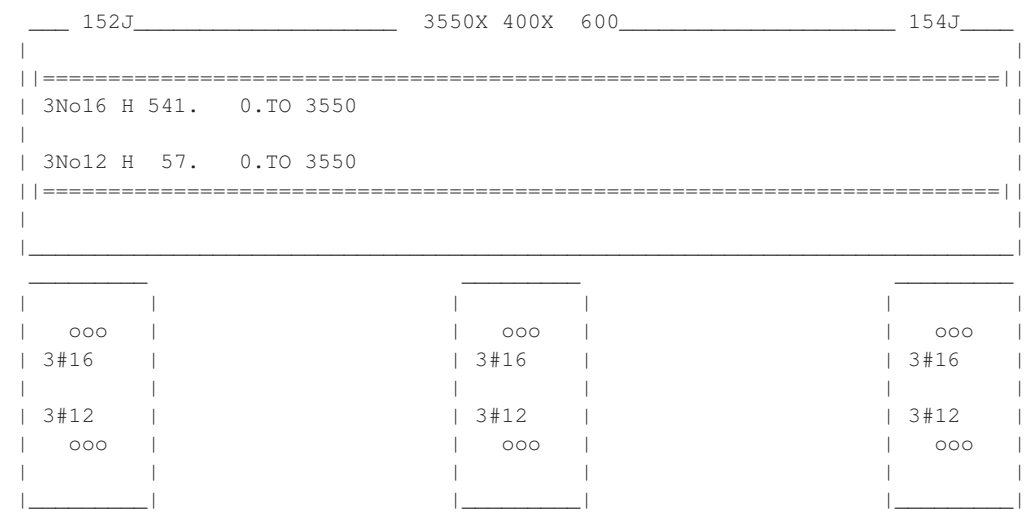
ACI 318-11 BEAM NO. 231 DESIGN RESULTS

LEN - 3550. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	3550.	YES	YES
2	541.	3 - 16MM	0.	3550.	YES	YES

B E A M N O. 231 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 9.23 KNS Vc= 184.99 KNS Vs= 0.00 KNS
Tu= 0.47 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 36.25 KNS Vc= 180.73 KNS Vs= 0.00 KNS
Tu= 0.47 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 232 DESIGN RESULTS

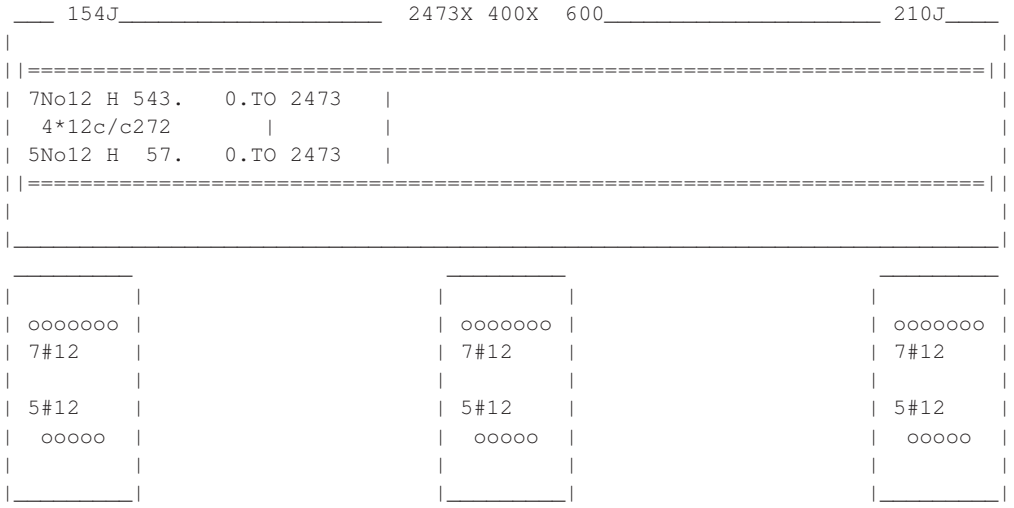
LEN - 2473. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	5 - 12MM	0.	2473.	YES	YES
2	543.	7 - 12MM	0.	2473.	YES	YES

B E A M N O. 232 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 84.61 KNS Vc= 190.24 KNS Vs= 0.00 KNS
Tu= 0.18 KN-MET Tc= 9.7 KN-MET Ts= 0.0 KN-MET LOAD 201
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 700. MM

AT END SUPPORT - Vu= 35.12 KNS Vc= 185.27 KNS Vs= 0.00 KNS
Tu= 0.11 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 239 DESIGN RESULTS

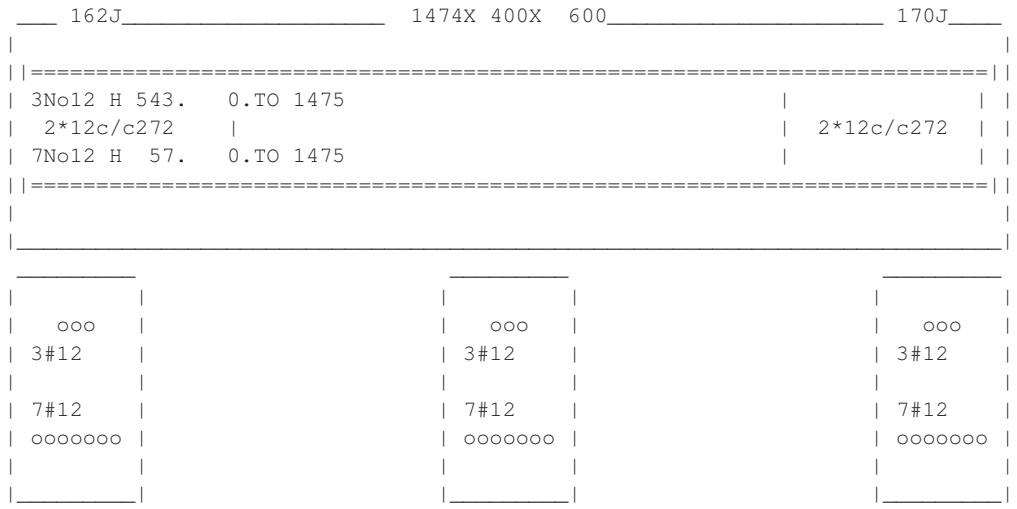
LEN - 1475. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	7 - 12MM	0.	1475.	YES	YES
2	543.	3 - 12MM	0.	1475.	YES	YES

B E A M N O. 239 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 81.42 KNS Vc= 185.73 KNS Vs= 0.00 KNS
Tu= 1.58 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 217
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 201. MM

AT END SUPPORT - Vu= 77.40 KNS Vc= 189.24 KNS Vs= 0.00 KNS
Tu= 1.34 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 201. MM



ACI 318-11 BEAM NO. 246 DESIGN RESULTS

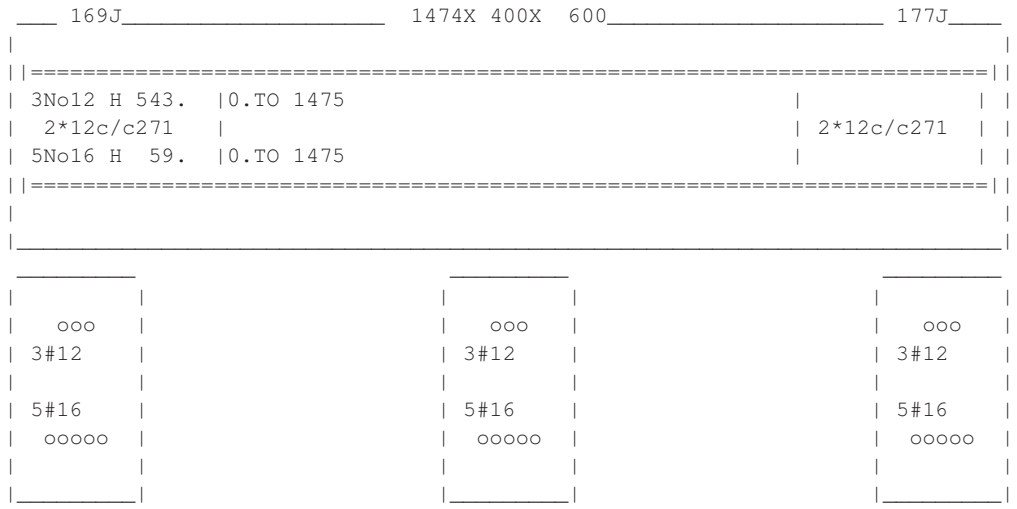
LEN - 1475. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	5 - 16MM	0.	1475.	YES	YES
2	543.	3 - 12MM	0.	1475.	YES	YES

B E A M N O. 246 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 90.12 KNS Vc= 188.34 KNS Vs= 0.00 KNS
Tu= 3.28 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 117
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 201. MM

AT END SUPPORT - Vu= 71.82 KNS Vc= 188.87 KNS Vs= 0.00 KNS
Tu= 2.73 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 271. MM C/C FOR 201. MM



ACI 318-11 BEAM NO. 254 DESIGN RESULTS

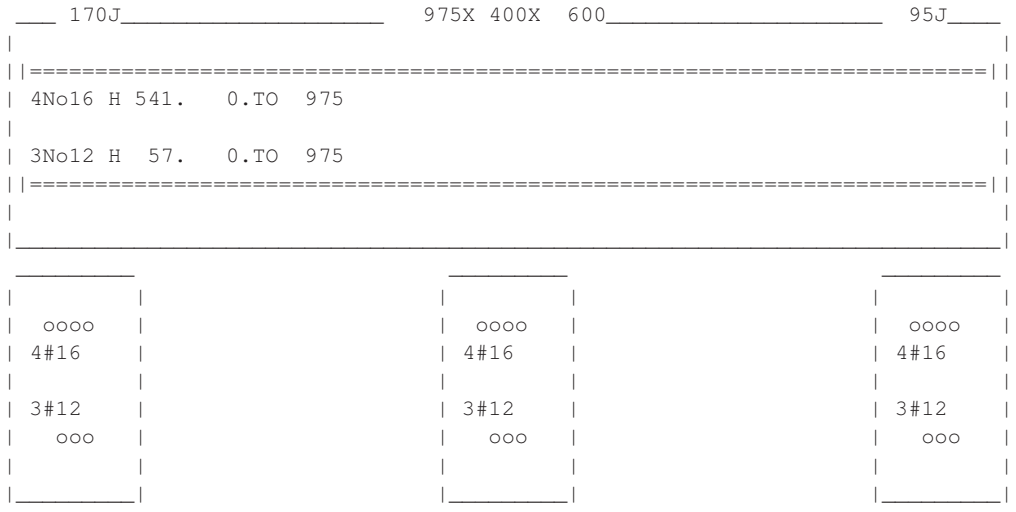
LEN - 975. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	975.	YES	YES
2	541.	4 - 16MM	0.	975.	YES	YES

B E A M N O. 254 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 254 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 254 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 261 DESIGN RESULTS

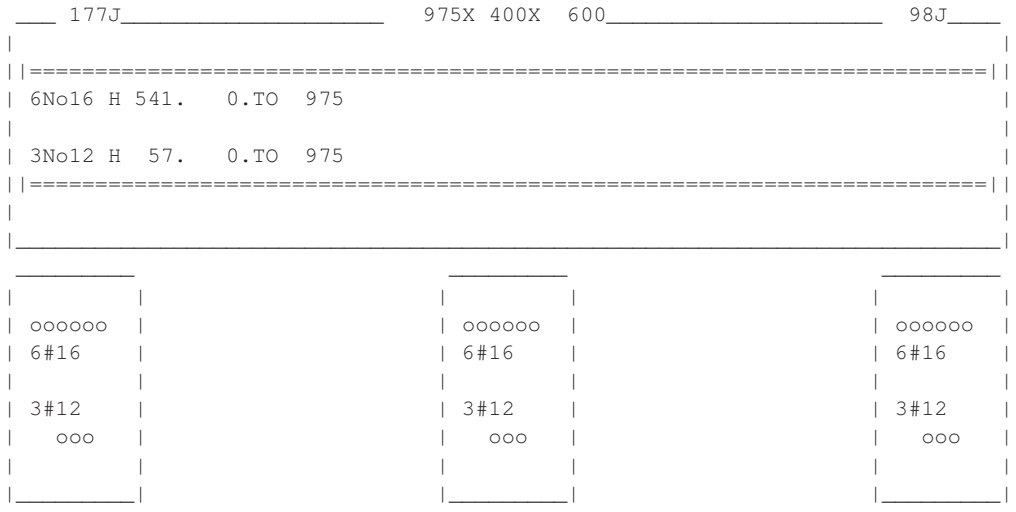
LEN - 975. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	975.	YES	YES
2	541.	6 - 16MM	0.	975.	YES	YES

B E A M N O. 261 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 261 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 261 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



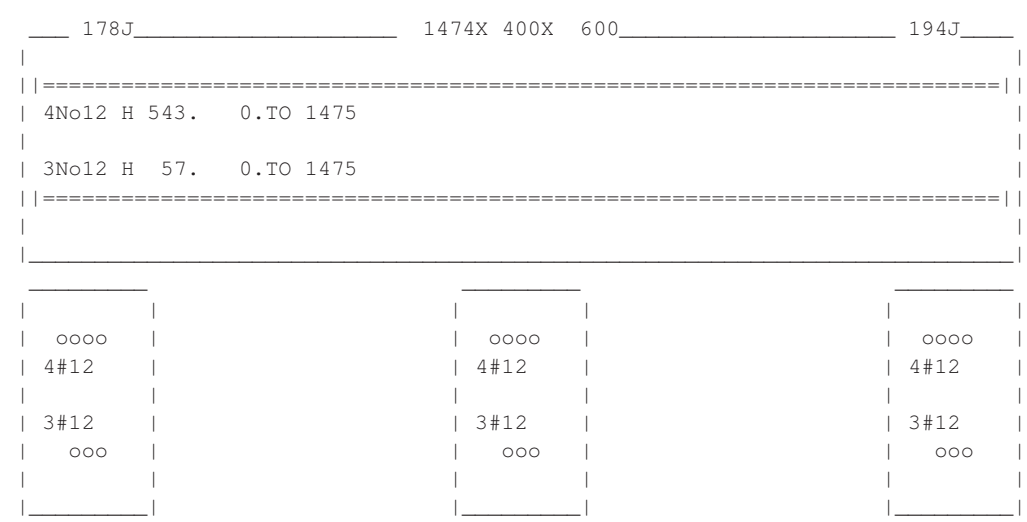
ACI 318-11 BEAM NO. 269 DESIGN RESULTS

LEN - 1475. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	1475.	YES	YES
2	543.	4 - 12MM	0.	1475.	YES	YES

B E A M N O. 269 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 40.38 KNS Vc= 189.64 KNS Vs= 0.00 KNS
Tu= 0.23 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 26.92 KNS Vc= 189.64 KNS Vs= 0.00 KNS
Tu= 0.23 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



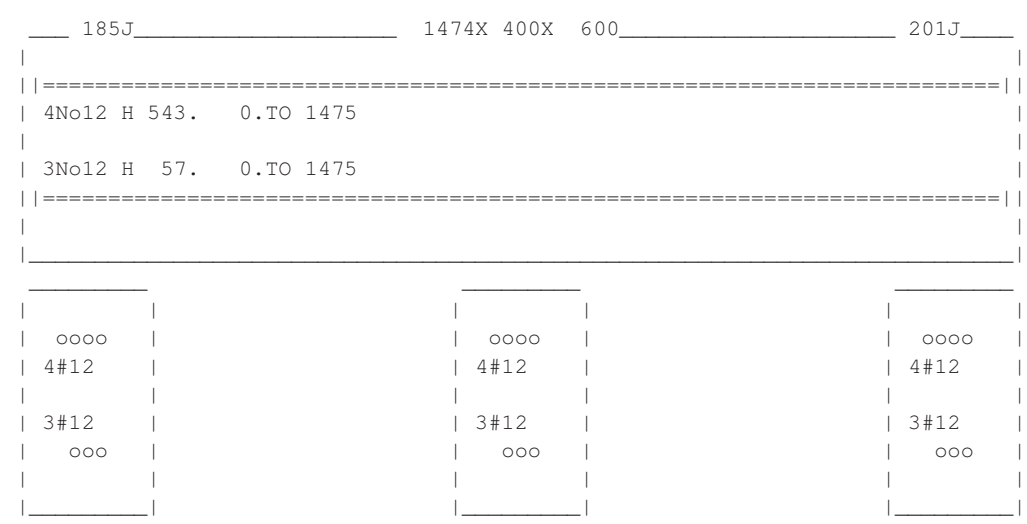
ACI 318-11 BEAM NO. 276 DESIGN RESULTS

LEN - 1475. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	1475.	YES	YES
2	543.	4 - 12MM	0.	1475.	YES	YES

B E A M N O. 276 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 24.31 KNS Vc= 189.24 KNS Vs= 0.00 KNS
Tu= 0.56 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.85 KNS Vc= 189.24 KNS Vs= 0.00 KNS
Tu= 0.56 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



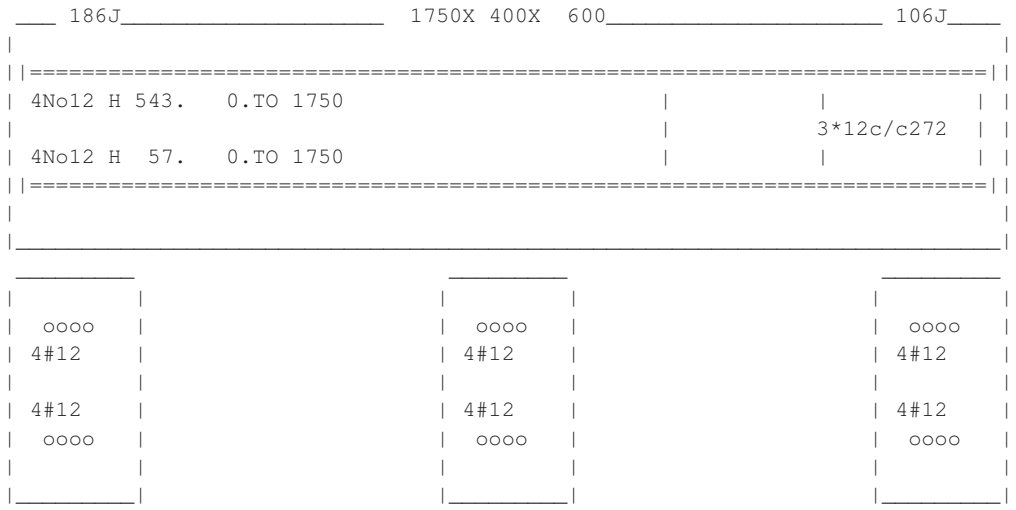
ACI 318-11 BEAM NO. 287 DESIGN RESULTS

LEN - 1750. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	4 - 12MM	0.	1750.	YES	YES
2	543.	4 - 12MM	0.	1750.	YES	YES

B E A M N O. 287 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 28.12 KNS Vc= 189.49 KNS Vs= 0.00 KNS
Tu= 0.76 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 71.20 KNS Vc= 189.39 KNS Vs= 0.00 KNS
Tu= 1.08 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 202
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 339. MM



ACI 318-11 BEAM NO. 294 DESIGN RESULTS

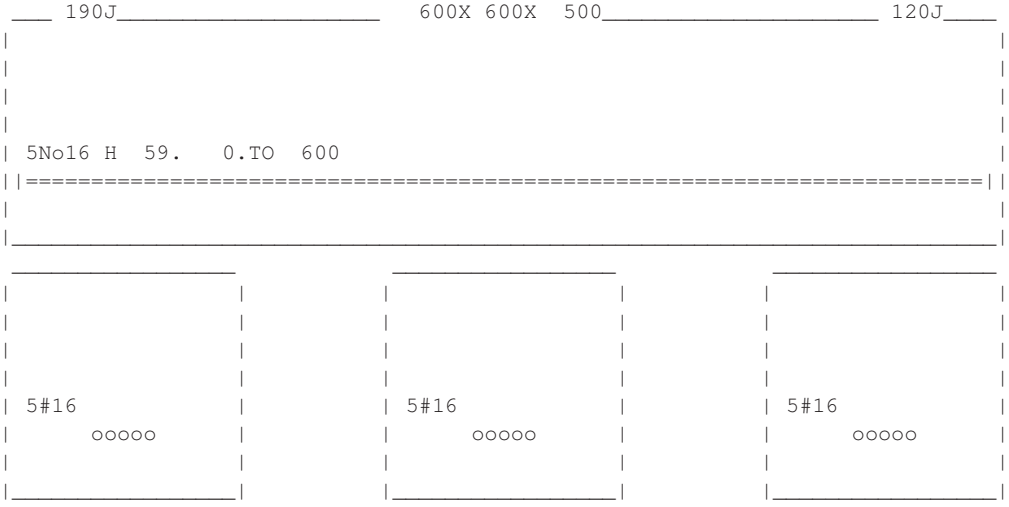
LEN - 600. MM FY - 414. FC - 28. MPA, SIZE - 600. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	5 - 16MM	0.	600.	YES	YES

B E A M N O. 294 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 294 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 294 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



ACI 318-11 BEAM NO. 295 DESIGN RESULTS

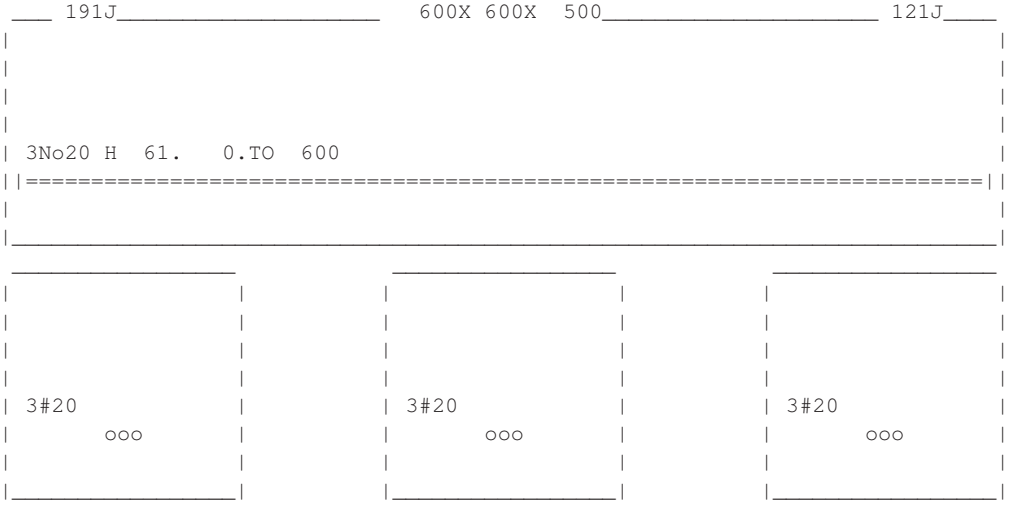
LEN - 600. MM FY - 414. FC - 28. MPA, SIZE - 600. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	61.	3 - 20MM	0.	600.	YES	YES

B E A M N O. 295 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 295 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 295 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



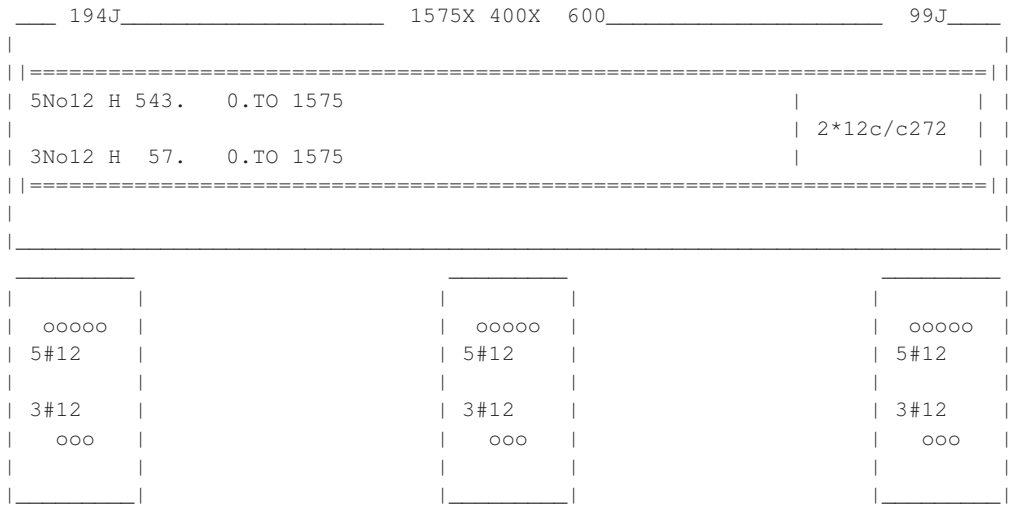
ACI 318-11 BEAM NO. 298 DESIGN RESULTS

LEN - 1575. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	1575.	YES	YES
2	543.	5 - 12MM	0.	1575.	YES	YES

B E A M N O. 298 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 25.94 KNS Vc= 189.64 KNS Vs= 0.00 KNS
Tu= 0.23 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 75.22 KNS Vc= 189.50 KNS Vs= 0.00 KNS
Tu= 0.48 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 203
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 251. MM



ACI 318-11 BEAM NO. 305 DESIGN RESULTS

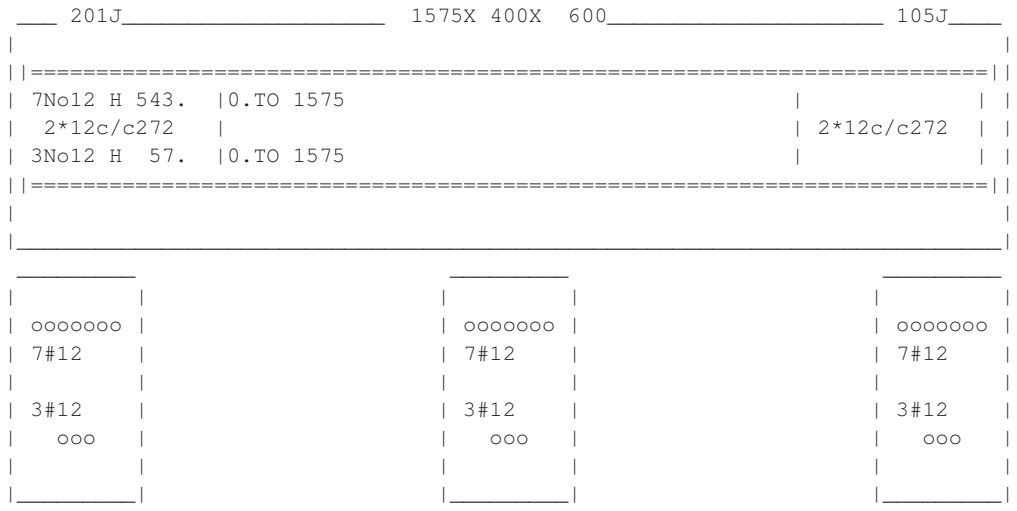
LEN - 1575. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	1575.	YES	YES
2	543.	7 - 12MM	0.	1575.	YES	YES

B E A M N O. 305 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 79.93 KNS Vc= 189.02 KNS Vs= 0.00 KNS
Tu= 1.05 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 203
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 251. MM

AT END SUPPORT - Vu= 78.98 KNS Vc= 189.17 KNS Vs= 0.00 KNS
Tu= 0.80 KN-MET Tc= 9.4 KN-MET Ts= 0.0 KN-MET LOAD 103
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 251. MM



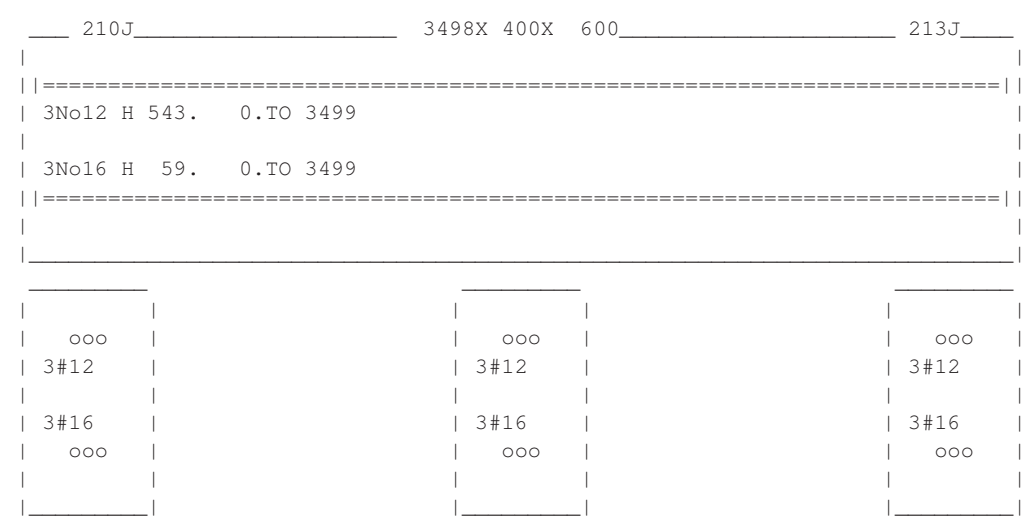
ACI 318-11 BEAM NO. 324 DESIGN RESULTS

LEN - 3499. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	3 - 16MM	0.	3499.	YES	YES
2	543.	3 - 12MM	0.	3499.	YES	YES

B E A M N O. 324 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.83 KNS Vc= 180.40 KNS Vs= 0.00 KNS
Tu= 0.12 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 36.70 KNS Vc= 190.36 KNS Vs= 0.00 KNS
Tu= 0.12 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 329 DESIGN RESULTS

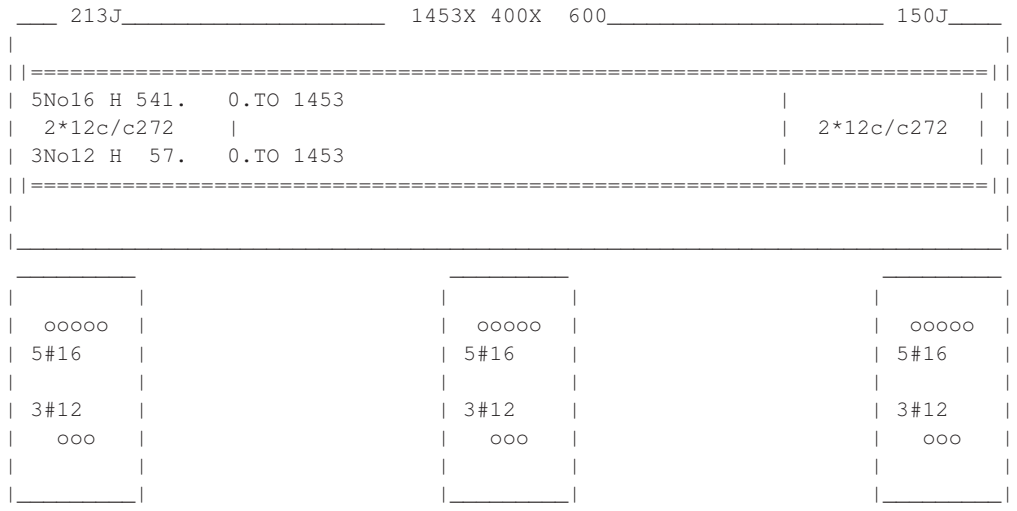
LEN - 1453. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	1453.	YES	YES
2	541.	5 - 16MM	0.	1453.	YES	YES

B E A M N O. 329 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 69.78 KNS Vc= 185.71 KNS Vs= 0.00 KNS
Tu= 0.18 KN-MET Tc= 9.7 KN-MET Ts= 0.0 KN-MET LOAD 103
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 190. MM

AT END SUPPORT - Vu= 70.83 KNS Vc= 183.88 KNS Vs= 0.00 KNS
Tu= 0.12 KN-MET Tc= 9.7 KN-MET Ts= 0.0 KN-MET LOAD 1
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 190. MM



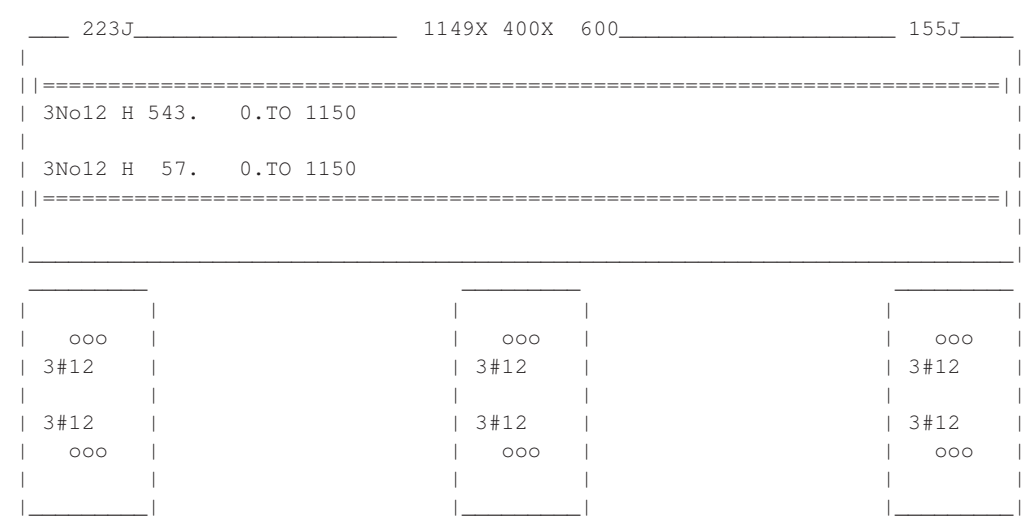
ACI 318-11 BEAM NO. 345 DESIGN RESULTS

LEN - 1150. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	1150.	YES	YES
2	543.	3 - 12MM	0.	1150.	YES	YES

B E A M N O. 345 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 24.72 KNS Vc= 199.88 KNS Vs= 0.00 KNS
Tu= 0.13 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 26.13 KNS Vc= 333.42 KNS Vs= 0.00 KNS
Tu= 0.13 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



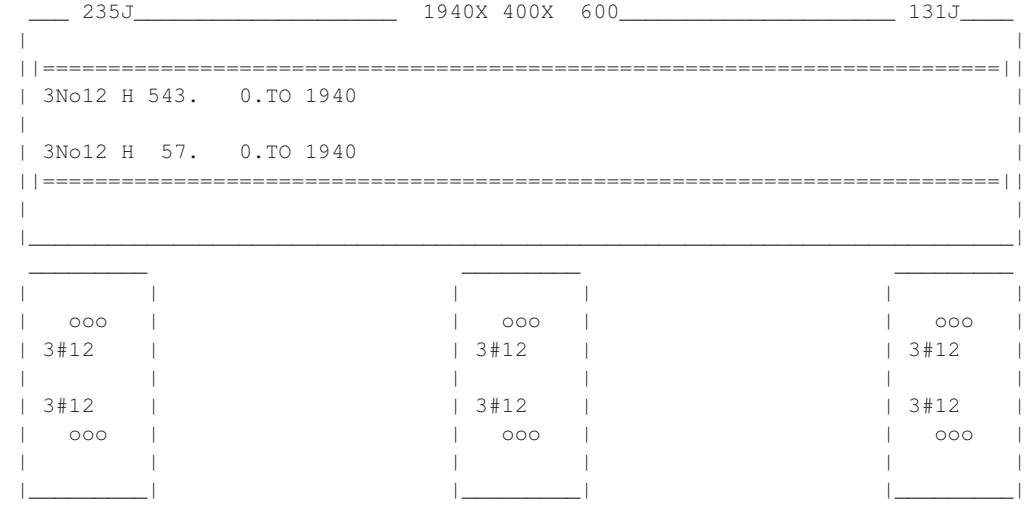
ACI 318-11 BEAM NO. 372 DESIGN RESULTS

LEN - 1940. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	1940.	YES	YES
2	543.	3 - 12MM	0.	1940.	YES	YES

B E A M N O. 372 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.89 KNS Vc= 182.81 KNS Vs= 0.00 KNS
Tu= 0.66 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 28.81 KNS Vc= 192.37 KNS Vs= 0.00 KNS
Tu= 0.66 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



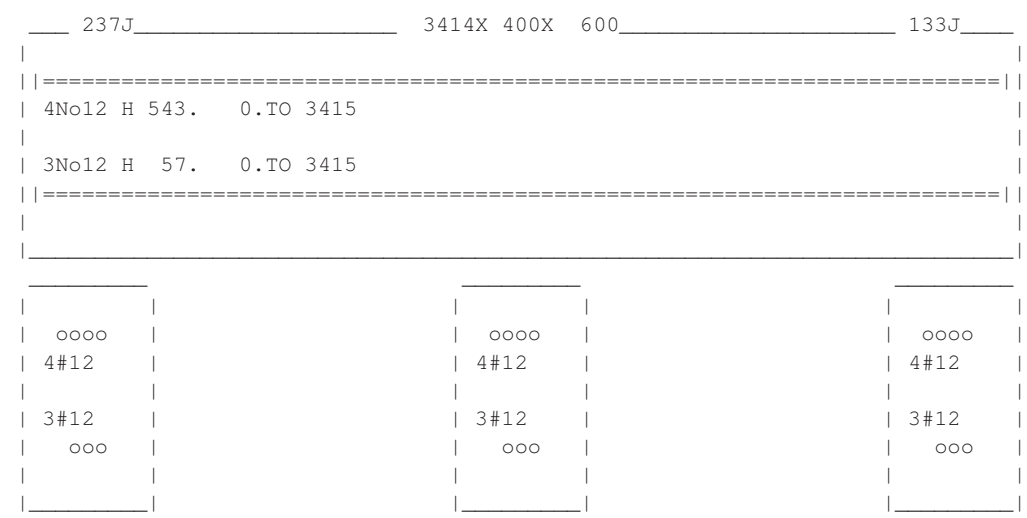
ACI 318-11 BEAM NO. 375 DESIGN RESULTS

LEN - 3415. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	3415.	YES	YES
2	543.	4 - 12MM	0.	3415.	YES	YES

B E A M N O. 375 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 5.97 KNS Vc= 180.88 KNS Vs= 0.00 KNS
Tu= 0.25 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 37.03 KNS Vc= 182.91 KNS Vs= 0.00 KNS
Tu= 0.25 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



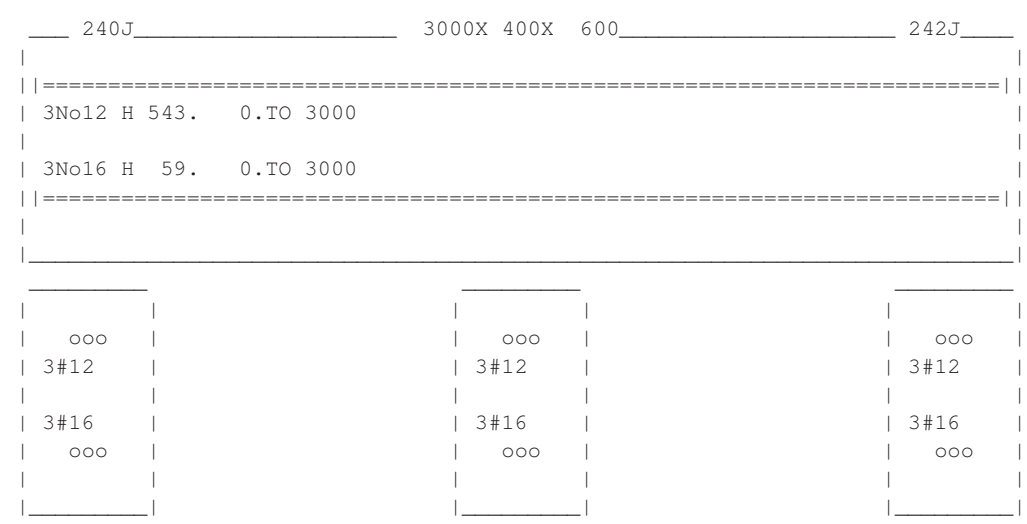
ACI 318-11 BEAM NO. 379 DESIGN RESULTS

LEN - 3000. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	3 - 16MM	0.	3000.	YES	YES
2	543.	3 - 12MM	0.	3000.	YES	YES

B E A M N O. 379 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 6.98 KNS Vc= 180.59 KNS Vs= 0.00 KNS
Tu= 0.06 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 28.40 KNS Vc= 188.60 KNS Vs= 0.00 KNS
Tu= 0.06 KN-MET Tc= 9.5 KN-MET Ts= 0.0 KN-MET LOAD 1
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 381 DESIGN RESULTS

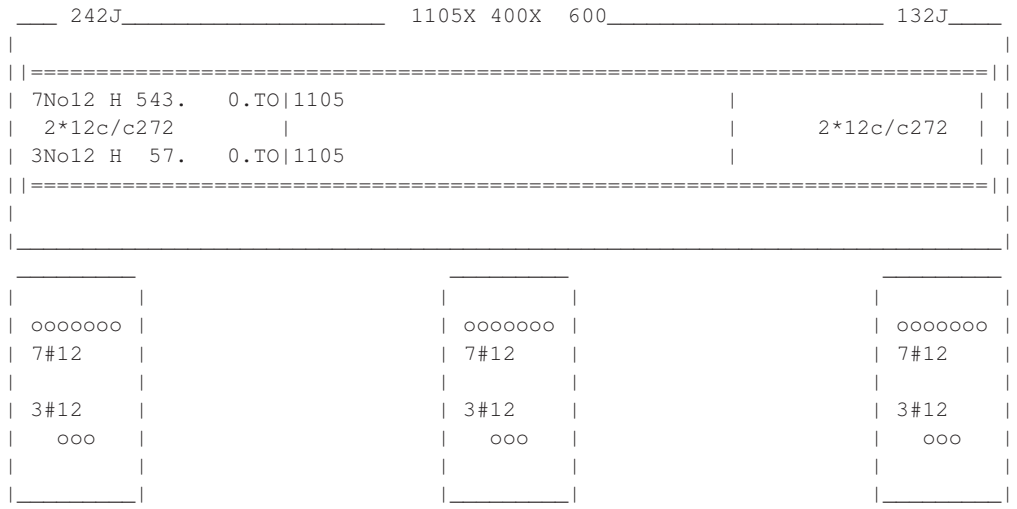
LEN - 1105. MM FY - 414. FC - 28. MPA, SIZE - 400. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	57.	3 - 12MM	0.	1105.	YES	YES
2	543.	7 - 12MM	0.	1105.	YES	YES

B E A M N O. 381 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 71.66 KNS Vc= 184.67 KNS Vs= 0.00 KNS
Tu= 0.83 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 102
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 16. MM

AT END SUPPORT - Vu= 72.25 KNS Vc= 184.32 KNS Vs= 0.00 KNS
Tu= 0.83 KN-MET Tc= 9.6 KN-MET Ts= 0.0 KN-MET LOAD 102
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 272. MM C/C FOR 16. MM



**********END OF BEAM DESIGN**********

489. DESIGN COLUMN 1 TO 17 20 TO 39 70 TO 86 89 TO 108 155 TO 170 185 TO 188 193 -
490. 194 TO 205 296 297 314 317 319 397 398

ACI 318-11 COLUMN NO. 1 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 9169.2 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 32 MM	2.681	9	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 2 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 3 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

32 - 12 MM 1.005 1 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 4 DESIGN RESULTS
==========

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 7430.4 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 20 MM	2.094	9	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 5 DESIGN RESULTS
==========

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
AREA OF STEEL REQUIRED = 1880.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
4 - 25 MM	1.227	10	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 6 DESIGN RESULTS
==========

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 7 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 8 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 9 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 6069.6 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 20 MM	1.745	9	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 10 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 11 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 12 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 13 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 14 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

32 - 12 MM 1.005 1 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 15 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 3625.2 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 10 STA 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 16 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 9295.2 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 32 MM 2.681 9 STA 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 17 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 20 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 21 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 22 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 23 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 24 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 25 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 26 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 27 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 28 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
AREA OF STEEL REQUIRED = 1913.6 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
4 - 25 MM	1.227	10	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 29 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 30 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 31 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 32 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 10706.4 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

24 - 25 MM 3.272 9 STA 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 33 DESIGN RESULTS
==========

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 3650.4 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	10	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 34 DESIGN RESULTS
==========

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 35 DESIGN RESULTS
==========

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 36 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 3625.2 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	215	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 37 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 38 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 39 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 9774.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 25 MM	2.727	9	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 70 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 71 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 72 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 73 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 74 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 75 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 76 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 77 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 78 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 79 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 80 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 81 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 82 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 3625.2 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	115	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 83 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 84 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 85 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 86 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 89 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 90 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 91 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 92 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 93 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 94 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 95 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 96 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 97 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 98 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 99 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 100 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 101 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 102 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 103 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 104 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 105 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 106 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 107 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 108 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 155 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 156 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 157 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 158 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 159 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 160 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 161 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 162 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 163 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 164 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 165 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 166 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 167 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 168 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 169 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 170 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 185 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 186 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 187 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 188 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 193 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 194 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 195 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 196 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 197 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 198 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 199 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 200 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 201 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 202 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 203 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 204 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 205 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 296 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 297 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 314 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 317 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 319 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 16 MM	1.005	1	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 256.00 MM

ACI 318-11 COLUMN NO. 397 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

32 - 12 MM 1.005 1 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

ACI 318-11 COLUMN NO. 398 DESIGN RESULTS

FY - 413.7 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

32 - 12 MM 1.005 1 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

491. END CONCRETE DESIGN
492. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 29,2020 TIME= 16:10:40 ****

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*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01			Outotec / L&T Revision No.:	0	1	2	
Date	30-05-2020	29-06-2020	28.11.2023			Date	30-05-2020	29-06-2020	28.11.2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT			Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENTS ON 23.06.2020	AS BUILT	
Prepared By	SNM	SNM	SNM			Prepared By	SNM	SNM	SNM	
Checked	SJM	SJM	SJM			Checked	SJM	SJM	SJM	
Approved	PRI	PRI	PRI			Approved	PRI	PRI	PRI	

ATTACHMENT 4 : COMMENT RESOLUTION LOG SHEET / COMPLIANCE REPORT

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: MAADEN-TRANSMIT-002660, 24 June 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 24 June 2020	Comment by	EPC Response DATE: 29 June 2020	Status	Remarks
			General Comments (cc)				
MD-216-3690-EG-CV-CAL-1371-C01	3690	1	Project design criteria says SBC304-2018	Bechtel / Ma'aden	Corrected	Closed	